

Sticky Takeya in $\mathbb{R}^4$ as a Contact–Symplectic Incidence Problem

Contact Geometry, Maslov Four-Cycles, and Lossless Edge Flow

Chenxi Cai

Chenxi.Cai@live.com

Abstract

We identify the space of oriented affine lines in $\mathbb{R}^4$ with the symplectic manifold T^*S^3 and marked line segments with Reeb intervals in its contactification $J^1(S^3)$. Euclidean points become Legendrian probes, so the Sticky Takeya problem is an exact contact incidence problem. The infinitesimal Euclidean front is governed by the matrix pencil $B + sA$. Its determinant satisfies

$$|\det(B + iA)|^2 = 1 - \|\omega|_V\|_{\text{op}}^2,$$

and front collapse is equivalent to simultaneous incidence of the tangent plane V with the full Lagrangian pencil $L_s = \{(a, -sa)\}$. This isolates three polarized Maslov degeneracies. At small scales, collision pairs converge to these Maslov incidences; nondegenerate four-cycles produce physical Lagrangian bushes, while persistent rank loss is ruled out by the polarized classification. A mass-preserving edge-flow argument then propagates the bush alternative across the full stopping tree without squaring the incoming mass. The key point is to exhaust every fixed-proportion route on its unused occurrence measure before passing to the next generation. The resulting absolute flows telescope with coefficient one, so the apparent C^L loss and the compensation-zone debt disappear. A deterministic subpower cap schedule gives a quadratic terminal tail. The same argument is linear under arbitrary source restrictions and carries affine fibre and carrier-tree marks as conditional variables; it therefore yields the uniform shaded, weighted finite-scale estimate required by marked applications.

Contents

1	Introduction	2
1.1	Main results	2
1.2	Proof architecture	3
1.3	Where the endpoint is won	3
1.4	Organization	3
2	The contact model of marked lines	3
3	Selector reduction and the regular regime	4
	The complete quantitative argument	5
4	Reeb-front pencils and polarized null geometry	5
4.1	Matrix pencils and front rank	5
4.2	Polarized normal forms	8
4.3	Rectifiable and selector reductions	9
5	Physical collision, recurrence, and entropy	11
5.1	One-scale collision geometry	11
5.2	Weighted graph regularization and radial recurrence	14
5.3	Entropy pruning and common-neighbor structure	16
5.4	Endpoint codes and the transition to a Maslov shell	18
6	Angular blow-up and Maslov four-cycles	20
6.1	Darboux blow-up and four-cycle extraction	20
6.2	Time synchronization and physical bush cells	23
6.3	Rank loss, interpolation, and packet reuse	27
6.4	Front seeds and Reeb-time completion	31

7	The codimension-two residual and polarized routing	35
7.1	Residual criteria and shell return	35
7.2	Anchored rank packets	37
7.3	Polarization and affine-offset control	42
7.4	Weighted slices and defect routing	45
7.5	Normal shear and direct packet bounds	52
7.6	Azimuth sectors and rooted contact stars	57
7.7	Two-generation flags and relative residual return	68
8	Bush families, compactness, and Frostman stopping	76
8.1	Flag amplification and reduction to bushes	76
8.2	Maximal bush extraction and vector Frostman escape	80
8.3	Coherent hairbrushes and re-collision	83
8.4	Fine Maslov returns and stopping leaves	90
8.5	Coherence across logarithmic blocks	96
8.6	Resolution shells and the density cocycle	103
9	Cross-generation edge-flow closure	107
9.1	Absolute occurrences and the edge-Carleson reduction	107
9.2	Angular and separated-time strict progress	110
9.3	Same-window amplification	113
9.4	Common-target Markovization	115
9.5	Flagged common-target routing	117
9.6	Cross-generation conservation and deterministic saturation	122
9.7	The marked source-hereditary interface	125
10	Discussion	135
A	Measurable disintegration and passive marks	135
B	Uniform finite-scale parameter hierarchy	136

1 Introduction

Let $\mathcal{L}^+(\mathbb{R}^4)$ denote the space of oriented affine lines. A full-direction family contains a line in every direction. Following the continuum formulation of stickiness [12, 13, 17], the extremal condition is that this family has packing dimension three, the smallest possible dimension compatible with surjectivity onto S^3 .

The purpose of this paper is to exploit the geometry of line space rather than replace it by an auxiliary finite incidence structure. The key observation is that

$$\mathcal{L}^+(\mathbb{R}^4) \cong T^*S^3$$

canonically as a symplectic manifold. Adding the marked position along a line produces the contact manifold $J^1(S^3)$. Translation along a line is its Reeb flow, and incidence with a Euclidean point is intersection with a distinguished Legendrian sphere; see [7, 9, 1] for the symplectic, contact, and wave-front background.

This reformulation separates the smooth and rectifiable regimes from the genuinely fractal one. It also identifies the exact degenerate object: not a small scalar symplectic defect, but a three-plane incident with every member of a one-parameter Lagrangian pencil. The same pencil reappears after blowing up small-angle collisions. Thus the infinitesimal and multiscale parts of the proof are governed by one symplectic object.

1.1 Main results

Let $\pi : T^*S^3 \rightarrow S^3$ be the bundle projection and let $q : J^1(S^3) \rightarrow T^*S^3$ forget the jet coordinate.

Definition 1.1 (Sticky Takeya datum). *A compact marked family $\Gamma \subset J^1(S^3)$ is a Sticky Takeya datum if*

$$\pi(q(\Gamma)) = S^3, \quad \dim_{\mathbb{P}} q(\Gamma) = 3,$$

and every line in $q(\Gamma)$ carries at least one marked unit interval represented in Γ .

The Euclidean union K_{Γ} is defined precisely in Section 2.

Theorem 1.2 (Sticky Takeya theorem in four dimensions). *For every compact Sticky Takeya datum Γ in $\mathbb{R}^4$,*

$\dim_{\mathbb{H}} K_{\Gamma} = 4.$

1.2 Proof architecture

The proof has four stages.

- (i) The contact model converts marked segments and Euclidean point incidence into Reeb intervals and Legendrian intersections.
- (ii) The Reeb-front polynomial $P_V(s) = \det(B + sA)$ detects infinitesimal spreading. If it vanishes identically, V is trapped by the full Maslov pencil and belongs to one of three polarized strata.
- (iii) In a slope chart, near-collisions have residual $r = \beta + s_*\alpha \in \alpha^\perp$. This two-dimensional normal residual is the exact quantity needed for a Frostman estimate. Angular collapse rescales back to the Maslov pencil; a nondegenerate four-cycle produces a physical point-probe bush.
- (iv) Bush returns are organized by an incoming edge-flow measure. Exhausting each hereditary Markov route makes the absolute flow telescope across generations. A deterministic cap schedule then leaves a quadratic terminal tail with no generation factor.

The symplectic argument is therefore not an interpretation added after the incidence proof. It classifies angular collapse, supplies the four-time obstruction, and keeps each finite physical bush iteration from losing a power of mass. The remaining summation is closed by the absolute Markov-flow identity in the final quantitative section.

1.3 Where the endpoint is won

Three points deserve to be visible before the technical argument begins. First, the relevant degeneracy is not the vanishing of one chosen symplectic scalar. It is the intrinsic condition that the tangent plane meet every member of the Lagrangian pencil L_s . This is why the matrix pencil and the contact blow-up lead to the same alternatives.

Second, the graph-theoretic four-cycle is used only as a sampling device. It is credited only after the contact equations identify a common Euclidean point and a common Reeb window. The resulting object is a physical Lagrangian bush, and its old collision partner and old Maslov flag remain attached to the source occurrence.

Third, iteration is performed on the ordered edge measure, not on a repeatedly normalized vertex measure. Each local alternative is exhausted on the unused part. Children are therefore fractional restrictions of one absolute source, and their masses telescope with coefficient one. This is the mechanism that closes both the compensation regime and the marked finite-scale interface.

1.4 Organization

The first two sections establish the contact model and the selector reduction. The six quantitative sections then follow the proof itself: front-null pencils, physical collision recurrence, angular Maslov extraction, polarized residual routing, bush stopping, and cross-generation edge-flow closure. The appendices contain only measurable disintegration and the uniform finite-scale parameter hierarchy.

2 The contact model of marked lines

Every oriented affine line has a unique representation

$$\ell_{\theta,y} = \{y + t\theta : t \in \mathbb{R}\}, \quad \theta \in S^3, \quad y \in \theta^\perp.$$

Using the Euclidean metric to identify $\theta^\perp$ with $T_\theta^*S^3$, write $p = y^\flat$. This gives a diffeomorphism

$$\mathcal{L}^+(\mathbb{R}^4) \longrightarrow T^*S^3, \quad \ell_{\theta,y} \longmapsto (\theta, p).$$

Let λ be the tautological one-form and $\omega = d\lambda$.

Adjoin the longitudinal variable z and set

$$J^1(S^3) = T^*S^3 \times \mathbb{R}_z, \quad \alpha = dz - \lambda.$$

The Reeb vector field is $R_\alpha = \partial_z$. The Euclidean evaluation map is

$$\mathcal{E}(\theta, p, z) = p^\sharp + z\theta. \tag{1}$$

Proposition 2.1 (Euclidean incidence form). *Under Eq. (1),*

$$\theta \cdot dx = dz - p \cdot d\theta = \alpha.$$

In particular, Reeb translation is Euclidean translation along the oriented line.

Proof. Differentiate $x = p^\sharp + z\theta$. Since $p^\sharp \cdot \theta = 0$, one has $\theta \cdot dp^\sharp = -p^\sharp \cdot d\theta = -p \cdot d\theta$. The identities $\theta \cdot \theta = 1$ and $\theta \cdot d\theta = 0$ give the claim. $\square$

For $\Gamma \subset J^1(S^3)$ define its unit Reeb saturation and Euclidean front by

$$\text{Sat}_{1/2}^R(\Gamma) = \{(\theta, p, z + t) : (\theta, p, z) \in \Gamma, |t| \leq \tfrac{1}{2}\}, \quad K_\Gamma = \mathcal{E}(\text{Sat}_{1/2}^R(\Gamma)).$$

For $x \in \mathbb{R}^4$, let

$$h_x(\theta) = x \cdot \theta, \quad \Lambda_x = j^1 h_x = \{(\theta, dh_x(\theta), h_x(\theta)) : \theta \in S^3\}.$$

Since $(j^1 h_x)^* \alpha = 0$, the sphere Λ_x is Legendrian.

Theorem 2.2 (Exact Legendrian incidence). *For every $x \in \mathbb{R}^4$,*

$$x \in K_\Gamma \iff \Lambda_x \cap \text{Sat}_{1/2}^R(\Gamma) \neq \emptyset.$$

Proof. If $x = p^\sharp + (z + t)\theta$, then $x \cdot \theta = z + t$ and the projection of x to $\theta^\perp$ is $p^\sharp$. Hence $p = dh_x(\theta)$ and the corresponding point of the Reeb interval lies on Λ_x . The converse is the same calculation in reverse. $\square$

Forgetting z turns Λ_x into the exact Lagrangian graph $L_x = \text{Graph}(dh_x) \subset T^*S^3$. The contact condition remembers the additional filter $|h_x(\theta) - z| \leq 1/2$. This primitive window is essential: unmarked Lagrangian incidence alone does not distinguish the chosen unit segment on a line.

3 Selector reduction and the regular regime

Compactness and measurable selection allow us to choose one marked line in every direction; we use the standard Borel uniformization framework of [14].

Proposition 3.1 (Borel selector reduction). *Every compact full-direction marked family contains a Borel subset meeting each direction fibre in exactly one point. If the original unmarked family has packing dimension three, so does the selected line graph. Its Euclidean front is contained in the original front.*

Proof. The compact fibre relation over the Polish base S^3 has a Borel selector. Its graph projects onto S^3 , so its packing dimension is at least three; containment in the original line family gives the reverse inequality. Removing segments can only shrink the Euclidean union. $\square$

Thus it is enough to study a Borel marked section $s(\theta) = (\theta, p(\theta), z(\theta))$ whose line-space graph has packing dimension three. If this section were C^1 , its Reeb sweep would be

$$F(\theta, t) = p^\sharp(\theta) + (z(\theta) + t)\theta.$$

Proposition 3.2 (Smooth selectors spread). *For a C^1 selector, the four-dimensional Jacobian of F is*

$$J_F(\theta, t) = \det(\nabla p^\sharp(\theta) + (z(\theta) + t)I).$$

For each fixed θ this is a monic cubic in t . Hence the front contains a nonempty open set.

Proof. For $v \in T_\theta S^3$, the tangential component of $dF(v)$ is $\nabla_v p^\sharp + (z + t)v$, while $\partial_t F = \theta$ is normal to $T_\theta S^3$. The determinant is therefore the displayed 3×3 determinant. It cannot vanish for all t , and the inverse function theorem applies at a rank-four point. $\square$

The difficult case is consequently neither smooth nor carried by a positive-mass regular rectifiable sheet. The next section identifies the exact tangent obstruction.

The complete quantitative argument

The contact model and selector reduction are now fixed. What follows is the complete quantitative proof, ordered by its actual geometric flow. The front-null pencil is treated first; physical collisions, angular Maslov extraction, polarized residual routing, bush stopping, and cross-generation conservation follow without a second abbreviated proof.

4 Reeb-front pencils and polarized null geometry

4.1 Matrix pencils and front rank

The singular-pencil terminology and canonical forms used below follow [10, 16, 4].

Theorem 4.1 (Exact Reeb-front rank criterion). *At Reeb time t , put*

$$s = z + t.$$

Then the Reeb-saturated four-plane

$$W \oplus \mathbb{R}R_\alpha$$

maps with Euclidean rank four exactly when

$$\boxed{P_V(z + t) \neq 0.} \tag{2}$$

Consequently a front-regular tangent plane has rank-four Reeb sweep for every t except at at most three values.

Proof. Modulo the Reeb image direction θ , the Euclidean differential sends

$$(a, b) \mapsto b + (z + t)a \in \theta^\perp.$$

Relative to the chosen basis of V , the tangential 3×3 matrix is therefore

$$B + (z + t)A.$$

The Reeb derivative itself maps to θ , which is transverse to $\theta^\perp$. Hence the full four-dimensional differential has rank four exactly when this tangential matrix is invertible. Since P_V has degree at most three, a nonzero P_V has at most three real roots. $\square$

Theorem 4.2 (Complex-evaluation identity). *For every three-plane V ,*

$$\boxed{|P_V(i)|^2 = 1 - \sigma_\omega(V)^2.} \tag{3}$$

Proof. Because the chosen frame of V is orthonormal,

$$A^T A + B^T B = I.$$

Therefore

$$\begin{aligned} (B + iA)^*(B + iA) &= B^T B + A^T A + i(B^T A - A^T B) \\ &= I - iC_V. \end{aligned}$$

Taking determinants gives

$$|P_V(i)|^2 = \det(I - iC_V).$$

A real 3×3 skew-symmetric matrix has eigenvalues

$$0, \quad i\sigma, \quad -i\sigma,$$

where

$$\sigma = \|C_V\|_{\text{op}}.$$

Hence

$$\det(I - iC_V) = 1 - \sigma^2.$$

$\square$

Theorem 4.3 (Matrix-pencil/Maslov incidence equivalence). *For every three-plane V ,*

$$\boxed{P_V(s) = 0 \iff V \cap L_s \neq \{0\}.} \quad (4)$$

Consequently,

$$\boxed{P_V \equiv 0 \iff V \cap L_s \neq \{0\} \text{ for every } s \in \mathbb{R}.} \quad (5)$$

Proof. A nonzero vector of V has coordinates

$$(Ac, Bc)$$

for some $c \neq 0$. It belongs to L_s exactly when

$$Bc = -sAc,$$

equivalently

$$(B + sA)c = 0.$$

Thus $V \cap L_s \neq \{0\}$ exactly when $\det(B + sA) = 0$. The second statement follows because a polynomial of degree at most three that vanishes for every real s is identically zero. $\square$

Theorem 4.4 (Rectifiable Reeb-front polynomial theorem). *Let*

$$R \subset J^1(S^3)$$

be countably 3-rectifiable with

$$0 < \mathcal{H}^3(R) < \infty.$$

Assume the line projection

$$q : R \rightarrow T^*S^3$$

has approximate rank three almost everywhere on a measurable subset $R_0 \subset R$ of positive $\mathcal{H}^3$ -measure. Let V_q be the projected approximate tangent three-plane.

If

$$P_{V_q} \not\equiv 0$$

on a positive-measure subset of R_0 , then

$$\boxed{\mathcal{L}^4\left(\mathcal{E}(\text{Sat}_{1/2}^R(R))\right) > 0.} \quad (6)$$

Proof. Cover R up to a null set by countably many Lipschitz parametrizations. On points where the approximate line projection has rank three, use projected tangent coordinates. By Theorem 4.1, for each front-regular tangent the Reeb-sweep differential has rank four for almost every Reeb time. Fubini therefore gives a positive-measure subset of parameter space on which the four-Jacobian is positive. The area formula then implies that the Euclidean image has positive four-dimensional Lebesgue measure. $\square$

Lemma 4.5 (No constant right kernel). *One has*

$$\boxed{\ker A \cap \ker B = \{0\}.} \quad (7)$$

Equivalently, the pencil $B + sA$ has no nonzero constant polynomial right-kernel vector.

Proof. If $c \in \ker A \cap \ker B$, then

$$\mathbb{V}c = 0.$$

Since the columns of $\mathbb{V}$ are a basis of V , the stacked matrix has full column rank and therefore $c = 0$. $\square$

Theorem 4.6 (Three-type Kronecker classification). *Let $V \subset \mathbb{R}_a^3 \oplus \mathbb{R}_b^3$ be a front-null three-plane. Let*

$$\varepsilon$$

be the unique right minimal index,

$$\eta$$

the unique left minimal index, and

$$r_0$$

the size of the regular Kronecker part. Then

$$\boxed{\varepsilon + \eta + r_0 = 2, \quad \varepsilon \geq 1,} \quad (8)$$

and consequently exactly one of the following occurs:

$$\boxed{(\varepsilon, \eta, r_0) \in \{(1, 0, 1), (1, 1, 0), (2, 0, 0)\}.} \quad (9)$$

Proof. Because the normal rank is two, the right and left nullities over $\mathbb{R}(s)$ are both one. Hence there is exactly one right singular block and exactly one left singular block. The total row and column counts are

$$\varepsilon + (\eta + 1) + r_0 = 3,$$

and

$$(\varepsilon + 1) + \eta + r_0 = 3,$$

which both reduce to

$$\varepsilon + \eta + r_0 = 2.$$

A right minimal index $\varepsilon = 0$ would give a nonzero constant right-kernel vector, excluded by Theorem 4.5. Thus $\varepsilon \geq 1$, and the three displayed nonnegative integer triples are the only possibilities. $\square$

Corollary 4.7 (Exact front-null condition in Kähler coordinates). *A maximally symplectic three-plane is front-null exactly when*

$$\boxed{(a \times b) \cdot c = (a \times b) \cdot d = 0.} \quad (10)$$

Proposition 4.8 (Forced common output plane). *If V is front-null and $\rho(V) > 0$, then*

$$\boxed{c, d \in W_V.} \quad (11)$$

Consequently

$$\boxed{\text{im}(B + sA) \subseteq W_V \quad \forall s \in \mathbb{R}.} \quad (12)$$

Proof. Since $a \times b \neq 0$, the vector $a \times b$ spans $W_V^\perp$. The front-null equations (10) say that both c and d are orthogonal to $a \times b$, hence belong to W_V . Every column of $B + sA$ is then in W_V . $\square$

Theorem 4.9 (Transverse branch is exactly K_{200}). *If V is front-null and $\rho(V) > 0$, then*

$$\boxed{V \in K_{200}.} \quad (13)$$

Proof. By Theorem 4.8, the left minimal index is zero. It remains to exclude right minimal index one.

Rotate the Kähler basis inside $\mathcal{C}_V$ so that $a \perp b$, and apply a diagonal orthogonal transformation so that

$$a = \alpha e_1, \quad b = \beta e_2, \quad \alpha, \beta > 0, \quad \alpha^2 + \beta^2 = 1.$$

Because $c, d \in W_V$ and w is Hermitian-orthogonal to u , there is a phase ϕ with

$$c = -\beta \sin \phi e_1 + \alpha \cos \phi e_2,$$

$$d = \beta \cos \phi e_1 + \alpha \sin \phi e_2.$$

Delete the zero third output row and take the cross product of the two remaining row vectors of the resulting 2×3 pencil. A polynomial generator $q(s)$ of the right kernel has third component

$$-\alpha\beta(1 + s^2).$$

Since $\alpha\beta \neq 0$, no real linear polynomial can divide all components of $q(s)$. A quadratic common divisor would force a constant right-kernel generator, excluded by Theorem 4.5. Thus the primitive right-kernel generator has degree two:

$$\varepsilon = 2.$$

With $\eta = 0$, the classification Theorem 4.6 gives K_{200} . $\square$

$$\boxed{V = \text{span}_{\mathbb{R}}\{(e_1, 0), (0, e_1), (\mu e_2, \nu e_2)\}.} \quad (14)$$

Theorem 4.10 (Front-null direction-rank law). *Every front-null three-plane satisfies*

$$\boxed{\text{rank}(d\pi_{\text{dir}}|_V) = \text{rank } A \leq 2.} \quad (15)$$

More precisely:

- (i) in $\mathbf{K}_{110}$, $\text{rank } A = \text{rank } B = 2$;
- (ii) in $\mathbf{K}_{200}$, $\text{rank } A = \text{rank } B = 2$;
- (iii) in $\mathbf{K}_{101}$, each of $\text{rank } A, \text{rank } B$ is one or two, with at least one equal to two.

Proof. The inequality follows already from the vanishing of the leading coefficient of $P_V(s)$. For the sharper statements use the adapted gauges.

In $\mathbf{K}_{110}$, A contains the independent horizontal vectors e_1, c and B contains e_1, d , while $c, d \neq 0$. In $\mathbf{K}_{200}$, the first two horizontal columns $a, -b$ are independent and the first two vertical columns b, a are independent. In $\mathbf{K}_{101}$, the gauge (14) gives

$$\text{rank } A = \begin{cases} 1, & \mu = 0, \\ 2, & \mu \neq 0, \end{cases} \quad \text{rank } B = \begin{cases} 1, & \nu = 0, \\ 2, & \nu \neq 0. \end{cases}$$

Since $\mu^2 + \nu^2 = 1$, they cannot both equal one for the same vanishing reason. $\square$

Theorem 4.11 (No countably rectifiable Sticky selector with null front). *Assume Λ_s is countably 3-rectifiable. Let*

$$\Gamma_s = \{(\theta, p(\theta), z(\theta)) : \theta \in S^3\}$$

be any Borel marked lift whose relevant rectifiable pieces admit the standard approximate tangent lifting. Then the Reeb-saturated Euclidean front cannot be four-dimensional null on all direction-carrying rectifiable mass. In particular, whenever the marked graph itself is countably 3-rectifiable,

$$\boxed{\mathcal{L}^4(K_{\Gamma_s}) > 0.} \quad (16)$$

Proof. Apply the area formula to the Lipschitz direction projection

$$\pi_{\text{dir}} : \Lambda_s \rightarrow S^3.$$

Since the selector graph meets every direction fibre exactly once, the multiplicity of the projection is one. Therefore

$$\int_{\Lambda_s} J_3(\pi_{\text{dir}}|_{T_x \Lambda_s}) d\mathcal{H}^3(x) = \mathcal{H}^3(S^3) > 0.$$

Hence the approximate direction differential has rank three on a subset of positive $\mathcal{H}^3$ -measure.

At every such point the horizontal tangent matrix A is invertible, so the leading coefficient of

$$P_V(s) = \det(B + sA)$$

is nonzero. Thus $P_V \neq 0$. For a countably rectifiable marked lift, the rectifiable Reeb-front polynomial theorem Theorem 4.4 then gives positive four-dimensional front measure. $\square$

4.2 Polarized normal forms

Definition 4.12 (Collision parameter and secant residual). *For $\alpha \neq 0$, define*

$$\boxed{s_*(\alpha, \beta) = -\frac{\alpha \cdot \beta}{|\alpha|^2},} \quad (17)$$

and

$$\boxed{r(\alpha, \beta) = \beta + s_* \alpha.} \quad (18)$$

Then

$$r \perp \alpha.$$

Lemma 4.13 (Exact Pythagorean collision identity). *For every $s \in \mathbb{R}$,*

$$|\beta + s\alpha|^2 = |r(\alpha, \beta)|^2 + |\alpha|^2 |s - s_*|^2. \quad (19)$$

Proof. Write

$$\beta = r - s_*\alpha.$$

Then

$$\beta + s\alpha = r + (s - s_*)\alpha,$$

and the two terms are orthogonal. $\square$

Proposition 4.14 (Exact collision-window formula). *For $\alpha \neq 0$, let s_* and r be Theorem 4.12. If*

$$|r| > \delta,$$

then

$$m_\delta^I(\alpha, \beta) = 0.$$

If

$$|r| \leq \delta,$$

then

$$m_\delta^I(\alpha, \beta) = \left| I \cap \left[s_* - \frac{\sqrt{\delta^2 - |r|^2}}{|\alpha|}, s_* + \frac{\sqrt{\delta^2 - |r|^2}}{|\alpha|} \right] \right|. \quad (20)$$

Proof. This is immediate from the Pythagorean identity (19). $\square$

Definition 4.15 (Normalized secant collision energy). *Define*

$$\mathfrak{E}_\delta^I(\mu) = \delta^{-3} \iint m_\delta^I(a - a', b - b') d\mu(a, b) d\mu(a', b'). \quad (21)$$

Lemma 4.16 (Universal pairwise collision-window bound). *For every pair with $\alpha \neq 0$,*

$$m_\delta^J(\alpha, \beta) \leq \frac{2\delta}{|\alpha|}. \quad (22)$$

Proof. By the exact Pythagorean formula

$$|\beta + s\alpha|^2 = |r|^2 + |\alpha|^2 |s - s_*|^2,$$

the collision set is either empty or contained in an interval of radius

$$\frac{\delta}{|\alpha|}$$

around s_* . $\square$

4.3 Rectifiable and selector reductions

Proposition 4.17 (Bush mass gives neighbourhood volume at every size). *Assume*

$$s_0 \in J_* \Subset I$$

and let

$$H = H_{(c_0, s_0)}(\rho).$$

Then

$$\mathcal{L}^4(N_\rho(K_b)) \geq |U| \sigma(H) \int_I |s - s_0|^3 ds. \quad (23)$$

In particular there is $c_{I, J_, U} > 0$ such that*

$$\mathcal{L}^4(N_\rho(K_b)) \geq c_{I, J_*, U} \sigma(H). \quad (24)$$

Proof. For $a \in H$, write

$$b(a) + s_0 a = c_0 + e(a), \quad |e(a)| \leq \rho.$$

At an arbitrary level $s \in I$,

$$b(a) + sa = c_0 + (s - s_0)a + e(a).$$

Hence the horizontal slice of $N_\rho(K_b)$ at height s contains

$$c_0 + (s - s_0)H.$$

Its three-dimensional Lebesgue measure is

$$|s - s_0|^3 |H| = |s - s_0|^3 |U| \sigma(H).$$

Integrate in s . The uniform bound follows because

$$s_0 \mapsto \int_I |s - s_0|^3 ds$$

has a positive minimum on the compact set $J_\star$. □

5 Physical collision, recurrence, and entropy

5.1 One-scale collision geometry

Proposition 5.1 (Ideal escape stays near the actual Kakeya slab). *For every i ,*

$$\boxed{E_i \subset N_\rho(K_b)}. \quad (25)$$

Proof. For $a \in H_i$, write

$$b(a) + s_i a = c_i + e_i(a), \quad |e_i(a)| \leq \rho.$$

Then

$$b(a) + sa = c_i + (s - s_i)a + e_i(a).$$

Thus the actual selected-line point at the same longitudinal coordinate s lies within ρ of the corresponding ideal escaped point. $\square$

Theorem 5.2 (Exact one-bush escape volume). *For every i ,*

$$\boxed{|E_i| = |U| \sigma(H_i) A_I(s_i), \quad A_I(s_i) := \int_I |s - s_i|^3 ds.} \quad (26)$$

Proof. At fixed $s \neq s_i$, the horizontal slice is

$$c_i + (s - s_i)H_i,$$

whose three-dimensional Lebesgue measure is

$$|s - s_i|^3 |H_i| = |U| |s - s_i|^3 \sigma(H_i).$$

The exceptional level $s = s_i$ has measure zero. Integrate in s . $\square$

$$\boxed{\Psi_{ij,s}(a) = \frac{c_i - c_j + (s - s_i)a}{s - s_j}.} \quad (27)$$

Theorem 5.3 (Exact pairwise escaped-overlap formula). *For every $i \neq j$,*

$$\boxed{|E_i \cap E_j| = |U| \int_I |s - s_i|^3 \mathfrak{R}_{ij}(s) ds.} \quad (28)$$

Proof. Fix s away from s_i, s_j . The s -slice of E_i is

$$c_i + (s - s_i)H_i.$$

A point in that slice also belongs to E_j exactly when its unique i -slope $a \in H_i$ satisfies

$$\Psi_{ij,s}(a) \in H_j.$$

Thus

$$E_i(s) \cap E_j(s) = c_i + (s - s_i) (H_i \cap \Psi_{ij,s}^{-1}(H_j)).$$

Its three-dimensional measure is

$$|s - s_i|^3 |U| \mathfrak{R}_{ij}(s).$$

Integrate over s . The two exceptional longitudinal levels are null. $\square$

$$\boxed{|[b(a) + sa] - [b(a') + sa']| \leq 2\rho.} \quad (29)$$

Theorem 5.4 (Two-time self-homothety). *Let*

$$D_{ij;s,t} = \Psi_{ij,s}^{-1} \circ \Psi_{ij,t}.$$

Then

$$\boxed{D_{ij;s,t} = \text{Dil}_{a_{ij}}(q_{ij}(s,t)).} \quad (30)$$

Moreover

$$\boxed{\bar{\sigma}(H_i \triangle D_{ij;s,t}(H_i)) \leq \Delta_{ij}(s) + |q_{ij}(s,t)|^3 \Delta_{ij}(t).} \quad (31)$$

Proof. The map identity is the two-time ratio formula from the preceding stage.

Put

$$A_s = \Psi_{ij,s}^{-1}(H_j), \quad A_t = \Psi_{ij,t}^{-1}(H_j).$$

By definition of $D_{ij;s,t}$,

$$D_{ij;s,t}(A_t) = A_s.$$

Therefore

$$\begin{aligned} \bar{\sigma}(H_i \triangle D_{ij;s,t}(H_i)) &\leq \bar{\sigma}(H_i \triangle A_s) + \bar{\sigma}(D_{ij;s,t}(A_t) \triangle D_{ij;s,t}(H_i)) \\ &= \Delta_{ij}(s) + |q_{ij}(s, t)|^3 \Delta_{ij}(t). \end{aligned}$$

□

Lemma 5.5 (Approximate translation invariance is quantitatively small). *Let*

$$H \subset U$$

have

$$m = \bar{\sigma}(H) > 0,$$

and let

$$v \in \mathbb{R}^3.$$

Then

$$\boxed{\frac{\bar{\sigma}(H \triangle (H + v))}{m} \geq \min \left\{ 1, \frac{|v|}{D_U} \right\}}, \quad (32)$$

where

$$D_U = \text{diam}(U).$$

Consequently, if the left side is at most $\varepsilon < 1$, then

$$\boxed{|v| \leq \varepsilon D_U}. \quad (33)$$

Proof. If $|v| > D_U$, then H and $H + v$ are disjoint, so the normalized symmetric difference equals $2m$, which is stronger than the claim.

Assume $0 < |v| \leq D_U$. Choose the least positive integer n such that

$$n|v| > D_U.$$

Then

$$H \cap (H + nv) = \emptyset,$$

so

$$\bar{\sigma}(H \triangle (H + nv)) = 2m.$$

By telescoping and translation invariance of Lebesgue measure,

$$\bar{\sigma}(H \triangle (H + nv)) \leq n \bar{\sigma}(H \triangle (H + v)).$$

Also

$$n \leq \frac{D_U}{|v|} + 1 \leq \frac{2D_U}{|v|}.$$

Hence

$$\bar{\sigma}(H \triangle (H + v)) \geq \frac{2m}{n} \geq m \frac{|v|}{D_U}.$$

□

Theorem 5.6 (Homothety commutator formula). *Let*

$$F = D_{a,q}, \quad G = D_{b,r}.$$

Then

$$\boxed{(GF)^{-1}(FG) = \tau_v, \quad v = \frac{(1-q)(1-r)}{qr}(a-b),} \quad (34)$$

where

$$\tau_v(x) = x + v.$$

Proof. The two compositions have the same scalar linear part qr :

$$\begin{aligned} FG(x) &= qrx + q(1-r)b + (1-q)a, \\ GF(x) &= qrx + r(1-q)a + (1-r)b. \end{aligned}$$

Their translation-part difference is

$$(1-q)(1-r)(a-b).$$

Applying $(GF)^{-1}$, whose linear part is $(qr)^{-1}$, gives (34). $\square$

Theorem 5.7 (Two-axis approximate homothety rigidity). *Let*

$$H \subset U, \quad m = \bar{\sigma}(H) > 0,$$

and suppose

$$F = D_{a,q}, \quad G = D_{b,r},$$

with

$$q, r \neq 0.$$

Assume

$$\boxed{\bar{\sigma}(H \triangle F(H)) \leq \varepsilon_F m, \quad \bar{\sigma}(H \triangle G(H)) \leq \varepsilon_G m.} \quad (35)$$

Then

$$\boxed{\bar{\sigma}(H \triangle (H + v)) \leq C(q, r)(\varepsilon_F + \varepsilon_G)m,} \quad (36)$$

where v is given by (34), and one may take

$$C(q, r) = |qr|^{-3} (2 + |q|^3 + |r|^3).$$

Consequently

$$\boxed{\min \left\{ 1, \frac{|(1-q)(1-r)||a-b|}{|qr|D_U} \right\} \leq C(q, r)(\varepsilon_F + \varepsilon_G).} \quad (37)$$

Proof. First compare the two compositions:

$$\begin{aligned} \bar{\sigma}(FG(H) \triangle GF(H)) &\leq \bar{\sigma}(FG(H) \triangle F(H)) + \bar{\sigma}(F(H) \triangle H) \\ &\quad + \bar{\sigma}(H \triangle G(H)) + \bar{\sigma}(G(H) \triangle GF(H)). \end{aligned}$$

The first term equals

$$|q|^3 \bar{\sigma}(G(H) \triangle H) \leq |q|^3 \varepsilon_G m,$$

and the last equals

$$|r|^3 \bar{\sigma}(H \triangle F(H)) \leq |r|^3 \varepsilon_F m.$$

Hence

$$\bar{\sigma}(FG(H) \triangle GF(H)) \leq (1 + |r|^3) \varepsilon_F m + (1 + |q|^3) \varepsilon_G m.$$

Apply $(GF)^{-1}$, whose absolute Jacobian is $|qr|^{-3}$. By Theorem 5.6, the two resulting sets are $H + v$ and H . This gives (36). Finally use Theorem 5.5. $\square$

Proposition 5.8 (Sign-reversal strong recurrence remains rigid). *Suppose a fixed direction block H_i admits two raw small-defect recurrences whose ratio satisfies*

$$q = -1 + O(\varepsilon),$$

and suppose two partner edges produce such sign-reversal self-homotheties with centers a, b . Then, provided the dilation ratios stay in a fixed compact neighbourhood of -1 , the translation-commutator theorem of the preceding stage still gives

$$\boxed{|a - b| \lesssim D_U \varepsilon.} \quad (38)$$

Proof. Near $q = r = -1$, the coefficient

$$\frac{(1-q)(1-r)}{qr}$$

in the commutator translation stays bounded away from zero, while all Jacobian constants remain bounded. Apply Theorem 5.7. $\square$

5.2 Weighted graph regularization and radial recurrence

$$\mathcal{O}_{ij}^{+, \tau_0} = |U| \int_{I_+} |s - s_i|^3 \sigma(\{a \in H_i : \Psi_{ij,s}(a) \in H_j, \quad |a - \Psi_{ij,s}(a)| \geq \tau_0\}) ds. \quad (39)$$

$$\mathcal{N}_{i_0}^{\tau_0}(a, s) = \sum_{j \in J} \mathbf{1}_{H_j}(\Psi_{i_0 j, s}(a)) \mathbf{1}_{[\tau_0, \infty)}(|a - \Psi_{i_0 j, s}(a)|). \quad (40)$$

Theorem 5.9 (Exact truncated star identity and common parameter set). *If*

$$\mathcal{O}_{i_0 j}^{+, \tau_0} \geq \theta R \quad (j \in J),$$

then

$$\int_{I_+} \int_{H_{i_0}} |s - s_{i_0}|^3 \mathcal{N}_{i_0}^{\tau_0}(a, s) da ds = \sum_{j \in J} \mathcal{O}_{i_0 j}^{+, \tau_0} \geq D \theta R. \quad (41)$$

Consequently there is a measurable set

$$Z_{i_0} \subset H_{i_0} \times I_+$$

such that

$$(\sigma \times ds)(Z_{i_0}) \gtrsim \theta R, \quad (42)$$

and

$$\mathcal{N}_{i_0}^{\tau_0}(a, s) \gtrsim \theta D \quad ((a, s) \in Z_{i_0}). \quad (43)$$

Proof. Expand (40) and apply Tonelli. Since $da = |U| d\sigma(a)$, each summand is exactly (39). This proves (41). On the outer wing the weight is bounded above and below, $0 \leq \mathcal{N}_{i_0}^{\tau_0} \leq D$, and $\sigma(H_{i_0}) \asymp R$. Thresholding at a sufficiently small multiple of θD gives (42)–(43). $\square$

Lemma 5.10 (Pointwise heavy longitudinal packet). *For every $(a, s) \in Z_{i_0}$, there exists a level interval $K(a, s) \in \mathcal{K}_\Delta$ such that*

$$\#\{j \in J^{\tau_0}(a, s) : s_j \in K(a, s)\} \gtrsim \theta D \Delta. \quad (44)$$

Proof. The sets in the braces partition $J^{\tau_0}(a, s)$ into $O(\Delta^{-1})$ level bins, while (43) gives $\#J^{\tau_0}(a, s) \gtrsim \theta D$. Pigeonhole. $\square$

$$c_j = q_{i_0} + s_j a^\sharp + e_j, \quad |e_j| \lesssim \eta. \quad (45)$$

Lemma 5.11 (Packet centers and packet teeth are close). *If $j, k \in J^{\tau_0}(a, s)$ and $s_j, s_k \in K(a, s)$, then*

$$|x_j - x_k| \lesssim d_0^{-1} \Delta + \eta, \quad (46)$$

and

$$|a_j(y) - a_k(y)| \lesssim d_0^{-1} \Delta + \eta. \quad (47)$$

Proof. Subtract (45) for j, k :

$$c_j - c_k = (s_j - s_k) a^\sharp + e_j - e_k.$$

Since $|s_j - s_k| \leq \Delta$, this proves (46). At the common escaped point,

$$a_j(y) - a_k(y) = \frac{(s_j - s_k) a_j(y) - h_{jk}}{s - s_k}. \quad (48)$$

The outer-wing denominator is uniformly bounded below, $a_j(y) \in U$, and (46) controls both $|s_j - s_k|$ and $|h_{jk}|$. This gives (47). $\square$

Theorem 5.12 (Clustered fixed-angle star: degree or angular cap). *A clustered fixed-angle star of degree D obeys one of the following alternatives:*

(i)

$$D \lesssim \theta^{-1} \Delta^{-1}, \quad (49)$$

(ii) there is a measurable set $Y_{i_0} \subset E_{i_0}^+$ such that

$$|Y_{i_0}| \gtrsim \theta R, \quad (50)$$

and every $y \in Y_{i_0}$ supports at least

$$\gtrsim \theta D \Delta \quad (51)$$

simultaneous neighbor teeth in one cap of radius

$$C(d_0^{-1} \Delta + \eta). \quad (52)$$

Proof. If the right side of (51) is smaller than a sufficiently large fixed constant, then (49) holds. Otherwise map Z_{i_0} into the central escaped bush by

$$\Phi_{i_0}(a, s) = (c_{i_0} + (s - s_{i_0})a, s).$$

This map is injective on the outer wing and has Jacobian $|s - s_{i_0}|^3 \asymp 1$, so its image Y_{i_0} satisfies (50). At each image point, choose the variable packet from Theorem 5.10; its cardinality is (51), and Theorem 5.11 puts all its teeth in the cap (52). $\square$

Theorem 5.13 (Far-chart packing-or-cap decomposition). *There is an edge-disjoint decomposition*

$$\mathcal{E}_{\theta, \tau_0}^{\text{far}} = \mathcal{E}_{\theta, \tau_0}^{\text{far, pack}} \dot{\cup} \bigcup_{\nu} \mathcal{S}_{\nu}^{\text{far, cap}} \quad (53)$$

such that

$$\#\mathcal{E}_{\theta, \tau_0}^{\text{far, pack}} \lesssim d_R^{-3} \theta^{-1} R^{-3} = \theta^{-1} R^{-3+o(1)}, \quad (54)$$

and every removed star satisfies (50)–(52) with

$$C(d_R + \eta_R) \longrightarrow 0. \quad (55)$$

Proof. For far edges, $|a_{ij}| \lesssim d_R^{-1}$. Partition this radial-center region into η_R -cubes. Their number is

$$O(d_R^{-3} \eta_R^{-3}) = O(d_R^{-1} R^{-1}).$$

Inside each cell graph, repeatedly remove the full star of a maximum-degree vertex whenever its degree exceeds

$$C\theta^{-1} \Delta_R^{-1} = C\theta^{-1} d_R^{-2}.$$

By Theorem 5.12, every removed star has the asserted cap output. The residual graph in one cell has at most

$$O(N\theta^{-1} d_R^{-2}) = O(\theta^{-1} d_R^{-2} R^{-2})$$

edges. Multiplication by the cell count gives

$$O(d_R^{-1} R^{-1}) O(\theta^{-1} d_R^{-2} R^{-2}) = O(d_R^{-3} \theta^{-1} R^{-3}),$$

proving (54). The cap radius is (52) with $d_0 = d_R$, $\Delta_R = d_R^2$, and $\eta_R = (R/d_R^2)^{1/3}$. $\square$

Lemma 5.14 (An oriented near-diagonal displacement star has close centers). *Suppose all edges of a star use one orientation and their ordered displacements lie in one ζ_R -cube. Then its neighbor centers satisfy*

$$|x_j - x_k| \lesssim d_R + \zeta_R. \quad (56)$$

At any common escaped point, the corresponding neighbor teeth satisfy the same bound.

Proof. For the orientation $i_0 < j$, one has $c_j = c_{i_0} - h_{i_0 j}$; for the opposite orientation, $c_j = c_{i_0} + h_{j i_0}$. Thus the horizontal neighbor diameter is $O(\zeta_R)$. Their levels are each within d_R of s_{i_0} , so their longitudinal diameter is at most $2d_R$. This proves the center bound. Apply the exact identity (48) and the outer-wing denominator bound to obtain the tooth bound. $\square$

Theorem 5.15 (Near-chart packing-or-cap decomposition). *There is an edge-disjoint decomposition*

$$\mathcal{E}_{\theta, \tau_0}^{\text{near}} = \mathcal{E}_{\theta, \tau_0}^{\text{near, pack}} \dot{\cup} \bigcup_{\nu} \mathcal{S}_{\nu}^{\text{near, cap}} \quad (57)$$

such that

$$\#\mathcal{E}_{\theta, \tau_0}^{\text{near, pack}} \lesssim \theta^{-1} R^{-3}, \quad (58)$$

and every removed star has a common physical region of volume $\gtrsim \theta R$ on which at least $\gtrsim \theta D$ simultaneous neighbor teeth lie in a cap of radius

$$C(d_R + R^{1/3}) \rightarrow 0. \quad (59)$$

Proof. Work in one displacement-cell graph. While a vertex has degree larger than $C\theta^{-1}$, choose a maximum-degree vertex and split its incident edges according to whether the vertex is the first or second endpoint in the fixed ordering. One orientation retains at least half the degree. Apply Theorem 5.9 to that oriented star. On its set Z_{i_0} , at least $\gtrsim \theta D$ neighbors occur simultaneously. Map this set into $E_{i_0}^+$ by the central escaped-bush parametrization; the image has volume $\gtrsim \theta R$. By Theorem 5.14, all simultaneous neighbor teeth lie in the cap (59). Record and remove the oriented star.

When the process stops, the residual graph in each cell has maximum degree $O(\theta^{-1})$, hence at most $O(N\theta^{-1})$ edges. There are $O(R^{-1})$ cells and $N \asymp R^{-2}$, giving (58). $\square$

5.3 Entropy pruning and common-neighbor structure

Lemma 5.16 (The fixed-angle collision time is nondegenerate). *For a fixed edge ij and fixed source direction a ,*

$$\partial_{\xi} \Psi_{ij, \xi}(a) = \frac{a - \Psi_{ij, \xi}(a)}{\xi - s_j}. \quad (60)$$

On the outer wing and the fixed-angle set,

$$c_{\tau_0} \leq |\partial_{\xi} \Psi_{ij, \xi}(a)| \leq C.$$

Moreover the trajectory lies on one affine line and is monotone there. Consequently the set of times for which $\Psi_{ij, \xi}(a)$ belongs to one direction R -cube has length $O_{\tau_0}(R)$.

Proof. Differentiate

$$\Psi_{ij, \xi}(a) = \frac{c_i - c_j + (\xi - s_i)a}{\xi - s_j}.$$

This gives

$$\partial_{\xi} \Psi_{ij, \xi}(a) = \frac{(s_i - s_j)a - (c_i - c_j)}{(\xi - s_j)^2} = \frac{a - \Psi_{ij, \xi}(a)}{\xi - s_j}.$$

The outer-wing denominator is bounded above and below, while the truncation gives $|a - \Psi_{ij, \xi}(a)| \geq \tau_0$. Also

$$\Psi_{ij, \xi}(a) - a = \frac{c_i - c_j - (s_i - s_j)a}{\xi - s_j},$$

so the trajectory has a fixed direction and a monotone scalar parameter. Its intersection with an R -cube therefore has time length $O_{\tau_0}(R)$. $\square$

Corollary 5.17 (One edge contributes only R^4 to one double direction code). *For every fixed edge ij ,*

$$(\sigma \otimes d\xi) \{(a, \xi) : a \in \tau, \Psi_{ij, \xi}(a) \in \tau', |a - \Psi_{ij, \xi}(a)| \geq \tau_0\} \lesssim_{\tau_0} R^4. \quad (61)$$

The same bound holds after imposing the two intercept cells and both center-level intervals.

Proof. For each fixed $a \in \tau$, Theorem 5.16 bounds the admissible ξ -length by CR . Integrate over the direction cube, whose σ -measure is $O(R^3)$. All remaining restrictions only decrease the measure. $\square$

Theorem 5.18 (A double graph code has bounded endpoints and R^4 mass). *For one nonempty double code $\kappa^{(2)}$, all source endpoints $x_i = (c_i, s_i)$ lie in one $O(R)$ -ball and all output endpoints $x_j = (c_j, s_j)$ lie in another $O(R)$ -ball. Hence an R -separated simple center graph contributes only $O(1)$ ordered edges to the code, and*

$$\boxed{\sum_{i,j} (\sigma \otimes d\xi) \left(\mathcal{W}_{ij}^{\tau_0} \cap \{\kappa^{(2)}\} \right) \lesssim_{\tau_0} R^4.} \quad (62)$$

Proof. Choose actual graph points $(a_0, b(a_0)) \in \chi$ and $(u_0, b(u_0)) \in \chi'$ occurring in the code, and representatives $s_0 \in K$, $t_0 \in K'$. Since $a \in H_{x_i}(R)$,

$$|c_i - (b(a) + s_i a)| \leq R.$$

All $a, b(a), s_i$ in the source code differ from $a_0, b(a_0), s_0$ by $O(R)$, so all x_i lie in an $O(R)$ -ball about $(b(a_0) + s_0 a_0, s_0)$. The same argument with $u \in H_{x_j}(R)$ places all x_j in an $O(R)$ -ball about $(b(u_0) + t_0 u_0, t_0)$.

Each ball contains $O(1)$ vertices of an R -separated family. Simplicity of the graph gives only $O(1)$ ordered edges between the two endpoint sets. Apply (61) to each edge and sum. $\square$

Lemma 5.19 (One physical edge has universal one-dimensional code valence). *Fix a physical edge ij and a source endpoint code $v = (\chi, K)$. The fixed-angle outer-wing incidences of this edge meet at most*

$$\boxed{B_R \lesssim R^{-1}} \quad (63)$$

output endpoint codes, with no goodness assumption on either direction cube.

Proof. The output level interval K' is determined by the fixed level s_j . For a representative source direction a_0 of τ , the curve

$$\xi \mapsto \Psi_{ij,\xi}(a_0)$$

is a line segment of bounded length inside the bounded direction domain U , by Theorem 5.16. It meets $O(R^{-1})$ direction R -cubes. For $a \in \tau$, the affine homothety formula for $\Psi_{ij,\xi}(a)$ has a uniformly bounded coefficient of a on the outer wing. Hence allowing all $a \in \tau$ only thickens that segment by $O(R)$, and the same $O(R^{-1})$ direction-cube bound remains valid.

Fix one output direction cube τ' met by this tube and choose $u_{\tau'} \in \tau'$. Every incidence counted for the physical endpoint j satisfies

$$|b(u) + s_j u - c_j| \lesssim R.$$

Since $u \in \tau'$ and the center levels remain in a fixed bounded interval,

$$|b(u) - (c_j - s_j u_{\tau'})| \lesssim R.$$

Thus all output graph points of this fixed edge above τ' meet only $O(1)$ intercept R -cubes ζ' , regardless of how many other selector-graph sheets lie above τ' . The fixed level s_j determines one interval K' . Multiplying the $O(R^{-1})$ direction-cube count by this $O(1)$ anchored intercept count proves (63). $\square$

Theorem 5.20 (Universal high endpoint-code degree gives a genuine physical star). *Let $\mathfrak{D}$ be any double-code family, and let $\deg_{\mathfrak{D}}(v)$ denote endpoint-code degree. If*

$$\deg_{\mathfrak{D}}(v) > L,$$

then one of the $O(1)$ physical centers localized by v is incident to at least

$$\boxed{\gtrsim RL} \quad (64)$$

distinct original fixed-angle edges. These edges form a genuine physical star and retain their outer-wing collision witnesses.

Proof. The proof of Theorem 5.18 shows that one endpoint code $v = (\chi, K)$ contains only $O(1)$ possible physical centers. For one such center i , every fixed physical edge ij accounts for at most B_R incident double codes by (63). Therefore the L code edges incident to v require at least cL/B_R distinct physical edges after pigeonholing the $O(1)$ centers. If v was the output endpoint, reverse the physical edges; the fixed-angle outer-wing relation and (63) are symmetric under this reversal. Use (63). $\square$

5.4 Endpoint codes and the transition to a Maslov shell

Lemma 5.21 (One double code determines its the preceding stage extraction cell). *All physical edges contained in one double code satisfy*

$$\boxed{\text{diam}\{a_{ij}\} \lesssim R d_R^{-2} \ll \eta_R \quad \text{in the far chart}, \quad \text{diam}\{h_{ij}\} \lesssim R \ll \zeta_R \quad \text{in the near chart.}} \quad (65)$$

Consequently one double code meets only $O(1)$ of the extraction cells used in Theorems 5.13 and 5.15.

Proof. By Theorem 5.18, the two endpoint codes localize $x_i = (c_i, s_i)$ and $x_j = (c_j, s_j)$ to respective $O(R)$ -balls. Hence the ordered displacement $h_{ij} = c_i - c_j$ varies by $O(R)$. In the far chart $|s_i - s_j| \geq d_R$, and the quotient map

$$(h, d) \mapsto h/d$$

has Lipschitz constant $O(d_R^{-2})$ on the bounded physical parameter region. Thus $a_{ij} = h_{ij}/(s_i - s_j)$ varies by $O(R d_R^{-2})$. Finally,

$$R d_R^{-2} = o\left((R/d_R^2)^{1/3}\right) = o(\eta_R), \quad R = o(R^{1/3}) = o(\zeta_R).$$

A set of diameter smaller than the grid scale meets only boundedly many half-open grid cells. $\square$

Corollary 5.22 (A nonsummable high code has the prescribed clustered/separated center split). *Let an endpoint code v satisfy $\deg(v) > AL_*$. Assign each incident double code to one of the $O(1)$ compatible the preceding stage extraction cells furnished by Theorem 5.21, and write $\deg_\alpha(v)$ for the resulting cell-degree. After adjusting absolute constants, at least one of the following holds:*

- (i) *some extraction cell satisfies $\deg_\alpha(v) > cL_*$, and the physical edges in that cell contain a clustered star of degree $\gtrsim D_0$;*
- (ii) *one of the $O(1)$ physical centers localized by v is the center of a genuine star containing at least cA physical edges whose radial centers, respectively displacements, occupy extraction-scale-separated cells.*

Equivalently,

$$\boxed{\max_\alpha \deg_\alpha(v) \gtrsim L_* \quad \text{or} \quad \#\{\alpha : \deg_\alpha(v) > 0\} \gtrsim A.} \quad (66)$$

Thus, after choosing any slowly divergent $A_R = R^{-o(1)}$, a fixed-angle contribution outside the dimension budget can leave the double-code stage only through the clustered alternative or through a high endpoint code meeting at least cA_R separated distinguished-center cells, exactly as prescribed in the preceding stage.

Proof. The cell-degrees sum to $\deg(v)$, up to the harmless bounded assignment multiplicity. If their maximum is $\gtrsim L_*$, then Theorems 5.19 and 5.20 give at least

$$RL_* \asymp D_0$$

distinct physical edges inside that single extraction cell. In the near chart first split the edges according to the two ordered displacement orientations, losing only an absolute factor. Otherwise division of $\deg(v) > AL_*$ by the maximum cell-degree gives at least cA occupied cells. For every incident double code choose one realizing ordered physical edge and assign the code to the unique half-open extraction cell containing that edge's distinguished center or displacement. Color the extraction grid by coordinate residues modulo 3; one color retains a fixed fraction and makes the chosen cells extraction-scale separated. Choose one realizing edge from each retained cell. Distinct retained cells give distinct physical edges. Their endpoint belonging to v is one of only $O(1)$ physical centers by Theorem 5.18; pigeonholing that bounded set leaves cA edges incident to one center. Reverse all edges first if v is their output endpoint, and in the near chart split the two ordered displacement orientations. Both operations lose only an absolute factor. $\square$

Lemma 5.23 (The distinguished center is selected by the contact secant). *Let a double-code incidence come from a physical edge ij , source direction a , output direction $u = \Psi_{ij,\xi}(a)$, and outer-wing parameter ξ . Then*

$$\boxed{\begin{aligned} b(u) - b(a) &= \xi(a - u) + O(R), \\ h_{ij} &= (\xi - s_j)u - (\xi - s_i)a, \\ a_{ij} &= a + \frac{\xi - s_j}{s_i - s_j}(u - a) \quad (|s_i - s_j| \geq d_R). \end{aligned}} \quad (67)$$

Consequently, on the fixed-angle piece $|a - u| \geq \tau_0$, the two endpoint graph cells determine ξ up to $O_{\tau_0}(R)$. Thus the separated-center alternative in Theorem 5.22 is not a free-center family: its center cells are the ones selected by the contact secants of the same graph incidences.

Proof. Choose errors e_i, e_j of size $O(R)$ so that

$$c_i = b(a) + s_i a + e_i, \quad c_j = b(u) + s_j u + e_j.$$

The collision relation

$$c_i + (\xi - s_i)a = c_j + (\xi - s_j)u$$

gives the first two identities in (67). Since $h_{ij} = c_i - c_j$, the second identity can also be written

$$h_{ij} = (s_i - s_j)a + (\xi - s_j)(u - a).$$

Division by $s_i - s_j$ gives the far-chart formula. Finally, taking the scalar product of the first identity with $a - u$ yields

$$\xi = \frac{(b(u) - b(a)) \cdot (a - u)}{|a - u|^2} + O_{\tau_0}(R),$$

which proves the asserted selection statement. □

6 Angular blow-up and Maslov four-cycles

6.1 Darboux blow-up and four-cycle extraction

$$\boxed{\iint_{|a-u|\leq\tau} \frac{d\sigma(a)d\sigma(u)}{|a-u|} \lesssim_U \tau^2.} \quad (68)$$

Proposition 6.1 (Critical mass commutes with the Darboux blow-up). *Let $q_R(P)$ be the collision mass assigned to P . Up to the fixed enlargement used to hold neighboring grid cells,*

$$\boxed{q_R(P) = m_P^2 \widehat{\tau}_R^{-1} M_{\epsilon_R}^{J,\text{sh}}(\mu_P^\#).} \quad (69)$$

On the density-good patches $m_P \asymp_U \widehat{\tau}_R^3$, which carry all but $o(1)$ of the direction mass after passing to a dyadic subsequence, this gives

$$\boxed{\kappa_R \asymp_U \sum_P m_P M_{\epsilon_R}^{J,\text{sh}}(\mu_P^\#).} \quad (70)$$

Thus κ_R is precisely the direction-mass average of the critical collision masses in the unit Darboux blow-ups.

Proof. Under the simultaneous dilation

$$(a - a_P, b - b_P) = \widehat{\tau}_R(\bar{a}, \bar{b}),$$

the collision inequality at thickness R becomes the same inequality at thickness ϵ_R , while the collision-time length is unchanged. Hence the normalized weight scales as

$$w_R^J = \widehat{\tau}_R^{-1} w_{\epsilon_R}^J.$$

Replacing each copy of $\sigma|_P$ by $m_P \sigma_P$ proves (69). Summing in P , dividing by $\widehat{\tau}_R^2$, and using $m_P/\widehat{\tau}_R^3 \asymp_U 1$ gives (70).

For the density-good assertion, use the dyadic martingale differentiation theorem on $\mathbf{1}_U$: the U -mass in dyadic cells whose relative density stays below any fixed positive threshold tends to zero. The bounded coloring and enlargement change only fixed constants. $\square$

Lemma 6.2 (Rescaled contact-secant identity). *Every physical re-collision represented in (67) satisfies*

$$\boxed{\Pi_\xi(\alpha_R, \beta_R) = \beta_R + \xi\alpha_R = O\left(\frac{R}{\tau_R}\right), \quad \Pi_\xi(\alpha, \beta) = \beta + \xi\alpha.} \quad (71)$$

Thus the rescaled secant lies within $O(R/\tau_R)$ of

$$L_\xi = \ker \Pi_\xi = \{(\alpha, -\xi\alpha) : \alpha \in \mathbb{R}^3\},$$

the Lagrangian member of the Reeb/Maslov pencil.

Proof. The first line of (67) is

$$b(u) - b(a) + \xi(u - a) = O(R).$$

Divide by τ_R . The description of the kernel is immediate, and it is Lagrangian because it is the graph of the symmetric map $-\xi I$. $\square$

$$\boxed{\text{supp } \mathfrak{q}_\infty \subset \{(a, \alpha, \beta, \xi) : (\alpha, \beta) \in L_\xi\}.} \quad (72)$$

$$\boxed{\rho^{\otimes 4} \{(a_0, a_1, a_2, a_3) : |\det(a_1 - a_0, a_2 - a_0, a_3 - a_0)| \leq \Delta\} \lesssim \Delta.} \quad (73)$$

$$|b_y - b_x + s_{xy}(a_y - a_x)| \leq \epsilon. \quad (74)$$

Theorem 6.3 (Quantitative Maslov four-cycle saving). *With*

$$e(h) = \iint h(x, y) d\nu(x) d\nu(y),$$

one has, for every $0 < \Delta < 1$,

$$\boxed{e(h)^4 \lesssim_{U,J} \Delta + \mathfrak{B}_{C_{U,J}\epsilon/\Delta}(h)}. \quad (75)$$

In particular, with $\Delta = \sqrt{\epsilon}$,

$$e(h) \lesssim_{U,J} \epsilon^{1/8} + \mathfrak{B}_{C_{U,J}\sqrt{\epsilon}}(h)^{1/4}.$$

Proof. Twice applying Cauchy–Schwarz to the common-neighbor kernel gives the four-cycle inequality

$$e(h)^4 \leq \int h_{01} h_{12} h_{23} h_{30} d\nu^4.$$

Split the integral according to whether the horizontal tetrahedron determinant is at most Δ . The first part is $O(\Delta)$ by (73).

For a nondegenerate four-cycle, orient its horizontal edge vectors cyclically as v_1, v_2, v_3, v_4 , so $v_1 + v_2 + v_3 + v_4 = 0$. Summing (74) around the cycle and subtracting the fourth edge label gives

$$(s_1 - s_4)v_1 + (s_2 - s_4)v_2 + (s_3 - s_4)v_3 = O(\epsilon).$$

The three vectors v_1, v_2, v_3 have bounded norm and determinant comparable, up to a fixed linear change, to the horizontal tetrahedron determinant. Cramer’s rule therefore yields

$$\max_{1 \leq i \leq 4} |s_i - s_4| \lesssim_{U,J} \frac{\epsilon}{\Delta}.$$

Choose $s = s_4$ and $c = b_{x_0} + sa_{x_0}$. Propagating the four approximate edge relations around the cycle gives

$$|b_{x_i} + sa_{x_i} - c| \lesssim_{U,J} \frac{\epsilon}{\Delta} \quad (0 \leq i \leq 3).$$

Thus every nondegenerate cycle is counted by $\mathfrak{B}_{C\epsilon/\Delta}(h)$, proving (75). $\square$

Theorem 6.4 (Quantitative Maslov common-neighbor dichotomy). *Under the hypotheses of Theorem 6.3, put*

$$\eta = e(h).$$

There are constants depending only on the bounded coordinate region and on J such that at least one of the following holds:

(i)

$$\boxed{\eta \lesssim \epsilon^{1/4}}. \quad (76)$$

(ii) *there are $s_0 \in J$ and $c_0 \in \mathbb{R}^3$ for which*

$$\boxed{\nu \{ (a, b) : |b + s_0 a - c_0| \leq C\epsilon\eta^{-4} \} \gtrsim \eta^3}. \quad (77)$$

Thus a collision graph of mass larger than the fourth-root error scale produces, at the same finite scale, one positive-mass approximate affine Lagrangian point-probe bush.

Proof. Write

$$K(p, q) = \int h(p, y) h(q, y) d\nu(y), \quad d\lambda_{p,q}(y) = h(p, y) h(q, y) d\nu(y).$$

The four-cycle integral is $\iint K(p, q)^2 d\nu(p) d\nu(q)$ and is at least η^4 . Choose

$$\Delta = c_* \eta^4$$

with $c_* > 0$ sufficiently small. By (73), the determinant-degenerate cycles have total mass at most $\eta^4/2$. Hence the nondegenerate common-neighbor pairs have total mass

$$G \gtrsim \eta^4. \quad (78)$$

For fixed p, q , let $G(p, q)$ be their contribution to G . Since

$$\iint K(p, q) d\nu(p) d\nu(q) = \int \left(\int h(x, y) d\nu(x) \right)^2 d\nu(y) \leq \iint h d\nu^2 = \eta,$$

there is a pair (p, q) with

$$\frac{G(p, q)}{K(p, q)} \gtrsim \eta^3. \quad (79)$$

For a nondegenerate pair of common neighbors y, y' , the Cramer-rule argument in the proof of Theorem 6.3 gives

$$|s_{py} - s_{py'}| \lesssim r, \quad r := C\epsilon\eta^{-4}.$$

If $r \gtrsim 1$, then (i) holds. Otherwise partition J into half-open intervals of length comparable to r . A good pair y, y' can occur only in the same interval or in a bounded number of neighboring intervals. Therefore

$$G(p, q) \lesssim K(p, q) \sup_I \lambda_{p,q} \{y : s_{py} \in I\},$$

where the supremum is over intervals I of length Cr . By (79), some such interval I_0 , centered at s_0 , has

$$\lambda_{p,q} \{y : s_{py} \in I_0\} \gtrsim \eta^3.$$

Because $0 \leq h \leq 1$, the same set has ν -measure at least $c\eta^3$. For every y in this set the edge relation between p and y gives

$$|b_y + s_0 a_y - (b_p + s_0 a_p)| \leq \epsilon + C_U |s_{py} - s_0| \lesssim r.$$

Taking $c_0 = b_p + s_0 a_p$ proves (ii). $\square$

Lemma 6.5 (A dyadic code-edge layer retains the joint-box mass). *For fixed L , let $\omega_L(e)$ be the total assigned incidence mass from $\mathcal{B}_L$ carried by the double-code edge e . Then*

$$0 \leq \omega_L(e) \lesssim R^4$$

and, with dyadic $R^3 \leq \lambda \lesssim 1$,

$$\boxed{\sum_{B \in \mathcal{B}_L} m_B \lesssim R^{-1-o(1)} + (\log R^{-1}) \max_{\lambda} \lambda R^4 |E(\mathcal{G}_{L,\lambda})|,} \quad (80)$$

where

$$E(\mathcal{G}_{L,\lambda}) = \{e : \lambda R^4 \leq \omega_L(e) < 2\lambda R^4\}.$$

Thus any contribution above the target $R^{-2+o(1)}$ survives in a simple code graph whose edges all have comparable positive incidence mass.

Proof. The half-open center and time assignments partition the incidence mass, so summing $\omega_L(e)$ over e gives the left side. The double-code capacity (62) gives the pointwise upper bound. There are at most N_R^2 possible code edges, where

$$N_R = |\mathcal{V}_R| \lesssim R^{-4-o(1)}.$$

Therefore the edges with $\omega_L(e) < R^7$ carry total mass at most

$$R^7 N_R^2 \lesssim R^{-1-o(1)},$$

which is already below the required $R^{-2+o(1)}$ budget. The remaining normalized weights $\omega_L(e)/R^4$ occupy $O(\log R^{-1})$ dyadic classes. Pigeonholing proves (80). $\square$

Lemma 6.6 (Power excess creates many code four-cycles). *Let $\mathcal{G}$ be any simple graph on the endpoint-code vertex set and put $N_R = |\mathcal{V}_R|$. If, for some $A \geq A_0$,*

$$|E(\mathcal{G})| \geq A N_R^{3/2},$$

then $\mathcal{G}$ contains at least

$$\boxed{c A^4 N_R^2} \quad (81)$$

unoriented simple four-cycles.

Proof. Write $E = |E(\mathcal{G})|$. If $d(x, y)$ denotes the number of common neighbors of x, y , then

$$\#C_4(\mathcal{G}_L) = \frac{1}{2} \sum_{\{x, y\}} \binom{d(x, y)}{2}.$$

Also

$$\sum_{\{x, y\}} d(x, y) = \sum_z \binom{\deg z}{2} \gtrsim E^2/N_R.$$

Twice using convexity, and taking A_0 large enough to absorb the linear terms in the binomial coefficients, gives

$$\#C_4(\mathcal{G}_L) \gtrsim E^4/N_R^4 \geq cA^4N_R^2.$$

□

Lemma 6.7 (A nondegenerate code four-cycle is one near-Lagrangian bush). *Let*

$$v_0 v_1 v_2 v_3 v_0$$

be a four-cycle of endpoint codes. Choose once and for all one graph representative

$$z_k = (a_k, b_k) \in \chi_k \cap G_b$$

for each $v_k = (\chi_k, K_k)$. Then every edge $v_k v_{k+1}$, with indices modulo four, has a collision label $\xi_k \in I + O(R)$ satisfying

$$\boxed{b_{k+1} - b_k + \xi_k(a_{k+1} - a_k) = O(R).} \quad (82)$$

If

$$|\det(a_1 - a_0, a_2 - a_1, a_3 - a_2)| \geq \Delta,$$

then there are $s_0 \in I + O(R)$ and $c_0 \in \mathbb{R}^3$ such that

$$\boxed{|b_k + s_0 a_k - c_0| \lesssim R/\Delta \quad (0 \leq k \leq 3).} \quad (83)$$

Thus every horizontally nondegenerate code four-cycle lies in one $O(R/\Delta)$ -approximate affine Lagrangian point-probe bush.

Proof. For an edge $v_k v_{k+1}$, choose one realizing physical incidence. Its actual graph endpoints belong to χ_k, χ_{k+1} , so replacing them by the fixed representatives changes the contact-secant identity (67) by $O(R)$. This proves (82).

Put $w_k = a_{k+1} - a_k$. Summing (82) around the cycle gives

$$\xi_0 w_0 + \xi_1 w_1 + \xi_2 w_2 + \xi_3 w_3 = O(R), \quad w_0 + w_1 + w_2 + w_3 = 0.$$

After subtracting ξ_3 times the second identity, Cramer's rule yields

$$\max_{0 \leq k \leq 3} |\xi_k - \xi_3| \lesssim R/\Delta.$$

Take $s_0 = \xi_3$ and $c_0 = b_0 + s_0 a_0$. Propagating (82) around the four edges proves (83). □

6.2 Time synchronization and physical bush cells

Lemma 6.8 (Four-time Maslov interpolation on one contact cycle). *For a code four-cycle as in Theorem 6.7, put*

$$d_k = (a_{k+1} - a_k, b_{k+1} - b_k) \in \mathbb{R}_a^3 \oplus \mathbb{R}_b^3 \quad (k \bmod 4).$$

Then

$$d_0 + d_1 + d_2 + d_3 = 0.$$

If $\text{Span}\{d_0, d_1, d_2\}$ has dimension at most two, the cycle already has full secant rank at most two. Otherwise let

$$V_C = \text{Span}\{d_0, d_1, d_2\}$$

and let $P_{V_C}(s)$ be its Reeb-front polynomial. For the four collision labels $\xi_0, \dots, \xi_3$,

$$\boxed{|P_{V_C}(\xi_k)| \lesssim_{\tau_0, I} R \quad (0 \leq k \leq 3).} \quad (84)$$

Consequently, if

$$\mathfrak{V}(\xi_0, \xi_1, \xi_2, \xi_3) := \prod_{0 \leq i < j \leq 3} |\xi_i - \xi_j| \geq v,$$

then

$$\boxed{\sup_{s \in J} |P_{V_C}(s)| \lesssim_{I, J, \tau_0} \frac{R}{v}.} \quad (85)$$

In particular Theorem 6.14 gives

$$|\det A_C| + |P_{V_C}(i)| \lesssim R/v,$$

so a four-time cycle with $R/v \rightarrow 0$ is quantitatively horizontal-rank deficient and maximally symplectic.

Proof. The telescoping identity gives $\sum_k d_k = 0$, hence $d_3 \in V_C$. Choose an orthonormal frame

$$T_C c = (A_C c, B_C c)$$

for V_C . Write $d_k = T_C c_k$. Fixed-angle separation gives

$$|c_k| = |d_k| \geq |a_{k+1} - a_k| \geq \tau_0.$$

By (82),

$$|(B_C + \xi_k A_C)c_k| = |b_{k+1} - b_k + \xi_k(a_{k+1} - a_k)| \lesssim R.$$

Thus the smallest singular value of $B_C + \xi_k A_C$ is $O_{\tau_0}(R)$. The other two singular values are uniformly bounded on the fixed interval containing the ξ_k 's, proving (84).

Lagrange interpolation of the cubic P_{V_C} at the four nodes ξ_k gives

$$P_{V_C}(s) = \sum_{k=0}^3 P_{V_C}(\xi_k) \prod_{\ell \neq k} \frac{s - \xi_\ell}{\xi_k - \xi_\ell}.$$

All numerator factors are bounded on J . Since all pairwise distances are also bounded above, the lower bound on their total product implies

$$\prod_{\ell \neq k} |\xi_k - \xi_\ell| \gtrsim_I v$$

for each k . Insert (84) to obtain (85), and then apply Theorem 6.14. $\square$

Lemma 6.9 (Two-time common neighbors are bushes or angular packets). *Fix two code vertices p, q and a family Y of their common neighbors. For each $y \in Y$ choose collision labels s_y on py and t_y on qy , so that*

$$\begin{aligned} b_y - b_p + s_y(a_y - a_p) &= O(R), \\ b_y - b_q + t_y(a_y - a_q) &= O(R). \end{aligned}$$

Suppose all s_y, t_y lie in the union of two intervals I_1, I_2 of length η_s , whose distance is at least $\delta_s \geq 4\eta_s$. Then Y is the union of four pieces with the following disposition:

- (i) *if $s_y, t_y \in I_\alpha$, the corresponding graph points y lie in one $O(R + \eta_s)$ -approximate Lagrangian bush at the representative time of I_α ;*
- (ii) *if $s_y \in I_\alpha, t_y \in I_\beta, \alpha \neq \beta$, their horizontal directions lie in one ball of radius*

$$\boxed{C_{I,U} \frac{R + \eta_s}{\delta_s^2}.} \quad (86)$$

Consequently any positive common-neighbor mass carried by two separated Reeb-time packets has a fixed positive fraction in a physical near-bush or in a shrinking angular packet.

Proof. If both labels lie in I_α , with representative s_α , the first edge equation gives

$$b_y + s_\alpha a_y = b_p + s_\alpha a_p + O(R + \eta_s).$$

Thus all such y 's belong to the displayed bush. The second equation shows, whenever the piece is nonempty, that the bush center defined from q differs from the one defined from p by the same error.

For the off-diagonal pieces, subtract the two edge equations to obtain

$$(t_y - s_y)a_y = b_q - b_p + t_y a_q - s_y a_p + O(R). \quad (87)$$

Here $|t_y - s_y| \geq \delta_s/2$. Replacing s_y, t_y by the representatives s_α, t_β changes the rational expression on the right of (87), after division by $t_y - s_y$, by at most

$$C_{I,U}(R + \eta_s)\delta_s^{-2}.$$

Hence every a_y lies in the asserted ball about the value obtained from (s_α, t_β) . There are two diagonal and two off-diagonal packet pairs, completing the four-piece decomposition. $\square$

Lemma 6.10 (Anchored time-packet trichotomy). *Let $Y = N(p) \cap N(q)$ be a common-neighbor set of cardinality T , equipped with the two edge labels s_y, t_y from Theorem 6.9. Fix*

$$R \leq \delta_s \ll 1, \quad \tau_1 > 0.$$

At least one of the following holds, with constants depending only on the bounded coordinate region and on τ_1 :

- (i) $|a_p - a_q| < \tau_1$;
- (ii) *at least cT^2 pairs $(y, y') \in Y^2$ have three labels among*

$$s_y, t_y, s_{y'}, t_{y'}$$

which are pairwise $c\delta_s$ -separated;

- (iii) *there are two intervals I_1, I_2 of length $C\delta_s$ such that*

$$\#\{y \in Y : s_y, t_y \in I_1 \cup I_2\} \geq cT.$$

In (iii), Theorem 6.9 sends a fixed fraction of the retained neighbors to one near-bush or to at most two angular packets.

Proof. Assume (i) fails. Call two neighbors y, y' compatible when their four labels can be covered by two intervals of length $10\delta_s$. An elementary one-dimensional packing argument says that an incompatible pair contains three $c\delta_s$ -separated labels. Hence, if a fixed positive fraction of Y^2 is incompatible, (ii) holds.

Otherwise discard at most a fixed small fraction of Y so that every remaining neighbor is compatible with a fixed positive fraction of the others. If one such y_0 has

$$|s_{y_0} - t_{y_0}| > 20\delta_s,$$

then compatibility with y_0 forces both labels of every compatible neighbor into the two $O(\delta_s)$ -intervals about s_{y_0} and t_{y_0} . This gives (iii).

It remains to consider the neighbors with $|s_y - t_y| \leq 20\delta_s$. Subtracting their two contact-edge equations as in (87), and putting $r_y = (s_y + t_y)/2$, gives

$$|b_q - b_p + r_y(a_q - a_p)| \lesssim R + \delta_s.$$

Since $|a_p - a_q| \geq \tau_1$, all such r_y 's lie in one interval of length $C_{\tau_1}(R + \delta_s) \lesssim_{\tau_1} \delta_s$. Thus their two labels lie in one $C_{\tau_1}\delta_s$ -interval. If they form a fixed fraction of Y , (iii) holds with the second interval empty. Otherwise the preceding off-diagonal argument applies to a fixed fraction of the remaining neighbors and again gives (iii). $\square$

Lemma 6.11 (Nondegenerate cycle density concentrates in one bush cell). *Fix $0 < \Delta \leq 1$, and let $\mathcal{Z}_\Delta$ be any family of code four-cycles satisfying the determinant lower bound in Theorem 6.7. Then there are $s_Q \in I + O(R)$, $c_Q \in \mathbb{R}^3$, and a set $\mathcal{W}_Q \subset \mathcal{V}_R$ such that every representative $z_v = (a_v, b_v)$, $v \in \mathcal{W}_Q$, satisfies*

$$|b_v + s_Q a_v - c_Q| \lesssim R/\Delta,$$

and

$$|\mathcal{W}_Q| \gtrsim |\mathcal{Z}_\Delta|^{1/4} \frac{R}{\Delta}. \quad (88)$$

Consequently, if the nondegenerate cycles form a fixed proportion of the output in (81), then

$$|\mathcal{W}_Q| \gtrsim AN_R^{1/2} \frac{R}{\Delta}. \quad (89)$$

Proof. The relevant (s, c) -parameters stay in a fixed bounded subset of $\mathbb{R}^4$. Cover that set by half-open cells of side comparable to $r = R/\Delta$. There are $O(r^{-4})$ cells. Assign each cycle in $\mathcal{Z}_\Delta$ to a cell containing one bush parameter supplied by Theorem 6.7. One cell Q receives at least

$$c|\mathcal{Z}_\Delta|r^4$$

cycles. All vertices of every assigned cycle satisfy the enlarged bush inequality for the center of Q . If their union has T vertices, it supports at most T^4 ordered four-cycles. Thus

$$T^4 \gtrsim |\mathcal{Z}_\Delta|r^4,$$

which proves (88). Insert (81) for the last assertion. $\square$

Corollary 6.12 (Bush-cell aggregation retains comparable edge mass). *Apply Theorem 6.11 inside one graph $\mathcal{G}_{L,\lambda}$. If a bush cell Q receives Z_Q four-cycles, and $\mathcal{E}_Q$ denotes the union of their code edges, then*

$$\boxed{\sum_{e \in \mathcal{E}_Q} \omega_L(e) \gtrsim \lambda R^4 Z_Q^{1/2}.} \quad (90)$$

In particular, if

$$|E(\mathcal{G}_{L,\lambda})| \geq AN_R^{3/2}$$

and a fixed proportion of the resulting four-cycles are Δ -nondegenerate, some $O(R/\Delta)$ -Lagrangian bush cell carries incidence mass at least

$$\boxed{c\lambda A^2 R^4 N_R (R/\Delta)^2.} \quad (91)$$

Since $N_R \gtrsim R^{-4}$, the last quantity is

$$\gtrsim \lambda A^2 R^2 \Delta^{-2}.$$

Proof. A simple four-cycle is determined, up to a fixed number of choices, by two of its opposite edges. Hence $Z_Q \lesssim |\mathcal{E}_Q|^2$, so

$$|\mathcal{E}_Q| \gtrsim Z_Q^{1/2}.$$

Every edge in the dyadic layer has mass at least λR^4 , proving (90). The proof of Theorem 6.11, combined with Theorem 6.6, gives

$$Z_Q \gtrsim A^4 N_R^2 (R/\Delta)^4,$$

which yields (91).

Finally, the projection of G_b onto U is all of U , so at least $c_U R^{-3}$ direction cubes contain an occupied graph product cell. The definition of $\mathcal{V}_R$ pairs every such graph cell with cR^{-1} level intervals. Thus $N_R \gtrsim R^{-4}$. $\square$

Proposition 6.13 (Double-cell de-multiplication produces a physical bush). *Let Q receive Z_Q nondegenerate four-cycles from one dyadic edge layer $\mathcal{G}_{L,\lambda}$, and let $\mathcal{E}_Q$ be the union of their code edges. Then there is a physical $O(R/\Delta)$ -bush H_Q satisfying*

$$\boxed{\sigma(H_Q) \gtrsim \lambda R^4 Z_Q^{1/4}.} \quad (92)$$

If Q is the cell supplied by power supersaturation, then

$$\boxed{\sigma(H_Q) \gtrsim \lambda A R^4 N_R^{1/2} \frac{R}{\Delta} \gtrsim \lambda A R^3 \Delta^{-1}.} \quad (93)$$

The same quantities are lower bounds, up to fixed constants, for $\mathcal{L}^4(N_{CR/\Delta}(K_b))$.

Proof. Let $\mathcal{X}_Q$ be the set of selector graph product cells χ occurring in endpoint codes incident to $\mathcal{E}_Q$, and put $D_Q = |\mathcal{X}_Q|$. Each χ supports only $O(R^{-1})$ endpoint codes (χ, K) . Since a simple graph on $O(D_Q R^{-1})$ vertices has at most $O(D_Q^2 R^{-2})$ edges,

$$D_Q \gtrsim R|\mathcal{E}_Q|^{1/2} \gtrsim RZ_Q^{1/4},$$

where the last inequality is the first step in the proof of Theorem 6.12.

Now fix $e \in \mathcal{E}_Q$ incident to a graph cell χ . The double code restricts its collision time to an interval of length $O(R)$, and contains only $O(1)$ physical edges. Therefore its lower mass bound

$$\omega_L(e) \geq \lambda R^4$$

forces the set of source directions in χ that actually support this code to have σ -mass at least $c\lambda R^3$. On the outer wing the source–output homothety Jacobian is bounded above and below, so the same conclusion holds at either endpoint of e .

Choose, for each $\chi \in \mathcal{X}_Q$, one incident edge and the corresponding positive-mass direction subset. Distinct half-open graph product cells give disjoint subsets of the selector graph, hence disjoint direction subsets because G_b is a graph over U . All their graph points are within $O(R/\Delta)$ of the affine Lagrangian leaf

$$b + s_Q a = c_Q$$

assigned to Q . Their union is therefore contained in one physical $O(R/\Delta)$ -bush and has mass at least

$$c\lambda R^3 D_Q \gtrsim \lambda R^4 Z_Q^{1/4},$$

proving (92). Insert the lower bound for Z_Q from the proof of Theorem 6.11 and use $N_R \gtrsim R^{-4}$ to obtain (93). The volume assertion is Theorem 4.17. $\square$

6.3 Rank loss, interpolation, and packet reuse

Lemma 6.14 (Uniform separation from total Maslov trapping). *For every nontrivial compact interval $J \subset \mathbb{R}$, there is $C_J < \infty$ such that*

$$|\det A| + |P_V(i)| \leq C_J \sup_{s \in J} |P_V(s)|. \quad (94)$$

Consequently, if $\sup_{s \in J} |P_V(s)| \leq \varepsilon$, then

$$|\det A| \lesssim_J \varepsilon, \quad \|\omega|_V\|_{\text{op}} \geq \sqrt{1 - C_J^2 \varepsilon^2}. \quad (95)$$

Moreover the distance of V , in the Grassmannian, from the front-null locus

$$\mathfrak{N} = \{W : P_W \equiv 0\} = \mathbf{K}_{101} \cup \mathbf{K}_{110} \cup \mathbf{K}_{200}$$

tends uniformly to zero with ε .

Proof. On the four-dimensional vector space of real polynomials of degree at most three, coefficient norm, evaluation at i , and the supremum norm on J are equivalent. The leading coefficient of P_V is $\det A$, proving (94). The complex-evaluation identity (3) gives

$$1 - \|\omega|_V\|_{\text{op}}^2 = |P_V(i)|^2,$$

and hence (95). For the last assertion, otherwise there would be $\varepsilon_n \downarrow 0$ and planes V_n staying a fixed distance from $\mathfrak{N}$. Compactness of the Grassmannian gives $V_n \rightarrow V$; continuity yields $P_V|_J = 0$, hence $P_V \equiv 0$, a contradiction. The equality with the three strata is the classification of the preceding flag construction. $\square$

Lemma 6.15 (Rank-adapted Maslov interpolation). *Let V have orthonormal frame matrices A, B , let $r \in \{0, 1, 2, 3\}$, and suppose*

$$\text{dist}(A, \{A_0 : \text{rank } A_0 \leq r\}) \leq \delta_A.$$

Let $s_0, \dots, s_r$ lie in a fixed bounded interval and satisfy

$$\prod_{0 \leq i < j \leq r} |s_i - s_j| \geq v$$

(with the empty product equal to one). If

$$|P_V(s_j)| \leq \epsilon \quad (0 \leq j \leq r),$$

then

$$\sup_{s \in J} |P_V(s)| \lesssim_{I,J} \frac{\epsilon + \delta_A}{v}. \quad (96)$$

Proof. Choose A_0 of rank at most r with $\|A - A_0\| \leq 2\delta_A$, and put

$$Q(s) = \det(B + sA_0).$$

Multilinearity of the determinant shows that Q has degree at most r , while its local Lipschitz dependence on the matrix entries gives

$$\sup_{s \in I \cup J} |P_V(s) - Q(s)| \lesssim_{I,J} \delta_A.$$

Hence $|Q(s_j)| \lesssim \epsilon + \delta_A$. Lagrange interpolation of the degree- r polynomial Q at $s_0, \dots, s_r$, followed by the boundedness of all numerator factors, gives

$$\sup_J |Q| \lesssim_{I,J} (\epsilon + \delta_A) v^{-1}.$$

Add the preceding approximation error. $\square$

Corollary 6.16 (Three-time trapping closes a horizontally planar code cycle). *Use the notation of Theorem 6.8 and suppose*

$$\mathfrak{g}_C := |d_0 \wedge d_1 \wedge d_2| \geq \gamma, \quad |\det(a_1 - a_0, a_2 - a_1, a_3 - a_2)| \leq \Delta.$$

If three of the four collision labels, say $\xi_{j_0}, \xi_{j_1}, \xi_{j_2}$, satisfy

$$\prod_{0 \leq p < q \leq 2} |\xi_{j_p} - \xi_{j_q}| \geq v,$$

then

$$\boxed{\sup_{s \in J} |P_{V_C}(s)| \lesssim \frac{R + (\Delta/\gamma)^{1/3}}{v}.} \quad (97)$$

Thus, provided the right side tends to zero, the cycle plane converges to the front-null, maximally symplectic Kähler locus. If $\mathfrak{g}_C < \gamma$, the full secant triple itself is quantitatively rank deficient.

Proof. Let C be the coordinate matrix determined by

$$[d_0 \ d_1 \ d_2] = T_C C.$$

Because T_C is orthonormal, $|\det C| = \mathfrak{g}_C$. Taking horizontal components gives

$$[a_1 - a_0, a_2 - a_1, a_3 - a_2] = A_C C,$$

and hence

$$|\det A_C| \leq \Delta/\gamma.$$

The distance from a 3×3 matrix to the rank-two locus is its smallest singular value, which is at most $|\det A_C|^{1/3}$. Use (84) at the three selected labels and apply Theorem 6.15 with $r = 2$. $\square$

Lemma 6.17 (A shared two-edge Maslov flag fixes the horizontal plane). *Let $u, v \in \mathbb{R}^3$ be linearly independent, let $s, t \in \mathbb{R}$, and put*

$$W = \text{Span}\{(u, -su), (v, -tv)\} \subset \mathbb{R}_a^3 \oplus \mathbb{R}_b^3.$$

If a three-plane $V \supset W$ is front-null, then

$$\boxed{V \subset E \oplus E, \quad E = \text{Span}\{u, v\} \subset \mathbb{R}^3.} \quad (98)$$

Equivalently, once two horizontally independent contact edges are fixed, every front-null three-dimensional extension has the same horizontal and vertical polarization plane.

Proof. Choose a third vector (w, z) so that

$$V = \text{Span}\{(u, -su), (v, -tv), (w, z)\}.$$

In this, not necessarily orthonormal, frame the Reeb-front determinant is a nonzero constant multiple of

$$\det((r-s)u, (r-t)v, z+rw) = (r-s)(r-t) [\det(u, v, z) + r \det(u, v, w)].$$

Front-nullity says that this polynomial vanishes identically. Hence

$$\det(u, v, z) = \det(u, v, w) = 0,$$

so $w, z \in E$. The first two spanning vectors already belong to $E \oplus E$, proving (98). $\square$

Corollary 6.18 (Quantitative coherence of anchored cycle planes). *Suppose a three-plane V contains unit-scale vectors*

$$d_1 = (u, \beta_1), \quad d_2 = (v, \beta_2)$$

such that

$$|u \times v| \geq \kappa, \quad |\beta_1 + su| + |\beta_2 + tv| \leq \varepsilon, \quad \sup_{r \in J} |P_V(r)| \leq \varepsilon.$$

Then

$$\boxed{\text{dist}_{\text{Gr}}(V, \{V' \subset E \oplus E : \dim V' = 3\}) \leq C_{I,J} \frac{\varepsilon}{\kappa^2}, \quad E = \text{Span}\{u, v\}.} \quad (99)$$

Thus all near-front-null cycle planes sharing the same well-conditioned two-edge anchor are quantitatively concentrated near one fixed polarized four-space $E \oplus E$; their horizontal directions lie near the same affine two-plane.

Proof. Choose a unit vector $d_3 = (w, z) \in V$ orthogonal to $\text{Span}\{d_1, d_2\}$. The three vectors d_1, d_2, d_3 have volume at least $\kappa\kappa$, since their first two horizontal components have cross product at least κ . Put

$$Q(r) = \det(\beta_1 + ru, \beta_2 + rv, z + rw).$$

Changing from this bounded basis to an orthonormal frame of V gives

$$\sup_{r \in J} |Q(r)| \lesssim \varepsilon.$$

Write $e_1 = \beta_1 + su$ and $e_2 = \beta_2 + tv$. Multilinearity and the bounds on all data give

$$\sup_{r \in J} |(r-s)(r-t)[\det(u, v, z) + r \det(u, v, w)]| \lesssim \varepsilon.$$

On the finite-dimensional space of polynomials of degree at most three, multiplication of a linear polynomial by the monic quadratic $(r-s)(r-t)$ has a uniformly bounded inverse on its image when s, t range in the fixed bounded interval. Hence

$$|\det(u, v, z)| + |\det(u, v, w)| \lesssim_{I,J} \varepsilon.$$

The distance of a vector $x \in \mathbb{R}^3$ from E is $|\det(u, v, x)|/|u \times v|$, so both w and z lie within $C_{I,J}\varepsilon/\kappa$ of E . The first two spanning vectors lie within ε of $E \oplus E$. Finally the condition number of $[d_1 \ d_2 \ d_3]$ is $O(\kappa^{-1})$, giving the Grassmannian bound $C_{I,J}\varepsilon/\kappa^2$. $\square$

Proposition 6.19 (Anchored determinant-small cycles form one geometric packet). *Let $Y = N(p) \cap N(q)$ have cardinality T , assume*

$$|a_p - a_q| \geq \tau_1 > 0,$$

and suppose all edges from p, q to Y belong to one weight layer with lower weight λR^4 . Let $\emptyset \neq \Gamma \subset \binom{Y}{2}$, put

$$Z_\Gamma := |\Gamma|, \quad m_\Gamma := \max\{1, Z_\Gamma/T\},$$

For every $\{y, y'\} \in \Gamma$, suppose the contact cycle

$$p, y, q, y', p$$

has horizontal determinant at most Δ , and three of its four collision labels have Vandermonde product at least v . For such a pair put

$$\mathfrak{g}(y, y') = |(z_y - z_p) \wedge (z_q - z_y) \wedge (z_{y'} - z_q)|.$$

For every $0 < \gamma, \kappa < 1$, there are $y_0 \in Y$ and $Y_0 \subset Y \setminus \{y_0\}$, with

$$|Y_0| \gtrsim m_\Gamma,$$

for which at least one of the following holds:

(i) *with*

$$F = \text{Span}\{z_{y_0} - z_p, z_q - z_{y_0}\} \subset \mathbb{R}^6,$$

one has $\dim F = 2$, $\dim \pi_a(F) \leq 2$, and every $y \in Y_0$ belongs to the same full-secant planar packet

$$\text{dist}(z_y, z_p + F) \lesssim \gamma/\kappa; \quad (100)$$

(ii) every $y \in Y_0$ belongs to the horizontal line packet

$$\text{dist}(a_y, \text{aff}\{a_p, a_q\}) \lesssim \kappa/\tau_1; \quad (101)$$

(iii) putting

$$E = \text{Span}\{a_{y_0} - a_p, a_q - a_{y_0}\},$$

one has

$$|(a_{y_0} - a_p) \times (a_q - a_p)| \geq \kappa,$$

and, for every $y \in Y_0$, the cycle secant plane $V_{y_0, y}$ satisfies

$$\boxed{\begin{aligned} \sup_{s \in J} |P_{V_{y_0, y}}(s)| &\lesssim \varepsilon_{\Delta, \gamma, v}, \\ \text{dist}_{\text{Gr}}(V_{y_0, y}, \{V' \subset E \oplus E : \dim V' = 3\}) &\leq C\varepsilon_{\Delta, \gamma, v}/\kappa^2, \\ \text{dist}(a_y, a_p + E) &\lesssim \varepsilon_{\Delta, \gamma, v}/\kappa^2, \end{aligned}} \quad (102)$$

where

$$\varepsilon_{\Delta, \gamma, v} = R + \frac{R + (\Delta/\gamma)^{1/3}}{v}. \quad (103)$$

In every case the retained endpoint codes de-multiply to an actual selector-direction subset H_{Y_0} of mass

$$\boxed{\sigma(H_{Y_0}) \gtrsim \lambda R^4 m_\Gamma.} \quad (104)$$

In every case this subset lies in the corresponding line or plane packet enlarged by $O(R)$.

Proof. Suppose first that at least a fixed fraction of the edges of Γ have $\mathfrak{g}(y, y') < \gamma$; denote that subgraph by Γ_{rk} . Put

$$Y_{\text{full-line}} = \{y \in Y : |(z_y - z_p) \wedge (z_q - z_y)| < \kappa\}.$$

If $|Y_{\text{full-line}}| \gtrsim m_\Gamma$, then, since $|z_q - z_p| \geq |a_q - a_p| \geq \tau_1$, all these graph points lie within $C\kappa/\tau_1$ of the affine line through z_p, z_q . Horizontal projection gives (ii). Otherwise deletion of $Y_{\text{full-line}}$ removes fewer than half the edges of Γ_{rk} . The remaining graph has a vertex y_0 with $\gtrsim m_\Gamma$ neighbors Y_0 . Its two base secants have wedge norm at least κ ; dividing the triple-volume inequality $\mathfrak{g}(y_0, y) < \gamma$ by that base area gives

$$\text{dist}(z_y, z_p + F) \lesssim \gamma/\kappa, \quad F = \text{Span}\{z_{y_0} - z_p, z_q - z_{y_0}\}.$$

This is (i).

Otherwise a subgraph $\Gamma_{\text{fr}} \subset \Gamma$, with $|\Gamma_{\text{fr}}| \gtrsim Z_\Gamma$, has full secant volume at least γ . Put

$$Y_{\text{line}} = \{y \in Y : |(a_y - a_p) \times (a_q - a_p)| < \kappa\}.$$

If $|Y_{\text{line}}| \gtrsim m_\Gamma$, the elementary point-to-line formula and $|a_q - a_p| \geq \tau_1$ give (ii). Otherwise, choosing the implicit small constant in this last alternative absolutely small, deletion of Y_{line} removes fewer than half the edges of Γ_{fr} , because it removes at most $|Y_{\text{line}}|T$ edges. The remaining graph has a vertex $y_0 \notin Y_{\text{line}}$ with $\gtrsim m_\Gamma$ neighbors (using one retained edge when $Z_\Gamma < T$); call that neighbor set Y_0 .

For every $y \in Y_0$, the cycle p, y_0, q, y, p has full secant volume at least γ , horizontal determinant at most Δ , and three collision labels with Vandermonde at least v . Therefore Theorem 6.16 gives the first line of (102). All these cycle planes contain the two fixed contact secants

$$z_{y_0} - z_p, \quad z_q - z_{y_0},$$

whose horizontal parts have cross product at least κ ; their contact errors are $O(R)$ by (82). Apply Theorem 6.18, absorbing the bounded normalization constants into its modulus, to obtain the second line. Since $z_y - z_q \in V_{y_0, y}$, horizontal projection gives the third line. This proves (iii).

It remains to verify that the packet has physical mass. Let $\mathcal{X}_0$ be the set of selector graph product cells underlying the endpoint codes in Y_0 . A fixed graph cell supports only $O(R^{-1})$ level codes, hence

$$|\mathcal{X}_0| \gtrsim R|Y_0| \gtrsim Rm_\Gamma.$$

For each $\chi \in \mathcal{X}_0$, choose one code y over χ and one of the weighted edges py or qy . Exactly as in Theorem 6.13, its weight lower bound λR^4 forces at least $c\lambda R^3$ selector-direction mass inside χ . Distinct half-open graph cells are disjoint on G_b , so summation proves (104). The graph cell diameter accounts for the final $O(R)$ enlargement. $\square$

Proposition 6.20 (Exact graph-cell reuse ledger for anchored Maslov packets). *Let a family of anchored packets in one λ -weight layer contain a total of S retained endpoint-code occurrences. For a packet Q , let $\mathcal{X}_Q$ be the distinct selector graph cells supporting its retained endpoint codes. Define*

$$r(\chi) = \#\{Q : \chi \in \mathcal{X}_Q\}, \quad \mathcal{R}_{\text{anc}} = \sum_{\chi} r(\chi)^2.$$

Then

$$\sum_{\chi} r(\chi) \gtrsim RS, \quad (105)$$

and the union H_{anc} of the corresponding actual selector-direction subsets obeys

$$\sigma(H_{\text{anc}}) \gtrsim \lambda R^5 \frac{S^2}{\mathcal{R}_{\text{anc}}}. \quad (106)$$

More explicitly, for every $K \geq 1$, at least one of the following holds:

(i) the cells with $r(\chi) \leq K$ carry a selector-direction union of mass

$$\sigma(H_{\text{anc}}^{\text{low}}) \gtrsim \lambda R^4 S / K; \quad (107)$$

(ii) some endpoint code belongs to at least cRK distinct anchored packets, and is incident to a code star which produces a genuine physical star of degree

$$D_{\text{phys}} \gtrsim R^{3/2} K^{1/2}. \quad (108)$$

Thus packet reuse is not a loss of information: its high part returns to a higher-order common contact star at one endpoint code.

Proof. A selector graph cell supports only $O(R^{-1})$ level codes. Thinning separately inside each packet therefore gives

$$|\mathcal{X}_Q| \gtrsim R |Y_Q|.$$

Summation proves (105). Each occupied graph cell contains, by the same weighted double-cell de-multiplication as in Theorem 6.13, an actual direction subset of mass at least $c\lambda R^3$. Distinct cells are disjoint on the selector graph. Cauchy–Schwarz gives

$$\#\{\chi : r(\chi) > 0\} \geq \frac{(\sum_{\chi} r(\chi))^2}{\sum_{\chi} r(\chi)^2},$$

which proves (106).

If at least half of $\sum r(\chi)$ lies on $r(\chi) \leq K$, those cells number at least cRS/K , and the preceding $c\lambda R^3$ mass per cell proves (107). Otherwise some cell has $r(\chi) > K$. At most $O(R^{-1})$ endpoint codes lie over that graph cell, so one endpoint code y occurs in at least cRK of the anchored packets. Their anchors are distinct unordered pairs $\{p, q\}$ with $p, q \in N(y)$; hence

$$\binom{\deg(y)}{2} \gtrsim RK, \quad \deg(y) \gtrsim (RK)^{1/2}.$$

Apply (64) to obtain (108). □

6.4 Front seeds and Reeb-time completion

Lemma 6.21 (Four-cluster Reeb-front seed). *Fix a compact interval J with nonempty interior and a bounded set $E \subset \mathbb{R}_a^3 \oplus \mathbb{R}_b^3$. For four points*

$$x_j = (a_j, b_j) \in E, \quad 0 \leq j \leq 3,$$

put

$$A_{\Delta} = [a_1 - a_0, a_2 - a_0, a_3 - a_0], \quad B_{\Delta} = [b_1 - b_0, b_2 - b_0, b_3 - b_0].$$

If $|\det A_{\Delta}| \geq \kappa > 0$, then

$$\sup_{s \in J} |\det [\Pi_s(x_1) - \Pi_s(x_0), \Pi_s(x_2) - \Pi_s(x_0), \Pi_s(x_3) - \Pi_s(x_0)]| \geq c_J \kappa. \quad (109)$$

There are constants $c_J > 0$ and $c'_J > 0$, independent of E and κ , such that

$$\left| \left\{ s \in J : \left| \det \left[\Pi_s(x_1) - \Pi_s(x_0), \Pi_s(x_2) - \Pi_s(x_0), \Pi_s(x_3) - \Pi_s(x_0) \right] \right| \geq c_J \kappa \right\} \right| \geq c'_J. \quad (110)$$

If a probability measure ν gives mass at least m to each of four sufficiently small balls of radius $r \leq c_{J,E} \kappa$ about the x_j , then for every such good s , the projected measure $\Pi_{s\#}\nu$ gives mass at least m to four disjoint output balls whose centers form a tetrahedron with smallest altitude at least $c_{J,E} \kappa$. Thus every horizontally nondegenerate four-cluster produces a definite coarse projection-entropy gain on a longitudinal set of fixed positive measure.

Proof. The determinant in (109) is

$$P_\Delta(s) = \det(B_\Delta + sA_\Delta),$$

a cubic polynomial whose leading coefficient is $\det A_\Delta$. Equivalence of polynomial norms on J gives

$$\|P_\Delta\|_{L^\infty(J)} \geq c_J \kappa.$$

Apply the degree-three Remez inequality. If

$$\left| \{s \in J : |P_\Delta(s)| < c_J^* \kappa\} \right| > |J| - c'_J$$

for a sufficiently small fixed $c_J^* > 0$, Remez would bound the supremum on all of J by $C_J c_J^* \kappa < c_J \kappa$, a contradiction. Choose $c'_J > 0$ first, then choose c_J^* accordingly and rename it as the constant in (110). This is stronger than the derivative estimate: no factor κ is lost in the length of the good-time set.

On that set the three projected difference vectors have determinant at least $c_J \kappa$ and, because E and J are bounded, norm at most $C_{J,E}$. The smallest altitude of the tetrahedron is therefore at least $c_{J,E} \kappa$. Input balls of radius $r \leq c_{J,E} \kappa$ project into disjoint output balls about the four vertices and retain their input mass. $\square$

Lemma 6.22 (Sticky graph mass contains a quantitative Reeb-front simplex). *Assume the quantitative packing-piece estimate*

$$N_r(G_b) \leq C_\varepsilon r^{-3-\varepsilon}.$$

Let $H \subset U$ be measurable with $\sigma(H) = m > 0$, and let

$$0 < r \leq c_U m.$$

Then there are graph points

$$z_j = (a_j, b(a_j)), \quad 0 \leq j \leq 3,$$

and graph product cells χ_j of diameter $O(r)$, containing z_j , such that

$$\left| \det(a_1 - a_0, a_2 - a_0, a_3 - a_0) \right| \gtrsim_U m, \quad \mu_H(\chi_j) \gtrsim_{\varepsilon, U} m r^{3+\varepsilon}, \quad (111)$$

where $\mu_H = (\text{id}, b)_\#(\sigma|_H)$. Consequently there is a set $G_H \subset J$ with $|G_H| \gtrsim_J 1$ such that, for every $s \in G_H$, the projected measure $\Pi_{s\#}\mu_H$ has four $O(r)$ -clusters, each of mass $\gtrsim_{\varepsilon, U} m r^{3+\varepsilon}$, whose centers form a tetrahedron of determinant $\gtrsim_U m$.

Proof. Use the half-open product grid of side r from Section 5. There are at most $C_\varepsilon r^{-3-\varepsilon}$ cells meeting G_b . Call a cell heavy when

$$\mu_H(\chi) \geq c_\varepsilon m r^{3+\varepsilon},$$

where c_ε is chosen so that the union of the nonheavy cells has μ_H -mass at most $m/2$. Let H_{hv} be the horizontal preimage of the heavy cells; then

$$\sigma(H_{\text{hv}}) \geq m/2.$$

We use the elementary maximal-simplex bound: every bounded measurable $F \subset \mathbb{R}^3$ of Lebesgue measure v contains four points whose tetrahedron determinant is at least $c_U v$. Indeed choose successively a diameter, a point of maximal distance from its line, and a point of maximal distance from the resulting plane. Up to a fixed enlargement, the set is contained in a box whose three side lengths are those successive widths; its volume is bounded by their product, which is comparable to the selected determinant.

Apply this bound to H_{hv} , recalling that σ is normalized Lebesgue measure on U . It gives four points with horizontal determinant at least $c_U m$. Moving each point by $O(r)$ changes that determinant by $O_U(r)$, so the choice $r \leq c_U m$ keeps the lower bound after replacing the points by representatives of their heavy graph cells. The four cells are necessarily distinct. Their defining heaviness gives the second bound in (111).

Apply Theorem 6.21 with horizontal tetrahedron constant $c_U m$. Its good-parameter set has measure at least $c'_J > 0$; product cells of diameter $O(r)$ are contained in $O(r)$ -balls, and every one retains its μ_H -mass under projection. This proves the final assertion. $\square$

Corollary 6.23 (Uniform longitudinal persistence of shrinking simplices). *Let $R_n \downarrow 0$. At the n -th scale suppose Theorem 6.21 applies with an arbitrary $\kappa_n > 0$, and choose the input cluster radius at most $c_{J,E}\kappa_n$. If $G_n \subset J$ is the corresponding set of good longitudinal parameters, then*

$$\left| \limsup_{n \rightarrow \infty} G_n \right| \geq c'_J.$$

Thus a fixed positive-measure set of Reeb parameters sees quantitatively nondegenerate projection seeds on infinitely many scales even when $\kappa_n \downarrow 0$.

Proof. The preceding lemma gives $|G_n| \geq c'_J$. Apply reverse Fatou to the bounded functions $\mathbf{1}_{G_n}$:

$$\int_J \limsup_n \mathbf{1}_{G_n} \geq \limsup_n \int_J \mathbf{1}_{G_n}.$$

□

Lemma 6.24 (Reeb-time integration lifts transverse Frostman measures). *Let*

$$K_s := \{\Pi_s(a, b(a)) : a \in U\}, \quad K := \{(y, s) : s \in J, y \in K_s\}.$$

Suppose $G \subset J$ has positive Lebesgue measure and, for almost every $s \in G$, there is a weakly measurable probability kernel $s \mapsto \nu_s$ carried by K_s such that, uniformly in s ,

$$\nu_s(B(y, \rho)) \leq C_\varepsilon \rho^{3-\varepsilon} \quad (y \in \mathbb{R}^3, 0 < \rho < 1). \quad (112)$$

Then the measure

$$\nu(A) := \frac{1}{|G|} \int_G \nu_s\{y : (y, s) \in A\} ds \quad (113)$$

is carried by K and satisfies

$$\boxed{\nu(B((y_0, s_0), \rho)) \leq \frac{2C_\varepsilon}{|G|} \rho^{4-\varepsilon}.} \quad (114)$$

Consequently $\dim_H K \geq 4 - \varepsilon$.

Proof. Only parameters $s \in G \cap (s_0 - \rho, s_0 + \rho)$ contribute to the ball on the left of (114). For each such parameter its transverse section lies in $B(y_0, \rho)$, so (112) bounds the conditional mass by $C_\varepsilon \rho^{3-\varepsilon}$. Integration over an interval of length at most 2ρ proves (114). The mass distribution principle gives the final claim. □

Corollary 6.25 (Exact remaining contact–symplectic front endpoint). *In both the fixed-angle low-reuse branch and the angular Maslov branch, Remez persistence supplies a set $G_\infty \subset J$ of fixed positive measure on which front seeds occur at arbitrarily small scales. To obtain $\dim_H K = 4$, it is therefore enough to upgrade the persistent cluster measures, on one positive-measure subset of G_∞ , to the transverse kernels (112) for every $\varepsilon > 0$. Once that purely spatial upgrade is proved, Theorem 6.24 performs the longitudinal and Hausdorff passage with no further loss.*

Definition 6.26 (Weighted Maslov residual content). *For $0 < \rho < 1$, define*

$$\boxed{\mathfrak{Z}_\rho^{J^+}(\mu) := \iint_{a \neq a'} \frac{\mathbf{1}_{\{s_*(a, a') \in J^+\}} \mathbf{1}_{\{|r(a, a')| \leq \rho\}}}{|a - a'|} d\sigma(a) d\sigma(a').} \quad (115)$$

Theorem 6.27 (Codimension-two Maslov residual criterion). *Assume that for some $0 < \eta < 2$,*

$$\boxed{\mathfrak{Z}_\rho^{J^+}(\mu) \leq C_\eta \rho^{2-\eta} \quad (0 < \rho < 1).} \quad (116)$$

Then, for every

$$1 < t < 3 - \eta,$$

the projected measures $\nu_s = (\Pi_s)_\# \mu$, with $I_t(\nu_s) := \iint |y - y'|^{-t} d\nu_s(y) d\nu_s(y')$, satisfy

$$\boxed{\int_J I_t(\nu_s) ds = \int_J \iint |\Pi_s(x) - \Pi_s(x')|^{-t} d\mu(x) d\mu(x') ds < \infty.} \quad (117)$$

Consequently

$$\dim_H K \geq 4 - \eta. \quad (118)$$

If (116) holds with exponent $2 - \eta$ for every $\eta > 0$, then $\dim_H K = 4$.

Proof. The Pythagorean collision identity gives

$$|\beta + s\alpha|^2 = |r|^2 + |\alpha|^2 |s - s_*|^2. \quad (119)$$

Let $d_0 = \text{dist}(J, \mathbb{R} \setminus J^+) > 0$. If $s_* \notin J^+$, then for $s \in J$,

$$|\beta + s\alpha| \geq d_0 |\alpha|,$$

so this part of the averaged energy is bounded by a constant multiple of

$$\iint |a - a'|^{-t} d\sigma(a) d\sigma(a'),$$

which is finite for $t < 3$ because the direction marginal is bounded Lebesgue measure in three dimensions.

If $s_* \in J^+$, extending the s -integral to $\mathbb{R}$ in (119) gives, for $r \neq 0$,

$$\int_J |\beta + s\alpha|^{-t} ds \leq C_t |\alpha|^{-1} |r|^{1-t}. \quad (120)$$

The hypothesis (116) forces the weighted pair measure of $\{s_* \in J^+, r = 0\}$ to vanish. On $0 < |r| \leq 1$, dyadic summation gives

$$\begin{aligned} & \iint_{\substack{s_* \in J^+ \\ 0 < |r| \leq 1}} |\alpha|^{-1} |r|^{1-t} d\sigma d\sigma' \\ & \lesssim_t \sum_{k \geq 0} 2^{(k+1)(t-1)} \mathfrak{Z}_{2^{-k}}^{J^+}(\mu) \\ & \lesssim_{t,\eta} \sum_{k \geq 0} 2^{-k(3-\eta-t)} < \infty. \end{aligned}$$

The region $|r| > 1$ is harmless. Tonelli's theorem and (120) prove (117).

Hence $I_t(\nu_s) < \infty$ for almost every $s \in J$, so $\dim_H K_s \geq t$ for almost every such s . The standard slicing theorem for the analytic Reeb front gives $\dim_H K \geq t + 1$. Letting $t \uparrow 3 - \eta$ proves (118); then let $\eta \downarrow 0$ for the final assertion. $\square$

7 The codimension-two residual and polarized routing

The energy, slicing, and Frostman principles are used in their standard geometric-measure form; see [8, 15].

7.1 Residual criteria and shell return

Proposition 7.1 (Collision energy is the discretized residual content). *Let $\mathfrak{E}_{\rho, \geq \tau}^L(\mu)$ denote Theorem 4.15 with the pair integral restricted to $|a - a'| \geq \tau$ and time interval L . Likewise append $\geq \tau$ to $\mathfrak{Z}$ when its pair integral is so restricted. There is $c = c(J^-, J, J^+) > 0$ such that, whenever $0 < \rho \leq c\tau$,*

$$c\rho^{-2}\mathfrak{Z}_{\rho/2, \geq \tau}^{J^-}(\mu) \leq \mathfrak{E}_{\rho, \geq \tau}^J(\mu) \leq C\rho^{-2}\mathfrak{Z}_{\rho, \geq \tau}^{J^+}(\mu). \quad (121)$$

Thus the codimension-two target $\mathfrak{Z}_\rho \lesssim \rho^{2-o(1)}$ is exactly the subpower collision-energy target $\mathfrak{E}_\rho \lesssim \rho^{-o(1)}$ on every fixed-angle shell.

Proof. If $|r| \leq \rho/2$, $s_* \in J^-$, and $|\alpha| \geq \tau$, then $\rho \leq c\tau$ and the separation of J^- from ∂J imply that the collision window in Theorem 4.14 lies inside J and has length at least $c\rho/|\alpha|$. Multiplication by the normalization ρ^{-3} proves the left inequality.

Conversely, a nonempty ρ -collision window forces $|r| \leq \rho$ and places s_* within $\rho/|\alpha| \leq \rho/\tau$ of J , hence in J^+ . The universal window bound Theorem 4.16 gives length at most $2\rho/|\alpha|$, proving the right inequality. $\square$

Lemma 7.2 (Exact angular-shell ledger for the residual content). *For a dyadic shell write*

$$\mathfrak{Z}_{\rho, \tau}^{J^+} := \iint_{\tau \leq |a - a'| < 2\tau} \frac{\mathbf{1}_{\{s_* \in J^+\}} \mathbf{1}_{\{|r| \leq \rho\}}}{|a - a'|} d\sigma(a) d\sigma(a').$$

For every $\rho \leq \tau_0 \leq 1$,

$$\mathfrak{Z}_\rho^{J^+} \lesssim_U \tau_0^2 + \sum_{\substack{\tau \text{ dyadic} \\ \tau_0 \leq \tau \lesssim 1}} \mathfrak{Z}_{\rho, \tau}^{J^+}. \quad (122)$$

Consequently, for a fixed $\eta > 0$, choose

$$\tau_0 = \rho^{1-\eta/4}.$$

Then the ultra-near term is $O(\rho^{2-\eta/2})$, and (493) follows if every remaining shell obeys

$$\kappa_{\rho, \tau} := \frac{\mathfrak{Z}_{\rho, \tau}^{J^+}}{\tau^2} \lesssim_\eta \frac{(\rho/\tau)^2 \rho^{-\eta}}{\log(2/\rho)}. \quad (123)$$

Proof. Discarding both indicators in the part $|a - a'| < 2\tau_0$ and applying the Newton estimate (68) bounds that part by $C_U \tau_0^2$. The complementary pairs belong to the displayed logarithmic family of dyadic shells, proving (122). With the stated choice of τ_0 , its first term is smaller than $\rho^{2-\eta}$. Summing (123) over $O(\log(2/\rho))$ shells proves the final claim. $\square$

Proposition 7.3 (Every sub-four-dimensional front forces a power-excess Maslov shell). *Assume*

$$\dim_H K \leq 4 - \delta$$

for some $\delta > 0$, and fix $0 < \eta < \delta$. Then there are scales $\rho_n \downarrow 0$, dyadic angular radii τ_n , and numbers $A_n \rightarrow \infty$ such that

$$\rho_n^{1-\eta/4} \leq \tau_n \lesssim 1, \quad \epsilon_n := \rho_n/\tau_n \longrightarrow 0, \quad (124)$$

and

$$\kappa_{\rho_n, \tau_n} \geq A_n \frac{\epsilon_n^2 \rho_n^{-\eta}}{\log(2/\rho_n)}. \quad (125)$$

In particular a Hausdorff counterexample cannot hide in the ultra-near diagonal: it creates a genuine Darboux-separated, power-excess Lagrangian-pencil collision shell.

Proof. If $\mathfrak{Z}_\rho^{J^+} \lesssim \rho^{2-\eta}$ held at all sufficiently small scales, then Theorem 6.27 would give

$$\dim_H K \geq 4 - \eta > 4 - \delta,$$

a contradiction. Hence, after passing to a sequence, there are $A_n \rightarrow \infty$ for which

$$\mathfrak{Z}_{\rho_n}^{J^+} \geq A_n \rho_n^{2-\eta}.$$

Apply Theorem 7.2 with $\tau_{0,n} = \rho_n^{1-\eta/4}$. Its ultra-near contribution is

$$O(\rho_n^{2-\eta/2}) = o(\rho_n^{2-\eta}),$$

so, after decreasing A_n by an absolute factor, one of the $O(\log(2/\rho_n))$ remaining shells satisfies

$$\mathfrak{Z}_{\rho_n, \tau_n}^{J^+} \geq A_n \frac{\rho_n^{2-\eta}}{\log(2/\rho_n)}.$$

Divide by τ_n^2 and use $\epsilon_n = \rho_n/\tau_n$ to obtain (125). Finally $\tau_n \geq \rho_n^{1-\eta/4}$ gives $\epsilon_n \leq \rho_n^{\eta/4} \rightarrow 0$, proving (124). $\square$

Corollary 7.4 (Counterexample-driven return and the exact exponent gap). *For every hypothetical Hausdorff deficit $\delta > 0$, the shells in Theorem 7.3 may be passed through the bounded density and weight decompositions of Theorems 6.1 and 6.5 and then tested by the quantitative Maslov alternatives of the preceding stage. The geometric outputs are:*

- (i) a threshold approximate Lagrangian bush with vanishing physical radius;
- (ii) a further angular Darboux collapse with explicit scale separation;
- (iii) a persistent nonplanar Reeb-front seed carrying the residual excess (125).

There is, however, one genuinely numerical alternative which must not be hidden. If every normalized patch falls into the present one-patch saving rather than a geometric output, then the shell may satisfy simultaneously

$$\boxed{A_n \frac{\epsilon_n^2 \rho_n^{-\eta}}{\log(2/\rho_n)} \lesssim \kappa_{\rho_n, \tau_n} \lesssim \epsilon_n^{1/4}.} \quad (126)$$

The interval in (126) can be nonempty. Therefore a residual-target violation is not automatically above the fourth-root threshold. A surviving counterexample must either realize the persistent seed output (iii), or remain in precisely this diffuse exponent-gap window. Closing the proof requires a scale-persistent improvement from the upper exponent $1/4$ to the normal-slice exponent $2 - o(1)$; merely reapplying the same four-cycle estimate does not supply it.

Lemma 7.5 (The residual is exactly a two-coordinate symplectic normal map). *Fix a dyadic shell $\tau \leq |\alpha| < 2\tau$, where $\alpha = a - a'$, $\beta = b(a) - b(a')$, and cover the sphere of directions $\hat{\alpha} = \alpha/|\alpha|$ by finitely many conic charts. On a chart choose a smooth orthonormal frame $e_1(\hat{\alpha}), e_2(\hat{\alpha})$ of $\alpha^\perp$, put*

$$d = (\alpha, \beta), \quad n_j = (e_j, 0) \in \mathbb{R}_a^3 \oplus \mathbb{R}_b^3,$$

and define

$$\Phi_{\tau,j}(a, a') := \tau^{-1} \omega(d, n_j), \quad \Phi_\tau = (\Phi_{\tau,1}, \Phi_{\tau,2}). \quad (127)$$

Then, for the Maslov closest-time residual

$$s_* = -\frac{\alpha \cdot \beta}{|\alpha|^2}, \quad r = \beta + s_* \alpha \in \alpha^\perp,$$

one has the exact identity

$$\boxed{\Phi_{\tau,j} = -\tau^{-1} e_j \cdot r, \quad |\Phi_\tau| = \frac{|r|}{\tau}.} \quad (128)$$

If $(\chi_\ell)_\ell$ is a smooth partition of unity subordinate to the conic charts and

$$d\Lambda_{\tau,\ell} := \tau^{-2} \chi_\ell(\hat{\alpha}) \mathbf{1}_{\{\tau \leq |\alpha| < 2\tau\}} \frac{d\sigma(a) d\sigma(a')}{|\alpha|}, \quad (129)$$

then $\Lambda_{\tau,\ell}(\mathbb{R}^6) \lesssim_U 1$ and, with $\epsilon = \rho/\tau$,

$$\boxed{\kappa_{\rho,\tau} = \sum_{\ell} \Lambda_{\tau,\ell} \{s_* \in J^+, |\Phi_{\tau}| \leq \epsilon\}}. \quad (130)$$

Thus the missing quadratic exponent is literally the two-dimensional small-ball exponent of the symplectic normal map Φ_{τ} ; it is not an exponent imported from Euclidean endpoint codes.

Proof. For the standard symplectic form

$$\omega((x, y), (x', y')) = x \cdot y' - x' \cdot y$$

one has

$$\omega(d, n_j) = -e_j \cdot \beta = -e_j \cdot r,$$

because $e_j \perp \alpha$. Since e_1, e_2 are an orthonormal basis of $\alpha^{\perp}$, this proves (128). The Newton bound gives

$$\iint_{\tau \leq |a-a'| < 2\tau} \frac{d\sigma(a)d\sigma(a')}{|a-a'|} \lesssim_U \tau^{-1} \tau^3 = \tau^2,$$

which proves the uniform mass bound in (129). Finally sum the partition of unity and use (128) in the definition $\kappa_{\rho,\tau} = \tau^{-2} \mathfrak{Z}_{\rho,\tau}^{J^+}$. $\square$

Proposition 7.6 (Conditional two-normal estimate closes one shell). *For a conic chart let*

$$\lambda_{\tau,\ell} := \mathbf{1}_{\{s_* \in J^+\}} \Lambda_{\tau,\ell}, \quad \Phi_{\tau} = (\phi_1, \phi_2).$$

Suppose the first marginal has density at most L_1 ,

$$(\phi_1)_{\#} \lambda_{\tau,\ell} \leq L_1 du,$$

and, in the disintegration

$$\lambda_{\tau,\ell} = \int \lambda_u d((\phi_1)_{\#} \lambda_{\tau,\ell})(u),$$

the conditional second marginals satisfy

$$(\phi_2)_{\#} \lambda_u \leq L_2 dv$$

for almost every u . Then

$$\boxed{\lambda_{\tau,\ell} \{|\Phi_{\tau}| \leq \epsilon\} \lesssim L_1 L_2 \epsilon^2}. \quad (131)$$

Consequently, if on every surviving chart

$$L_1 L_2 \lesssim_{\eta} \frac{\rho^{-\eta}}{\log(2/\rho)}, \quad (132)$$

then (123) holds.

Proof. The Euclidean ball $\{|\Phi_{\tau}| \leq \epsilon\}$ is contained in $\{|\phi_1| \leq \epsilon, |\phi_2| \leq \epsilon\}$. Disintegrating first in ϕ_1 , the conditional estimate contributes at most $2L_2\epsilon$; integrating over $[-\epsilon, \epsilon]$ contributes at most $2L_1\epsilon$. This proves (131). Sum over the fixed finite chart family and apply (130). $\square$

7.2 Anchored rank packets

Corollary 7.7 (Every residual excess creates a high-density symplectic normal flag). *Let*

$$\kappa_{\ell} := \lambda_{\tau,\ell} \{|\Phi_{\tau}| \leq \epsilon\}, \quad I_{\epsilon} = [-\epsilon, \epsilon],$$

and use the disintegration of Theorem 7.6. Define the two successive normalized concentrations

$$D_1 := \frac{(\phi_1)_{\#} \lambda_{\tau,\ell}(I_{\epsilon})}{2\epsilon}, \quad D_2 := \operatorname{ess\,sup}_{u \in I_{\epsilon}} \frac{(\phi_2)_{\#} \lambda_u(I_{\epsilon})}{2\epsilon}. \quad (133)$$

Then

$$\boxed{D_1 D_2 \geq \frac{\kappa_{\ell}}{4\epsilon^2}}. \quad (134)$$

In particular, one chart of every shell in Theorem 7.3 satisfies

$$D_1 D_2 \gtrsim \frac{A_n \rho_n^{-\eta}}{\log(2/\rho_n)}, \quad \max\{D_1, D_2\} \gtrsim \left(\frac{A_n \rho_n^{-\eta}}{\log(2/\rho_n)} \right)^{1/2}. \quad (135)$$

Thus the exponent-gap branch is not diffuse in both symplectic normal directions: it carries a quantitatively divergent one-dimensional normal concentration after at most one conditioning.

Proof. The ball $\{|\Phi_\tau| \leq \epsilon\}$ lies inside the rectangle $\{\phi_1 \in I_\epsilon, \phi_2 \in I_\epsilon\}$. Therefore

$$\kappa_\ell \leq \int_{I_\epsilon} (\phi_2)_\# \lambda_u(I_\epsilon) d((\phi_1)_\# \lambda_{\tau, \ell})(u) \leq 4\epsilon^2 D_1 D_2,$$

which proves (134). There are only finitely many conic charts, so (130) supplies one with $\kappa_\ell \gtrsim \kappa_{\rho_n, \tau_n}$. Substitute (125) and use $\max\{D_1, D_2\} \geq (D_1 D_2)^{1/2}$. $\square$

Proposition 7.8 (Arbitrary-threshold routing of a normalized Maslov graph). *Use the normalized collision graph $0 \leq h \leq 1$ and probability measure ν from Theorem 6.3, and put*

$$M = e(h) = \iint h(x, y) d\nu(x) d\nu(y).$$

For a four-cycle x_0, x_1, x_2, x_3, x_0 , write

$$\mathcal{D}(x_0, x_1, x_2, x_3) = |\det(a_1 - a_0, a_2 - a_0, a_3 - a_0)|.$$

For every threshold $0 < \epsilon \leq c\Delta < 1$, at least one of the following holds:

(i) there are $s_0 \in J$ and $c_0 \in \mathbb{R}^3$ such that

$$\nu\{(a, b) : |b + s_0 a - c_0| \leq C\epsilon/\Delta\} \gtrsim M^3; \quad (136)$$

(ii) the determinant-small four-cycle content satisfies

$$\int h_{01} h_{12} h_{23} h_{30} \mathbf{1}_{\{\mathcal{D} \leq \Delta\}} d\nu^4 \gtrsim M^4. \quad (137)$$

Thus the fourth-root estimate in (76) is only what results from discarding the determinant-small cycles by their ambient sublevel measure. If those cycles are retained, no lower threshold on M is needed to obtain a bush-or-rank-loss alternative.

Proof. Twice applying Cauchy–Schwarz gives total four-cycle content at least M^4 . Split this content at $\mathcal{D} = \Delta$. If the determinant-small part carries at least half, then (137) holds.

Otherwise let $G(p, q)$ be the determinant-nondegenerate contribution of the two common neighbors of the anchors p, q , and let

$$K(p, q) = \int h(p, y) h(q, y) d\nu(y).$$

Then $\iint G(p, q) d\nu(p) d\nu(q) \gtrsim M^4$, whereas

$$\iint K(p, q) d\nu(p) d\nu(q) = \int \left(\int h(x, y) d\nu(x) \right)^2 d\nu(y) \leq M.$$

Hence some anchor pair satisfies $G(p, q)/K(p, q) \gtrsim M^3$. For every retained common-neighbor pair, the Cramer-rule calculation in Theorem 6.3 gives

$$|s_{py} - s_{py'}| \lesssim \epsilon/\Delta.$$

Partitioning J into intervals of this length and repeating the last pigeonhole in the proof of Theorem 6.4 produces one interval carrying $\gtrsim M^3$ of the common-neighbor measure. Since $h \leq 1$, its underlying ν -mass is at least the same size, and the contact-edge equation places it in the bush (136). $\square$

Proposition 7.9 (Bounded-density cost of every anchored Maslov rank packet). *Work in a normalized density-good Darboux patch, so the horizontal marginal of the selector measure has density at most C_U on a fixed bounded region. Use the hypotheses and notation of Theorem 6.19, but denote its horizontal noncollinearity threshold by $0 < \theta < 1$. If the de-multiplied output H_{Y_0} has direction mass*

$$h := \sigma(H_{Y_0}),$$

then the three alternatives of that proposition imply, respectively,

full-secant plane : $h \lesssim_U R + \gamma/\theta,$ horizontal line : $h \lesssim_U (R + \theta/\tau_1)^2,$ polarized plane : $h \lesssim_U R + \varepsilon_{\Delta, \gamma, v}/\theta^2.$	(138)
--	-------

Consequently put

$$\delta_{\Delta, \tau_1, v} := \Delta^{2/15} \tau_1^{-14/15} v^{-2/5}.$$

Provided the following thresholds lie in $(0, 1)$, choose

$$\theta = \Delta^{1/15} \tau_1^{8/15} v^{-1/5}, \quad \gamma = \Delta^{1/5} \tau_1^{-2/5} v^{-3/5}. \quad (139)$$

Then the alternative-independent estimate is

$h \lesssim_U \delta_{\Delta, \tau_1, v} + R \Delta^{-2/15} \tau_1^{-16/15} v^{-3/5}.$	(140)
--	-------

In particular, if

$$R \leq \Delta^{4/15} \tau_1^{2/15} v^{1/5},$$

then

$\lambda R^4 m_\Gamma \lesssim_U h \lesssim_U \Delta^{2/15} \tau_1^{-14/15} v^{-2/5}.$	(141)
--	-------

Thus a determinant-small anchored family cannot carry substantial de-multiplied direction mass at arbitrarily small determinant threshold.

Proof. If $F \subset \mathbb{R}^6$ is the full-secant plane in (100), then $\dim \pi_a(F) \leq 2$. Hence the horizontal projection of H_{Y_0} lies in an $O(R + \gamma/\theta)$ -neighborhood of an affine subspace of dimension at most two. Its intersection with the fixed bounded direction region has Lebesgue measure $O_U(R + \gamma/\theta)$. The bounded horizontal density proves the first line of (138).

In the horizontal-line alternative, the corresponding tubular neighborhood has three-dimensional Lebesgue measure $O_U((R + \theta/\tau_1)^2)$, proving the second line. In the polarized alternative, (102) places the directions within $O(R + \varepsilon_{\Delta, \gamma, v}/\theta^2)$ of the affine two-plane $a_p + E$, proving the third line.

Now use (103). Under (139),

$$\frac{\gamma}{\theta} = \frac{\theta^2}{\tau_1^2} = \delta_{\Delta, \tau_1, v}, \quad \frac{(\Delta/\gamma)^{1/3}}{v\theta^2} = \delta_{\Delta, \tau_1, v}.$$

The remaining R -term in the polarized alternative is

$$O\left(R \Delta^{-2/15} \tau_1^{-16/15} v^{-3/5}\right),$$

which proves (140). Under the displayed condition on R , this term is at most $\delta_{\Delta, \tau_1, v}$. Combine the result with the lower mass bound (104) to obtain (141). $\square$

Lemma 7.10 (Weighted cycle-relative anchored time trichotomy). *Fix anchors p, q , let $\lambda = \lambda_{p, q}$ be the common-neighbor measure of total mass $T > 0$, and let $\Gamma \subset Y \times Y$ be symmetric with*

$$Z := (\lambda \times \lambda)(\Gamma) > 0.$$

Give every $y \in Y$ its two contact labels s_y, t_y , and fix $R \leq \delta_s < 1$ and $0 < \tau_1 < 1$. At least one of the following holds:

(i) $|a_p - a_q| < \tau_1;$

(ii) a subfamily $\Gamma_{\text{sep}} \subset \Gamma$, of product measure

$$(\lambda \times \lambda)(\Gamma_{\text{sep}}) \gtrsim Z, \quad (142)$$

has three pairwise $c\delta_s$ -separated labels among $s_y, t_y, s_{y'}, t_{y'}$ for every $(y, y') \in \Gamma_{\text{sep}}$;

(iii) there are two intervals $I_1, I_2 \subset J$, each of length at most $C_{U, \tau_1}(R + \delta_s)$, and a set $Y_0 \subset Y$ such that

$$\lambda(Y_0) \gtrsim Z/T, \quad s_y, t_y \in I_1 \cup I_2 \quad (y \in Y_0). \quad (143)$$

In case (iii), if the intervals are not quantitatively separated, their union gives an $O_{U, \tau_1}(R + \delta_s)$ -bush. Otherwise the pointwise argument of Theorem 6.9 partitions Y_0 into four sets, one of ν -mass $\gtrsim Z/T$, which is a bush or an angular packet of radius

$$O_{U, \tau_1}\left(\frac{R + \delta_s}{g^2}\right), \quad g := \text{dist}(I_1, I_2). \quad (144)$$

Proof. Assume (i) fails. Call a pair (y, y') incompatible when its four labels cannot be covered by two intervals of length $10\delta_s$. The elementary one-dimensional packing argument from Theorem 6.10 shows that every incompatible pair contains three pairwise $c\delta_s$ -separated labels. If the incompatible part of Γ has measure at least $Z/4$, then (ii) holds.

Otherwise the compatible-edge subgraph Γ_c has measure at least $3Z/4$. Put

$$Y_{\text{far}} = \{y : |s_y - t_y| > 100\delta_s\}, \quad d_c(y) = \lambda\{y' : (y, y') \in \Gamma_c\}.$$

If some $y_0 \in Y_{\text{far}}$ has $d_c(y_0) \geq Z/(8T)$, compatibility forces both labels of every compatible neighbor of y_0 into the two intervals of length $10\delta_s$ about its two labels. These intervals are separated by more than eight times their length, and the neighbor set proves (iii).

Suppose no such far vertex exists. Then

$$\int_{Y_{\text{far}}} d_c(y) d\lambda(y) < Z/8.$$

By symmetry, compatible edges with at least one far endpoint have measure below $Z/4$. Hence the close-close compatible edges have measure at least $Z/2$, and some close vertex has a set of close compatible neighbors of λ -mass at least $Z/(2T)$. For every close vertex subtract the two contact-edge equations and put $r_y = (s_y + t_y)/2$. Since $|a_p - a_q| \geq \tau_1$,

$$|b_q - b_p + r_y(a_q - a_p)| \lesssim R + \delta_s$$

places all r_y 's in one interval of length $C_{U, \tau_1}(R + \delta_s)$. Hence both labels of the selected close-neighbor set lie in one enlarged interval, proving (143).

If the two intervals in (iii) are not quantitatively separated, their union is one enlarged time interval and the first contact-edge equation gives a bush. If they are separated, the four pointwise cases in Theorem 6.9 give a bush or an angular packet with the stated bounds. Since $d\lambda = h(p, y)h(q, y)d\nu(y) \leq d\nu(y)$, passing to underlying direction mass loses nothing. $\square$

Proposition 7.11 (Weighted continuous anchored rank packet). *Fix anchors p, q with $|a_p - a_q| \geq \tau_1$, and let*

$$d\lambda_{p,q}(y) := h(p, y)h(q, y)d\nu(y), \quad T := \lambda_{p,q}(Y) > 0. \quad (145)$$

Let $\Gamma \subset Y \times Y$ be a symmetric measurable family of anchored cycles, and put

$$Z := (\lambda_{p,q} \times \lambda_{p,q})(\Gamma) > 0, \quad m_\Gamma := Z/T.$$

Assume every pair $(y, y') \in \Gamma$ has horizontal determinant at most Δ , and three of its four contact labels have Vandermonde product at least v . Then, for every $0 < \gamma, \theta < 1$, there are $y_0 \in Y$ and a measurable set $Y_0 \subset Y$ such that

$$\nu(Y_0) \geq \lambda_{p,q}(Y_0) \gtrsim \frac{Z}{T} = m_\Gamma, \quad (146)$$

and one of the three geometric conclusions in Theorem 6.19 holds on Y_0 , with its parameter κ there replaced by θ and with exactly the same errors Eqs. (100) to (102).

Proof. The proof of Theorem 6.19 uses only deletion and average degree, so it extends to the finite measure $\lambda_{p,q}$. For completeness, split Γ according to whether the full secant volume $\mathbf{g}(y, y')$ is below γ , retain a part of product measure at least $Z/2$, and rename it Γ . If the retained part is the low-volume part, define

$$Y_{\text{full-line}} = \{y : |(z_y - z_p) \wedge (z_q - z_y)| < \theta\}.$$

If $\lambda_{p,q}(Y_{\text{full-line}}) \gtrsim Z/T$, this set gives the line alternative. Otherwise the cycles incident to it have product measure at most $2T\lambda_{p,q}(Y_{\text{full-line}})$, so deletion retains a fixed fraction of the relevant cycle mass. Weighted average degree then selects y_0 whose retained neighbor set has $\lambda_{p,q}$ -mass $\gtrsim Z/T$; division of the triple-volume bound by the base two-volume gives the full-secant plane alternative.

If instead the retained part is the full-secant-volume part, repeat the same argument with

$$Y_{\text{line}} = \{y : |(a_y - a_p) \times (a_q - a_p)| < \theta\}.$$

Either this set has $\lambda_{p,q}$ -mass $\gtrsim Z/T$, giving the horizontal-line alternative, or weighted average degree fixes y_0 and a neighbor set of that mass on which the pointwise three-time interpolation and shared-flag argument from Theorem 6.19 gives the polarized-plane alternative. Finally $h(p, y)h(q, y) \leq 1$ implies $\nu(Y_0) \geq \lambda_{p,q}(Y_0)$. $\square$

Corollary 7.12 (Determinant-small Maslov mass yields an actual direction packet). *Assume the determinant-small output (137) occurs, and that a fixed fraction of that cycle content has anchors satisfying $|a_p - a_q| \geq \tau_1$ and has three-label Vandermonde product at least v . Then one anchored rank-loss packet has actual normalized direction mass*

$$\boxed{h \gtrsim M^3.} \quad (147)$$

Consequently the bounded-density estimate (140) gives

$$\boxed{M^3 \lesssim_U \Delta^{2/15} \tau_1^{-14/15} v^{-2/5} + R \Delta^{-2/15} \tau_1^{-16/15} v^{-3/5}.} \quad (148)$$

If $R \leq \Delta^{4/15} \tau_1^{2/15} v^{1/5}$, this simplifies to

$$\boxed{M \lesssim_U \Delta^{2/45} \tau_1^{-14/45} v^{-2/15}.} \quad (149)$$

Proof. Let $Z(p, q)$ be the determinant-small common-neighbor pair content for the anchors p, q , and let $T(p, q) = \lambda_{p,q}(Y)$. The retained fixed fraction gives

$$\iint Z(p, q) d\nu(p) d\nu(q) \gtrsim M^4,$$

whereas the calculation in Theorem 7.8 gives

$$\iint T(p, q) d\nu(p) d\nu(q) \leq M.$$

Therefore some retained anchor pair obeys

$$\frac{Z(p, q)}{T(p, q)} \gtrsim M^3.$$

Apply Theorem 7.11 and then Theorem 7.9. Taking a cube root gives (149). $\square$

Theorem 7.13 (Exact local routing below the fourth-root scale). *Fix*

$$0 < \epsilon \leq c\Delta < 1, \quad \epsilon \leq \delta_s < 1, \quad 0 < \tau_1 < 1.$$

For a normalized Maslov graph of edge mass M , at least one of the following holds:

- (i) an $O(\epsilon/\Delta)$ -Lagrangian bush has direction mass $\gtrsim M^3$;
- (ii) a common-neighbor family of direction mass $\gtrsim M^3$ has an anchor pair satisfying $|a_p - a_q| < \tau_1$, and hence enters the next angular Darboux refinement;
- (iii) a two-time packet of direction mass $\gtrsim M^3$ lies in intervals of width $O_{U, \tau_1}(\epsilon + \delta_s)$; if their gap is g , it gives either one $O_{U, \tau_1}(\epsilon + \delta_s + g)$ -bush or an angular packet of radius $O_{U, \tau_1}((\epsilon + \delta_s)/g^2)$;

(iv) a separated-time anchored rank packet has direction mass $h \gtrsim M^3$ and obeys (148), with R there read as ϵ ; in the physical-error range of (149), one has

$$M \lesssim_U \Delta^{2/45} \tau_1^{-14/45} v^{-2/15}, \quad v \gtrsim \delta_s^3.$$

This alternative holds without assuming $M \gtrsim \epsilon^{1/4}$. Thus every local patch in the former fourth-root exponent gap is now routed to a physical bush, a deeper angular packet, or a quantitatively mass-capped Maslov rank packet.

Proof. Apply Theorem 7.8. Its first output is (i). In its determinant-small output, write $Z(p, q)$ and $T(p, q)$ as in the proof of Theorem 7.12. Split the total Z -mass according to whether $|a_p - a_q| < \tau_1$. One part carries a fixed fraction. Since

$$\iint T(p, q) d\nu(p) d\nu(q) \leq M,$$

that part contains anchors with

$$Z(p, q)/T(p, q) \gtrsim M^3.$$

If the close-anchor part was selected, then $Z(p, q) \leq T(p, q)^2$ implies $T(p, q) \gtrsim M^3$, giving (ii).

Otherwise apply Theorem 7.10, with $R = \epsilon$, to the determinant-small cycle family for the selected separated anchors. Its time-concentrated output is (iii) and retains direction mass $\gtrsim Z/T \gtrsim M^3$. Its separated-time output has $v \gtrsim \delta_s^3$; apply Theorems 7.9 and 7.11 to obtain (iv). $\square$

Corollary 7.14 (The weighted two-time output has no scale-persistent drift). *Consider a sequence of outputs (iii) in Theorem 7.13 with $\epsilon_n \rightarrow 0$ and choose $\delta_{s,n} \rightarrow 0$. After passing to a subsequence, the same direction-mass scale $\gtrsim M_n^3$ enters one of the following:*

- (i) the interval gap $g_n \rightarrow 0$, and their union is one Lagrangian bush of radius $O_{U, \tau_1}(\epsilon_n + \delta_{s,n} + g_n) \rightarrow 0$;
- (ii) $g_n \geq g_0 > 0$, and (144) is a shrinking angular packet.

Thus the weighted time-packet passage preserves the Z/T mass scale and introduces no third longitudinal-drift alternative.

7.3 Polarization and affine-offset control

Lemma 7.15 (Transverse Maslov polarizations supply the second horizontal codimension). *Let $\Pi_1, \Pi_2 \subset \mathbb{R}^3$ be affine two-planes with unit normals n_1, n_2 , and assume*

$$|n_1 \times n_2| \geq \vartheta > 0.$$

For every bounded direction region U and $0 < w < 1$,

$$|U \cap N_w(\Pi_1) \cap N_w(\Pi_2)| \lesssim_U \frac{w^2}{\vartheta}. \quad (150)$$

Consequently, if a selector-direction subset H has density bounded by C_U and is carried by two anchored Maslov packets of thickness w whose polarization planes have Grassmannian distance at least $c\vartheta$, then

$$\sigma(H) \lesssim_U \frac{w^2}{\vartheta}. \quad (151)$$

This is the missing quadratic horizontal factor on every transversely reused graph cell.

Proof. After a rigid motion, the two plane conditions are

$$|n_1 \cdot x - c_1| \leq w, \quad |n_2 \cdot x - c_2| \leq w.$$

Use coordinates along n_1, n_2 , and along the intersection direction $n_1 \times n_2$. The Jacobian of this coordinate map is $|n_1 \times n_2| \geq \vartheta$. The first two coordinates range over intervals of length $O(w)$, while boundedness of U bounds the third. This proves (150); the horizontal density bound gives (151). $\square$

Corollary 7.16 (Weighted summation of the transverse high-reuse cells). *Use the reuse multiplicities $r(\chi)$ of Theorem 6.20. Let $\mathcal{C}_{\text{tr}}(K, \vartheta)$ be a family of pairwise disjoint graph cells such that $r(\chi) \geq K$ and two packets through every χ have polarization planes separated by at least $c\vartheta$. If all cells and packet tubes have common direction thickness $O(w)$, then their actual direction union satisfies*

$$\boxed{\sigma(H_{\text{tr}}) \lesssim_U \frac{w^2}{K\vartheta} \sum_{\chi \in \mathcal{C}_{\text{tr}}(K, \vartheta)} r(\chi).} \quad (152)$$

Thus the transverse quadratic factor is summed with the reciprocal reuse gain K^{-1} , not with the total number of ambient graph cells.

Proof. The direction cube underlying a reused graph cell lies, after a fixed enlargement by its own diameter, in the two transverse plane tubes. Hence Theorem 7.15 bounds its actual selector-direction mass by $C_U w^2 / \vartheta$. The graph cells are disjoint, and

$$\#\mathcal{C}_{\text{tr}}(K, \vartheta) \leq K^{-1} \sum_{\chi \in \mathcal{C}_{\text{tr}}(K, \vartheta)} r(\chi).$$

Multiplication gives (152). $\square$

Lemma 7.17 (A polarized cycle packet controls the vertical affine offset). *In alternative (iii) of Theorem 6.19, with its noncollinearity parameter denoted by θ , every retained neighbor also satisfies*

$$\boxed{\text{dist}(b_y, b_p + E) \lesssim_{U, I, J} \varepsilon_{\Delta, \gamma, v} / \theta^2.} \quad (153)$$

Hence a coherent polarized packet is contained, up to the displayed error, in one affine phase-space four-plane

$$(a_p, b_p) + (E \oplus E).$$

Proof. The cycle secant $z_y - z_p$ belongs to $V_{y_0, y}$. By (102), that three-plane lies within $C\varepsilon_{\Delta, \gamma, v} / \theta^2$ in the Grassmannian of a three-plane contained in $E \oplus E$. All secants stay in a fixed bounded region, so

$$\text{dist}(z_y - z_p, E \oplus E) \lesssim_{U, I, J} \varepsilon_{\Delta, \gamma, v} / \theta^2.$$

Take the vertical component. $\square$

Lemma 7.18 (A common direction cell aligns coherent horizontal plane offsets). *Write affine polarization planes as*

$$\Pi_i = \{a : n_i \cdot a = c_i\}, \quad |n_i| = 1.$$

Assume n_i all lie within ϑ of one unit normal n_∞ , and a direction cube $Q \subset U$ of diameter $O(w)$ meets every tube $N_w(\Pi_i)$. Then

$$\boxed{|c_i - c_j| \lesssim_U w + \vartheta \quad (i, j).} \quad (154)$$

Thus reuse of one graph cell by coherently polarized packets automatically aligns their affine horizontal offsets at the geometric resolution of that cell.

Proof. Choose $a_i \in Q$ with $|n_i \cdot a_i - c_i| \leq w$. Since Q has diameter $O(w)$,

$$\begin{aligned} |c_i - c_j| &\leq |c_i - n_i \cdot a_i| + |n_i \cdot (a_i - a_j)| + |(n_i - n_j) \cdot a_j| + |n_j \cdot a_j - c_j| \\ &\lesssim_U w + \vartheta. \end{aligned}$$

This is (154). $\square$

Corollary 7.19 (A common product cell aligns both phase offsets). *For a coherently polarized packet put*

$$c_i^a = n_i \cdot a_{p_i}, \quad c_i^b = n_i \cdot b_{p_i}.$$

Suppose a graph product cell of diameter $O(w)$ meets packet (i) inside the affine phase-plane tube supplied by Theorem 7.17, of additional thickness e_i . If $|n_i - n_\infty| \leq \vartheta$, then all packets through that cell satisfy

$$\boxed{|c_i^a - c_j^a| + |c_i^b - c_j^b| \lesssim_U w + \vartheta + e_i + e_j.} \quad (155)$$

Proof. Apply the calculation in Theorem 7.18 first to the horizontal coordinates of a point in the common product cell and then to its vertical coordinates. The latter application uses (153); the two packet errors contribute $e_i + e_j$. $\square$

Lemma 7.20 (One-dimensional Reeb-window sum for parallel phase offsets). *Let ζ be a finite measure of total mass m on a fixed bounded subset of the phase-offset plane $\mathbb{R}_q \times \mathbb{R}_p$. Assume its q -marginal satisfies*

$$\zeta_q(I) \leq D|I| \quad \text{for every interval } I \text{ with } |I| \geq W. \quad (156)$$

Then, for every fixed bounded Reeb interval J ,

$$\boxed{\int_J \iint \mathbf{1}_{\{|(p-p') + s(q-q')| \leq W\}} d\zeta(q, p) d\zeta(q', p') ds \lesssim_J DmW \log(2/W).} \quad (157)$$

This estimate is purely the contact/Reeb window calculation for the normal phase-offset pair; it uses no Fourier or spectral argument.

Proof. For $|q - q'| < W$, the pair mass is at most

$$\int \zeta_q([q - W, q + W]) d\zeta(q, p) \lesssim DmW,$$

and the time interval has bounded length. On the dyadic shell $r \leq |q - q'| < 2r$, $r \geq W$, the collision-time window has length at most $C_J W/r$. On the other hand (156) bounds the pair mass of that shell by $CDrm$. Hence every dyadic shell contributes $O_J(DmW)$; there are $O(\log(2/W))$ shells. $\square$

Lemma 7.21 (Pointwise Reeb degree of one phase offset). *Under the hypotheses of Theorem 7.20, every fixed $\xi = (q, p)$ satisfies*

$$\boxed{\int_J \int \mathbf{1}_{\{|(p-p') + s(q-q')| \leq W\}} d\zeta(q', p') ds \lesssim_J DW \log(2/W).} \quad (158)$$

Consequently the two coordinate marginals of the normal-window pair measure are each dominated by $C_J DW \log(2/W) \zeta$, not merely by the trivial measure $|J|m\zeta$.

Proof. For $|q - q'| < W$, the q' -mass is at most CDW , and the time interval has bounded length. On $r \leq |q - q'| < 2r$, $r \geq W$, the q' -mass is at most CDr , while the admissible time interval has length at most $C_J W/r$. Each dyadic shell therefore costs $O_J(DW)$; summing the shells proves (158). The marginal statement follows by integrating the pointwise estimate against the first, respectively second, copy of ζ . $\square$

Lemma 7.22 (Exact normal–tangential factorization in a Maslov polarization). *Fix an oriented horizontal two-plane $E \subset \mathbb{R}^3$, let n be its unit normal, and write every phase point uniquely as*

$$a = x + qn, \quad b = y + pn, \quad x, y \in E, \quad q, p \in \mathbb{R}. \quad (159)$$

In these coordinates the canonical symplectic form and the incidence Lagrangian split as

$$\omega = \omega_E + dq \wedge dp, \quad L_s = \{(u, -su) : u \in E\} \oplus \{(rn, -srn) : r \in \mathbb{R}\}.$$

For two phase points $z = (a, b)$, $z' = (a', b')$, put

$$T_s(z, z') = (y - y') + s(x - x') \in E, \quad N_s(z, z') = (p - p') + s(q - q') \in \mathbb{R}.$$

Then

$$\boxed{|\Pi_s(z) - \Pi_s(z')|^2 = |T_s(z, z')|^2 + |N_s(z, z')|^2.} \quad (160)$$

Consequently, for every $W > 0$,

$$\mathbf{1}_{\{|T_s| \leq W/\sqrt{2}\}} \mathbf{1}_{\{|N_s| \leq W/\sqrt{2}\}} \leq \mathbf{1}_{\{|\Pi_s(z) - \Pi_s(z')| \leq W\}} \leq \mathbf{1}_{\{|T_s| \leq W\}} \mathbf{1}_{\{|N_s| \leq W\}}. \quad (161)$$

This is the intrinsic two-stage form of the coherent Maslov collision: the scalar factor is the Reeb-normal window, while the remaining factor is the tangential front collision in $E \oplus E$.

Proof. The orthogonal splitting $\mathbb{R}^3 = E \oplus \mathbb{R}n$ gives

$$(b - b') + s(a - a') = T_s(z, z') + N_s(z, z')n.$$

Pythagoras proves (160); the two indicator inequalities follow immediately. $\square$

7.4 Weighted slices and defect routing

Lemma 7.23 (Weighted finite-scale q -slice retention). *Let G be one quantitative Sticky packing piece with selector measure μ , bounded direction density, and*

$$N_r(G) \leq C_\zeta r^{-3-\zeta}. \quad (162)$$

Let Γ be a finite nonnegative measure on $G \times G \times J$, of total mass $M > 0$, such that both graph-coordinate marginals are dominated by $L\mu$. For a dyadic q -interval I of length r , let $\mathcal{N}_I(r)$ be the number of full (x, q, y, p) product cells of side r which meet G over I . There is a collection $\mathcal{I}_{\text{good}}$ for which

$$\Gamma\{q(z), q(z') \text{ both lie in good intervals}\} \geq \frac{1}{2}M \quad (163)$$

and every $I \in \mathcal{I}_{\text{good}}$ obeys

$$\mathcal{N}_I(r) \lesssim_{\zeta, U} \frac{L}{M} r^{-2-\zeta}. \quad (164)$$

The number of tangential (x, y) -cells occurring over such an interval is bounded by the same right-hand side.

Proof. A fixed orthogonal change of coordinates changes covering numbers only by a dimensional constant, so (162) gives

$$\sum_I \mathcal{N}_I(r) \lesssim_\zeta r^{-3-\zeta}.$$

Call I bad when the right side of (164), with a sufficiently large fixed implicit constant, is exceeded. Then

$$\#\{I : I \text{ bad}\} \lesssim_U \frac{M}{Lr}.$$

Bounded direction density gives $\mu\{q \text{ lies in a bad interval}\} \lesssim_U r \#\{I : I \text{ bad}\}$, and the choice of the implicit constant makes this at most $M/(4L)$. Domination of the first Γ -marginal shows that pairs whose first coordinate is bad have mass at most $M/4$; the same argument applies to the second coordinate. Removing both sets proves (163). Projection of a full product cell to (x, y) meets only boundedly many tangential cells, which proves the last assertion. $\square$

Lemma 7.24 (Collision-preserving multiscale vertical-entropy pruning). *Retain the hypotheses and notation of Theorem 7.23. For a dyadic direction cube $A \subset \mathbb{R}_a^3$ of side r , let*

$$v_r(A) := \#\{B : (A \times B) \cap G \neq \emptyset\},$$

where B ranges over the dyadic intercept cubes of side r . Fix $0 < W < 1$, let

$$\mathcal{D}_W = \{r : r \text{ dyadic}, W \leq r \leq 1\}, \quad \ell_W = 1 + \#\mathcal{D}_W \lesssim \log(2/W),$$

and let $0 \leq \Theta \leq \Gamma$ be any submeasure of total mass $H > 0$. There is a Borel endpoint set $G_{\Theta, W} \subset G$ such that

$$\Theta(G_{\Theta, W} \times G_{\Theta, W} \times J) \geq \frac{1}{2}H \quad (165)$$

and, for every $z = (a, b(a)) \in G_{\Theta, W}$ and every $r \in \mathcal{D}_W$,

$$v_r(A_r(a)) \lesssim_{\zeta, U} \frac{L}{H} \ell_W r^{-\zeta}. \quad (166)$$

Here $A_r(a)$ is the direction r -cube containing a .

Moreover, fix a polarized normal coordinate interval I of length W . For every $r \in \mathcal{D}_W$ and every tangential direction r -cube $X \subset E_x$, the part of $G_{\Theta, W}$ with $q \in I$ and $x \in X$ meets at most

$$\lesssim_{\zeta, U} \frac{L}{H} \ell_W r^{-\zeta} \quad (167)$$

tangential intercept r -cubes in E_y . Thus the same estimate controls the number of tangential contact-front tube branches above X at every intermediate scale, not only the total number of W -cells in one q -slice.

Proof. The product-grid form of (162) gives

$$\sum_A v_r(A) \lesssim_{\zeta,U} r^{-3-\zeta}.$$

For each $r \in \mathcal{D}_W$, call A bad when

$$v_r(A) > C_{\zeta,U} \frac{L}{H} \ell_W r^{-\zeta},$$

with the constant chosen below. The number of bad cubes is at most

$$\lesssim_{\zeta,U} \frac{H}{L\ell_W} r^{-3}.$$

Bounded direction density therefore gives

$$\mu\{a \text{ lies in a bad } r\text{-cube}\} \lesssim_{\zeta,U} \frac{H}{L\ell_W}.$$

After increasing $C_{\zeta,U}$, domination of either coordinate marginal of Γ by $L\mu$, together with $\Theta \leq \Gamma$, makes the Θ -mass for which a prescribed endpoint is bad at a prescribed scale at most $H/(4\ell_W)$. Sum over the two endpoints and all scales in $\mathcal{D}_W$. Removing their union costs at most $H/2$, proving (165) and (166).

If $q \in I$ and $r \geq W$, then I meets only $O(1)$ dyadic q -intervals of length r . Likewise $X \times I$ is contained in $O(1)$ full direction r -cubes. Above each one, projection of a full intercept cube in (y, p) meets only $O(1)$ tangential y -cubes. Hence (166) implies (167). $\square$

Proposition 7.25 (Collision-preserving total Maslov-defect ledger). *Normalize the coherent selector occurrence measure to total mass one, and absorb D , reuse caps, and logarithms into $W^{-o(1)}$. Pass to a scale subsequence on which*

$$\|\Lambda_W\| = W^{1+\beta+o(1)}, \quad \frac{\mathcal{C}_W}{\|\Lambda_W\|} = W^{\gamma+o(1)}, \quad \chi := \beta + \gamma, \quad (168)$$

where $\beta, \gamma \geq 0$. Then

$$\boxed{\mathcal{C}_W = W^{1+\chi+o(1)}}. \quad (169)$$

Consequently $\chi \geq 2 - o(1)$ is already the desired full collision bound. If $0 \leq \chi < 2$, apply Theorem 7.24 to $\Theta_W \leq \Lambda_W$. At least half of the actual collision mass survives on an endpoint set for which, simultaneously for every dyadic $W \leq r \leq 1$,

$$\boxed{v_r(A_r(a)) \leq W^{-\chi-o(1)} r^{-\zeta}}. \quad (170)$$

Inside every normal W -slab, the same bound holds for tangential contact-front branches above one tangential direction r -cube. Moreover each endpoint projection of the retained collision measure requires direction support of mass at least

$$\boxed{\gtrsim W^{\chi+o(1)}}. \quad (171)$$

Thus the collision-preserving branch tax and the endpoint-support mass are reciprocal at the power scale. The exponent χ , not β alone, is the invariant defect carried by the self-selected Maslov diagonal.

Proof. The first identity is multiplication in (168). The pointwise Reeb-degree estimate gives both coordinate marginals of Λ_W , and hence of Θ_W , the domination

$$(\Theta_W)_1, (\Theta_W)_2 \leq L\nu, \quad L = W^{1-o(1)}.$$

Use $H = \mathcal{C}_W = W^{1+\chi+o(1)}$ in (166); this gives (170). The tangential assertion is (167). Finally, if S supports either endpoint marginal of the retained measure, then

$$\frac{1}{2} \mathcal{C}_W \leq (\Theta_W)_i(S) \leq L\nu(S).$$

Hence $\nu(S) \gtrsim \mathcal{C}_W/L = W^{\chi+o(1)}$, proving (171). $\square$

Proposition 7.26 (The total-defect layer re-enters the Maslov four-cycle routing). *Work on a normalized angular shell $c_0 \leq |a - a'| \leq C_0$ and normalize $\nu(G) = 1$. Choose a fixed C_{c_0} larger than the maximum W -collision-window length divided by W , and define*

$$h_W(z, z') := \frac{1}{C_{c_0} W} \int_J \mathbf{1}_{\{|N_s(z, z')| \leq W\}} \mathbf{1}_{\{|T_s(z, z')| \leq W\}} ds. \quad (172)$$

Then h_W is a symmetric graphon with $0 \leq h_W \leq 1$, and its edge mass is

$$M_W := e(h_W) = \frac{C_W}{C_{c_0} W}. \quad (173)$$

Every edge with $h_W(z, z') > 0$ has a measurable label $s_{zz'} \in J$ satisfying

$$|(b - b') + s_{zz'}(a - a')| \leq \sqrt{2}W. \quad (174)$$

Consequently, for every threshold $W \leq c\Delta < 1$, the arbitrary-threshold Maslov routing Theorem 7.8 gives at least one of:

(i) an $O(W/\Delta)$ -approximate Lagrangian point-probe bush of ν -mass

$$\gtrsim M_W^3; \quad (175)$$

(ii) determinant-small Maslov four-cycle content

$$\int (h_W)_{01}(h_W)_{12}(h_W)_{23}(h_W)_{30} \mathbf{1}_{\{\mathcal{D} \leq \Delta\}} d\nu^4 \gtrsim M_W^4. \quad (176)$$

On a total-defect level $\mathcal{C}_W = W^{1+\chi+o(1)}$, these outputs are respectively

$$\boxed{\text{bush mass} \gtrsim W^{3\chi+o(1)}, \quad \text{rank-loss four-cycle content} \gtrsim W^{4\chi+o(1)}}. \quad (177)$$

Thus the collision-preserving correction does not create a new analytic branch: it returns the entire bad layer, with its exact defect exponent, to the manuscript's existing Lagrangian-bush/Maslov-rank-loss alternative.

Proof. For a fixed pair on the normalized angular shell, $s \mapsto (b - b') + s(a - a')$ has speed at least c_0 . The set on which both indicators in (172) equal one lies inside a full-vector collision interval of length at most $C_{c_0}W$. Hence $0 \leq h_W \leq 1$; symmetry is immediate. Tonelli gives (173).

The nonempty compact collision-time section of an edge admits a Borel choice of $s_{zz'}$; alternatively, choose the midpoint of its interval. The orthogonal factorization (160) gives (174). The horizontal marginal of ν has the bounded density required in Theorem 7.8. Apply that proposition with collision error $\sqrt{2}W$, absorbing the fixed factor into c . This proves (175) and (176). Finally $M_W = \mathcal{C}_W / (C_{c_0}W) = W^{\chi+o(1)}$, which yields (177). $\square$

Proposition 7.27 (Total defect extracts a quantitative common contact-star layer). *Work on a normalized angular shell $c_0 \leq |a - a'| \leq C_0$, normalize $\nu(G) = 1$, and define the full polarized collision degree*

$$d_W(z) := \int_J \int \mathbf{1}_{\{|N_s(z, z')| \leq W\}} \mathbf{1}_{\{|T_s(z, z')| \leq W\}} d\nu(z') ds. \quad (178)$$

Then $\int d_W d\nu = \mathcal{C}_W$ and $d_W(z) \leq L$, where $L \lesssim DW \log(2/W)$. There are a dyadic $h \in [\mathcal{C}_W, CL]$ and a Borel center set

$$S_h = \{z : h \leq d_W(z) < 2h\}$$

such that, with $\ell_{W,C} = 1 + \log(2L/\mathcal{C}_W)$,

$$\boxed{\nu(S_h) \gtrsim \frac{\mathcal{C}_W}{h \ell_{W,C}}}. \quad (179)$$

For every $z \in S_h$, there is a Reeb time $s_z \in J$ for which the simultaneous neighbor set obeys

$$\boxed{\nu\{z' : |N_{s_z}(z, z')| \leq W, |T_{s_z}(z, z')| \leq W\} \gtrsim_J h}. \quad (180)$$

All selected lines in this set pass through the $O(W)$ -ball about the same front point $(\Pi_{s_z}(z), s_z)$; hence its geometric support is a W -scale common contact star carrying ν -weight $\gtrsim h$. If ν is an occurrence measure rather than the unrepeated selector measure, conversion of this weight to actual direction mass still requires the existing reuse de-multiplication. Its time-swept set of distinct ν -leaves satisfies

$$\boxed{\nu\{z' : \exists s \in J, |N_s(z, z')| \leq W, |T_s(z, z')| \leq W\} \gtrsim_{c_0, J} \frac{h}{W}}. \quad (181)$$

Under the power normalization of Theorem 7.25, one may write

$$h = W^{1+\theta+o(1)} \quad \text{for some} \quad 0 \leq \theta \leq \chi. \quad (182)$$

The three retained quantities then have the exact power ledger

<i>center ν-mass</i>	<i>one-time star ν-weight</i>	<i>swept leaf ν-mass</i>
$W^{\chi-\theta+o(1)}$	$W^{1+\theta+o(1)}$	$W^{\theta+o(1)}$

(183)

If the occurrence measure is dominated by the unrepeated selector measure in the form $\nu \leq K\sigma$, then the corresponding actual direction masses are bounded below by the three entries in (183) divided by K . In particular, when the coherent reuse cap is subpower, the passage from the weighted star to a physical star loses no power of W . Thus a total defect $\chi < 2$ is already concentrated on a quantitative family of weighted common contact stars. The final step below de-multiplies the θ -layer through the coherent reuse ledger and either cross an existing Lagrangian-bush removal threshold or sum the remaining weighted Maslov incidences.

Proof. The identity $\int d_W d\nu = C_W$ is Tonelli's theorem. The pointwise bound follows from $d_W \leq d_W^{\text{normal}}$ and Theorem 7.21. Degrees below $C_W/2$ contribute at most $C_W/2$ because $\nu(G) = 1$. Dyadically decompose the remaining interval $[C_W/2, L]$. One level contributes at least $cC_W/\ell_{W,C}$ to $\int d_W d\nu$; since $d_W < 2h$ on that level, (179) follows.

For fixed $z \in S_h$, put

$$F_z(s) = \nu\{z' : |N_s(z, z')| \leq W, |T_s(z, z')| \leq W\}.$$

Then $\int_J F_z(s) ds = d_W(z) \geq h$, so averaging in s proves (180). The two inequalities imply $|\Pi_{s_z}(z) - \Pi_{s_z}(z')| \leq \sqrt{2}W$, giving the common front point.

For each fixed pair in the normalized angular shell, the full collision-time set has length at most $C_{c_0}W$: indeed $s \mapsto (b - b') + s(a - a')$ has speed at least c_0 . If $\mathcal{N}_W(z)$ denotes the swept neighbor set in (181), then

$$h \leq d_W(z) \leq C_{c_0}W\nu(\mathcal{N}_W(z)),$$

which proves that estimate. Finally $C_W = W^{1+\chi+o(1)} \leq h \leq L = W^{1-o(1)}$ gives (182); substituting it into (179), (180), and (181) yields (183). $\square$

Lemma 7.28 (Polarized branch coloring into direction-separated tangential families). *Discretize a collision-preserved endpoint set at scale W , using full product cells in (x, q, y, p) . Fix a normal q -interval I of length W . Every occupied labeled cell determines a tangential W -tube in $E \times J$,*

$$\mathbb{T}_\epsilon = \{(u, s) : |u - (y_\chi + sx_\chi)| \leq CW\},$$

while retaining its normal (q, p) -label. Suppose that, above every tangential direction W -cell, there are at most B such labeled tubes with $q \in I$. Then this labeled tube family is the union of at most CB subfamilies, each of which has essentially W -separated tangential directions. Summing over all normal q -intervals in a bounded region gives at most

$$\boxed{J_W \lesssim BW^{-1}} \quad (184)$$

direction-separated tangential color classes.

Proof. Split the tangential direction grid into a fixed number of congruence classes so that distinct cells of one class have cW -separated centers. Above one direction cell enumerate the at most B labeled tubes and give the k -th tube color k , together with the congruence color of its direction cell. A color class contains at most one tube above each direction cell and its directions are essentially W -separated. There are $O(B)$ colors for one I , and $O(W^{-1})$ normal intervals. $\square$

Lemma 7.29 (Dense-shading multiplicity bound in one tangential color). *Fix $\varepsilon > 0$. Let $\eta = \eta(\varepsilon) > 0$ be supplied by the three-dimensional shaded Keakeya theorem of Guth–Wang–Zahl [11]. Let $\mathbb{T}$ be one essentially direction-separated family of unit W -tubes in a bounded subset of $E \times J$, and put*

$$m(u, s) = \sum_{T \in \mathbb{T}} \mathbf{1}_T(u, s).$$

For a dyadic multiplicity level $X_k = \{2^k \leq m < 2^{k+1}\}$, give each tube the induced shading $Y_k(T) = T \cap X_k$. Write

$$\mathbb{T}_{k, \text{dense}} = \{T : |Y_k(T)| \geq W^\eta |T|\}.$$

Then at least one of the following holds:

- (i) *the dense tubes carry at least half of the incidence on X_k , and*

$$\boxed{2^k \lesssim_\varepsilon W^{-\varepsilon};} \quad (185)$$

- (ii) *more than half of that incidence is carried by tube shadings satisfying*

$$\boxed{|Y_k(T)| < W^\eta |T|.} \quad (186)$$

Consequently, if alternative (ii) does not carry a fixed portion of any multiplicity level, then

$$\boxed{\int m^2 \lesssim_\varepsilon W^{-\varepsilon} \int m.} \quad (187)$$

Proof. An essentially direction-separated tube family has $\Delta_{\max}(\mathbb{T}) \lesssim 1$. In alternative (i), restrict to $\mathbb{T}_{k, \text{dense}}$ with the shadings Y_k . The shaded three-dimensional Keakeya theorem gives

$$|X_k| \geq |U(\mathbb{T}_{k, \text{dense}}, Y_k)| \gtrsim_\varepsilon W^\varepsilon \#\mathbb{T}_{k, \text{dense}} |T|.$$

On the other hand, the dense incidence is at least a fixed multiple of $2^k |X_k|$ and at most $\#\mathbb{T}_{k, \text{dense}} |T|$. Combining the inequalities proves (185). If alternative (ii) never carries a fixed portion, dyadic summation gives

$$\int m^2 \asymp \sum_k 2^k \int_{X_k} m \lesssim_\varepsilon W^{-\varepsilon} \sum_k \int_{X_k} m,$$

which is (187). $\square$

Proposition 7.30 (The dense tangential contribution closes every total defect below one half). *Use the collision-preserved endpoint set in Theorem 7.25, and let ν_* be the restriction of ν to that set. At scale W , replace ν_* by its weighted occupied product cells. After discarding endpoint weight $o(C_W)$ and making a logarithmic dyadic refinement, the cell weights are constant on each weight layer, every weight is at most $C_U W^3$, and there are only $W^{-o(1)}$ relevant layers. The discarded endpoint weight removes only $o(C_W)$ from the collision measure.*

Suppose that, in every relevant weight layer, color class, and multiplicity level, the dense alternative (i) of Theorem 7.29 holds. If the vertical branch bound is B , then

$$\boxed{\int_J \iint \mathbf{1}_{\{|T_s(z, z')| \leq W\}} d\nu_*(z) d\nu_*(z') ds \lesssim_\varepsilon B W^{2-\varepsilon-o(1)} \nu_*(G).} \quad (188)$$

For the collision-preserving branch $B = W^{-\chi-\zeta-o(1)}$, this is

$$\boxed{\lesssim W^{2-\chi-\varepsilon-\zeta-o(1)}.} \quad (189)$$

Since the retained full collision mass is at least $\frac{1}{2} C_W = W^{1+\chi+o(1)}$, every level with

$$\boxed{\chi < \frac{1}{2}} \quad (190)$$

is impossible in the dense-shading branch: choose $\varepsilon, \zeta > 0$ with $\varepsilon + \zeta < 1 - 2\chi$. Therefore every hypothetical bad level with $\chi < 1/2$ is forced into the short-Reeb-shading alternative (186).

Proof. Let $\mathcal{Q}_W$ be the occupied full product cells and put $w_{\mathfrak{c}} = \nu_*(\mathfrak{c})$. Bounded horizontal density gives

$$\sum_{\mathfrak{c} \text{ above } A} w_{\mathfrak{c}} \lesssim_U W^3$$

for every full direction W -cell A , so in particular $w_{\mathfrak{c}} \lesssim_U W^3$. Since $\#\mathcal{Q}_W \lesssim_{\zeta} W^{-3-\zeta}$, weights below a sufficiently large fixed power of W have total mass $o(\mathcal{C}_W)$; choose that power after fixing the total-defect level. Because the collision kernel is bounded by one and $\nu(G) = 1$, deleting endpoint mass e deletes at most $O_J(e)$ collision mass. The remaining weights have logarithmically many dyadic levels.

For one weight level of size w , let

$$F(u, s) = \sum_{\mathfrak{c}} w \mathbf{1}_{\mathbb{T}_{\mathfrak{c}}}(u, s).$$

By Theorem 7.28, write its labeled tubes as the union of $J_W \lesssim BW^{-1}$ direction-separated color classes and write $F = \sum_{j=1}^{J_W} F_j$. On the dense-shading branch, Theorem 7.29 gives

$$\int F_j^2 \lesssim_{\varepsilon} w W^{-\varepsilon} \int F_j.$$

Pointwise Cauchy–Schwarz across the colors, the weight bound $w \lesssim W^3$, and $\sum_j \int F_j \lesssim W^2 \nu_*(G)$ give

$$\int F^2 \leq J_W \sum_j \int F_j^2 \lesssim_{\varepsilon} (BW^{-1}) W^3 W^{-\varepsilon} W^2 \nu_*(G) = BW^{4-\varepsilon} \nu_*(G).$$

The logarithmic sum over weight layers is absorbed by $W^{-o(1)}$.

If $|T_s(z, z')| \leq W$, the two enlarged tangential tube cross-sections at time s have intersection area $\gtrsim W^2$. After replacing points by their product-cell representatives, which changes all radii by a fixed factor,

$$\int_J \iint \mathbf{1}_{\{|T_s| \leq W\}} d\nu_* d\nu'_* ds \lesssim W^{-2} \int_{E \times J} F^2.$$

This proves (188). Substitute the branch bound from (170) to obtain (189). Comparing it with the retained mass (165) gives the last assertion. $\square$

Corollary 7.31 (Mass-sensitive dense cross-collision bound). *Let $\nu_1, \nu_2 \leq \nu$ be two endpoint restrictions of masses m_1, m_2 . Assume their scale- W tube families obey the same vertical branch cap B , and that all relevant weight, color, and multiplicity levels lie in the dense alternative of Theorem 7.29. Then*

$$\boxed{\int_J \iint \mathbf{1}_{\{|T_s(z, z')| \leq W\}} d\nu_1(z) d\nu_2(z') ds \lesssim_{\varepsilon} BW^{2-\varepsilon-o(1)} (m_1 m_2)^{1/2}.} \quad (191)$$

Proof. Let F_1, F_2 be the weighted tangential tube functions used in the proof of Theorem 7.30. The tube-intersection comparison and Cauchy–Schwarz give

$$\text{left side of (191)} \lesssim W^{-2} \int F_1 F_2 \leq W^{-2} \|F_1\|_2 \|F_2\|_2.$$

The mass-sensitive form of the preceding proof gives $\|F_i\|_2^2 \lesssim BW^{4-\varepsilon-o(1)} m_i$. After replacing ε by $\varepsilon/2$, the displayed estimate follows. $\square$

Lemma 7.32 (Normal-compatible collision shading regularization). *Discretize the retained measure Θ_W^* at scale W , put $\nu^* = \nu|_{G_{\Theta, W}}$, and let $w_{\mathfrak{c}} = \nu^*(\mathfrak{c})$ be the endpoint-cell weights. For two labeled cells define the measurable edge-time density*

$$g_{\mathfrak{c}\mathfrak{d}}(s) := \iint_{\mathfrak{c} \times \mathfrak{d}} \mathbf{1}_{\{|N_s(z, z')| \leq W\}} \mathbf{1}_{\{|T_s(z, z')| \leq W\}} d\nu^*(z) d\nu^*(z'). \quad (192)$$

Thus

$$\boxed{\mathcal{C}_W^* = \sum_{\mathfrak{c}, \mathfrak{d}} \int g_{\mathfrak{c}\mathfrak{d}}(s) ds, \quad 0 \leq g_{\mathfrak{c}\mathfrak{d}}(s) \leq w_{\mathfrak{c}} w_{\mathfrak{d}}.} \quad (193)$$

After deleting $o(\mathcal{C}_W^*)$, there is a decomposition into $W^{-o(1)}$ nonnegative symmetric pieces. One piece, of mass $\mathcal{C}_{\alpha} \gtrsim W^{o(1)} \mathcal{C}_W^*$, has two endpoint families $\mathcal{V}_{\alpha, 1}, \mathcal{V}_{\alpha, 2}$ and numbers $m_i, h_i, \lambda_i > 0$, $i = 1, 2$, with the following properties: throughout this lemma, $\asymp_{W^{o(1)}}$ denotes two-sided comparison with losses bounded by $W^{-o(1)}$.

(i) the endpoint weight and integrated degree are biregular at the power scale,

$$\boxed{\sum_{\mathbf{c} \in \mathcal{V}_{\alpha,i}} w_{\mathbf{c}} \asymp_{W^{o(1)}} m_i, \quad \int_J D_{\alpha,i,\mathbf{c}}(s) ds \asymp_{W^{o(1)}} h_i, \quad m_i h_i \asymp_{W^{o(1)}} \mathcal{C}_{\alpha};} \quad (194)$$

(ii) for every retained endpoint cell $\mathbf{c}$, its active time set

$$S_{\alpha,i,\mathbf{c}} = \{s : D_{\alpha,i,\mathbf{c}}(s) > 0\}$$

has $|S_{\alpha,i,\mathbf{c}}| \asymp_{W^{o(1)}} \lambda_i$, and on a subset of comparable measure

$$\boxed{D_{\alpha,i,\mathbf{c}}(s) \asymp_{W^{o(1)}} \frac{h_i}{\lambda_i};} \quad (195)$$

(iii) the time-saturated tangential shading

$$Y_{\alpha,i}(\mathbb{T}_{\mathbf{c}}) := \{(u, s) \in \mathbb{T}_{\mathbf{c}} : s \in S_{\alpha,i,\mathbf{c}}\} \quad (196)$$

has density $\asymp_{W^{o(1)}} \lambda_i$ inside its tube.

(iv) the decomposition is simultaneous with the tangential color and multiplicity levels used in Theorem 7.29: for every selected edge-time, the midpoint of the two enlarged tangential tube cross-sections lies in fixed dyadic multiplicity levels for the two endpoint colors. Thus, when $\lambda_i \geq W^\eta$, every retained tube on side i belongs to the dense class for that selected multiplicity level.

Here $D_{\alpha,i,\mathbf{c}}(s)$ is the total opposite-endpoint weight joined to $\mathbf{c}$ by the selected piece at time s , including the selected dyadic edge-density factor. In particular the dichotomy

$$\boxed{\lambda_1, \lambda_2 \geq W^\eta \quad \text{or} \quad \min\{\lambda_1, \lambda_2\} < W^\eta} \quad (197)$$

is a decomposition of the actual normal-compatible collision mass, not of an enlarged unweighted tangential overlap.

Proof. Identity (193) is Tonelli's theorem, and the upper bound follows by deleting the two indicators. Discard endpoint cells of sufficiently small weight exactly as in the proof of Theorem 7.30; their total collision contribution is $o(\mathcal{C}_W^*)$. On the remaining finite grid write

$$r_{\mathbf{cd}}(s) = \frac{g_{\mathbf{cd}}(s)}{w_{\mathbf{c}} w_{\mathbf{d}}} \in [0, 1].$$

The part where $r_{\mathbf{cd}} < W^A$, with A chosen after the total-defect level, has mass at most $W^A |J|$ and is negligible. Dyadically partition the two endpoint weights and the remaining values of $r_{\mathbf{cd}}$. This costs only a power of $\log(2/W)$.

For every nonzero edge-time, choose the midpoint of the two tangential front centers; after a fixed enlargement it lies in both tangential tubes. Record the dyadic multiplicities of the two endpoint color classes at that point. This adds only logarithmically many choices, without selecting one of the power-many colors. For each resulting piece define its weighted time degree at either endpoint by summing the opposite endpoint weights times the selected r -level. Dyadically partition the positive degrees, then the integrals of those degrees along each endpoint, and finally the measures of their time supports after the selected tangential multiplicity restriction. Remove endpoint classes carrying less than a fixed logarithmic fraction of the piece and repeat once from the opposite side. The usual two-sided degree pruning leaves a subpiece with a logarithmic fraction of the mass on which both endpoint degrees have the stated lower and upper bounds. Summing an endpoint's integrated degree against its cell weight gives the piece mass; this proves (194). The degree and support-size pigeonholes give (195). A unit tangential W -tube has cross-sectional area $\asymp W^2$, so the time-saturated set (196) has volume $\asymp W^2 \lambda_i$, proving its density statement. There are only logarithmically many choices at each stage, which is $W^{-o(1)}$; one piece carries the asserted mass. $\square$

Proposition 7.33 (Reeb-time support converts collision degree into physical bush volume). *Let one piece of Theorem 7.32 have endpoint data (m_i, h_i, λ_i) , $i = 1, 2$, and suppose $\nu \leq K\sigma$. For either endpoint orientation and every retained endpoint cell $\mathbf{c}$, there is a time $s_{i,\mathbf{c}} \in J$ and an actual selector-direction set $H_{i,\mathbf{c}}$ which is a physical CW-bush at the corresponding front point and obeys*

$$\boxed{\sigma(H_{i,\mathbf{c}}) \gtrsim \frac{h_i}{K\lambda_i} W^{o(1)}}. \quad (198)$$

Consequently

$$\boxed{\mathcal{L}^4(N_{CW}(K_b)) \gtrsim \frac{h_i}{K\lambda_i} W^{o(1)} \quad (i = 1, 2).} \quad (199)$$

This holds for dense and sparse pieces alike; no summation over center cells and no escaped-bush overlap estimate are required.

Proof. For every retained endpoint cell, the biregularization gives a subset of its active time set on which

$$D_{\alpha,i,\epsilon}(s) \asymp_{W^{o(1)}} \frac{h_i}{\lambda_i}.$$

Choose one such time. Since the selected edge density is at most one, the actual opposite-endpoint occurrence weight of the corresponding neighbor set is at least the left side. Division by the reuse cap gives (198). The two polarized inequalities in (160) place every leaf front in the same $O(W)$ -ball; the diameter of a phase-space cell changes the constant only. Apply the arbitrary-mass Reeb escape formula Theorem 4.17 at radius CW to obtain (199). $\square$

7.5 Normal shear and direct packet bounds

Proposition 7.34 (A q -local dense collision gains the missing color factor). *Let a dense piece of Theorem 7.32 have endpoint masses m_1, m_2 , and suppose its collision-preserving vertical branch cap is B . Partition the normal direction coordinate q into half-open intervals of length W , and let $\mathcal{C}_{\text{loc}}$ be the part of the actual normal-compatible collision mass joining cells whose q -intervals have distance $O(W)$. Then*

$$\boxed{\mathcal{C}_{\text{loc}} \lesssim_{\epsilon} B W^{3-\epsilon-o(1)} (m_1 m_2)^{1/2}.} \quad (200)$$

The extra factor W , compared with (191), comes from retaining one normal-offset slab before coloring the tangential tubes.

Proof. For one q -interval I , let $F_{i,I}$ be the weighted tangential tube function of endpoint side i , restricted to the selected cell-weight and multiplicity layer. Inside I , Theorem 7.28 uses $O(B)$ colors, not the global $O(BW^{-1})$ colors. On one color the dense-shading estimate and the cell-weight bound $w_{\epsilon} \lesssim W^3$ give

$$\|F_{i,I,j}\|_2^2 \lesssim_{\epsilon} W^{3-\epsilon-o(1)} \int F_{i,I,j}.$$

Since a tangential unit W -tube has volume $\asymp W^2$,

$$\sum_j \int F_{i,I,j} \lesssim W^2 m_{i,I},$$

where $m_{i,I}$ is the endpoint mass over I . Pointwise Cauchy–Schwarz over the $O(B)$ colors therefore yields

$$\|F_{i,I}\|_2^2 \lesssim_{\epsilon} B W^{5-\epsilon-o(1)} m_{i,I}.$$

For a pair I, I' at distance $O(W)$, Cauchy–Schwarz between the two endpoint sides and the standard tangential tube-intersection comparison give

$$\mathcal{C}_{\text{loc}}(I, I') \lesssim W^{-2} \int F_{1,I} F_{2,I'} \lesssim_{\epsilon} B W^{3-\epsilon-o(1)} (m_{1,I} m_{2,I'})^{1/2}.$$

Every I has only $O(1)$ admissible I' . Sum and apply discrete Cauchy–Schwarz, using $\sum_I m_{i,I} = m_i$, to prove (200). Dropping the normal indicator and the selected edge-density factor only enlarges the expression, so the estimate applies to the actual cellwise collision density (192). $\square$

Corollary 7.35 (Below unit defect, every dense survivor has genuine normal shear). *Let $1/2 \leq \chi < 1$, take*

$$B = W^{-\chi-\zeta-o(1)}, \quad m_i = W^{\chi-\theta_i+o(1)}, \quad 0 \leq \theta_i \leq \chi,$$

and choose $\epsilon, \zeta > 0$ with $\epsilon + \zeta < 2(1 - \chi)$. Then

$$\boxed{\mathcal{C}_{\text{loc}} \lesssim W^{3-(\theta_1+\theta_2)/2-\epsilon-\zeta-o(1)} = o(W^{1+\chi+o(1)}).} \quad (201)$$

Hence a dense retained collision layer of mass $W^{1+\chi+o(1)}$ cannot remain in $|q - q'| = O(W)$. After a logarithmic dyadic refinement, a subpower fraction of that mass lies on

$$\boxed{r \leq |q - q'| < 2r, \quad W \lesssim r \leq 1.} \quad (202)$$

Proof. Substitute the displayed powers in (200). Since $(\theta_1 + \theta_2)/2 \leq \chi$, the exponent on the right of (201) is at least $3 - \chi - \varepsilon - \zeta > 1 + \chi$. This proves the strict power saving. The complement of the local part is covered by $O(\log(2/W)) = W^{-o(1)}$ dyadic q -separation annuli, so one annulus retains a subpower fraction. $\square$

Proposition 7.36 (A normal-shear annulus labels the full collision windows). *Apply two-sided degree regularization once more inside the annulus (202), absorbing logarithms into $W^{-o(1)}$. For a retained endpoint cell on side i , let h_i be its integrated annular collision degree and λ_i its active-time support length. Its swept opposite-endpoint selector-direction set $L_{i,c}$ satisfies the fixed-angle bound*

$$\sigma(L_{i,c}) \gtrsim \frac{h_i}{KW} W^{o(1)}. \quad (203)$$

The active time set requires at least

$$N_{\text{edge},i} \gtrsim \max \left\{ 1, \frac{\lambda_i}{W} \right\} W^{o(1)} \quad (204)$$

distinct full collision windows of length $O(W)$. In addition, every such edge between phase offsets (q, p) and (q', p') has a normal phase

$$s_N := -\frac{p - p'}{q - q'}, \quad |s_e - s_N| \lesssim \frac{W}{r}, \quad (205)$$

where s_e is any selected full collision time for that edge.

If

$$h_i = W^{1+\theta_i+o(1)}, \quad \lambda_i = W^{\ell_i+o(1)}, \quad r = W^{\rho+o(1)}, \quad 0 \leq \rho \leq 1,$$

then

$$\sigma(L_{i,c}) \gtrsim W^{\theta_i+o(1)}, \quad N_{\text{edge},i} \gtrsim \max\{1, W^{\ell_i-1+o(1)}\}, \quad |s_e - s_N| \lesssim W^{1-\rho+o(1)}. \quad (206)$$

Thus the annular survivor is a family of fixed-angle contact edges with power-counted Reeb windows and an additional normal-symplectic phase label; the latter, rather than a weaker window count, is the new information supplied by the q -shear.

Proof. On the normalized angular shell, every full-vector collision section of one edge has length $O(W)$. Fubini therefore gives

$$h_i \lesssim W \nu(L_{i,c}) W^{-o(1)},$$

and division by the reuse cap proves (203). The same $O(W)$ length bound covers an active set of measure $\lambda_i W^{o(1)}$; a greedy disjoint-subinterval selection proves (204). On the annulus $|q - q'| \asymp r$, the normal component of the polarized contact equation is

$$|(p - p') + s_e(q - q')| \leq CW.$$

Division by $|q - q'|$ proves (205). Substitution of the three power parameters and $K = W^{-o(1)}$ gives (206). $\square$

Proposition 7.37 (Mesoscopic q -near collisions retain the color gain). *Under the hypotheses of Theorem 7.34, fix $W \leq R_q \leq 1$, and let $C_{\leq R_q}$ be the actual normal-compatible collision mass on $|q - q'| \leq CR_q$. Then*

$$C_{\leq R_q} \lesssim_\varepsilon B R_q W^{2-\varepsilon-o(1)} (m_1 m_2)^{1/2}. \quad (207)$$

For $R_q = W$ this is (200); for $R_q < 1$ it retains the precise fraction R_q of the global q -color count.

Proof. Partition the bounded q -range into half-open intervals I of length R_q . Such an interval contains $O(R_q/W)$ scale- W normal direction slabs. Apply Theorem 7.28 in each of them: the tangential tubes over I use at most $O(BR_q/W)$ direction-separated colors. The one-color calculation in the proof of Theorem 7.34 is unchanged,

$$\|F_{i,I,j}\|_2^2 \lesssim_\varepsilon W^{5-\varepsilon-o(1)} m_{i,I,j}.$$

Pointwise Cauchy-Schwarz over $O(BR_q/W)$ colors gives

$$\|F_{i,I}\|_2^2 \lesssim_\varepsilon B R_q W^{4-\varepsilon-o(1)} m_{i,I}.$$

Only $O(1)$ pairs of coarse q -intervals can satisfy $|q - q'| \leq CR_q$. Apply Cauchy-Schwarz between the endpoint functions, multiply by the tangential intersection factor W^{-2} , and sum the coarse interval pairs. This proves (207). $\square$

Lemma 7.38 (A marked edge costs the two regularized graphon degrees). *Let $0 \leq h \leq 1$ be a bipartite graphon on $X_1 \times X_2$, with row and column degrees bounded by D_1, D_2 . If $0 \leq h_m \leq h$ has edge mass*

$$M_m := \iint h_m(x, y) d\nu_1(x) d\nu_2(y),$$

then the alternating four-cycle mass whose first edge is marked obeys

$$\boxed{\int h_m(x_0, y_1) h(x_2, y_1) h(x_2, y_3) h(x_0, y_3) d\nu_1(x_0) d\nu_2(y_1) d\nu_1(x_2) d\nu_2(y_3) \leq M_m D_1 D_2.} \quad (208)$$

Consequently cycles having at least one marked edge have mass at most $4M_m D_1 D_2$.

Proof. For fixed x_0, y_1, x_2 ,

$$\int h(x_2, y_3) h(x_0, y_3) d\nu_2(y_3) \leq D_1.$$

Integrating $h(x_2, y_1)$ in x_2 contributes at most D_2 . The remaining integral is M_m , proving (208). Sum over the four possible marked positions. $\square$

Proposition 7.39 (Mesoscopic normal phases survive on every rank-loss cycle below $3/5$). *Apply the arbitrary-threshold Maslov routing after passing to the two-sided degree block; equivalently symmetrize that bipartite block on two disjoint copies. Put*

$$\bar{\theta} = \frac{\theta_1 + \theta_2}{2}, \quad M_W = W^{\chi+o(1)}, \quad D_i = W^{\theta_i+o(1)}.$$

Suppose the determinant-small output has four-cycle mass $\gtrsim M_W^4$. Let $R_q = W^\alpha$, where

$$\boxed{0 < \alpha < 1, \quad 1 + \alpha + \bar{\theta} > 4\chi.} \quad (209)$$

Then, after deleting $o(M_W^4)$ and making four logarithmic dyadic refinements, every edge of the retained rank-loss cycles satisfies

$$\boxed{r_e \leq |q - q'| < 2r_e, \quad R_q \lesssim r_e \leq 1,} \quad (210)$$

and carries a normal phase label

$$\boxed{s_{N,e} := -\frac{p - p'}{q - q'}, \quad |s_e - s_{N,e}| \lesssim \frac{W}{r_e} \leq CW^{1-\alpha}.} \quad (211)$$

The condition (209) can always be met when

$$\boxed{\frac{1}{2} \leq \chi < \frac{3}{5}.} \quad (212)$$

Indeed the dense wedge gives $\bar{\theta} \geq 1 - \chi - o(1)$, so it is enough to choose

$$5\chi - 2 < \alpha < 1.$$

Proof. Normalize (207) by the universal collision-window factor W . With $B = W^{-\chi-\zeta-o(1)}$, $R_q = W^\alpha$, and $(m_1 m_2)^{1/2} = W^{\chi-\bar{\theta}+o(1)}$, the marked q -near graphon edge mass satisfies

$$M_{\leq R_q} \lesssim W^{1+\alpha-\bar{\theta}-\varepsilon-\zeta-o(1)}.$$

Apply Theorem 7.38. Since $D_1 D_2 = W^{2\bar{\theta}+o(1)}$, all rank-loss cycles containing a q -near edge have mass at most

$$\boxed{\lesssim W^{1+\alpha+\bar{\theta}-\varepsilon-\zeta-o(1)}.} \quad (213)$$

Choose $\varepsilon, \zeta > 0$ smaller than the strict gap in (209). Then (213) is $o(W^{4\chi+o(1)})$, so those cycles may be deleted. Dyadically partition the four remaining q -separations; one four-tuple of annuli retains a subpower fraction. Equation (205) gives (211). Finally $\bar{\theta} \geq 1 - \chi - o(1)$ reduces the strict inequality in (209) to $\alpha > 5\chi - 2 + o(1)$, which admits an $\alpha < 1$ precisely below $\chi = 3/5$. $\square$

Lemma 7.40 (The anchored horizontal determinant gives the packet thickness directly). *Use the polarized-plane alternative of Theorems 6.19 and 7.11, and denote its horizontal noncollinearity threshold by θ_{hr} . Put*

$$u = a_{y_0} - a_p, \quad v = a_q - a_{y_0}, \quad E = \text{Span}\{u, v\}.$$

Then every retained $y \in Y_0$ satisfies

$$\boxed{\text{dist}(a_y, a_p + E) \leq \frac{\Delta}{|u \times v|} \lesssim \frac{\Delta}{\theta_{\text{hr}}}.} \quad (214)$$

Consequently the actual selector directions obtained after graph-cell de-multiplication lie in the

$$\boxed{t_{\text{dir}} \lesssim W + \frac{\Delta}{\theta_{\text{hr}}}} \quad (215)$$

neighborhood of $a_p + E$. Unlike the full phase-space conclusion (102), this horizontal estimate contains neither γ^{-1} nor v^{-1} .

Proof. For the anchored cycle p, y_0, q, y, p , the determinant-small hypothesis is

$$|\det(u, v, a_y - a_q)| \leq \Delta.$$

Moreover

$$u \times (a_q - a_p) = u \times (u + v) = u \times v,$$

so the noncollinearity condition in the polarized alternative gives $|u \times v| \geq \theta_{\text{hr}}$. The point-to-plane formula therefore yields

$$\text{dist}(a_y - a_q, E) = \frac{|\det(u, v, a_y - a_q)|}{|u \times v|} \leq \frac{\Delta}{\theta_{\text{hr}}}.$$

Since $a_q - a_p = u + v \in E$, this is exactly (214). The final $O(W)$ is the graph-cell diameter already present in the de-multiplication conclusion of Theorem 6.19. $\square$

Lemma 7.41 (The contact equations give the same direct vertical thickness). *Under Theorem 7.40, suppose every retained neighbor y is joined to the anchor p by a selected contact edge*

$$|(b_y - b_p) + s_{py}(a_y - a_p)| \leq CW, \quad s_{py} \in J.$$

Then

$$\boxed{\text{dist}((a_y, b_y), (a_p, b_p) + (E \oplus E)) \lesssim_J W + \frac{\Delta}{\theta_{\text{hr}}}.} \quad (216)$$

This full phase-space packet estimate uses only the horizontal determinant and the original contact edge equation; it is independent of γ , the number of separated labels, and their Vandermonde product.

Proof. By (214), choose $e_y \in E$ such that

$$|a_y - a_p - e_y| \lesssim \Delta/\theta_{\text{hr}}.$$

Because E is linear and s_{py} stays in the fixed interval J ,

$$\begin{aligned} \text{dist}(b_y - b_p, E) &\leq |(b_y - b_p) + s_{py}(a_y - a_p)| + |s_{py}| \text{dist}(a_y - a_p, E) \\ &\lesssim_J W + \Delta/\theta_{\text{hr}}. \end{aligned}$$

Combine the horizontal and vertical bounds. $\square$

Proposition 7.42 (Anchored determinant-small cycles form a contact-polarized packet without a time-separation hypothesis). *Fix anchors p, q with $|a_p - a_q| \geq \tau_1$, let*

$$d\lambda_{p,q}(y) = h(p, y)h(q, y) d\nu(y), \quad T = \lambda_{p,q}(Y) > 0,$$

and let $\Gamma \subset Y \times Y$ be a symmetric measurable family of anchored cycles of $(\lambda_{p,q} \times \lambda_{p,q})$ -mass $Z > 0$. Assume only that every cycle in Γ has horizontal determinant at most Δ ; impose no condition on its four contact labels. For every $0 < \theta_{\text{hr}} < 1$, there are y_0 and a measurable $Y_0 \subset Y$ with

$$\boxed{\nu(Y_0) \geq \lambda_{p,q}(Y_0) \gtrsim Z/T} \quad (217)$$

for which one of the following holds:

- (i) its directions lie in an $O(\theta_{\text{hr}}/\tau_1)$ -tube about the horizontal line through a_p, a_q ;
- (ii) for $E = \text{Span}\{a_{y_0} - a_p, a_q - a_{y_0}\}$, all graph points in Y_0 lie in the contact-polarized phase-space packet

$$\boxed{N_{C(W+\Delta/\theta_{\text{hr}})}((a_p, b_p) + (E \oplus E))}. \quad (218)$$

Proof. Split Y by

$$|(a_y - a_p) \times (a_q - a_p)| < \theta_{\text{hr}}.$$

If this set has $\lambda_{p,q}$ -mass $\gtrsim Z/T$, the point-to-line formula and $|a_q - a_p| \geq \tau_1$ give alternative (i). Otherwise the Γ -edges incident to it have product measure at most $2T$ times its measure. Choosing the implicit constant small enough, deletion retains a fixed fraction of Z . Weighted average degree then fixes y_0 and a neighbor set Y_0 of $\lambda_{p,q}$ -mass $\gtrsim Z/T$, with

$$|(a_{y_0} - a_p) \times (a_q - a_p)| \geq \theta_{\text{hr}}.$$

The determinant hypothesis and Theorem 7.40 give the horizontal E -packet, while Theorem 7.41 gives the vertical E -packet. This proves (ii). Finally $h(p, y)h(q, y) \leq 1$ gives (217). $\square$

Lemma 7.43 (The anchor-line azimuth has bounded marginal density). *Fix distinct anchor directions $a_p, a_q \in U$, put $w = a_q - a_p$, and let*

$$\pi_{w^\perp} : \mathbb{R}^3 \longrightarrow w^\perp$$

be orthogonal projection. Away from the affine line $L_{pq} = a_p + \mathbb{R}w$, define

$$\varphi_{pq}(a) := \mathbb{P}(\pi_{w^\perp}(a - a_p)) \in \mathbb{RP}^1.$$

If the direction marginal has density at most C_U , then every projective interval $\Omega \subset \mathbb{RP}^1$ of length $\delta \leq 1$ satisfies

$$\boxed{\nu\{a \in U \setminus L_{pq} : \varphi_{pq}(a) \in \Omega\} \lesssim_U \delta}. \quad (219)$$

The constant is uniform in the anchor line.

Proof. Use cylindrical coordinates about L_{pq} . Inside the fixed bounded region U , the longitudinal coordinate and radius range over bounded intervals, while the angular coordinate belongs to a set of length $O(\delta)$. The cylindrical Jacobian is the radius, whose integral is uniformly bounded. This gives the Lebesgue-volume estimate $O_U(\delta)$; apply the bounded horizontal density. $\square$

Lemma 7.44 (The horizontal four-cycle determinant is an anchor-azimuth area). *For fixed anchors p, q , put*

$$w = a_q - a_p, \quad u_y = \pi_{w^\perp}(a_y - a_p).$$

Then every anchored cycle p, y, q, y', p satisfies the exact identity

$$\boxed{\det(a_y - a_p, a_q - a_y, a_{y'} - a_q) = \det(a_y - a_p, w, a_{y'} - a_p)}, \quad (220)$$

and hence

$$\boxed{|\det(a_y - a_p, a_q - a_y, a_{y'} - a_q)| = |w| |u_y| |u_{y'}| |\sin(\varphi_{pq}(a_y) - \varphi_{pq}(a_{y'}))|}. \quad (221)$$

If

$$|(a_y - a_p) \times w|, |(a_{y'} - a_p) \times w| \geq \theta_{\text{az}},$$

then determinant at most Δ implies

$$\boxed{d_{\mathbb{RP}^1}(\varphi_{pq}(a_y), \varphi_{pq}(a_{y'})) \lesssim_U \frac{\Delta}{\theta_{\text{az}}^2}}. \quad (222)$$

Proof. Write $a_y - a_p = u$ and $a_{y'} - a_p = v$. Multilinearity gives

$$\det(u, w - u, v - w) = \det(u, w, v),$$

which is (220). Components of u, v parallel to w do not contribute, proving (221). Moreover

$$|u_y| = \frac{|(a_y - a_p) \times w|}{|w|}, \quad |u_{y'}| = \frac{|(a_{y'} - a_p) \times w|}{|w|}.$$

The anchor directions remain in a fixed bounded chart, so

$$|w| |u_y| |u_{y'}| \gtrsim_U \theta_{\text{az}}^2.$$

For a sufficiently small right side, sine is comparable to projective angular distance; for a larger right side (222) is trivial after changing the constant. $\square$

Proposition 7.45 (Direct bound for the complete anchored determinant-small graph). *Use the anchored graphon notation*

$$d\lambda_{p,q}(y) = h(p, y)h(q, y) d\nu(y), \quad T = \lambda_{p,q}(Y),$$

and let $\Gamma \subset Y \times Y$ be symmetric, of $(\lambda_{p,q} \times \lambda_{p,q})$ -mass Z , with horizontal determinant at most Δ on every pair. Assume $|a_p - a_q| \geq \tau_1$. Then, for every $0 < \theta_{\text{az}} < 1$,

$$\frac{Z}{T} \lesssim_U \left(\frac{\theta_{\text{az}}}{\tau_1} \right)^2 + \frac{\Delta}{\theta_{\text{az}}^2}. \quad (223)$$

With

$$\theta_{\text{az}} = (\Delta \tau_1^2)^{1/4},$$

provided this lies in $(0, 1)$, one obtains

$$\frac{Z}{T} \lesssim_U \Delta^{1/2} \tau_1^{-1}. \quad (224)$$

This estimate controls the entire anchored cycle graph and does not first select one average-degree packet.

Proof. Let

$$Y_{\text{line}} = \{y : |(a_y - a_p) \times (a_q - a_p)| < \theta_{\text{az}}\}.$$

The point-to-line formula and $|a_q - a_p| \geq \tau_1$ give

$$\lambda_{p,q}(Y_{\text{line}}) \leq \nu(Y_{\text{line}}) \lesssim_U (\theta_{\text{az}}/\tau_1)^2.$$

Pairs with at least one endpoint in this set have product measure at most

$$2T \lambda_{p,q}(Y_{\text{line}}).$$

On the complementary pair set, Theorem 7.44 places y' , for each fixed y , in a projective angular interval of length

$$O_U(\Delta/\theta_{\text{az}}^2).$$

Since $\lambda_{p,q} \leq \nu$, Theorem 7.43 bounds the $\lambda_{p,q}$ -mass of this interval by the same quantity. Integration in y therefore bounds the far-far pair mass by

$$C_U T \Delta / \theta_{\text{az}}^2.$$

Add the two contributions and divide by T , proving (223). The displayed parameter choice balances $\theta_{\text{az}}^2/\tau_1^2$ with $\Delta/\theta_{\text{az}}^2$. $\square$

7.6 Azimuth sectors and rooted contact stars

Lemma 7.46 (Sector-square form of the anchored azimuth estimate). *Retain the hypotheses and notation of Theorem 7.45. Put*

$$Y_{\text{line}}(p, q; \theta_{\text{az}}) = \{y : |(a_y - a_p) \times (a_q - a_p)| < \theta_{\text{az}}\}, \quad L_{p,q} = \lambda_{p,q}(Y_{\text{line}}),$$

and suppose

$$\delta_{\text{az}} := C_U \frac{\Delta}{\theta_{\text{az}}^2} \leq C_U.$$

There is a bounded-overlap family $\mathcal{J}_{p,q}$ of projective intervals of length $O_U(\delta_{\text{az}})$ such that, on writing

$$A_{p,q,\omega} = \{y \notin Y_{\text{line}} : \varphi_{pq}(a_y) \in \omega\}, \quad m_{p,q,\omega} = \lambda_{p,q}(A_{p,q,\omega}),$$

one has

$$Z(p, q) \lesssim T(p, q) L_{p,q} + \sum_{\omega \in \mathcal{J}_{p,q}} m_{p,q,\omega}^2. \quad (225)$$

In particular, without replacing either term by a density cap, one has the following exact alternative:

$$L_{p,q} \gtrsim Z(p, q)/T(p, q) \quad \text{or} \quad \sum_{\omega \in \mathcal{J}_{p,q}} m_{p,q,\omega}^2 \gtrsim Z(p, q). \quad (226)$$

The first output is the horizontal anchor-line packet itself; the second retains all azimuth packet occurrences and their quadratic masses.

Proof. Partition $\mathbb{RP}^1$ into half-open intervals of length comparable to δ_{az} , and enlarge each interval by a fixed number of neighbors. The enlarged family has bounded overlap. By (222), every determinant-small pair with both endpoints outside Y_{line} belongs to one enlarged interval. Its product mass is therefore at most a constant multiple of $\sum_{\omega} m_{p,q,\omega}^2$. Pairs with at least one endpoint in Y_{line} have mass at most $2T(p,q)L_{p,q}$. This proves (225); splitting according to which displayed contribution is at least a fixed fraction of $Z(p,q)$ proves (226). $\square$

Lemma 7.47 (Every azimuth sector is a contact-polarized packet). *For $\omega \in \mathcal{J}_{p,q}$, choose a representative projective ray $e_{\omega} \subset (a_q - a_p)^{\perp}$, and set*

$$E_{p,q,\omega} = \text{Span}\{a_q - a_p, e_{\omega}\} \subset \mathbb{R}^3.$$

Then

$$\text{dist}(a_y, a_p + E_{p,q,\omega}) \lesssim_U \delta_{\text{az}} \quad (y \in A_{p,q,\omega}). \quad (227)$$

If, as supplied by the normalized contact graphon, every retained edge (p, y) has a selected Reeb label $s_{py} \in J$ satisfying

$$|(b_y - b_p) + s_{py}(a_y - a_p)| \leq CW,$$

then the same sector obeys the full phase-space estimate

$$\boxed{\text{dist}((a_y, b_y), (a_p, b_p) + (E_{p,q,\omega} \oplus E_{p,q,\omega})) \lesssim_{U,J} W + \delta_{\text{az}}.} \quad (228)$$

Thus every summand $m_{p,q,\omega}^2$ in (225) is the pair mass of one explicitly identified contact-symplectic polarization occurrence.

Proof. The component of $a_y - a_p$ parallel to $a_q - a_p$ already lies in $E_{p,q,\omega}$. Its orthogonal component has projective direction within $O_U(\delta_{\text{az}})$ of e_{ω} , and its length is uniformly bounded because the direction chart is bounded. This proves (227). Since $E_{p,q,\omega}$ is linear and s_{py} remains in the fixed interval J ,

$$\begin{aligned} \text{dist}(b_y - b_p, E_{p,q,\omega}) &\leq |(b_y - b_p) + s_{py}(a_y - a_p)| \\ &\quad + |s_{py}| \text{dist}(a_y - a_p, E_{p,q,\omega}) \\ &\lesssim_{U,J} W + \delta_{\text{az}}. \end{aligned}$$

Combine the horizontal and vertical estimates. This is precisely the contact lift used in Theorem 7.41, now applied sector by sector rather than after an average-degree selection. $\square$

Lemma 7.48 (The sector-square ledger has an exact rooted contact load). *Let $0 \leq h \leq 1$ be the biregular bipartite Maslov graphon on $X_1 \times X_2$, with row and column degrees bounded by D_1, D_2 . Anchor on $p, q \in X_1$, and use the sector sets of Theorem 7.46. Thus*

$$d\lambda_{p,q}(y) = h(p, y)h(q, y) d\nu_2(y), \quad m_{p,q,\omega} = \lambda_{p,q}(A_{p,q,\omega}).$$

Define the rooted sector load on X_2 by

$$\mathcal{R}_2(y) := \iint_{X_1 \times X_1} \sum_{\omega \in \mathcal{J}_{p,q}} \mathbf{1}_{A_{p,q,\omega}}(y) h(p, y) h(q, y) m_{p,q,\omega} d\nu_1(p) d\nu_1(q). \quad (229)$$

Then the sector-square mass is rooted exactly:

$$\boxed{\int_{X_2} \mathcal{R}_2(y) d\nu_2(y) = \iint_{X_1 \times X_1} \sum_{\omega \in \mathcal{J}_{p,q}} m_{p,q,\omega}^2 d\nu_1(p) d\nu_1(q) = \mathfrak{S}_{\text{az}}.} \quad (230)$$

Moreover, if

$$d_2(y) := \int_{X_1} h(x, y) d\nu_1(x),$$

then one has the pointwise weighted-reuse estimate

$$\boxed{\mathcal{R}_2(y) \lesssim D_1 d_2(y)^2 \leq D_1 D_2^2.} \quad (231)$$

The symmetric assertion holds after interchanging the two endpoint sides.

Proof. The first identity is Tonelli:

$$\int_{X_2} \mathbf{1}_{A_{p,q,\omega}}(y) h(p, y) h(q, y) d\nu_2(y) = m_{p,q,\omega}.$$

For the pointwise estimate, put

$$c(p, q) := \int_{X_2} h(p, y') h(q, y') d\nu_2(y').$$

For fixed p, q, y , bounded overlap of the enlarged azimuth intervals gives

$$\sum_{\omega} \mathbf{1}_{A_{p,q,\omega}}(y) m_{p,q,\omega} \lesssim c(p, q).$$

Also $c(p, q) \leq \min\{\deg(p), \deg(q)\} \leq D_1$. Hence

$$\mathcal{R}_2(y) \lesssim D_1 \iint h(p, y) h(q, y) d\nu_1(p) d\nu_1(q) = D_1 d_2(y)^2,$$

which proves (231). $\square$

Proposition 7.49 (Occurrence-dependent lightness is low rooted load or a common contact star). *For every occurrence $Q = (p, q, \omega)$, let $B_Q \subset A_{p,q,\omega}$ be an arbitrary measurable subset; in particular B_Q may be the part of the packet lying in a q -slab declared light only after conditioning on Q . Define its weighted paired content by*

$$\mathcal{L}_2(\mathcal{B}) := \iint_{X_1 \times X_1} \sum_{\omega \in \mathcal{S}_{p,q}} m_{p,q,\omega} \lambda_{p,q}(B_Q) d\nu_1(p) d\nu_1(q). \quad (232)$$

Let $m_2 = \nu_2(\text{supp } d_2)$. For every $K > 0$,

$$\mathcal{L}_2(\mathcal{B}) \leq K m_2 + \int_{\{\mathcal{R}_2 > K\}} \mathcal{R}_2(y) d\nu_2(y). \quad (233)$$

Every root in the second term has normalized graphon degree

$$d_2(y) \gtrsim \left(\frac{K}{D_1} \right)^{1/2}. \quad (234)$$

More importantly, the sector label is not lost. For every such root there are anchors p, q and a sector ω with $y \in A_{p,q,\omega}$, $h(p, y) h(q, y) > 0$, and

$$m_{p,q,\omega} \gtrsim \frac{\mathcal{R}_2(y)}{d_2(y)^2} \gtrsim \frac{K}{D_2^2}. \quad (235)$$

Thus a high root is a common contact star carrying one actual anchor pair together with a same-azimuth $E_{p,q,\omega} \oplus E_{p,q,\omega}$ packet; it is not merely a vertex of large unlabeled degree. For the normalized contact graphon h_W , such a root therefore produces, by the original Reeb-label averaging, a one-time common contact star of occurrence weight

$$\gtrsim_J W \left(\frac{K}{D_1} \right)^{1/2}, \quad (236)$$

and a swept neighbor set of occurrence mass

$$\gtrsim \left(\frac{K}{D_1} \right)^{1/2}. \quad (237)$$

After the existing occurrence de-multiplication, these are precisely the common-contact-star inputs to the Reeb-time/Lagrangian-bush alternative of Theorems 7.27 and 7.33.

Proof. Root (232) at its restricted endpoint y . The resulting density $\mathcal{R}_{2,\mathcal{B}}(y)$ satisfies

$$0 \leq \mathcal{R}_{2,\mathcal{B}}(y) \leq \mathcal{R}_2(y)$$

because $B_Q \subset A_{p,q,\omega}$. On $\{\mathcal{R}_2 \leq K\}$, its integral is at most Km_2 ; on the complement it is at most the second term of (233). This proves that formula. Equation (234) follows immediately from (231). For the flagged assertion, the total anchor-sector weight at a fixed root is bounded by

$$\iint \sum_{\omega} \mathbf{1}_{A_{p,q,\omega}}(y) h(p,y) h(q,y) d\nu_1(p) d\nu_1(q) \lesssim d_2(y)^2$$

because the enlarged sector family has bounded overlap. The definition (229) is the weighted average of $m_{p,q,\omega}$ against this measure. Some contributing anchor-sector flag therefore satisfies

$$m_{p,q,\omega} \gtrsim \mathcal{R}_2(y)/d_2(y)^2.$$

Use $\mathcal{R}_2(y) > K$ and $d_2(y) \leq D_2$.

For h_W , undoing the normalization by the universal collision-window length W makes the full collision-time degree comparable to $Wd_2(y)$. Averaging that degree over the fixed Reeb interval gives a time at which the simultaneous contact neighbor mass is $\gtrsim_J Wd_2(y)$. Conversely, each swept neighbor contributes to the collision-time integral for at most $O(W)$, so its union has mass $\gtrsim d_2(y)$. Substitute (234). This is the pointwise form of the argument in Theorem 7.27; the cited de-multiplication and bush map supply the final routing statement. $\square$

Corollary 7.50 (Rank-loss mass forces high rooted contact stars in every non-line orientation). *On a total-defect biregular layer, write*

$$M = W^{\chi+o(1)}, \quad m_i = W^{\chi-\theta_i+o(1)}, \quad D_i = W^{\theta_i+o(1)}.$$

Suppose that, after anchoring on the side opposite X_i , an occurrence-dependent subfamily of the non-line sector ledger retains content $\gtrsim M^4$. Fix $\eta > 0$ and set

$$K_i = W^{3\chi+\theta_i+\eta+o(1)}.$$

Then its low-root part on side i is

$$\boxed{K_i m_i = W^{4\chi+\eta+o(1)} = o(M^4)}. \quad (238)$$

Consequently the surviving high-root part returns, in every endpoint orientation satisfying this non-line hypothesis, to a normalized common-contact star with the corresponding degree bound

$$\boxed{\begin{aligned} d_1 &\gtrsim W^{(3\chi+\theta_1-\theta_2+\eta)/2+o(1)}, \\ d_2 &\gtrsim W^{(3\chi+\theta_2-\theta_1+\eta)/2+o(1)}. \end{aligned}} \quad (239)$$

The associated one-time star weights are the displayed quantities multiplied by W ; their swept leaf masses have no additional W -loss. In the same orientations, the retained flagged contact packets satisfy

$$\boxed{\begin{aligned} m_{\text{flag},1} &\gtrsim W^{3\chi-\theta_1+\eta+o(1)}, \\ m_{\text{flag},2} &\gtrsim W^{3\chi-\theta_2+\eta+o(1)}. \end{aligned}} \quad (240)$$

Proof. Apply Theorem 7.49 with X_2 rooted and $K = K_2$. Equation (238) shows that the low-root term cannot carry a fixed portion of M^4 , so (234) gives the second line of (239). If the reversed anchoring also remains in its non-line sector branch, repeat with $K = K_1$ to obtain the first line. If it does not, that portion has already entered the anchor-line angular-collapse branch. The time and swept-mass statements are (236)–(237). Finally (235) gives $m_{\text{flag},i} \gtrsim K_i/D_i^2$, which is (240). $\square$

Lemma 7.51 (A high root retains a weighted family of Maslov flags). *Use the X_2 -rooted notation of Theorem 7.48. For fixed y , put*

$$d\eta_y(p, q, \omega) := \mathbf{1}_{A_{p,q,\omega}}(y) h(p,y) h(q,y) d\nu_1(p) d\nu_1(q),$$

where the bounded-overlap sector index is counted as part of the measure space. Then

$$\int m_{p,q,\omega} d\eta_y(p, q, \omega) = \mathcal{R}_2(y), \quad \|\eta_y\| \lesssim d_2(y)^2. \quad (241)$$

If $\mathcal{R}_2(y) > 0$, there is a dyadic number H_y satisfying

$$\boxed{\frac{\mathcal{R}_2(y)}{C d_2(y)^2} \lesssim H_y \leq D_1} \quad (242)$$

and a flag family

$$\mathcal{F}_y(H_y) := \{(p, q, \omega) : H_y \leq m_{p,q,\omega} < 2H_y\}$$

such that

$$\boxed{H_y \eta_y(\mathcal{F}_y(H_y)) \gtrsim \frac{\mathcal{R}_2(y)}{1 + \log(2 + D_1 d_2(y)^2 / \mathcal{R}_2(y))}.} \quad (243)$$

Every flag in this family consists of the same root y , two actual contact neighbors p, q , and an H_y -mass common-neighbor set lying in the corresponding $E_{p,q,\omega} \oplus E_{p,q,\omega}$ contact-polarized packet. The symmetric statement holds with the endpoint sides reversed.

Proof. The first equality is the definition (229); bounded overlap of $\mathcal{F}_{p,q}$ gives the estimate for $\|\eta_y\|$. Since $m_{p,q,\omega} \leq c(p, q) \leq D_1$, all packet masses lie below D_1 . Set

$$H_0 = \frac{\mathcal{R}_2(y)}{2C d_2(y)^2},$$

with C larger than the implicit constant in (241). The flags with packet mass below H_0 contribute at most $\mathcal{R}_2(y)/2$. The interval $[H_0, D_1]$ has

$$O(1 + \log(2 + D_1/H_0))$$

dyadic layers. One layer retains the reciprocal of this logarithm of the remaining first moment. On that layer $m_{p,q,\omega} < 2H_y$, so its contribution is at most $2H_y \eta_y(\mathcal{F}_y(H_y))$. This proves (242)–(243). $\square$

Corollary 7.52 (The rooted M^4 output enters the original graph-cell reuse ledger). *Suppose a high-root portion of the non-line sector ledger carries a fixed part of M^4 . After dyadic decomposition of the root load, the flag mass H_y , and the original graphon edge weights, there is a layer losing only $W^{-o(1)}$ on which*

$$\boxed{\int H_y \eta_y(\mathcal{F}_y(H_y)) d\nu_i(y) \gtrsim M^4 W^{o(1)}}. \quad (244)$$

After the manuscript's scale- W graph-cell de-multiplication, the measure $\eta_y(\mathcal{F}_y(H_y))$ is the anchor-pair occurrence weight and H_y is the retained endpoint-packet mass. Hence (244) is the weighted continuous counterpart of the input S in Theorem 6.20; it retains the root, both contact edges, the azimuth sector, and the polarization plane.

Proof. Integrate (243) over the high-root set. All logarithms are $W^{-o(1)}$ on a fixed power layer. The root loads and H_y occupy only $W^{-o(1)}$ dyadic classes, and the graphon construction already includes the edge-weight regularization. Pigeonhole these classes. Scale- W de-multiplication changes the resulting occurrence ledger only by the cell factors already recorded in Theorem 6.20; no new normalization of a packet or root is performed. $\square$

Proposition 7.53 (Weighted continuous high-reuse contact-star trichotomy). *Fix a root $y \in X_2$, and let $\mathcal{F} \subset \mathcal{F}_y(H)$ be a symmetric measurable flag family from Theorem 7.51. Put*

$$d\mu_y(p) := h(p, y) d\nu_1(p), \quad g_y(p, q) := \sum_{\omega} \mathbf{1}_{\mathcal{F}}(p, q, \omega), \quad P := \iint g_y(p, q) d\mu_y(p) d\mu_y(q).$$

The bounded-overlap sector construction gives $0 \leq g_y \lesssim 1$. Assume the normalized angular shell supplies

$$\tau_0 \leq |a_p - a_y| \leq C_U$$

for all participating contact neighbors. For every $B \geq 1$ at least one of the following holds.

(i) The μ_y -mass S of the neighbor support used by g_y satisfies

$$\boxed{S \gtrsim BP^{1/2}}. \quad (245)$$

(ii) There are one contact neighbor p_0 and a measurable fiber Q of contact neighbors with

$$\boxed{\mu_y(Q) \gtrsim B^{-1}P^{1/2}}. \quad (246)$$

For every $W \ll \zeta \leq \kappa \ll 1$, that fiber has at least one of:

- (a) an angular subset $Q_{\text{ang}} \subset Q$ of mass $\mu_y(Q_{\text{ang}}) \gtrsim \mu_y(Q)$, whose projective directions $a_q - a_y$ lie in one $O_{\tau_0}(\kappa)$ -cap;
- (b) a subset $Q_{\text{nd}} \subset Q$ of comparable mass for which

$$\boxed{(\mu_y \times \mu_y) \left\{ (q_0, q_1) \in Q_{\text{nd}}^2 : |\det(a_{p_0} - a_y, a_{q_0} - a_y, a_{q_1} - a_y)| \geq \zeta \right\} \gtrsim \mu_y(Q)^2}; \quad (247)$$

this is the positive pair-mass tetrahedral input to the four-cluster Reeb-front seed;

- (c) a subset $Q_{\text{pol}} \subset Q$ of mass $\mu_y(Q_{\text{pol}}) \gtrsim \mu_y(Q)$ and a horizontal plane $E \subset \mathbb{R}^3$ such that

$$\boxed{\begin{aligned} \text{dist}(a_q - a_y, E) &\lesssim \zeta/\kappa, \\ \text{dist}(b_q - b_y, E) &\lesssim_J W + \zeta/\kappa \end{aligned} \quad (q \in Q_{\text{pol}})}. \quad (248)$$

Thus this entire substar lies in one $O_J(W + \zeta/\kappa)$ -neighborhood of $z_y + (E \oplus E)$.

Every retained pair (p, q) still carries an azimuth-sector common-neighbor packet of mass $\asymp H$; none of the alternatives forgets that Maslov flag.

Proof. Let

$$N = \left\{ p : \int g_y(p, q) d\mu_y(q) > 0 \right\}, \quad S = \mu_y(N).$$

If (245) fails, averaging the fiber degree over N gives some $p_0 \in N$ for which

$$\int g_y(p_0, q) d\mu_y(q) \gtrsim P/S \gtrsim B^{-1}P^{1/2}.$$

Because $g_y \lesssim 1$, its underlying neighbor fiber has the mass in (246).

Put $\alpha_q = a_q - a_y$. If a fixed fraction of this fiber satisfies

$$|\alpha_{p_0} \times \alpha_q| < \kappa,$$

the fixed lower radial bounds place that part in one projective $O_{\tau_0}(\kappa)$ -cap, giving (a). Otherwise retain Q_{far} of comparable mass on which $|\alpha_{p_0} \times \alpha_q| \geq \kappa$. If the determinant-large pair set in Q_{far}^2 has a fixed fraction of the product mass, take $Q_{\text{nd}} = Q_{\text{far}}$ and obtain (247).

In the remaining case, Fubini selects $q_0 \in Q_{\text{far}}$ such that a comparable-mass set $Q_{\text{pol}} \subset Q_{\text{far}}$ satisfies

$$|\det(\alpha_{p_0}, \alpha_{q_0}, \alpha_q)| < \zeta.$$

For $E = \text{Span}\{\alpha_{p_0}, \alpha_{q_0}\}$, the point-to-plane formula and $|\alpha_{p_0} \times \alpha_{q_0}| \geq \kappa$ give the first line of (248). Each edge yq has a selected Reeb label $s_{yq} \in J$ with

$$|(b_q - b_y) + s_{yq}(a_q - a_y)| \lesssim W.$$

Since E is linear and J is bounded, the vertical estimate follows. The packet flag of each pair was part of $\mathcal{F}$ throughout. $\square$

Corollary 7.54 (Power ledger for the continuous flagged trichotomy). *Root on side i , with the anchors on side j , and use the high-root threshold*

$$K_i = W^{3\chi + \theta_i + \eta + o(1)}$$

from Theorem 7.50. On a dyadic flag layer write

$$H_i = W^{\phi_i + o(1)}.$$

Then

$$\boxed{\theta_j \leq \phi_i \leq 3\chi - \theta_i + \eta,} \quad (249)$$

and the corresponding anchor-pair mass P_i obeys

$$\boxed{H_i P_i \gtrsim W^{3\chi + \theta_i + \eta + o(1)}, \quad P_i \gtrsim W^{3\chi + \theta_i - \phi_i + \eta + o(1)}. } \quad (250)$$

Consequently Theorem 7.53 gives either a neighbor-support star of mass $\gtrsim BP_i^{1/2}$, or a fixed-anchor substar of mass

$$\boxed{q_i \gtrsim B^{-1} W^{(3\chi + \theta_i - \phi_i + \eta)/2 + o(1)}} \quad (251)$$

which is angular, tetrahedrally nondegenerate, or contact-polarized as in (248).

Proof. The lower bound in (242), with $\mathcal{R}_i \geq K_i$ and $d_i \leq D_i$, gives $H_i \gtrsim K_i/D_i^2 = W^{3\chi - \theta_i + \eta + o(1)}$, which is the upper bound for ϕ_i . The common-neighbor mass is at most the degree D_j of either anchor, giving $\phi_i \geq \theta_j$. Equation (243) gives (250) after absorbing its logarithm into $W^{o(1)}$. Apply (246). $\square$

Lemma 7.55 (Continuous Reeb-time synchronization of a contact-neighbor star). *Let $N \subset X_j$ be a contact-neighbor set of a root $y \in X_i$, measured by*

$$d\mu_y(x) = h_W(x, y) d\nu_j(x), \quad \mu_y(N) = S.$$

For every $W \leq \delta_s \leq 1$, there are a time $s_0 \in J$ and a subset $N_{s_0} \subset N$ such that

$$\boxed{\nu_j(N_{s_0}) \geq \mu_y(N_{s_0}) \gtrsim_J \delta_s S,} \quad (252)$$

and all its selected lines form an $O_J(W + \delta_s)$ -Lagrangian point-probe bush centered at the front point $(b_y + s_0 a_y, s_0)$.

Proof. Choose measurably one contact label $s_{xy} \in J$ for every edge with $h_W(x, y) > 0$. Partition J into $O_J(\delta_s^{-1})$ half-open intervals and retain one containing μ_y -mass $\gtrsim_J \delta_s S$. If s_0 is its midpoint, then

$$|(b_x - b_y) + s_0(a_x - a_y)| \lesssim_J W + \delta_s$$

on the retained set. This is exactly the asserted point-probe bush, and $\mu_y \leq \nu_j$ proves (252). $\square$

Lemma 7.56 (A continuous contact-plane substar returns to one angular cap). *Let Q be the polarized substar in (248), of μ_y -mass q , and assume*

$$W + \zeta/\kappa \leq \varpi \ll 1.$$

Then one projective cap of aperture $O_{U, \tau_0}(\varpi)$ contains a contact-neighbor subset of mass

$$\boxed{\mu_y(Q_{\text{cap}}) \gtrsim_{U, \tau_0} \varpi q.} \quad (253)$$

Proof. After radial normalization, the first line of (248) places all directions in an $O_{U, \tau_0}(\varpi)$ -neighborhood of $\mathbb{P}(E) \subset \mathbb{RP}^2$. Cover this projective circle by $O_{U, \tau_0}(\varpi^{-1})$ caps of aperture $O_{U, \tau_0}(\varpi)$, using half-open cells to avoid overcounting, and pigeonhole with the measure μ_y . $\square$

Proposition 7.57 (Adaptive continuous routing of every high-reuse root). *In Theorem 7.53, fix a desired bush mass $0 < \mathfrak{b} \leq 1$, a time width $W \leq \delta_s \ll 1$, and parameters*

$$W \ll \zeta \leq \kappa \ll 1, \quad W + \zeta/\kappa \leq \varpi \ll 1.$$

Set

$$\boxed{B_{\text{time}} := \max \left\{ 1, \frac{\mathfrak{b}}{\delta_s P^{1/2}} \right\}. } \quad (254)$$

Then every high-reuse root has at least one of:

- (i) an $O_J(W + \delta_s)$ -Lagrangian bush of actual direction mass

$$\boxed{\gtrsim_J \mathfrak{b};} \quad (255)$$

(ii) a projective angular cap of aperture $O_{U,\tau_0}(\max\{\kappa, \varpi\})$ carrying contact-neighbor mass

$$\boxed{\gtrsim \varpi B_{\text{time}}^{-1} P^{1/2}}, \quad (256)$$

(iii) a tetrahedrally nondegenerate substar whose pair mass is

$$\boxed{\gtrsim B_{\text{time}}^{-2} P}. \quad (257)$$

Thus the continuous high-reuse branch has exactly the three exits prescribed by the original contact-symplectic proof route; the contact-plane output is not a fourth remainder.

Proof. Apply Theorem 7.53 with $B = B_{\text{time}}$. In its large-support alternative, Theorem 7.55 gives bush mass

$$\gtrsim_J \delta_s B_{\text{time}} P^{1/2} \geq \mathfrak{b}.$$

In the fixed-anchor alternative, the fiber mass is $q \gtrsim B_{\text{time}}^{-1} P^{1/2}$. Its direct angular branch gives at least this mass in an $O(\kappa)$ -cap. Its contact-plane branch gives $\gtrsim \varpi q$ in an $O(\varpi)$ -cap by Theorem 7.56. The weaker common bound is (256). Finally (247) gives pair mass $\gtrsim q^2 \gtrsim B_{\text{time}}^{-2} P$, proving (257). $\square$

Corollary 7.58 (Power form at the rank-loss bush scale). *In the setting of Theorem 7.54, take $\mathfrak{b} = M^3 = W^{3\chi+o(1)}$, write $\delta_s = W^{u+o(1)}$, and suppress subpower angular factors. If $B_{\text{time}} > 1$, the two non-bush outputs in Theorem 7.57 have respectively*

$$\boxed{\begin{array}{l} \text{angular-star mass} \gtrsim W^{u+\theta_i-\phi_i+o(1)}, \\ \text{tetrahedral pair mass} \gtrsim W^{2u+2\theta_i-2\phi_i+o(1)}. \end{array}} \quad (258)$$

If $B_{\text{time}} = 1$, both lower bounds improve to the corresponding powers of $P_i^{1/2}$ and P_i from (250).

Proof. When $B_{\text{time}} > 1$,

$$B_{\text{time}}^{-1} P_i^{1/2} = \frac{\delta_s P_i}{M^3}.$$

Substitute $P_i \gtrsim W^{3\chi+\theta_i-\phi_i+o(1)}$ from (250); square for the seed term. If $B_{\text{time}} = 1$, use (256)–(257) directly. $\square$

Lemma 7.59 (A surviving rooted load occupies a quantitative root set). *Root the sector-square ledger on side i , with anchors on side j , and let $\mathcal{H}_i$ be any root set carrying load*

$$\int_{\mathcal{H}_i} \mathcal{R}_i(y) d\nu_i(y) \gtrsim M^4.$$

Then

$$\boxed{\nu_i(\mathcal{H}_i) \gtrsim \frac{M^4}{D_j D_i^2}}. \quad (259)$$

On a total-defect biregular layer this becomes

$$\boxed{\nu_i(\mathcal{H}_i) \gtrsim W^{4\chi-\theta_j-2\theta_i+o(1)}}. \quad (260)$$

Proof. The symmetric form of (231) gives $\mathcal{R}_i(y) \lesssim D_j D_i^2$ pointwise. Integrate this bound over $\mathcal{H}_i$, compare with M^4 , and substitute the degree powers. $\square$

Proposition 7.60 (Every non-bush high-reuse output is an actual persistent front seed). *Suppose a root enters alternative (ii) or (iii) of Theorem 7.57. Then its fiber contains an unrepeated selector-direction set H_y of mass*

$$\boxed{\mathfrak{m}_y := \sigma(H_y) \gtrsim \varpi B_{\text{time}}^{-1} P^{1/2}}. \quad (261)$$

For every $0 < r \leq c_U \mathfrak{m}_y$, the selector graph over H_y contains four graph r -clusters satisfying

$$\boxed{|\det(a_1 - a_0, a_2 - a_0, a_3 - a_0)| \gtrsim_U \mathfrak{m}_y, \quad \mu_{H_y}(\chi_k) \gtrsim_{\varepsilon, U} \mathfrak{m}_y r^{3+\varepsilon}}. \quad (262)$$

Moreover a set $G_y \subset J$, of measure bounded below independently of $\mathfrak{m}_y, r$, sees the corresponding four separated Reeb-front clusters. Along any scale sequence, a fixed positive-measure limsup set of Reeb times sees such seeds infinitely often.

Proof. In the angular alternative, (256) already gives a contact-neighbor set of μ_y -mass at least the right side of (261). Since $\mu_y = h_W(\cdot, y)\nu_j \leq \nu_j$, its underlying unrepeat selector-direction mass is at least the same; trim it by nonatomicity. In the tetrahedral alternative the fixed-anchor fiber itself has mass $\gtrsim B_{\text{time}}^{-1}P^{1/2}$, which is stronger because $\varpi \leq 1$. Apply Theorem 6.22 to H_y , obtaining (262). Uniform Remez persistence is (110); the scale-limsup assertion is Theorem 6.23. $\square$

Lemma 7.61 (Exact two-root identity for a reused contact leaf). *Let x be a selector graph point joined by normalized contact edges to two roots y, y' . Choose edge labels $s, t \in J$ so that*

$$|(b_x - b_y) + s(a_x - a_y)| \lesssim W, \quad |(b_x - b_{y'}) + t(a_x - a_{y'})| \lesssim W.$$

Then

$$(b_y + sa_y) - (b_{y'} + ta_{y'}) = (s - t)a_x + O(W). \quad (263)$$

Consequently, for every $W \leq \delta_s \leq 1$, exactly one of the following estimates applies:

(i) if $|s - t| \leq \delta_s$, then the two roots have a secondary contact collision,

$$|(b_y - b_{y'}) + s(a_y - a_{y'})| \lesssim_J W + \delta_s; \quad (264)$$

(ii) if $|s - t| > \delta_s$, then the shared leaf direction lies in the radial Darboux fiber

$$\left| a_x - \frac{(b_y + sa_y) - (b_{y'} + ta_{y'})}{s - t} \right| \lesssim \frac{W}{\delta_s}. \quad (265)$$

Proof. The two edge equations say

$$b_x + sa_x = b_y + sa_y + O(W), \quad b_x + ta_x = b_{y'} + ta_{y'} + O(W).$$

Subtract them to obtain (263). In the close-label case, add $(t - s)a_{y'}$ and use boundedness of the direction chart. In the separated-label case, divide by $s - t$. $\square$

Proposition 7.62 (The seed-overlap second moment has only collision and angular sources). *Let $(\mathcal{H}, \kappa)$ be a finite root measure, and let every root $y \in \mathcal{H}$ carry a measurable contact-neighbor seed set H_y with selected edge labels $s_y(x) \in J$. Define the leaf multiplicity*

$$\mathcal{M}(x) := \int_{\mathcal{H}} \mathbf{1}_{H_y}(x) d\kappa(y).$$

Then

$$\int \mathcal{M}(x) d\nu(x) = \int_{\mathcal{H}} \nu(H_y) d\kappa(y), \quad (266)$$

and, for every $W \leq \delta_s \leq 1$,

$$\int \mathcal{M}(x)^2 d\nu(x) = \mathcal{O}_{\text{rec}}(\delta_s) + \mathcal{O}_{\text{rad}}(\delta_s), \quad (267)$$

where $\mathcal{O}_{\text{rec}}$ is supported on root pairs satisfying (264), and $\mathcal{O}_{\text{rad}}$ is supported on shared leaves satisfying (265). In particular,

$$\nu\left(\bigcup_{y \in \mathcal{H}} H_y\right) \geq \frac{(\int_{\mathcal{H}} \nu(H_y) d\kappa(y))^2}{\mathcal{O}_{\text{rec}}(\delta_s) + \mathcal{O}_{\text{rad}}(\delta_s)}. \quad (268)$$

Thus failure of transverse de-multiplication is not an arbitrary overlap phenomenon: it is either another collision layer of the original contact graph or a radial/angular Darboux concentration.

Proof. Equation (266) is Tonelli. Expand

$$\int \mathcal{M}^2 d\nu = \iint_{\mathcal{H} \times \mathcal{H}} \nu(H_y \cap H_{y'}) d\kappa(y) d\kappa(y').$$

On every shared leaf split the two selected labels at $|s_y(x) - s_{y'}(x)| = \delta_s$, and apply Theorem 7.61; this defines the two nonnegative terms in (267). Cauchy-Schwarz on $\text{supp } \mathcal{M}$ gives (268). $\square$

Lemma 7.63 (Existence of a fixed-angle root re-collision is paid by the normalized contact graph). *Fix $\tau_0 > 0$. Let $J \in J^+$, $0 < R_s \ll \tau_0$, and let y, y' satisfy*

$$\tau_0 \leq |a_y - a_{y'}| \leq C_U.$$

If for some $s_0 \in J$,

$$|(b_y - b_{y'}) + s_0(a_y - a_{y'})| \leq R_s,$$

then the enlarged collision window obeys

$$|\{s \in J^+ : |(b_y - b_{y'}) + s(a_y - a_{y'})| \leq 2R_s\}| \gtrsim_U R_s. \quad (269)$$

Consequently, if

$$h_{2R_s}^{J^+}(y, y') := \mathbf{1}_{\{\tau_0 \leq |a_y - a_{y'}| \leq C_U\}} \frac{1}{C_{U, \tau_0} R_s} \int_{J^+} \mathbf{1}_{\{|(b_y - b_{y'}) + s(a_y - a_{y'})| \leq 2R_s\}} ds,$$

then

$$\boxed{\mathbf{1}_{\{\exists s_0 \in J : |(b_y - b_{y'}) + s_0(a_y - a_{y'})| \leq R_s\}} \lesssim_{U, \tau_0} h_{2R_s}^{J^+}(y, y').} \quad (270)$$

Proof. For

$$|s - s_0| \leq c_U R_s,$$

the change in the collision vector is at most $|s - s_0||a_y - a_{y'}| \leq R_s$ after choosing c_U small. Since $J \in J^+$, this interval lies in J^+ for all sufficiently small R_s . It has length comparable to R_s , proving (269). The fixed-shell lower cutoff τ_0 permits a normalizing constant C_{U, τ_0} for which the graph is bounded by one; division by that normalization gives (270). $\square$

Proposition 7.64 (Quantitative fixed-angle charge for the close-label seed overlap). *Use Theorem 7.62, put*

$$R_s = C_J(W + \delta_s),$$

and split its close-label root pairs at the fixed normalized cutoff $|a_y - a_{y'}| = \tau_0$, where $R_s \ll \tau_0$. If the root measure κ is a submeasure of the retained selector measure, then

$$\boxed{\mathcal{O}_{\text{rec}}(\delta_s) \lesssim \iint_{|a_y - a_{y'}| < \tau_0} d\kappa(y) d\kappa(y') + \iint h_{2R_s}^{J^+}(y, y') d\kappa(y) d\kappa(y').} \quad (271)$$

The first term is passed wholly to the next angular Darboux patch; the second is the edge mass of the same normalized Maslov collision graph at the coarser physical scale R_s . In particular the close-label overlap introduces no independent occurrence coefficient.

Proof. On the near-angle set discard the shared-leaf indicator and bound it by one. On the complementary set, (264) gives existence of an R_s -collision at a time in J . Discarding the shared-leaf indicator and applying Theorem 7.63 gives the second term. $\square$

Lemma 7.65 (Separated Reeb labels lie in one affine radial Darboux plane). *Let $U \in \mathbb{R}^3$ and $J \in \mathbb{R}$ be bounded, and keep the root coordinates in one bounded contact chart. For fixed roots y, y' , put*

$$\mathcal{D}_{\delta_s} := \{(s, t) \in J^2 : |s - t| \geq \delta_s\}, \quad \Phi_{y, y'}(s, t) := \frac{(b_y + sa_y) - (b_{y'} + ta_{y'})}{s - t}.$$

Then the full image of $\mathcal{D}_{\delta_s}$ is contained in the fixed affine plane

$$\boxed{\Pi_{y, y'} := a_y + \text{span}\{b_y - b_{y'}, a_y - a_{y'}\}.} \quad (272)$$

If $0 < W \leq \delta_s \leq 1$ and $\rho = W/\delta_s$, then

$$\boxed{|U \cap N_{C\rho}(\Phi_{y, y'}(\mathcal{D}_{\delta_s}))| \lesssim_{U, J, C} \rho = W\delta_s^{-1}.} \quad (273)$$

The estimate is uniform in the two roots. Thus the apparent four-variable radial fiber in (265) is one contact-polarized plane and pays one full direction codimension with no parameter-covering loss.

Proof. Write $u = s - t$. The contact numerator gives the exact homothety identity

$$\Phi_{y,y'}(s, t) = a_y + \frac{b_y - b_{y'}}{u} + \frac{t}{u}(a_y - a_{y'}).$$

This proves (272). The intersection of a bounded subset of $\mathbb{R}^3$ with the $C\rho$ -neighborhood of an affine plane has volume $O_{U,C}(\rho)$. This remains true, with a stronger bound, if the displayed span has dimension zero or one. Hence (273) follows. $\square$

Proposition 7.66 (The separated-label seed overlap pays the radial Darboux codimension). *Under the hypotheses of Theorem 7.62, assume that the horizontal marginal of ν has density at most C_U . Then*

$$\mathcal{O}_{\text{rad}}(\delta_s) \lesssim_{U,J} \|\kappa\|^2 W \delta_s^{-1}. \quad (274)$$

In particular, for $\delta_s = W^\alpha$ with $0 < \alpha < 1$,

$$R_s \asymp W^\alpha, \quad \frac{W}{\delta_s} = W^{1-\alpha}, \quad \mathcal{O}_{\text{rad}}(\delta_s) \lesssim \|\kappa\|^2 W^{1-\alpha}. \quad (275)$$

Both geometric scales vanish and the radial overlap has a strict power gain.

Proof. For a fixed root pair (y, y') , every shared leaf counted by $\mathcal{O}_{\text{rad}}(\delta_s)$ has selected labels in $\mathcal{D}_{\delta_s}$, and (265) places its direction in the $C\rho$ -neighborhood of $\Phi_{y,y'}(\mathcal{D}_{\delta_s})$. The bounded horizontal density and Theorem 7.65 bound its ν -mass by $O_{U,J}(W\delta_s^{-1})$. Integrate this uniform bound against $d\kappa(y)d\kappa(y')$. Substituting $\delta_s = W^\alpha$ gives (275). $\square$

Proposition 7.67 (One-scale contact-symplectic closure of the seed second moment). *Keep the hypotheses of Theorems 7.62, 7.64 and 7.66. Write*

$$K := \|\kappa\| > 0, \quad A := \int_{\mathcal{H}} \nu(H_y) d\kappa(y), \quad \bar{m} := A/K,$$

and let $0 < Q \leq 1$. There is a constant $c_{U,J} > 0$ such that, whenever

$$W\delta_s^{-1} \leq c_{U,J} \frac{\bar{m}^2}{Q}, \quad (276)$$

at least one of the following holds:

- (i) the seed directions de-multiply at the requested mass,

$$\nu\left(\bigcup_{y \in \mathcal{H}} H_y\right) \geq Q; \quad (277)$$

- (ii) one root-centered angular Darboux patch carries mass

$$\kappa\{y : |a_y - a_{y_0}| < \tau_0\} \gtrsim_{U,J} K \frac{\bar{m}^2}{Q} \quad (278)$$

for some $y_0 \in \mathcal{H}$;

- (iii) for $\hat{\kappa} := K^{-1}\kappa$, the fixed-angle coarser-scale Maslov graph has edge mass

$$M_{R_s} := \iint h_{2R_s}^{J+}(y, y') d\hat{\kappa}(y) d\hat{\kappa}(y') \gtrsim_{U,J} \frac{\bar{m}^2}{Q}. \quad (279)$$

For every $2R_s \leq c\Delta < 1$, this last graph enters Theorem 7.8: it produces either an $O(R_s/\Delta)$ -Lagrangian point-probe bush of $\hat{\kappa}$ -mass $\gtrsim M_{R_s}^3$, or determinant-small Maslov four-cycle content $\gtrsim M_{R_s}^4$.

Thus a failed seed aggregation has no set-theoretic remainder: after the radial Darboux codimension is absorbed, it is exactly an angular change of chart or a quantified return to the manuscript's original bush/rank-loss route.

Proof. Put

$$\mathcal{A}_{\tau_0} := \iint_{|a_y - a_{y'}| < \tau_0} d\kappa(y) d\kappa(y'), \quad \mathcal{E}_{R_s} := \iint h_{2R_s}^{J^+}(y, y') d\kappa(y) d\kappa(y').$$

If (277) fails, then (268) gives

$$\mathcal{O}_{\text{rec}}(\delta_s) + \mathcal{O}_{\text{rad}}(\delta_s) > \frac{A^2}{Q}.$$

Use Theorems 7.64 and 7.66. After choosing the constant in (276) below their fixed implicit constants, the radial term accounts for at most a fixed small portion of A^2/Q . Hence

$$\mathcal{A}_{\tau_0} + \mathcal{E}_{R_s} \gtrsim_{U,J} \frac{A^2}{Q}. \quad (280)$$

If the first summand carries a fixed portion, Fubini gives a root y_0 for which

$$\kappa\{y : |a_y - a_{y_0}| < \tau_0\} \gtrsim \mathcal{A}_{\tau_0}/K \gtrsim K\bar{m}^2/Q,$$

which is (278). Otherwise divide the second summand by K^2 to obtain (279). The angular cutoff built into $h_{2R_s}^{J^+}$ makes it a symmetric graphon with values in $[0, 1]$. Its positive edges have selected contact labels with error $O(R_s)$, so Theorem 7.8 applies with collision scale $O(R_s)$ and gives the stated bush/rank-loss alternative. $\square$

7.7 Two-generation flags and relative residual return

Corollary 7.68 (The input-overlap Maslov–radial balance occurs at the one-third Reeb scale). *Take*

$$\boxed{\delta_s = W^{1/3}, \quad R_s \asymp W^{1/3}, \quad \frac{W}{\delta_s} = W^{2/3}.} \quad (281)$$

Suppose the angular output in (278) is passed to the angular Darboux branch, and decompose the coarse graph dyadically by its actual edge mass. Every coarse layer with $M_{R_s} > R_s^{2-o(1)}$ is a total-defect layer $\chi_s < 2$ and re-enters the existing Lagrangian-bush/Maslov-rank-loss route. On the complementary layers, Theorems 7.64 and 7.66 and (268) give

$$\boxed{\nu\left(\bigcup_{y \in \mathcal{H}} H_y\right) \gtrsim \min\left\{1, \bar{m}^2 W^{-2/3+o(1)}\right\},} \quad (282)$$

up to the angular piece already routed away. In particular, every seed layer with

$$\boxed{\bar{m}^2 \gtrsim W^{2/3+o(1)}} \quad (283)$$

passes the input-mass gate in (299); the relative residual gate is addressed separately by Theorem 7.73.

Proof. At $\delta_s = W^{1/3}$, Theorem 7.66 gives $\mathcal{O}_{\text{rad}} \lesssim K^2 W^{2/3}$. A coarse graph outside the bad total-defect layers has edge mass at most $R_s^{2-o(1)} = W^{2/3-o(1)}$, and hence contributes at most $K^2 W^{2/3-o(1)}$. Insert these two bounds into (268), using $A = K\bar{m}$; the factor K^2 cancels and yields (282). Equation (283) makes its right side subpower. $\square$

Corollary 7.69 (Power audit of the high-reuse flag layer). *Root on the side i for which $\theta_i \leq \theta_j$. Then (249) gives $\phi_i \geq \theta_j \geq \theta_i$. At the choice $\delta_s = W^{1/3}$:*

(i) *if $B_{\text{time}} > 1$, then*

$$\bar{m}^2 \gtrsim W^{2/3+2\theta_i-2\phi_i+o(1)} \geq W^{2/3+o(1)}, \quad (284)$$

so Theorem 7.68 closes the input-side seed overlap;

(ii) *if $B_{\text{time}} = 1$, put*

$$p_i := 3\chi + \theta_i - \phi_i + \eta. \quad (285)$$

Then $\bar{m}^2 \gtrsim P_i \gtrsim W^{p_i+o(1)}$, so every layer with $p_i \leq 2/3$ also passes the input-overlap gate. The only flag-power region not settled by the one-third balance is

$$\boxed{B_{\text{time}} = 1, \quad 3\chi + \theta_i - \phi_i + \eta > \frac{2}{3}, \quad \theta_i \leq \theta_j \leq \phi_i.} \quad (286)$$

Thus neither the amplified high-reuse branch nor any flag layer of pair mass at least $W^{2/3+o(1)}$ remains in the global frontier.

Proof. When $B_{\text{time}} > 1$, the first line of (258), together with (261), gives $\bar{m} \gtrsim W^{1/3+\theta_i-\phi_i+o(1)}$; square it and use $\phi_i \geq \theta_i$. When $B_{\text{time}} = 1$, Theorem 7.57 and (250) give $\bar{m}^2 \gtrsim P_i \gtrsim W^{p_i+o(1)}$. Apply (283). $\square$

Proposition 7.70 (The residual flag wedge retains its full packet-occurrence measure). *Work on a dyadically regularized residual layer (286). Let $\mathcal{H}_i$ have root mass K_i , assume*

$$\eta_y(\mathcal{F}_y(H_i)) \asymp P_i \quad (y \in \mathcal{H}_i),$$

For $f = (p, q, \omega)$, retain its actual common-neighbor density

$$w_f(z) := \mathbf{1}_{A_{p,q,\omega}}(z)h(p,z)h(q,z), \quad H_i \leq \int w_f d\nu_i < 2H_i.$$

On the flag space put

$$d\Lambda(y, f) := d\nu_i(y)d\eta_y(f), \quad \|\Lambda\| \asymp K_i P_i,$$

and define the packet-occurrence density on the selector graph by

$$\mathcal{N}_{\text{flag}}(z) := \int w_f(z) d\Lambda(y, f).$$

Then

$$\boxed{\int \mathcal{N}_{\text{flag}} d\nu_i \asymp K_i H_i P_i,} \quad (287)$$

and

$$\boxed{\int \mathcal{N}_{\text{flag}}^2 d\nu_i = \iint \left(\int w_f(z) w_{f'}(z) d\nu_i(z) \right) d\Lambda(y, f) d\Lambda(y', f').} \quad (288)$$

Consequently

$$\boxed{\nu_i(\{z : \mathcal{N}_{\text{flag}}(z) > 0\}) \gtrsim \frac{(K_i H_i P_i)^2}{\int \mathcal{N}_{\text{flag}}^2 d\nu_i}.} \quad (289)$$

In particular, the factor $H_i P_i$ from (244) is present before any root normalization and may not be replaced by the smaller single-fiber mass $P_i^{1/2}$.

Moreover $\mathcal{N}_{\text{flag}}$ is exactly the root-restricted square load

$$\boxed{\mathcal{N}_{\text{flag}}(z) = \iint \sum_{\omega} w_{p,q,\omega}(z) \left(\int_{\mathcal{H}_i} w_{p,q,\omega}(y) d\nu_i(y) \right) d\nu_j(p) d\nu_j(q) \leq \mathcal{R}_i(z).} \quad (290)$$

Equality with the full rooted load holds when the root restriction is removed; on the retained layer the displayed domination is all that is used. Hence

$$\boxed{\mathcal{N}_{\text{flag}}(z) \lesssim D_j d_i(z)^2 \lesssim D_j D_i^2.} \quad (291)$$

Proof. Tonelli and $H_i \leq \int w_f d\nu_i < 2H_i$ give (287). Expanding the square and applying Tonelli again gives (288). Cauchy–Schwarz on the support of $\mathcal{N}_{\text{flag}}$ proves (289). Finally substitute the definition $d\eta_y(p, q, \omega) = w_{p,q,\omega}(y) d\nu_j(p) d\nu_j(q)$, interchange the y and (p, q, ω) integrals, and obtain (290). The pointwise rooted load bound (231) gives (291). $\square$

Lemma 7.71 (Coherent flag reuse re-roots without losing occurrence mass). *For a selector point z on side i , define the full pair mass of the same flag-packet layer by*

$$P_i^{(H)}(z) := \iint \sum_{\omega: H_i \leq m_{p,q,\omega} < 2H_i} w_{p,q,\omega}(z) d\nu_j(p) d\nu_j(q). \quad (292)$$

Then the lifted occurrence density satisfies the pointwise bound

$$\boxed{\mathcal{N}_{\text{flag}}(z) \leq 2H_i P_i^{(H)}(z).} \quad (293)$$

For every threshold $0 < L \leq 1$, at least one of the following holds:

(i) the low-reuse packet union has actual selector-direction mass

$$\boxed{\nu_i\{0 < \mathcal{N}_{\text{flag}}, P_i^{(H)} \leq L\} \gtrsim \frac{K_i P_i}{L};} \quad (294)$$

(ii) the re-rooted high-pair set retains lifted load

$$\boxed{\int_{\{P_i^{(H)} > L\}} \mathcal{R}_i(z) d\nu_i(z) \gtrsim K_i H_i P_i.} \quad (295)$$

Thus high coherent reuse is not a new multiplicity: it is the original rooted load on a root set whose flag-pair mass has increased to at least L .

Proof. Using (290) before the final domination, the inner restricted root mass satisfies

$$\int_{\mathcal{H}_i} w_{p,q,\omega}(y) d\nu_i(y) \leq m_{p,q,\omega} < 2H_i.$$

Integration in (p, q, ω) proves (293). Split the first moment (287) over $P_i^{(H)} \leq L$ and $P_i^{(H)} > L$. If the first part carries at least half, then $\mathcal{N}_{\text{flag}} \leq 2H_i L$ there, so its support has mass at least

$$\frac{cK_i H_i P_i}{2H_i L} \gtrsim \frac{K_i P_i}{L},$$

which is (294). Otherwise the second part carries half. Since $\mathcal{N}_{\text{flag}} \leq \mathcal{R}_i$, this gives (295). $\square$

Corollary 7.72 (The high coherent-reuse half reaches the one-third input threshold). *Take $L = W^{2/3+o(1)}$. In alternative (ii) of Theorem 7.71, dyadic regularization on the set $P_i^{(H)} > L$ produces a rooted flag layer with pair mass at least $W^{2/3+o(1)}$, while retaining a fixed portion of $K_i H_i P_i$. Its non-bush seed mass is at least $(P_i^{(H)})^{1/2} \geq W^{1/3+o(1)}$, so Theorem 7.68 closes its input-side overlap. Consequently the coherent survivor may be assumed to satisfy the genuine low-reuse output*

$$\boxed{\nu_i\{0 < \mathcal{N}_{\text{flag}}, P_i^{(H)} \leq W^{2/3+o(1)}\} \gtrsim K_i P_i W^{-2/3-o(1)}.} \quad (296)$$

On the total-defect power layer, this lower bound is

$$\boxed{\gtrsim W^{7\chi-\theta_j-\theta_i-\phi_i+\eta-2/3+o(1)}.} \quad (297)$$

Proof. Only the last displayed power requires verification. Multiply the root-support power (260) by $P_i W^{-2/3} = W^{3\chi+\theta_i-\phi_i+\eta-2/3+o(1)}$. $\square$

Proposition 7.73 (Input de-multiplication reaches the Frostman target only through relative Maslov energy). *Let H be a measurable subset of the selector graph, let $\mu_H := \mathbf{1}_H \mu$, and put $m := \mu_H(H)$. For a half-open grid $\mathcal{Q}_r$ of physical r -cubes, define*

$$S_r(s) := \sum_{Q \in \mathcal{Q}_r} \mu_H(\Pi_s^{-1}(Q))^2.$$

Then

$$\boxed{\int_J N_r(\Pi_s(H)) ds \gtrsim_J \frac{m^2}{mr^3 + r \mathfrak{Z}_{Cr, \geq r}^{J+}(\mu_H)}.} \quad (298)$$

Consequently the two conditions

$$\boxed{m = r^{o(1)}, \quad \mathfrak{Z}_{Cr, \geq r}^{J+}(\mu_H) \lesssim m^2 r^{2-o(1)}} \quad (299)$$

imply

$$\boxed{\int_J N_r(\Pi_s(H)) ds \gtrsim r^{-3+o(1)}.} \quad (300)$$

Thus a subpower input-direction union is necessary to pay the near-direction diagonal, but it is not by itself a projection theorem: the separated pairs still require the quadratic relative Maslov residual bound in (299).

Proof. For every s , Cauchy–Schwarz over the occupied output cubes gives

$$m^2 \leq N_r(\Pi_s(H)) S_r(s).$$

Cauchy–Schwarz in s , or the harmonic–arithmetic mean inequality, therefore yields

$$\int_J N_r(\Pi_s(H)) ds \gtrsim_J \frac{m^2}{\int_J S_r(s) ds}.$$

Two points counted in the same output cube satisfy $|\Pi_s(z) - \Pi_s(z')| \leq Cr$. On $|a - a'| < r$, bounded horizontal density gives pair mass $O_U(mr^3)$. On $|a - a'| \geq r$, a nonempty collision window forces $s_* \in J^+$, $|r(z, z')| \leq Cr$, and has length at most $C_J r/|a - a'|$. Hence

$$\int_J S_r(s) ds \lesssim_{U,J} mr^3 + r \mathfrak{Z}_{Cr, \geq r}^+(\mu_H).$$

This proves (298); substitution of (299) gives (300). $\square$

Lemma 7.74 (Coarse old flags and fine new collisions freeze the input mass). *First fix one coarse scale $W > 0$ and one coarse flag output H of mass $m > 0$. As the new collision scale $r \downarrow 0$, every finite constant depending only on this frozen coarse layer, including $m^{\pm 1}$ and any fixed W -power, is $r^{-o(1)}$. Thus a residual estimate proved uniformly for all sufficiently small r , after freezing W, H , may absorb all such coarse constants and still feed the Hausdorff Frostman criterion.*

For the scale-sequence version, let $W_n \downarrow 0$, and for each n let H_n be a measurable selector subset produced at the coarse flag scale W_n , with mass

$$0 < m_n := \mu(H_n) \leq 1.$$

There is a fine collision scale $r_n \downarrow 0$ such that

$$\boxed{0 < r_n \leq \min\{W_n^n, m_n^n\}.} \quad (301)$$

Consequently

$$\boxed{m_n = r_n^{o(1)}, \quad r_n = o(W_n^A) \quad \text{for every fixed } A > 0.} \quad (302)$$

Thus the first condition in (299) does not require the coarse lower bound (297) to have exponent zero. It is paid by separating the new Maslov-collision scale from the old flag scale. This is a diagonal choice of two scales, not an intrinsic recursive construction. By itself the diagonal scale-sequence conclusion gives only a scale-wise/Minkowski output; a Hausdorff conclusion still requires the fixed-coarse-layer, all-fine-scales form above or the scale-coherent kernels in Theorems 6.24 and 6.25.

Proof. The fixed-coarse-layer assertion follows because the logarithm of every such constant is fixed while $|\log r| \rightarrow \infty$. For the sequence assertion, choose any positive r_n satisfying (301). Then

$$\frac{|\log m_n|}{|\log r_n|} \leq \frac{1}{n},$$

with the ratio interpreted as zero when $m_n = 1$. This proves $m_n = r_n^{o(1)}$. Likewise

$$\frac{r_n}{W_n^A} \leq W_n^{n-A} \rightarrow 0$$

for each fixed A , proving the second assertion. $\square$

Principle 7.75 (The final low-reuse task is a relative residual estimate, not a direction-union estimate). *For the Hausdorff endpoint, first freeze one sufficiently fine coarse low-reuse output H from (296), and then apply Theorem 7.73 uniformly over all sufficiently small new collision scales r . The mass $m = \mu(H) > 0$ is then a coarse constant and hence $m = r^{o(1)}$. For a scale-sequence/Minkowski statement one may instead use the diagonal scales from Theorem 7.74. In either case the preceding flag arguments remove radial, transverse-plane, and high coherent occurrence losses, and reduce the proof to the second condition in (299). The exact remaining estimate is therefore*

$$\boxed{\mathfrak{Z}_{Cr, \geq r}^+(\mu_H) \lesssim \mu(H)^2 r^{2-o(1)}.} \quad (303)$$

This is the relative form of the manuscript’s original codimension-two Maslov target (493). No Euclidean dimension credit is taken from the flag occurrence bounds before (303) is established on the fixed-coarse/all-fine scale family required by the Hausdorff endpoint.

Proposition 7.76 (Failure of the relative residual bound re-enters the normalized Maslov graph). *Let $J^- \in J \in J^+$, work on a normalized fixed-angle Darboux shell, and let H_{low} be the low-reuse output of (296). Put*

$$m := \mu(H_{\text{low}}), \quad \hat{\mu} := m^{-1} \mu_{H_{\text{low}}}.$$

If, along a scale sequence $r \downarrow 0$,

$$\mathfrak{Z}_{r/2, \geq \tau_0}^{J^-}(\mu_{H_{\text{low}}}) \geq A_r m^2 r^{2-\eta}, \quad A_r \longrightarrow \infty, \quad (304)$$

then the normalized collision graph

$$h_r(z, z') := \mathbf{1}_{\{\tau_0 \leq |a-a'| \leq C_U\}} \frac{1}{C_{\tau_0} r} \int_J \mathbf{1}_{\{|\Pi_s(z) - \Pi_s(z')| \leq r\}} ds \quad (305)$$

on $(H_{\text{low}}, \hat{\mu})$ has edge mass

$$M_r := e(h_r) \gtrsim A_r r^{2-\eta}. \quad (306)$$

For every $r \leq c\Delta < 1$, Theorem 7.8 therefore yields either

- (i) *an $O(r/\Delta)$ -Lagrangian point-probe bush of normalized $\hat{\mu}$ -mass $\gtrsim M_r^3$; or*
- (ii) *determinant-small Maslov four-cycle content $\gtrsim M_r^4$ inside the low old-flag-reuse set.*

No factor of m^{-1} remains in (306); the required quadratic factor m^2 in the relative residual target is exactly canceled by probability normalization.

Proof. The collision-window lower bound in (121), applied to $\hat{\mu}$, gives

$$e(h_r) \gtrsim \mathfrak{Z}_{r/2, \geq \tau_0}^{J^-}(\hat{\mu}) = m^{-2} \mathfrak{Z}_{r/2, \geq \tau_0}^{J^-}(\mu_{H_{\text{low}}}).$$

Insert (304). The fixed-angle cutoff makes $0 \leq h_r \leq 1$ after choosing C_{τ_0} , so Theorem 7.8 applies and gives the two stated outputs. $\square$

Principle 7.77 (The last unclosed incidence is an old-flag/new-cycle compatibility). *The bush output of Theorem 7.76 is already a physical exit. Hence a failure of (303) may be assumed to produce a determinant-small Maslov four-cycle entirely inside H_{low} . Every vertex of that cycle still carries the old azimuth-Maslov packet flags from Theorem 7.70, while the re-rooted old-flag pair mass is bounded by $W^{2/3+o(1)}$ on the low-reuse set. The final compatibility estimate must therefore bound a new Maslov four-cycle decorated by an old common-neighbor flag, with the old flag occurrence kept below the one-third threshold. This is a concrete six-point contact incidence; it is not a generic projection or set-overlap problem.*

Lemma 7.78 (Two-generation polarization comparison at the decorated cycle vertex). *Let an old flag packet lie in a phase-plane tube*

$$(\Pi_{\text{old}}^a \times \Pi_{\text{old}}^b)_{t_{\text{old}}} = (z_{\text{old}} + E_{\text{old}} \oplus E_{\text{old}})_{t_{\text{old}}},$$

and let the determinant-small new Maslov cycle routing place its decorated vertex in

$$(\Pi_{\text{new}}^a \times \Pi_{\text{new}}^b)_{t_{\text{new}}} = (z_{\text{new}} + E_{\text{new}} \oplus E_{\text{new}})_{t_{\text{new}}}.$$

Suppose both packets meet one graph product cell of diameter $O(r)$, and let $n_{\text{old}}, n_{\text{new}}$ be the corresponding horizontal unit normals. Put

$$T := t_{\text{old}} + t_{\text{new}} + r.$$

For every $T \leq \vartheta \leq 1$, exactly one of the following geometric estimates applies:

- (i) *if $|n_{\text{old}} \times n_{\text{new}}| \geq \vartheta$, then the possible decorated selector directions occupy volume and selector mass at most*

$$\lesssim_U \frac{(t_{\text{old}} + r)(t_{\text{new}} + r)}{\vartheta}; \quad (307)$$

- (ii) *if $|n_{\text{old}} - n_{\text{new}}| < \vartheta$, then their horizontal and vertical affine phase offsets satisfy*

$$|c_{\text{old}}^a - c_{\text{new}}^a| + |c_{\text{old}}^b - c_{\text{new}}^b| \lesssim_U T + \vartheta. \quad (308)$$

Consequently both generations lie, on the common cell and up to thickness $O_U(T + \vartheta)$, in one coherent polarization $z_ + (E \oplus E)$.*

Proof. The horizontal projection of the common graph cell meets both affine plane tubes. For transverse normals, the two defining normal coordinates have Jacobian $|n_{\text{old}} \times n_{\text{new}}|$, while the remaining direction coordinate ranges in a fixed bounded interval. This gives (307); bounded horizontal density gives the selector-mass version. In the coherent case, apply the common-cell offset calculation of Theorem 7.18 to the horizontal coordinates and Theorem 7.19 to the vertical coordinates, adding the two packet thicknesses and the cell diameter. This proves (308). $\square$

Lemma 7.79 (A coherent new Maslov plane makes its opposite anchors q -local). *Fix the old coherent polarization $E_* \subset \mathbb{R}^3$, let n_* be its unit normal, and write*

$$\mathbf{q}_*(a) := n_* \cdot a.$$

For an anchored new rank-loss cycle p, y_0, q, y, p , let

$$E_{\text{new}} = \text{Span}\{a_{y_0} - a_p, a_q - a_{y_0}\},$$

and choose the sign of its unit normal n_{new} so that $n_{\text{new}} \cdot n_ \geq 0$. If*

$$|n_{\text{new}} - n_*| \leq \vartheta,$$

then

$$|\mathbf{q}_*(a_q) - \mathbf{q}_*(a_p)| \lesssim_U \vartheta. \quad (309)$$

Conversely, if the left side is at least R_q , then

$$|n_{\text{new}} \times n_*| \gtrsim_U R_q. \quad (310)$$

Thus coherence of the old and new Maslov generations is not merely a statement about two plane parameters: it forces the opposite anchor pair of the new four-cycle onto the old Reeb-normal diagonal.

Proof. Put

$$u = a_{y_0} - a_p, \quad v = a_q - a_{y_0}.$$

Then $u, v \in E_{\text{new}}$ and $a_q - a_p = u + v \in E_{\text{new}}$. Therefore

$$\begin{aligned} |\mathbf{q}_*(a_q) - \mathbf{q}_*(a_p)| &= |n_* \cdot (a_q - a_p)| \\ &= |(n_* - n_{\text{new}}) \cdot (a_q - a_p)| \lesssim_U \vartheta, \end{aligned}$$

which proves (309). For the converse, the norm of the restriction of $a \mapsto n_* \cdot a$ to E_{new} is $|n_{\text{new}} \times n_*|$. Test it on $(a_q - a_p)/|a_q - a_p|$ and use the bounded direction chart. This is also the opposite-anchor form of Theorem 9.48. $\square$

Lemma 7.80 (Weighted q -local opposite-anchor four-cycle bound). *Let $0 \leq h \leq 1$ be a bipartite graphon on probability spaces (X, μ_X) and (Y, μ_Y) , and write*

$$M := \iint h(x, y) d\mu_X(x) d\mu_Y(y).$$

Assume the X -degrees obey

$$d_X(x) := \int h(x, y) d\mu_Y(y) \leq D_X$$

and that a scalar coordinate $\mathbf{q}_ : X \rightarrow \mathbb{R}$ has the resolution- r_0 Frostman bound*

$$(\mathbf{q}_*)_{\#} \mu_X(I) \leq L_q(|I| + r_0) \quad \text{for every interval } I. \quad (311)$$

Define the common-neighbor codegree

$$\mathsf{T}_X(x, x') := \int h(x, y) h(x', y) d\mu_Y(y).$$

Then, for every $R \geq 0$,

$$\iint_{\{|\mathbf{q}_*(x) - \mathbf{q}_*(x')| \leq R\}} \mathsf{T}_X(x, x')^2 d\mu_X(x) d\mu_X(x') \lesssim D_X L_q(R + r_0) M. \quad (312)$$

The left side is exactly the weighted four-cycle mass whose opposite X -anchors are q -local. In particular the estimate keeps the full graphon edge weights; it does not replace the collision graph by its support.

Proof. Since

$$\mathsf{T}_X(x, x') \leq \min\{d_X(x), d_X(x')\} \leq D_X,$$

one has $\mathsf{T}_X^2 \leq D_X \mathsf{T}_X$. By Tonelli and (311),

$$\begin{aligned} & \iint_{\{|q_*(x) - q_*(x')| \leq R\}} \mathsf{T}_X(x, x') d\mu_X(x) d\mu_X(x') \\ &= \int_Y \iint_{\{|q_*(x) - q_*(x')| \leq R\}} h(x, y) h(x', y) d\mu_X(x) d\mu_X(x') d\mu_Y(y) \\ &\leq CL_q(R + r_0) \int_Y \int_X h(x, y) d\mu_X(x) d\mu_Y(y) \\ &= CL_q(R + r_0)M. \end{aligned}$$

Multiply by D_X . Expanding T_X^2 gives the alternating product of the four edge weights, proving the four-cycle interpretation. $\square$

Proposition 7.81 (The coherent two-generation four-cycle pays an anchor-diagonal debt). *Apply the normalized Maslov graph construction on the final occurrence layer G_* , but keep the occurrence variables $(\mathbf{f}, z)$ and their weights from Theorem 9.46. After the collision-preserving two-sided degree regularization, view the graph as bipartite on two copies and write its edge mass and first-side degree cap as M and D_X . Suppose the normalized q -marginal satisfies (311).*

First remove the physical-bush output of the Maslov graph routing, then route its determinant-small four-cycles by Theorem 7.42. Apart from the close-anchor and horizontal-line destinations, compare the resulting E_{new} with the old flag plane E_ . For every angular threshold ϑ , the total four-cycle mass routed to*

$$|n_{\text{new}} - n_*| \leq \vartheta$$

is bounded by

$$\boxed{\mathcal{Z}_{\text{coh}}(\vartheta) \lesssim D_X L_q(\vartheta + r_0)M.} \quad (313)$$

Every complementary packet has $|n_{\text{new}} \times n_| \gtrsim \vartheta$ and therefore returns to the transverse old/new-polarization intersection ledger of Theorem 7.78.*

Consequently, if the rank-loss alternative carries four-cycle mass $\mathcal{Z}_{\text{rk}} \gtrsim M^4$ and the coherent class carries a fixed portion of it, then

$$\boxed{M^3 \lesssim D_X L_q(\vartheta + r_0).} \quad (314)$$

On a fine power layer

$$M = r^{\chi+o(1)}, \quad D_X = r^{\theta+o(1)}, \quad L_q = r^{-o(1)}, \quad \vartheta = r^{\alpha+o(1)}, \quad r_0 \lesssim \vartheta,$$

a coherent survivor must therefore obey

$$\boxed{\alpha + \theta \leq 3\chi + o(1).} \quad (315)$$

This is the exact anchor-diagonal debt left by the coherent old-flag/new-cycle geometry. It is the necessary region used in the later conditional tangential estimate (487).

Proof. For every coherent new packet, Theorem 7.79 places its opposite anchors in

$$|q_*(a_p) - q_*(a_q)| \lesssim_U \vartheta.$$

The rank-loss cycles routed to that packet form a subfamily of all four-cycles with such q -local opposite anchors. Apply Theorem 7.80 with $R = C_U \vartheta$, proving (313). If the normals are not coherent, the second part of Theorem 7.79 and the transverse tube-intersection calculation in Theorem 7.78 give the stated return.

If the coherent part is $\gtrsim M^4$, compare that lower bound with (313) and divide by $M > 0$, obtaining (314). Substitution of the displayed power parameterization gives (315). $\square$

Lemma 7.82 (A prescribed old-flag polarization cap costs one angular factor). *Keep the flag layer of Theorem 7.70. For $f = (p, q, \omega)$, let $E_f = E_{p,q,\omega}$ be the contact-polarization plane from Theorem 7.47, and let $[n_f] \in \mathbb{RP}^2$ be its unoriented normal. For a prescribed $[n] \in \mathbb{RP}^2$ and $0 < \vartheta \leq 1$, define*

$$\mathcal{N}_{\text{flag}}^{[n], \vartheta}(z) := \int \mathbf{1}_{\{d_{\mathbb{RP}^2}([n_f], [n]) \leq \vartheta\}} w_f(z) d\Lambda(y, f).$$

If the horizontal marginal of ν_j has density bounded by C_U , then

$$\boxed{\mathcal{N}_{\text{flag}}^{[n],\vartheta}(z) \lesssim_U H_i \vartheta \quad \text{for every } z.} \quad (316)$$

The estimate is uniform in the prescribed plane and retains both old contact-edge weights and the root occurrence weight.

Proof. Use the definition of the occurrence measure and integrate the root variable first:

$$\begin{aligned} \mathcal{N}_{\text{flag}}^{[n],\vartheta}(z) &= \iint \sum_{\omega} \mathbf{1}_{\{d_{\mathbb{RP}^2}([n_{p,q,\omega}], [n]) \leq \vartheta\}} w_{p,q,\omega}(z) \\ &\quad \times \left(\int_{\mathcal{H}_i} w_{p,q,\omega}(y) d\nu_i(y) \right) d\nu_j(p) d\nu_j(q). \end{aligned}$$

On the retained flag layer the parenthesized factor is at most $m_{p,q,\omega} < 2H_i$. Moreover

$$a_q - a_p \in E_{p,q,\omega}.$$

Thus the angular-cap condition implies

$$|n \cdot (a_q - a_p)| \lesssim_U \vartheta.$$

For fixed p , the allowed q 's lie in a direction slab of width $O_U(\vartheta)$, whose ν_j -mass is $O_U(\vartheta)$. For fixed p, q, z , bounded overlap of the azimuth intervals leaves only $O_U(1)$ values of ω for which $w_{p,q,\omega}(z) > 0$. Delete the remaining factors, which lie in $[0, 1]$, integrate first in q and then in p , and multiply by $2H_i$. This proves (316). $\square$

8 Bush families, compactness, and Frostman stopping

The terms bush and hairbrush refer to the classical Kakeya mechanisms of [6, 18]; the sticky implementation follows the geometric line developed in [12, 17].

8.1 Flag amplification and reduction to bushes

Corollary 8.1 (Conditional flag normals remain angularly diffuse on the occurrence layer). *Disintegrate the normalized occurrence measure over its physical marginal:*

$$d\hat{\rho}_*(\mathfrak{f}, z) = d\kappa_z(\mathfrak{f}) d\hat{\mu}_*(z), \quad \hat{\mu}_* := \frac{\mathbf{1}_{G_*} \mathcal{N}_{\text{flag}} d\nu_i}{\|\rho_*\|}.$$

Then, for $z \in G_*$,

$$\kappa_z \{ \mathfrak{f} : d_{\mathbb{RP}^2}([n_f], [n]) \leq \vartheta \} \lesssim_U \min \left\{ 1, \frac{H_i}{\lambda_*} \vartheta \right\}. \quad (317)$$

On the fixed-coarse/all-fine scale family,

$$\frac{H_i}{\lambda_*} = r^{-o(1)}. \quad (318)$$

The same conclusion holds on the diagonal pairs of Theorem 7.74.

Proof. The conditional measure is

$$d\kappa_z(\mathfrak{f}) = \mathcal{N}_{\text{flag}}(z)^{-1} w_f(z) d\Lambda(y, f).$$

Divide (316) by $\mathcal{N}_{\text{flag}}(z) \geq \lambda_*$, proving (317). For a frozen coarse layer, H_i/λ_* is a fixed finite constant and hence is $r^{-o(1)}$ as $r \downarrow 0$. For the diagonal choice, all retained quantities lie on fixed W -power layers and $|\log W|/|\log r| \rightarrow 0$, exactly as in Theorem 9.47. $\square$

Proposition 8.2 (Four old flags at the free cycle endpoint exclude the fully coherent subunit-defect return). *Let h_r be the normalized fine-scale Maslov graphon on the occurrence probability space $(\mathfrak{f}, z, \hat{\rho}_*)$. Its kernel depends only on the physical endpoints z, z' ; write its edge mass as*

$$M = r^{\chi+o(1)}.$$

Consider the determinant-small four-cycles returned by the collision-preserving Maslov routing, and write an ordered cycle as

$$p, y, q, y', p.$$

Restrict to the non-line packet output and, before sampling any old flag marks over the free endpoint y' , use the physical variables p, q, y to define

$$E_{\text{new}} := \text{Span}\{a_y - a_p, a_q - a_y\}.$$

The determinant-small condition places y' in the associated direct packet, while E_{new} and its normal are independent of the old flags sampled over y' . Extend the physical cycle conditionally by four independent samples from $\kappa_{y'}$. Let $\mathcal{Z}_{\text{all-coh}}(\vartheta)$ be the portion on which all four sampled old-flag normals lie within ϑ of the new normal $[n_{\text{new}}]$. Then

$$\mathcal{Z}_{\text{all-coh}}(\vartheta) \lesssim_U \left(\frac{H_i}{\lambda_*} \vartheta \right)^4. \quad (319)$$

Consequently, if this fully coherent class carries a fixed portion of a rank-loss output of mass $\gtrsim M^4$, then

$$M \lesssim_U \frac{H_i}{\lambda_*} \vartheta. \quad (320)$$

In the fixed-coarse/all-fine regime, with $\vartheta = r^{\alpha+o(1)}$, a fully coherent survivor must satisfy

$$\chi \geq \alpha - o(1). \quad (321)$$

In particular, every total-defect layer with $\chi < 1$ admits parameters

$$\chi < \alpha < d - \beta < 1, \quad \Delta = r^d, \quad \theta_{\text{hr}} = r^\beta,$$

for which the direct new-packet thickness obeys

$$t_{\text{new}} \lesssim r + \frac{\Delta}{\theta_{\text{hr}}} = o(r^\alpha) = o(\vartheta). \quad (322)$$

The fully coherent rank-loss return is then impossible by (321); every remaining old/new comparison contains a ϑ -transverse flag at the free endpoint y' and returns, with the positive fine-scale thickness ratio $t_{\text{new}}/\vartheta = o(1)$, to the transverse-polarization ledger. The close-anchor, horizontal-line, and prior physical-bush outputs retain their existing destinations.

Proof. Condition on the four physical vertices p, y, q, y' . The new plane E_{new} is already fixed by p, q, y , while the four old flags over y' are independent under $\kappa_{y'}^{\otimes 4}$. Apply (317) four times with the prescribed normal $[n_{\text{new}}]$. Their conditional product is at most

$$C_U \left(\frac{H_i}{\lambda_*} \vartheta \right)^4.$$

Integrate the physical four-cycle kernel, which is nonnegative and bounded by one. Since each $\kappa_{y'}$ has total mass one, the fourfold conditional extension leaves the unmarked four-cycle mass unchanged. This proves (319). Comparison with a coherent lower bound $\gtrsim M^4$ and taking fourth roots gives (320); now use (318) to obtain (321).

If $\chi < 1$, choose $\alpha \in (\chi, 1)$, then choose $d \in (\alpha, 1)$ and $\beta \in (0, d - \alpha)$. Since $d < 1$, one has $r \ll \Delta$, as required by the arbitrary-threshold Maslov routing. Since both 1 and $d - \beta$ exceed α , (322) follows from Theorem 7.40. The coherent necessary condition $\chi \geq \alpha - o(1)$ contradicts the strict choice $\alpha > \chi$. On the complement, at least one of the four old-flag normals over y' is separated from n_{new} by ϑ . Choose the first such flag and integrate the remaining conditional flag variables to one. The unequal-thickness intersection estimate of Theorems 7.78 and 9.49 supplies the factor $t_{\text{new}}/\vartheta = o(1)$ in the genuinely free y' -variable, with the frozen old thickness absorbed into the coarse constant. $\square$

Lemma 8.3 (Conditional tensoring of old flags is lossless for every physical collision kernel). *Use the disintegration from Theorem 8.1,*

$$d\widehat{\rho}_*(\mathbf{f}, z) = d\kappa_z(\mathbf{f}) d\widehat{\mu}_*(z).$$

For an integer $k \geq 1$, define the k -flag extension

$$d\widehat{\rho}_*^{[k]}(\mathbf{f}_1, \dots, \mathbf{f}_k, z) := \left(\prod_{j=1}^k d\kappa_z(\mathbf{f}_j) \right) d\widehat{\mu}_*(z). \quad (323)$$

Then $\widehat{\rho}_*^{[k]}$ is a probability measure with physical marginal $\widehat{\mu}_*$. For every $m \geq 1$ and every nonnegative kernel $K(z_1, \dots, z_m)$ depending only on the physical endpoints,

$$\int K(z_1, \dots, z_m) \prod_{\ell=1}^m d\widehat{\rho}_*^{[k]}(\mathbf{f}_\ell, z_\ell) = \int K(z_1, \dots, z_m) \prod_{\ell=1}^m d\widehat{\mu}_*(z_\ell). \quad (324)$$

In particular the fine Maslov graph edge mass, its four-cycle mass, and every collision-preserving restriction are unchanged by the lift. No new geometric incidence is assumed: the k marks are independent samples from the actual conditional old-flag occurrence law at the same physical point.

Proof. For every physical point z , κ_z is a probability measure. Integrating the product in $(\mathbf{f}_1, \dots, \mathbf{f}_k)$ therefore gives one. This proves the physical marginal identity, and applying it independently at the m physical endpoints proves (324). $\square$

Proposition 8.4 (Eight old flags at the free cycle endpoint close the coherent defect range below two). *Lift the normalized fine Maslov graphon h_r to $\widehat{\rho}_*^{[8]}$ by Theorem 8.3; write its unchanged edge mass as*

$$M = r^{\chi+o(1)}.$$

Write an ordered determinant-small cycle as

$$p, y, q, y', p.$$

Before the flag lift, use the physical variables p, q, y to set

$$E_{\text{new}} = \text{Span}\{a_y - a_p, a_q - a_y\};$$

the determinant-small condition places the remaining endpoint y' in the corresponding direct packet, while E_{new} is independent of the old flags sampled over y' . Let $\mathcal{Z}_{\text{eight-coh}}(\vartheta)$ be the cycle mass for which all eight old flag normals sampled conditionally over this free endpoint lie within ϑ of $[n_{\text{new}}]$. Then

$$\mathcal{Z}_{\text{eight-coh}}(\vartheta) \lesssim_U \left(\frac{H_i}{\lambda_*} \vartheta \right)^8. \quad (325)$$

If this class carries a fixed portion of a rank-loss output of mass $\gtrsim M^4$, it follows that

$$M \lesssim_U \left(\frac{H_i}{\lambda_*} \right)^2 \vartheta^2. \quad (326)$$

Thus, in the fixed-coarse/all-fine regime, a fully coherent survivor at $\vartheta = r^{\alpha+o(1)}$ must satisfy

$$\chi \geq 2\alpha - o(1). \quad (327)$$

For every bad total-defect exponent $\chi < 2$, choose

$$\frac{\chi}{2} < \alpha < d - \beta < 1, \quad \vartheta = r^\alpha, \quad \Delta = r^d, \quad \theta_{\text{hr}} = r^\beta. \quad (328)$$

Then the direct new-packet thickness is $o(\vartheta)$, the fully coherent output contradicts the eight-flag power-region inequality above, and every complementary packet has at least one of the eight old flags transverse to the new polarization at the free endpoint y' . It therefore returns with a positive fine-scale thickness gain to Theorem 7.78. Hence no determinant-small, non-line, fully coherent rank-loss return survives at any exponent $\chi < 2$.

Proof. Condition on p, q, y, y' , using the displayed ordering only to designate y' as the free packet endpoint. The new normal is already fixed by p, q, y . The eight old flags over y' are independent under $\kappa_{y'}^{\otimes 8}$. By (317), their simultaneous coherence has conditional probability at most

$$C_U \left(\frac{H_i}{\lambda_*} \vartheta \right)^8.$$

Integrate the physical four-cycle kernel. It is nonnegative and bounded by one, so (325) follows. The tensor invariance (324) ensures that the rank-loss lower bound remains $\gtrsim M^4$. Compare the two bounds and take fourth roots to obtain (326); then use (318) to get (327).

If $\chi < 2$, the interval $(\chi/2, 1)$ is nonempty. Choose α there, then $d \in (\alpha, 1)$ and $\beta \in (0, d - \alpha)$. Exactly as in (322),

$$t_{\text{new}} \lesssim r + r^{d-\beta} = o(r^\alpha).$$

The strict inequality $2\alpha > \chi$ contradicts (327). On the complement, one of the eight actual old flags over the free endpoint y' is ϑ -transverse to the new packet. Choose the first such flag, integrate the other seven conditional probability variables to one, and apply the unequal-thickness transverse intersection bound in the genuinely free y' -variable, as recorded in Theorems 7.78 and 9.49. Since $t_{\text{new}}/\vartheta = o(1)$ and the old layer is frozen, this gives the required positive fine-scale gain. $\square$

Corollary 8.5 (Finite flag amplification excludes coherent rank loss at any prescribed mesoscopic threshold). *Let*

$$M = r^{\chi+o(1)}, \quad 0 \leq \chi < 2,$$

and prescribe any $0 < d < 1$, so that $\Delta = r^d$. Choose an integer

$$k > \frac{4\chi}{d},$$

then choose

$$\boxed{\frac{4\chi}{k} < \alpha < d - \beta < d, \quad \vartheta = r^\alpha, \quad \theta_{\text{hr}} = r^\beta.} \quad (329)$$

On the k -flag extension $\hat{\rho}_*^{[k]}$, a fully coherent rank-loss output of mass $\gtrsim M^4$ would force

$$\boxed{M \lesssim_U \left(\frac{H_i}{\lambda_*} \vartheta \right)^{k/4}, \quad \chi \geq \frac{k\alpha}{4} - o(1),} \quad (330)$$

which contradicts (329). Since

$$t_{\text{new}} \lesssim r + r^{d-\beta} = o(r^\alpha),$$

every complementary rank-loss packet has a transverse old flag at the free endpoint and pays the fine-scale thickness gain. Thus, after the close-anchor, horizontal-line, and transverse ledgers are removed, the arbitrary-threshold Maslov routing produces a physical $O(r^{1-d})$ -Lagrangian bush for every prescribed $d \in (0, 1)$.

Proof. By Theorem 8.3, the k -flag lift leaves the physical four-cycle mass unchanged. Applying (317) independently to the k flags over the free endpoint gives

$$\mathcal{Z}_{k\text{-coh}}(\vartheta) \lesssim_U \left(\frac{H_i}{\lambda_*} \vartheta \right)^k.$$

Comparison with M^4 , followed by fourth roots, proves (330). The parameter inequalities give both its contradiction and $t_{\text{new}} = o(\vartheta)$. The first transverse flag is then treated exactly as in the proof of Theorem 8.4. $\square$

Corollary 8.6 (The final compatibility is either transverse-paid or one coherent Maslov diagonal). *After dyadic regularization of $t_{\text{old}}, t_{\text{new}}$ and a slowly vanishing choice $\vartheta = W^{o(1)}$, every old-flag/new-cycle contribution from Theorem 7.77 has one of two destinations:*

- (i) *it pays the transverse factor $(t_{\text{old}} + r)(t_{\text{new}} + r)/\vartheta$ from (307);*
- (ii) *all retained old and new packet data lie in one cell-locally aligned $E \oplus E$ polarization, and the new collision kernel factors exactly as*

$$\boxed{\mathbf{1}_{\{|\Pi_s(z) - \Pi_s(z')| \leq r\}} \leq \mathbf{1}_{\{|N_s(z, z')| \leq r\}} \mathbf{1}_{\{|T_s(z, z')| \leq r\}}.} \quad (331)$$

Hence the sole coherent remainder is the relative tangential diagonal conditioned by the normal Reeb window, now for the low old-flag-reuse measure. This is precisely the symplectic normal/tangential interface of Theorem 7.22; no direct use of Theorem 6.17 is made without its containment hypothesis. Moreover, after collision-preserving degree regularization, its rank-loss return is subject to the quantitative anchor-diagonal constraint (314); any four-cycle mass exceeding that bound is already in the transverse or physical-output alternatives. The lossless eight-flag extension at the free endpoint is stronger still. By Theorem 8.3, it leaves the physical collision graph and its four-cycle lower bound unchanged, while Theorem 8.4 removes the fully coherent rank-loss return for every bad fine total-defect exponent $\chi < 2$. Thus failure of (487) has no independent coherent incidence destination: it returns to a transverse old/new polarization, a horizontal-line or close-anchor chart, or a physical Lagrangian bush.

Proposition 8.7 (Eight-flag amplification reduces every bad coherent rank-loss return to a physical bush). *Fix one coarse low-reuse occurrence layer G_* , normalize its physical occurrence marginal, and let $r \downarrow 0$ range over fine collision scales. Suppose that collision-preserving pruning and two-sided degree regularization give a normalized Maslov graph with*

$$M = r^{\chi+o(1)}, \quad 0 \leq \chi < 2.$$

Choose

$$\frac{\chi}{2} < \alpha < d - \beta < 1, \quad \vartheta = r^\alpha, \quad \Delta = r^d, \quad \theta_{\text{hr}} = r^\beta.$$

Then the arbitrary-threshold routing has one of the following four destinations:

- (i) *a close-anchor/angular or horizontal-line packet;*
- (ii) *a transverse old/new polarization packet, with thickness gain*

$$\frac{t_{\text{new}}}{\vartheta} \lesssim r^{1-\alpha} + r^{d-\beta-\alpha} = o(1);$$

- (iii) a physical $O(r^{1-d})$ -Lagrangian point-probe bush of normalized occurrence mass $\gtrsim M^3$;
- (iv) a fully coherent determinant-small rank-loss packet.

The fourth alternative is impossible. Hence, after the first two destinations have been paid in their weighted ledgers, every remaining bad collision layer produces the physical bush in (iii).

Writing $m_* = \nu_i(G_*) > 0$, that bush has unrepeated selector-direction mass $\gtrsim_{G_*} m_* M^3$, and the exact Reeb escape map gives

$$\boxed{\mathcal{L}^4(N_{C_{G_*} r^{1-d}}(K_b)) \gtrsim_{G_*} m_* M^3 = r^{3\chi+o(1)}}. \quad (332)$$

Proof. Apply Theorems 7.8 and 7.26. Its first output is (iii). Split its determinant-small output by the anchored horizontal geometry. The close-anchor and horizontal-line pieces give (i); the remaining cycles lie in the direct contact-polarized packet of Theorem 7.42. Order such a cycle as p, y, q, y', p , with y' the free packet endpoint, and lift that endpoint to eight conditional old flags by Theorem 8.3. The fully coherent part is impossible by Theorem 8.4. On its complement, the first transverse flag at y' gives (ii) through Theorem 7.78; the displayed gain is the direct packet thickness divided by ϑ .

On the frozen occurrence layer, $\lambda_* \leq \mathcal{N}_{\text{flag}} < 2\lambda_*$, so the normalized physical occurrence marginal is comparable, with constants depending only on G_* , to $m_*^{-1}\nu_i|_{G_*}$. Therefore a normalized bush of mass $\gtrsim M^3$ contains actual selector-direction mass $\gtrsim_{G_*} m_* M^3$. Its radius is $O(r/\Delta) = O(r^{1-d})$. The inclusion of its ideal Reeb escape in the corresponding neighborhood of K_b , followed by the one-bush volume formula Theorems 5.1 and 5.2, proves (332). $\square$

8.2 Maximal bush extraction and vector Frostman escape

Proposition 8.8 (Maximal extraction gives a paid induced branch or an M -mass bush family). *Keep the notation of Theorem 8.7, fix any $d \in (0, 1)$, and choose the finite flag number and angular parameters from Theorem 8.5. Let h be the normalized Maslov graph and suppose*

$$e(h) = M = r^{\chi+o(1)}, \quad 0 \leq \chi < 2,$$

Then at least one of the following holds:

- (i) *there is an induced measurable subgraph A whose absolute edge mass is at least $M/2$ and whose arbitrary-threshold routing places a fixed part of its normalized rank-loss output in a close-anchor/angular, horizontal-line, or transverse-polarization ledger;*
- (ii) *there is a finite family of pairwise disjoint selector sets*

$$H_1, \dots, H_N \subset G_*, \quad N \lesssim M^{-3},$$

such that every H_j is an $O(r^{1-d})$ -Lagrangian point-probe bush and

$$\boxed{\sum_{j=1}^N \hat{\mu}_*(H_j) \gtrsim M}. \quad (333)$$

Every H_j in (ii) may be chosen with the conditional determinant-nondegenerate origin witness of Theorem 9.21; the origin-witness kernel has physical marginal $\hat{\mu}_|_{H_j}$ and survives every later restriction.*

In the second alternative, let E_j be the corresponding ideal Reeb escape, trimmed to a fixed outer Reeb wing, and use the original selector-direction normalization on G_ . Then*

$$\boxed{\sum_{j=1}^N |E_j| \gtrsim_{G_*} m_* M}. \quad (334)$$

Writing

$$\mathcal{O}_{\text{bush}} := \sum_{1 \leq j < k \leq N} |E_j \cap E_k|,$$

the exact escaped second moment gives

$$\boxed{\mathcal{L}^4(N_{C_{G_*} r^{1-d}}(K_b)) \gtrsim_{G_*} \frac{(m_* M)^2}{m_* M + \mathcal{O}_{\text{bush}}}}. \quad (335)$$

Thus the loss from M to M^3 is not intrinsic to the physical output. After maximal de-multiplication, the only remaining loss is the overlap of the actual Reeb-escaped bushes.

Proof. Work with the normalized physical occurrence marginal $\hat{\mu}_*$. Make a finite maximal selection of disjoint bush leaf sets. If at some stage the complement A has induced edge mass

$$\iint_{A \times A} h d\hat{\mu}_* d\hat{\mu}_* \geq \frac{M}{2},$$

normalize $\hat{\mu}_*|_A$. Its normalized edge mass is at least $M/2$. The same free-endpoint conditional flag law is available on A , and the exponent of its edge mass is at most $\chi + o(1)$. Apply the arbitrary-threshold routing and Theorem 8.5. If a paid geometric output occurs, alternative (i) holds and the selection stops. Otherwise use the origin-marked form Theorem 9.21 of the $O(r^{1-d})$ -bush. Its unnormalized $\hat{\mu}_*$ -mass is at least

$$\hat{\mu}_*(A) \left(\frac{M/2}{\hat{\mu}_*(A)^2} \right)^3 \gtrsim M^3.$$

Assume alternative (i) never occurs. Consequently a maximal bush family is finite and has $N \lesssim M^{-3}$. Its complement has induced edge mass below $M/2$. If $U = \bigcup_j H_j$, then $0 \leq h \leq 1$ gives

$$\frac{M}{2} < e(h) - e(h|_{U^c \times U^c}) \leq 2\hat{\mu}_*(U).$$

This proves (333).

The dyadic occurrence bounds on G_* convert the normalized mass in (333) into original selector-direction mass $\gtrsim_{G_*} m_* M$. The H_j are disjoint, and the outer-wing one-bush Jacobian is bounded above and below. The one-bush volume formula therefore gives (334). Finally put

$$\mathcal{M}_{\text{bush}} := \sum_j \mathbf{1}_{E_j}.$$

Then

$$\int \mathcal{M}_{\text{bush}} = \sum_j |E_j|, \quad \int \mathcal{M}_{\text{bush}}^2 = \sum_j |E_j| + 2\mathcal{O}_{\text{bush}}.$$

Since every $E_j \subset N_{C_{G_*} r^{1-d}}(K_b)$, Cauchy–Schwarz proves (335). $\square$

Principle 8.9 (The remaining escaped overlap is again a contact–Maslov collision). *For every pair $j \neq k$, the exact affine homothety formula Theorem 5.3 writes*

$$|E_j \cap E_k| = |U| \int |s - s_j|^3 \sigma \left(H_j \cap \Psi_{j,k,s}^{-1}(H_k) \right) ds.$$

Every direction pair counted by this integral obeys the secondary physical collision (29) at thickness $O(r^{1-d})$. Splitting those pairs by their angular separation gives exactly two destinations: the fixed-angle part is a coarser Reeb-window Maslov graph, and the near-angle part is an angular Darboux refinement. Hence the denominator $\mathcal{O}_{\text{bush}}$ in (335) contains no unstructured set-overlap term. Its weighted summation across the maximal family, with the old flag occurrence measures retained, is the contact-overlap estimate treated below.

Theorem 8.10 (A direction–Frostman Lagrangian bush escapes to a front–Frostman measure). *Let $r_n \downarrow 0$. For each n , suppose $H_n \subset U$ is an ε_n -approximate Lagrangian point-probe bush centered at (c_n, s_n) , where*

$$\varepsilon_n \downarrow 0, \quad s_n \in J \Subset I,$$

and let η_n be the normalized selector-direction measure on H_n . Assume that for some $0 < \delta < 3$, some $\rho_n \downarrow 0$ with $\varepsilon_n \leq \rho_n$, and one constant C_δ ,

$$\boxed{\eta_n(B(a, R)) \leq C_\delta R^{3-\delta} \quad (a \in \mathbb{R}^3, \rho_n \leq R \leq 1).} \quad (336)$$

Then the actual Sticky Reeb front carries a nonzero $(4 - \delta)$ -Frostman measure. In particular,

$$\boxed{\dim_H K_b \geq 4 - \delta.} \quad (337)$$

If such bushes exist for every $\delta > 0$, then $\dim_H K_b = 4$.

Proof. Choose once and for all an interval $I_{\text{out}} \Subset I$ with positive length and

$$\text{dist}(I_{\text{out}}, J) > 0.$$

For $a \in H_n$, write the actual selector front as

$$F_n(a, s) := (b(a) + sa, s).$$

The approximate bush relation gives, uniformly for $s \in I_{\text{out}}$,

$$|b(a) + sa - (c_n + (s - s_n)a)| \lesssim \varepsilon_n. \quad (338)$$

Let

$$\Lambda_n := (F_n)_\# \left(\eta_n \otimes \frac{\mathbf{1}_{I_{\text{out}}} ds}{|I_{\text{out}}|} \right).$$

These are probability measures supported on K_b . Fix a physical ball $B((y_0, s_0), R)$ with $R \geq \rho_n$. Only Reeb parameters in an interval of length $O(R)$ contribute. At each such parameter, (338) and $\text{dist}(I_{\text{out}}, J) > 0$ place every contributing direction in a ball of radius

$$\lesssim R + \varepsilon_n \lesssim R.$$

Therefore (336) gives

$$\Lambda_n(B((y_0, s_0), R)) \lesssim_{\delta, I, J} R^{4-\delta} \quad (R \geq \rho_n). \quad (339)$$

The front lies in a fixed compact region after the standing chart localization, so a subsequence of Λ_n converges weakly to a probability measure Λ supported on K_b . For every fixed $R > 0$, eventually $R \geq \rho_n$. Choose a continuous cutoff equal to one on a given closed R -ball and supported in its $(R + \zeta)$ -enlargement. Weak convergence and (339) give an upper bound $C(R + \zeta)^{4-\delta}$; let $\zeta \downarrow 0$. Thus Λ is $(4 - \delta)$ -Frostman, proving (337). $\square$

Theorem 8.11 (Reeb-coherent vector Frostman escape for a family of Lagrangian bushes). *For every n , let*

$$\{H_{n,j} : 1 \leq j \leq N_n\}$$

be pairwise disjoint selector-direction sets. Suppose $H_{n,j}$ is an ε_n -approximate Lagrangian point-probe bush centered at $(c_{n,j}, s_{n,j})$, where

$$\varepsilon_n \downarrow 0, \quad s_{n,j} \in J \Subset I.$$

Put

$$q_n := \sum_{j=1}^{N_n} \sigma(H_{n,j}) > 0.$$

Choose a fixed outer interval $I_{\text{out}} \Subset I$ with $\text{dist}(I_{\text{out}}, J) > 0$, and define

$$\mathbf{a}_{n,j}(y, s) := \frac{y - c_{n,j}}{s - s_{n,j}} \quad (y \in \mathbb{R}^3, s \in I_{\text{out}}).$$

Assume that for some $0 < \delta < 3$, some $\rho_n \downarrow 0$ with $\varepsilon_n \leq \rho_n$, and one constant C_δ ,

$$\sup_{\substack{y \in \mathbb{R}^3 \\ s \in I_{\text{out}}}} \frac{1}{q_n} \sum_{j=1}^{N_n} \sigma(H_{n,j} \cap B(\mathbf{a}_{n,j}(y, s), R)) \leq C_\delta R^{3-\delta} \quad (\rho_n \leq R \leq 1). \quad (340)$$

Then K_b carries a nonzero $(4 - \delta)$ -Frostman measure and

$$\dim_H K_b \geq 4 - \delta.$$

Thus (340), rather than a separate Frostman bound on one arbitrarily selected bush, is the exact Reeb-coherent physical aggregation condition for the maximal family in Theorem 8.8.

Proof. Define a probability measure on the actual front by

$$\Lambda_n := \frac{1}{q_n |I_{\text{out}}|} \sum_{j=1}^{N_n} (F|_{H_{n,j} \times I_{\text{out}}})_{\#} (\sigma|_{H_{n,j}} \otimes ds), \quad F(a, s) := (b(a) + sa, s).$$

Fix a physical R -ball centered at (y_0, s_0) , with $R \geq \rho_n$. Only a Reeb interval of length $O(R)$ contributes. At each contributing s , the approximate bush relation and $\text{dist}(I_{\text{out}}, J) > 0$ place every contributing direction in

$$B(\mathbf{a}_{n,j}(y_0, s), C(R + \varepsilon_n)) \subset B(\mathbf{a}_{n,j}(y_0, s), C'R).$$

Therefore (340), with the same physical y_0, s for all components, gives after summing in j , dividing by q_n , and integrating the $O(R)$ -length time interval

$$\Lambda_n(B((y_0, s_0), R)) \lesssim_{\delta, I, J} R^{4-\delta}.$$

Weak compactness and the continuous-cutoff argument from Theorem 8.10 produce a $(4 - \delta)$ -Frostman weak limit supported on K_b . $\square$

8.3 Coherent hairbrushes and re-collision

Corollary 8.12 (Failure of bush-family escape is a Reeb-coherent hairbrush concentration). *Fix $0 < \delta < 3$. If no subsequence of the maximal bush families satisfies (340) with this δ , then there are a subsequence, $A_n \rightarrow \infty$, and radii*

$$\varepsilon_n \leq R_n \downarrow 0,$$

and points $y_n \in \mathbb{R}^3$, $s_n^{\text{out}} \in I_{\text{out}}$ such that, with

$$a_{n,j} := \mathbf{a}_{n,j}(y_n, s_n^{\text{out}}),$$

$$\boxed{\sum_{j=1}^{N_n} \sigma(H_{n,j} \cap B(a_{n,j}, R_n)) > A_n q_n R_n^{3-\delta}.} \quad (341)$$

The caps retain their bush-center labels, selected Reeb windows, and conditional old-flag laws. More importantly, all cap centers are generated by the same physical outer-wing point (y_n, s_n^{out}) . Thus the sole non-Frostman output of the maximal physical family is a genuine Reeb-coherent multi-bush hairbrush, not an arbitrary simultaneous cap choice.

Proof. Take the contrapositive of (340) with constants $A_n \rightarrow \infty$. The left side of (341) is at most q_n ; hence

$$R_n^{3-\delta} < A_n^{-1},$$

which forces $R_n \rightarrow 0$. $\square$

Corollary 8.13 (The M^{-4} debt is exactly a lower bound for the bad hairbrush scale). *For the maximal families in alternative (ii) of Theorem 8.8, every Reeb-coherent failure (341) satisfies*

$$\boxed{R_n^\delta \gtrsim_{U, G_*} A_n \frac{q_n}{N_n} \gtrsim_{U, G_*} A_n M_n^4.} \quad (342)$$

Equivalently, at every radius satisfying

$$R^\delta \leq c_U q_n / N_n, \quad (343)$$

the Reeb-coherent family Frostman bound holds automatically. Thus the factor M_n^{-4} in (463) is not an artifact of selecting one component: it is the exact family cardinality-to-mass scale barrier.

Proof. Every direction cap of radius R has normalized Lebesgue mass $O_U(R^3)$. Hence, uniformly in the common outer point,

$$\sum_{j=1}^{N_n} \sigma(H_{n,j} \cap B(\mathbf{a}_{n,j}(y, s), R)) \lesssim_U N_n R^3.$$

Comparison with (341) gives $A_n q_n R_n^{3-\delta} \lesssim_U N_n R_n^3$, which is the first inequality in (342). Use $q_n \gtrsim_{G_*} M_n$ and $N_n \lesssim M_n^{-3}$ for the second. The same cap-volume bound divided by q_n proves (343). $\square$

Lemma 8.14 (A Reeb-coherent cap family is one exact re-collision cocycle orbit). *Fix one outer point $(y, s) \in \mathbb{R}^3 \times I_{\text{out}}$. For every pair of distinct bush labels j, k ,*

$$\boxed{\Psi_{jk,s}(\mathbf{a}_{n,j}(y, s)) = \mathbf{a}_{n,k}(y, s).} \quad (344)$$

Moreover, uniformly for $s \in I_{\text{out}}$,

$$\Psi_{jk,s}(B(\mathbf{a}_{n,j}(y, s), R)) \subset B(\mathbf{a}_{n,k}(y, s), C_{I,J}R). \quad (345)$$

If $|u - s| \leq R$ and the cap on the left meets U , the same uniform bound gives

$$\Psi_{jk,u}(B(\mathbf{a}_{n,j}(y, s), R)) \subset B(\mathbf{a}_{n,k}(y, s), C_{I,J,U}R). \quad (346)$$

Thus the caps in (341) are not independently centered angular patches: they are a single orbit of the exact contact re-collision maps.

Proof. By (27),

$$\begin{aligned} \Psi_{jk,s}(\mathbf{a}_{n,j}(y, s)) &= \frac{c_{n,j} - c_{n,k} + (s - s_{n,j})(y - c_{n,j})/(s - s_{n,j})}{s - s_{n,k}} \\ &= \frac{y - c_{n,k}}{s - s_{n,k}} = \mathbf{a}_{n,k}(y, s). \end{aligned}$$

The linear part of $\Psi_{jk,s}$ is the scalar $(s - s_{n,j})/(s - s_{n,k})$. Since $s \in I_{\text{out}}$, $s_{n,j}, s_{n,k} \in J$, and $\text{dist}(I_{\text{out}}, J) > 0$, its absolute value is uniformly bounded. This proves (345). For $|u - s| \leq R$, subtract the two explicit affine formulas and use that every cap meeting U has bounded center. The coefficient denominators remain separated from zero, so the center error is $O_{I,J,U}(R)$, proving (346). $\square$

Lemma 8.15 (The common-hairbrush blow-up lands exactly in the Lagrangian pencil). *Suppress n , write*

$$s_0 := s_n^{\text{out}}, \quad y_0 := y_n, \quad d_j := s_0 - s_{n,j}, \quad a_j := \frac{y_0 - c_{n,j}}{d_j},$$

and put

$$a = a_j + R_n \xi, \quad s = s_0 + R_n \tau.$$

The ideal outer-wing front of the j -th bush, translated by (y_0, s_0) and dilated by R_n^{-1} , is exactly

$$\boxed{R_n^{-1}(c_{n,j} + (s - s_{n,j})a - y_0) = d_j \xi + \tau a_j + R_n \tau \xi.} \quad (347)$$

For two labels $j \neq k$, directions $a_j + R_n \xi$ and $a_k + R_n \eta$ re-collide at the scaled time τ if and only if

$$\boxed{\beta_{jk} + \tau \alpha_{jk} + R_n \tau (\xi - \eta) = 0, \quad \alpha_{jk} := a_j - a_k, \quad \beta_{jk} := d_j \xi - d_k \eta.} \quad (348)$$

Consequently every ideal blow-up limit $R_n \rightarrow 0$ obeys

$$\boxed{\beta_{jk} + \tau \alpha_{jk} = 0, \quad (\alpha_{jk}, \beta_{jk}) \in L_\tau.} \quad (349)$$

Thus a Reeb-coherent Frostman failure returns, under its canonical physical blow-up, to the same Maslov/Lagrangian incidence pencil used in the angular-collapse routing. For the actual selector front, the rescaled error relative to (347) is at most

$$\varepsilon_n / R_n.$$

Hence (349) is an exact actual-front limit when $\varepsilon_n / R_n \rightarrow 0$; without that separation one obtains the corresponding finite-scale Maslov incidence at thickness $O(\varepsilon_n / R_n)$, and no zero-error limit is claimed.

Proof. The identity $c_{n,j} = y_0 - d_j a_j$ gives

$$c_{n,j} + (s_0 + R_n \tau - s_{n,j})(a_j + R_n \xi) - y_0 = R_n(d_j \xi + \tau a_j) + R_n^2 \tau \xi,$$

which is (347). Equality of the two rescaled horizontal fronts is exactly (348). Letting $R_n \rightarrow 0$ gives (349), and the last membership is the definition of the Lagrangian graph $L_\tau = \{(\alpha, \beta) : \beta + \tau \alpha = 0\}$. $\square$

Proposition 8.16 (A Reeb-coherent bad cap family merges into one larger physical Lagrangian bush). *For the caps in (341), put*

$$A_{n,j} := H_{n,j} \cap B(a_{n,j}, R_n), \quad S_n := \sum_j \sigma(A_{n,j}),$$

and

$$H_n^{\text{hair}} := \bigcup_{j=1}^{N_n} A_{n,j}.$$

Then the disjointness of the $H_{n,j}$ gives

$$\boxed{\sigma(H_n^{\text{hair}}) = S_n > A_n q_n R_n^{3-\delta}.} \quad (350)$$

Moreover H_n^{hair} is one $C_{I,J}(R_n + \varepsilon_n)$ -Lagrangian point-probe bush centered at the actual outer-wing point (y_n, s_n^{out}) . Since $\varepsilon_n \leq R_n$,

$$\boxed{|b(a) + s_n^{\text{out}} a - y_n| \lesssim_{I,J} R_n \quad (a \in H_n^{\text{hair}}).} \quad (351)$$

Consequently the exact Reeb escape map gives the physical credit

$$\boxed{\mathcal{L}^4(N_{C_{I,J}R_n}(K_b)) \gtrsim_{I,J,U} S_n > A_n q_n R_n^{3-\delta}.} \quad (352)$$

Thus a failure of the Reeb-coherent family Frostman bound is itself a larger physical bush, not merely an angular label attached to the old bush family.

Proof. For $a \in A_{n,j}$, the ε_n -bush relation and the definition of $a_{n,j}$ give

$$\begin{aligned} |b(a) + s_n^{\text{out}} a - y_n| &\leq |b(a) - (c_{n,j} - s_{n,j} a)| + |c_{n,j} + (s_n^{\text{out}} - s_{n,j})a - y_n| \\ &\leq \varepsilon_n + |s_n^{\text{out}} - s_{n,j}| |a - a_{n,j}| \\ &\lesssim_{I,J} \varepsilon_n + R_n. \end{aligned}$$

This proves (351); (350) is the cap concentration inequality. Because $I_{\text{out}} \Subset I$, one can choose, uniformly in s_n^{out} , a Reeb subinterval of fixed positive length inside I and a fixed positive distance from s_n^{out} . Apply Theorems 5.1 and 5.2 to the merged bush on that interval. Its ideal escape has volume comparable to S_n and lies in the $C_{I,J}R_n$ -neighborhood of the actual front, proving (352). $\square$

Corollary 8.17 (At the hairbrush scale the physical product collision is already large). *There is an interval $W'_n \subset I$ of length $c_{I,J}R_n$, containing or adjacent to s_n^{out} , such that*

$$\boxed{\int_{W'_n} \iint_{H_n^{\text{hair}} \times H_n^{\text{hair}}} \mathbf{1}_{\{|\Pi_u(z(a)) - \Pi_u(z(a'))| \leq c_{I,J,U} R_n\}} d\sigma(a) d\sigma(a') du \gtrsim_{I,J,U} R_n S_n^2.} \quad (353)$$

Hence the ideal-coupling thickening issue does not occur at the coarse hairbrush scale R_n ; it arises only when one asks for a collision-preserving transfer to a scale $o(R_n)$.

Proof. By (351), every point $\Pi_{s_n^{\text{out}}}(z(a))$, $a \in H_n^{\text{hair}}$, lies in one $O(R_n)$ -ball. Since U is bounded, the same holds with a fixed enlargement for $|u - s_n^{\text{out}}| \leq c_{I,J}R_n$. The collision indicator is therefore one on the full product for every $u \in W'_n$. Integrate its product mass S_n^2 over W'_n . $\square$

Corollary 8.18 (Family Frostman failure reduces to one physical-bush Frostman problem). *Let*

$$\eta_n^{\text{hair}} := S_n^{-1} \sigma|_{H_n^{\text{hair}}}.$$

For each fixed $0 < \delta < 3$, exactly one of the following occurs after passing to a subsequence:

- (i) the measures η_n^{hair} satisfy the uniform direction-Frostman hypothesis of Theorem 8.10 above the bush scale $C_{I,J}R_n$, and therefore $\dim_H K_b \geq 4 - \delta$;
- (ii) there are $\tilde{A}_n \rightarrow \infty$, radii $\tilde{R}_n \geq C_{I,J}R_n$ with $\tilde{R}_n \downarrow 0$, and direction caps such that

$$\boxed{\eta_n^{\text{hair}}(B(\tilde{a}_n, \tilde{R}_n)) > \tilde{A}_n \tilde{R}_n^{3-\delta}.} \quad (354)$$

The restricted direction set in (ii) remains one $C_{I,J}R_n$ -bush centered at (y_n, s_n^{out}) and retains all original bush labels and old conditional Maslov flags.

Proof. Apply Theorems 8.10 and 9.40 to the merged bushes from Theorem 8.16. Restricting directions does not alter the pointwise bush inequality or the conditional flag laws. $\square$

Corollary 8.19 (Merging removes the artificial component-selection factor M^{-4}). *In alternative (ii) of Theorem 8.18, put*

$$\tilde{p}_n := \sigma\left(H_n^{\text{hair}} \cap B(\tilde{a}_n, \tilde{R}_n)\right).$$

Then

$$\tilde{p}_n > \tilde{A}_n S_n \tilde{R}_n^{3-\delta}. \quad (355)$$

After translating and dilating this cap by $\tilde{R}_n^{-1}$, its normalized direction measure has density

$$\tilde{D}_{n,\text{cap}} \lesssim_U \frac{\tilde{R}_n^3}{\tilde{p}_n} \lesssim_U \frac{\tilde{R}_n^\delta}{\tilde{A}_n S_n}. \quad (356)$$

Necessarily

$$\tilde{R}_n^\delta \gtrsim_U \tilde{A}_n S_n. \quad (357)$$

Thus the M_n^{-4} factor in (463) is only the cost of selecting one original component. After the Reeb-coherent caps are merged, the correct remaining normalization debt is S_n^{-1} , with no bush-cardinality factor.

Proof. Equation (355) is (354) multiplied by S_n . The affine rescaling calculation of Theorem 9.41 gives (356). Since a probability measure supported in a fixed bounded three-dimensional ball has density norm bounded below by a positive dimensional constant, (357) follows from (356). $\square$

Lemma 8.20 (Inside one Reeb hairbrush, clustered homothety centers force a rank-one direction locus). *Keep the notation of Theorem 8.15. If $s_{n,j} \neq s_{n,k}$, denote the distinguished homothety center of the pair by*

$$a_{jk}^{\text{hom}} := \frac{c_{n,j} - c_{n,k}}{s_{n,j} - s_{n,k}}.$$

Then

$$a_{jk}^{\text{hom}} = \frac{d_k a_k - d_j a_j}{d_k - d_j}, \quad d_j(a_j - a^\#) - d_k(a_k - a^\#) = -(d_k - d_j)(a_{jk}^{\text{hom}} - a^\#). \quad (358)$$

Let $0 < R, \eta \leq 1$, and let $G = (V, E)$ be a connected graph of bush labels with at least one edge such that

$$|s_{n,j} - s_{n,k}| \geq \delta_s, \quad |a_{jk}^{\text{hom}} - a^\#| \leq \eta \quad (jk \in E). \quad (359)$$

If the graph diameter is at most L , there is a vector $v_G \in \mathbb{R}^3$ such that

$$\text{dist}\left(a_j, \left\{a^\# + \frac{v_G}{d} : d \in s_0 - J\right\}\right) \lesssim_{I,J} L\eta \quad (j \in V). \quad (360)$$

In particular, if all caps $B(a_j, R)$, $j \in V$, meet the bounded direction chart U , then

$$\sum_{j \in V} \sigma(H_{n,j} \cap B(a_j, R)) \lesssim_{I,J,U,\delta_s} (R + L\eta)^2. \quad (361)$$

Thus homothety-center clustering inside a common Reeb hairbrush is already a quantitative direction-rank-loss branch; it is not an additional unstructured fixed-angle remainder.

Proof. Since $c_{n,j} = y_0 - d_j a_j$ and $s_{n,j} = s_0 - d_j$, direct substitution gives the first identity in (358); rearranging gives the second. Hence the quantities

$$v_j := d_j(a_j - a^\#)$$

differ by $O_{I,J}(\eta)$ across every edge. Fix a root j_0 and put $v_G = v_{j_0}$. Summation along a path of length at most L , followed by division by $|d_j| \gtrsim_{I,J} 1$, proves (360).

The parameter set $s_0 - J$ is a compact interval separated from zero. Under the cluster hypotheses above, the fact that the caps meet U and that one edge has $|d_j - d_k| \geq \delta_s$ bounds $a^\#$ and v_G in terms of I, J, U, δ_s . Therefore the curve $d \mapsto a^\# + v_G/d$ has uniformly bounded length. Its $(R + L\eta)$ -neighborhood in $\mathbb{R}^3$ has volume $O_{I,J,U,\delta_s}((R + L\eta)^2)$. The sets $H_{n,j}$ are pairwise disjoint, so their cap masses sum to at most the normalized Lebesgue measure of that tube. This proves the claimed cluster cap-mass bound. $\square$

Corollary 8.21 (A heavy Reeb-hairbrush star cannot have one clustered homothety center). *Fix a root label j_0 and a set of leaves $\mathcal{N}$ satisfying*

$$|s_{n,j_0} - s_{n,k}| \geq \delta_s \quad (k \in \mathcal{N}).$$

If all $a_{j_0 k}^{\text{hom}}$, $k \in \mathcal{N}$, lie in one ball of radius η , then

$$\boxed{\sum_{k \in \mathcal{N} \cup \{j_0\}} \sigma(H_{n,k} \cap B(a_k, R)) \lesssim_{I,J,U,\delta_s} (R + \eta)^2.} \quad (362)$$

Consequently every star whose retained cap mass is larger than the right side contains two leaves whose distinguished homothety centers are η -separated. This is the geometric separated-center input for the next old flag-polarization comparison; no transverse flag conclusion is asserted before that comparison is made.

Proof. Apply Theorem 8.20 to the star graph, whose diameter is at most two. The final assertion is the contrapositive. $\square$

Proposition 8.22 (A coherent bad cap family forces exact local re-collision energy). *Keep the cap notation $A_{n,j}, S_n$ from Theorem 8.16. For all large n , choose an interval $W_n \subset I_{\text{out}}$ of length comparable to R_n , having s_n^{out} as an endpoint or midpoint, and define the localized ideal outer-wing pieces*

$$\mathcal{E}_{n,j} := \{(c_{n,j} + (u - s_{n,j})a, u) : a \in A_{n,j}, u \in W_n\}.$$

Then all $\mathcal{E}_{n,j}$ lie in one $O(R_n)$ -ball around (y_n, s_n^{out}) ,

$$|\mathcal{E}_{n,j}| \asymp_{I,J} |U| R_n \sigma(A_{n,j}),$$

and

$$\boxed{\sum_{j \neq k} \int_{W_n} |u - s_{n,j}|^3 \sigma(A_{n,j} \cap \Psi_{jk,u}^{-1}(A_{n,k})) du \gtrsim_{I,J} \frac{S_n^2}{R_n^2} - R_n S_n.} \quad (363)$$

In particular, the bad-cap inequality gives the explicit lower input

$$S_n > A_n q_n R_n^{3-\delta} \quad (364)$$

to the exact the preceding stage re-collision ledger. The first term on the right of (363) dominates the diagonal precisely when $S_n \gg R_n^3$, equivalently when the coherent hairbrush has genuine angular density above ambient cap volume.

Proof. For $a \in A_{n,j}$, the defining identity $c_{n,j} + (s_n^{\text{out}} - s_{n,j})a_{n,j} = y_n$ gives

$$|c_{n,j} + (u - s_{n,j})a - y_n| \lesssim_{I,J,U} R_n \quad (u \in W_n).$$

This proves the common-ball containment. The map

$$(a, u) \mapsto (c_{n,j} + (u - s_{n,j})a, u)$$

is injective and has Jacobian $|u - s_{n,j}|^3 \asymp_{I,J} 1$ on $A_{n,j} \times W_n$, proving the asserted volume formula. Cauchy-Schwarz inside the common four-dimensional $O(R_n)$ -ball yields

$$\sum_{j,k} |\mathcal{E}_{n,j} \cap \mathcal{E}_{n,k}| \gtrsim_{I,J} \frac{(R_n S_n)^2}{R_n^4} = \frac{S_n^2}{R_n^2}.$$

The diagonal is $\sum_j |\mathcal{E}_{n,j}| \lesssim_{I,J} |U| R_n S_n$. For $j \neq k$, the localized version of the exact pair-overlap calculation in Theorem 5.3 gives

$$|\mathcal{E}_{n,j} \cap \mathcal{E}_{n,k}| = |U| \int_{W_n} |u - s_{n,j}|^3 \sigma(A_{n,j} \cap \Psi_{jk,u}^{-1}(A_{n,k})) du.$$

Subtract the diagonal and divide by $|U|$. $\square$

Proposition 8.23 (Dense coherent hairbrush energy forces high Reeb contact-label reuse). *For $a \in A_{n,j}$ and $u \in W_n$, define the exact target-label multiplicity*

$$\mathbf{m}_{n,j}(a, u) := \sum_{k \neq j} \mathbf{1}_{A_{n,k}}(\Psi_{jk,u}(a)). \quad (365)$$

Every term counted by $\mathbf{m}_{n,j}(a, u)$ is a distinct bush label whose ideal Reeb front passes through the same point

$$(c_{n,j} + (u - s_{n,j})a, u).$$

Moreover the off-diagonal energy in (363) has the exact representation

$$\mathcal{R}_n := \sum_{j \neq k} \int_{W_n} |u - s_{n,j}|^3 \sigma(A_{n,j} \cap \Psi_{jk,u}^{-1}(A_{n,k})) du \asymp_{I,J} \sum_j \int_{W_n} \int_{A_{n,j}} \mathbf{m}_{n,j}(a, u) d\sigma(a) du. \quad (366)$$

For every $B \geq 1$, the contribution of $\{\mathbf{m}_{n,j} < B\}$ to the right side is at most

$$BR_n S_n. \quad (367)$$

Consequently, whenever $S_n/R_n^3 \rightarrow \infty$, a fixed portion of $\mathcal{R}_n$ is carried by

$$\mathbf{m}_{n,j}(a, u) \gtrsim_{I,J} \frac{S_n}{R_n^3} \gtrsim A_n q_n R_n^{-\delta}. \quad (368)$$

Thus the high-density half of Theorem 9.38 is an actual common Reeb contact-star reuse event, with all bush labels still present.

Proof. For fixed j, u , Tonelli gives

$$\int_{A_{n,j}} \mathbf{m}_{n,j}(a, u) d\sigma(a) = \sum_{k \neq j} \sigma(A_{n,j} \cap \Psi_{jk,u}^{-1}(A_{n,k})).$$

On the fixed outer wing, $|u - s_{n,j}|^3 \asymp_{I,J} 1$, proving (366). On the set where $\mathbf{m}_{n,j} < B$, delete all other restrictions and use $|W_n| \asymp R_n$ and $\sum_j \sigma(A_{n,j}) = S_n$; this gives (367).

If $S_n/R_n^3 \rightarrow \infty$, (363) gives $\mathcal{R}_n \gtrsim S_n^2/R_n^2$. Choose a sufficiently small fixed $c_{I,J} > 0$ and put $B = c_{I,J} S_n/R_n^3$. Then (367) is at most half of $\mathcal{R}_n$, so the complementary high-reuse set carries a fixed portion. Finally (364) gives the final comparison in (368) after decreasing the implicit constant. $\square$

Lemma 8.24 (Ideal re-collision energy is the fine product-collision limit). *For $u \in W_n$, write*

$$T_{n,j,u}(a) := c_{n,j} + (u - s_{n,j})a$$

and define

$$\mathcal{C}_{n,\rho}^{\text{id}} := \sum_{j \neq k} \int_{W_n} \iint_{A_{n,j} \times A_{n,k}} \mathbf{1}_{\{|T_{n,j,u}(a) - T_{n,k,u}(a')| \leq \rho\}} d\sigma(a) d\sigma(a') du.$$

For every fixed n ,

$$\lim_{\rho \downarrow 0} \rho^{-3} \mathcal{C}_{n,\rho}^{\text{id}} \asymp_{I,J,U} \mathcal{R}_n. \quad (369)$$

Moreover

$$|T_{n,j,u}(a) - T_{n,k,u}(a')| \leq \rho \implies |\Pi_u(z(a)) - \Pi_u(z(a'))| \leq \rho + 2\varepsilon_n. \quad (370)$$

Consequently, if a scale $\rho_n = o(R_n)$ can be chosen in the differentiation regime of (369) with $\varepsilon_n = o(\rho_n)$, then (457) holds with $\tilde{r}_n = 2\rho_n$. The only missing issue in that transfer is uniform separation between the bush-error scale and the direction-set differentiation scale.

Proof. Let $f_{n,j,u}$ be the density of $(T_{n,j,u})_{\#}(\sigma|_{A_{n,j}})$ with respect to three-dimensional Lebesgue measure. On the outer wing,

$$f_{n,j,u}(x) = \frac{1}{|U| |u - s_{n,j}|^3} \mathbf{1}_{T_{n,j,u}(A_{n,j})}(x).$$

If

$$\phi_\rho(x) := \frac{\mathbf{1}_{B(0,\rho)}(x)}{|B(0,\rho)|},$$

then the normalized (j, k, u) -summand of $\rho^{-3}\mathcal{C}_{n,\rho}^{\text{id}}$ is, up to the dimensional ball constant,

$$\int f_{n,j,u}(x)(f_{n,k,u} * \phi_\rho)(x) dx.$$

The approximate-identity theorem gives convergence in L^2 to $\int f_{n,j,u}f_{n,k,u}$. The L^2 norms are bounded on the outer wing by constants times $\sigma(A_{n,j})^{1/2}$ and $\sigma(A_{n,k})^{1/2}$, so dominated convergence applies in u ; the label family is finite for fixed n . Finally the exact change of variables from Theorem 5.3 identifies

$$\int f_{n,j,u}f_{n,k,u} \asymp_{I,J,U} \sigma\left(A_{n,j} \cap \Psi_{jk,u}^{-1}(A_{n,k})\right).$$

Summation proves (369). Each actual bush point differs from its ideal point by at most ε_n , so the triangle inequality proves (370) and the last assertion. $\square$

Proposition 8.25 (Finite-scale cell transfer: product collision, contact-star reuse, or a density budget). *Fix n , a scale*

$$\varepsilon_n \leq r \leq R_n,$$

and a half-open direction grid $\mathcal{Q}_r$ of side r . Put

$$w_{n,k,Q} := \sigma(A_{n,k} \cap Q),$$

and call (k, Q) ω -heavy when

$$w_{n,k,Q} \geq \omega r^3, \quad 0 < \omega \leq C_U.$$

Define the reverse exact label multiplicity

$$\mathbf{m}_{n,k}^{\text{rev}}(a', u) := \sum_{j \neq k} \mathbf{1}_{A_{n,j}}(\Psi_{kj,u}(a')). \quad (371)$$

For every $B \geq 1$, at least one of the following holds:

(i) *a fixed portion of $\mathcal{R}_n$ lands in ω -heavy target cells, and the actual finite-thickness product collision obeys*

$$\boxed{\begin{aligned} \sum_{j \neq k} \int_{W_n} \iint_{A_{n,j} \times A_{n,k}} \mathbf{1}_{\{|\Pi_u(z(a)) - \Pi_u(z(a'))| \leq C_{I,J,U} r\}} d\sigma(a) d\sigma(a') du \\ \gtrsim_{I,J,U} \omega r^3 \mathcal{R}_n; \end{aligned}} \quad (372)$$

(ii) *a fixed portion of $\mathcal{R}_n$ lands where*

$$\boxed{\mathbf{m}_{n,k}^{\text{rev}}(a', u) \geq B}, \quad (373)$$

which is a common ideal Reeb contact star rooted at the target endpoint;

(iii) *the complete ideal energy satisfies*

$$\boxed{\mathcal{R}_n \lesssim_{I,J,U} B N_n \omega R_n^4}. \quad (374)$$

No limiting differentiation scale is used in this trichotomy.

Proof. For a target cell Q , set

$$E_{n,jk,Q}(u) := \{a \in A_{n,j} : \Psi_{jk,u}(a) \in A_{n,k} \cap Q\}.$$

If $a \in E_{n,jk,Q}(u)$ and $a' \in A_{n,k} \cap Q$, then the ideal source point equals the ideal target point generated by $\Psi_{jk,u}(a)$, while

$$|T_{n,k,u}(a') - T_{n,k,u}(\Psi_{jk,u}(a))| \lesssim_{I,J,U} r.$$

The two actual points introduce at most $2\varepsilon_n \leq 2r$ additional error. Hence the left side of (372) is bounded below by

$$\sum_{j \neq k} \sum_{Q \in \mathcal{Q}_r} \int_{W_n} w_{n,k,Q} \sigma(E_{n,jk,Q}(u)) du. \quad (375)$$

On the heavy cells, $w_{n,k,Q} \geq \omega r^3$. Since the outer-wing Jacobians are bounded above and below, a fixed portion of $\mathcal{R}_n$ in those cells proves (372).

It remains to estimate the part landing in light target cells and satisfying $\mathbf{m}_{n,k}^{\text{rev}} < B$. Change variables $a' = \Psi_{jk,u}(a)$. All Jacobians are uniformly comparable on the outer wing, so this energy is bounded by a constant times

$$\sum_k \int_{W_n} \int_{\substack{a' \in A_{n,k} \cap Q \\ Q \text{ light}}} \mathbf{m}_{n,k}^{\text{rev}}(a', u) d\sigma(a') du.$$

Each $A_{n,k}$ lies in a direction ball of radius R_n , which meets $O_U((R_n/r)^3)$ grid cells. Therefore its total mass in light cells is at most

$$C_U (R_n/r)^3 \omega r^3 \lesssim_U \omega R_n^3.$$

Using $\mathbf{m}_{n,k}^{\text{rev}} < B$, summing the N_n target labels, and integrating over $|W_n| \asymp R_n$ gives $C_{I,J,U} B N_n \omega R_n^4$. If neither of the first two alternatives carries a fixed portion of $\mathcal{R}_n$, this proves (374). $\square$

8.4 Fine Maslov returns and stopping leaves

Corollary 8.26 (The balanced finite-cell choice removes the unknown differentiation threshold). *Assume S_n/R_n^3 is larger than a sufficiently large constant and set*

$$B_n := c_{I,J} \frac{S_n}{R_n^3}, \quad \omega_n := c_{I,J,U} \frac{S_n}{N_n R_n^3},$$

with the constants chosen sufficiently small. Then alternative (iii) of Theorem 8.25 is impossible. Consequently, at every scale $\varepsilon_n \leq r \leq R_n$, either

$$\boxed{C_{n,r}^{\text{phys}} \gtrsim_{I,J,U} \frac{S_n}{N_n R_n^3} r^3 \mathcal{R}_n \gtrsim_{I,J,U} \frac{S_n^3}{N_n R_n^5} r^3,} \quad (376)$$

where $C_{n,r}^{\text{phys}}$ denotes the physical product-collision expression in (372), or a fixed portion of $\mathcal{R}_n$ is carried by reverse contact stars of label degree

$$\boxed{\mathbf{m}_{n,k}^{\text{rev}} \gtrsim_{I,J} \frac{S_n}{R_n^3}.} \quad (377)$$

Moreover the Frostman-failure lower bound and maximal-family estimates give

$$\boxed{\omega_n \gtrsim_{I,J,U,G_*} A_n M_n^4 R_n^{-\delta}.} \quad (378)$$

Thus the finite-scale transfer loses exactly the average relative density of one original labeled cap; if that density is too small, the complementary output is high Reeb contact-star reuse.

Proof. By (363) and $S_n/R_n^3 \gg 1$,

$$\mathcal{R}_n \gtrsim_{I,J} \frac{S_n^2}{R_n^2}.$$

With the displayed choices,

$$B_n N_n \omega_n R_n^4 = c_{I,J,U}^2 \frac{S_n^2}{R_n^2}.$$

Choose the product of the constants below the implicit constant in the preceding energy lower bound. Then (374) cannot hold. The two remaining alternatives give (376) and (377). Finally

$$\frac{S_n}{N_n R_n^3} > A_n \frac{q_n}{N_n} R_n^{-\delta} \gtrsim_{G_*} A_n M_n^4 R_n^{-\delta}$$

by (364), $q_n \gtrsim_{G_*} M_n$, and $N_n \lesssim M_n^{-3}$. $\square$

Proposition 8.27 (The fixed-angle physical branch is a normalized fine Maslov graph). *Let*

$$d\hat{\mu}_n^{\text{hair}}(z(a)) := S_n^{-1} \mathbf{1}_{H_n^{\text{hair}}}(a) d\sigma(a),$$

and fix $\tau_0 > 0$. Suppose that, for some $\varepsilon_n \leq r \leq R_n$, a fixed portion of the physical product collision in (372) is carried by pairs satisfying $|a - a'| \geq \tau_0$. Define

$$h_{n,r}(z(a), z(a')) := \mathbf{1}_{\{|a-a'| \geq \tau_0\}} \frac{1}{C_{\tau_0} r} \int_{W_n} \mathbf{1}_{\{|\Pi_u(z(a)) - \Pi_u(z(a'))| \leq Cr\}} du, \quad (379)$$

where C_{τ_0} is chosen sufficiently large. Then $0 \leq h_{n,r} \leq 1$, every edge of this graph is an $O(r)$ -thickened Maslov incidence, and its normalized edge mass satisfies

$$\widetilde{M}_{n,r} := \iint h_{n,r}(z, z') d\widehat{\mu}_n^{\text{hair}}(z) d\widehat{\mu}_n^{\text{hair}}(z') \gtrsim_{\tau_0, I, J, U} \omega r^2 \frac{\mathcal{R}_n}{S_n^2}. \quad (380)$$

In the dense regime $S_n/R_n^3 \gg 1$, this gives

$$\widetilde{M}_{n,r} \gtrsim_{\tau_0, I, J, U} \omega \left(\frac{r}{R_n} \right)^2. \quad (381)$$

For the balanced choice of Theorem 8.26,

$$\widetilde{M}_{n,r} \gtrsim_{\tau_0, I, J, U, G_*} A_n M_n^4 R_n^{-\delta} \left(\frac{r}{R_n} \right)^2. \quad (382)$$

The conditional old Maslov flags on the original occurrences lift to this graph without additional mass loss. If the required fixed-angle portion is absent, the discarded collision mass is instead concentrated in the τ_0 -Darboux branch.

Proof. For fixed a, a' , the difference

$$\Pi_u(z(a)) - \Pi_u(z(a'))$$

is affine in u , with u -slope $a - a'$. Hence, on $|a - a'| \geq \tau_0$, the set of $u \in W_n$ for which its norm is at most Cr has length $O_{\tau_0}(r)$. Enlarging C_{τ_0} therefore gives $h_{n,r} \leq 1$.

The collision relation itself is

$$(b(a) - b(a')) + u(a - a') = O(r),$$

so each edge lies in the $O(r)$ -neighborhood of the Maslov plane L_u . After restricting to the assumed fixed-angle portion, dividing the physical product-collision lower bound in Theorem 8.25 by the endpoint normalization S_n^2 and the time-window normalization $C_{\tau_0}r$ gives

$$\widetilde{M}_{n,r} \gtrsim_{\tau_0, I, J, U} \frac{\omega r^3 \mathcal{R}_n}{r S_n^2},$$

which is the asserted fine Maslov edge-mass bound. The local-energy bound $\mathcal{R}_n \gtrsim S_n^2/R_n^2$ proves its scale form. The average-cell-density estimate then proves the balanced fine-edge form. Finally, the finite-cell product consists of actual selector endpoints $z(a), z(a')$; the second endpoint need not equal the exact transported endpoint $z(\Psi_{jk,u}(a))$. Nevertheless each endpoint carries its own conditional probability law $\kappa_{z(a)}$ or $\kappa_{z(a')}$. Integrating any fixed finite tensor product of those laws gives 1, by Theorem 8.3. Preservation on the hairbrush energy is Theorem 9.37. Therefore decorating the physical edge measure by the old flags costs no additional mass factor. $\square$

Corollary 8.28 (At the bush scale the fixed-angle Maslov return has constant edge mass). *Fix $\tau_0 > 0$. If a fixed portion of the coarse physical collision from Theorem 8.17 is carried by $|a - a'| \geq \tau_0$, then*

$$\iint h_{n,R_n}^{\text{coarse}}(z, z') d\widehat{\mu}_n^{\text{hair}}(z) d\widehat{\mu}_n^{\text{hair}}(z') \gtrsim_{\tau_0, I, J, U} 1, \quad (383)$$

where

$$h_{n,R_n}^{\text{coarse}}(z(a), z(a')) := \mathbf{1}_{\{|a-a'| \geq \tau_0\}} \frac{1}{C_{\tau_0} R_n} \int_{W'_n} \mathbf{1}_{\{|\Pi_u(z(a)) - \Pi_u(z(a'))| \leq C_{I,J,U} R_n\}} du$$

and $0 \leq h_{n,R_n}^{\text{coarse}} \leq 1$. Hence the density factor ω_n is a strict fine-scale transfer cost; it is absent at the intrinsic bush scale. If the fixed-angle portion is absent, a fixed portion of the coarse collision lies in the τ_0 -Darboux branch.

Proof. On $|a - a'| \geq \tau_0$, the collision-time window has length $O_{\tau_0}(R_n)$, so the kernel is bounded by 1. Divide the fixed-angle part of (353) by $R_n S_n^2$ and use $\sigma(H_n^{\text{hair}}) = S_n$. $\square$

Corollary 8.29 (The exact two-scale scalar frontier after the Maslov return). *Assume the fixed-angle branch of Theorem 8.27, use the balanced cell density ω_n , and suppose that the resulting merged bush has a further*

bad direction cap of radius $\tilde{R}_n$, excess $\tilde{A}_n$, and normalized density constant $\tilde{D}_{n,\text{cap}}$ from Theorem 8.19. Then the quotient of the next cap-density tax by the available normalized Maslov edge mass obeys

$$\boxed{\frac{\tilde{D}_{n,\text{cap}}}{\tilde{M}_{n,r}} \lesssim_{\tau_0, I, J, U, G_*} \frac{\tilde{R}_n^\delta R_n^\delta}{\tilde{A}_n S_n A_n M_n^4} \left(\frac{R_n}{r}\right)^2.} \quad (384)$$

At the intrinsic bush scale, if the fixed-angle alternative of Theorem 8.28 holds, the sharper quotient is

$$\boxed{\frac{\tilde{D}_{n,\text{cap}}}{\tilde{M}_{n,R_n}^{\text{coarse}}} \lesssim_{\tau_0, I, J, U} \frac{\tilde{R}_n^\delta}{\tilde{A}_n S_n}.} \quad (385)$$

Thus the finite-cell and cap-normalization steps introduce no hidden factor: their complete two-scale loss is the right side of (384). The already proved estimates make this loss subpower whenever that displayed right side is $r^{-o(1)}$. Without a scale-coherent relation among $r, R_n, \tilde{R}_n, A_n, \tilde{A}_n, M_n, S_n$, however, (384) alone does not imply such absorption.

Proof. Divide (356) by (381) to obtain

$$\frac{\tilde{D}_{n,\text{cap}}}{\tilde{M}_{n,r}} \lesssim \frac{\tilde{R}_n^\delta}{\tilde{A}_n S_n \omega_n} \left(\frac{R_n}{r}\right)^2.$$

Now use the average-cell-density estimate, equivalently $\omega_n^{-1} \lesssim A_n^{-1} M_n^{-4} R_n^\delta$. For the coarse two-scale frontier, divide the merged-bush cap-density bound by the coarse Maslov edge bound. $\square$

Proposition 8.30 (Intrinsic-scale Maslov renormalization preserves a fixed mass fraction). *Assume the fixed-angle alternative of Theorem 8.28, and suppose the close-anchor, horizontal-line, and transverse old/new-polarization outputs are routed through their already assigned ledgers. Fix $d \in (0, 1)$. Then, after passing to a subsequence, either one of those paid outputs occurs, or there are pairwise disjoint sets*

$$H_{n,1}^{(1)}, \dots, H_{n,N_n^{(1)}}^{(1)} \subset H_n^{\text{hair}}$$

such that

$$\boxed{N_n^{(1)} \lesssim_{\tau_0, I, J, U, d} 1, \quad \sum_{\ell=1}^{N_n^{(1)}} \sigma(H_{n,\ell}^{(1)}) \gtrsim_{\tau_0, I, J, U, d} S_n,} \quad (386)$$

and every $H_{n,\ell}^{(1)}$ is an $O(R_n^{1-d})$ -Lagrangian point-probe bush. All conditional old-flag laws are retained.

Consequently, for every fixed $0 < \delta < 3$, either this new family satisfies the vector Frostman hypothesis of Theorem 8.11, with the roles of the two separated Reeb windows J and I_{out} interchanged, along a subsequence and $\dim_H K_b \geq 4 - \delta$, or it has a further Reeb-coherent hairbrush concentration at radii

$$R_n^{1-d} \lesssim \hat{R}_n \downarrow 0.$$

The latter concentration again merges into one physical bush by Theorem 8.16. This entire intrinsic-scale return uses neither ω_n^{-1} nor the cap-density factor $\tilde{D}_{n,\text{cap}}$.

Proof. Let $h_n = h_{n,R_n}^{\text{coarse}}$ and use the probability measure $\hat{\mu}_n^{\text{hair}}$. By (383), its edge mass M_n^{hair} is bounded below by a fixed positive constant. In particular

$$M_n^{\text{hair}} = R_n^{o(1)},$$

so it lies in the exponent range $\chi = 0 < 2$ of Theorem 8.5. Apply the graph-theoretic maximal extraction proof of Theorem 8.8 with collision scale R_n and threshold $\Delta_n = R_n^d$. Unless one of the stated paid ledgers occurs, it gives

$$N_n^{(1)} \lesssim (M_n^{\text{hair}})^{-3} \lesssim 1, \quad \sum_{\ell} \hat{\mu}_n^{\text{hair}}(H_{n,\ell}^{(1)}) \gtrsim M_n^{\text{hair}} \gtrsim 1,$$

and bush error $O(R_n/\Delta_n) = O(R_n^{1-d})$. Multiplying the last mass inequality by S_n proves (386). The lossless conditional tensor identity Theorem 8.3 retains the old flags.

The new bush centers lie in the compact window containing $W'_n \subset I_{\text{out}}$. Apply Theorem 8.11 after relabeling that window as the center window and choosing a compact subinterval of the original J as the next outer window. Their separation is bounded below by a fixed positive constant. If the vector Frostman hypothesis fails, the proof of Theorem 8.12 supplies $\hat{R}_n \geq cR_n^{1-d}$ tending to zero and a common outer-point cap family whose outer level is back in J . Theorem 8.16 merges it. No direction cap is normalized before the coarse Maslov graph is formed, which is why neither of the two displayed density factors appears. $\square$

Corollary 8.31 (One alternating return forces a two-probe angular lock). *Keep the non-Frostman alternative in Theorem 8.30. Let*

$$(y_n^{(0)}, s_n^{(0)}) \in \mathbb{R}^3 \times I_{\text{out}}$$

be the center of the incoming merged bush H_n^{hair} , and let

$$(y_n^{(1)}, s_n^{(1)}) \in \mathbb{R}^3 \times J$$

be the common outer point which centers the returned merged hairbrush $\widehat{H}_n^{\text{hair}} \subset H_n^{\text{hair}}$. Then

$$\widehat{H}_n^{\text{hair}} \subset B\left(\frac{y_n^{(1)} - y_n^{(0)}}{s_n^{(1)} - s_n^{(0)}}, C_{I,J}(R_n + \widehat{R}_n)\right) \subset B(a_n^\#, C_{I,J,d}\widehat{R}_n). \quad (387)$$

Consequently, for every fixed $\tau_0 > 0$, all pairs in $\widehat{H}_n^{\text{hair}} \times \widehat{H}_n^{\text{hair}}$ lie in the τ_0 -angular Darboux branch for large n . In particular an indefinitely repeated fixed-angle hairbrush-renormalization chain is impossible: after one alternating-window return, the next geometric destination is necessarily an angular Darboux chart.

More quantitatively, if a blow-up at direction scale $T_n \downarrow 0$ has a noncollapsed horizontal limit, then

$$T_n \lesssim_{I,J} R_n + \widehat{R}_n \asymp_{I,J,d} \widehat{R}_n. \quad (388)$$

Thus the alternating return cannot hide a third, substantially larger noncollapsed angular scale. In particular there is no noncollapsed angular blow-up for which the inherited physical error is $o(T_n)$; the remaining cap is at the physical resolution scale and cannot be credited with the error-free tangent Maslov relation without a further weighted coarse/fine argument.

Proof. Every $a \in \widehat{H}_n^{\text{hair}} \subset H_n^{\text{hair}}$ obeys the inherited point-probe relation

$$|b(a) + s_n^{(0)}a - y_n^{(0)}| \lesssim_{I,J} R_n.$$

The second hairbrush merge gives

$$|b(a) + s_n^{(1)}a - y_n^{(1)}| \lesssim_{I,J} \widehat{R}_n.$$

Subtracting and using $\text{dist}(J, I_{\text{out}}) > 0$ proves the first containment in (387). Since $\widehat{R}_n \gtrsim R_n^{1-d} \geq R_n$, the second containment follows. Its diameter tends to zero, which proves the fixed-angle assertion.

For the last claim, write $a = a_n + T_n\xi$ on a putative noncollapsed blow-up. After subtracting the two point-probe equations and dividing by T_n , the variation in ξ is $O((R_n + \widehat{R}_n)/T_n)$. If $(R_n + \widehat{R}_n)/T_n \rightarrow 0$, every limiting horizontal secant vanishes, contradicting noncollapse. This proves (388). $\square$

Proposition 8.32 (A resolution-scale angular cap is either a good leaf or a tax-free fine return). *Let $H \subset G_*$ be any fixed measurable selector set of mass*

$$p := \sigma(H) > 0$$

contained in one direction ball of radius $0 < T < 1$, and retain at every point of H the conditional old-flag laws inherited from G_ . Then exactly one of the following applies.*

(i) *the relative residual estimate*

$$3_{Cr, \geq r}^{J^+}(\sigma|_H) \lesssim p^2 r^{2-o(1)}$$

holds uniformly for all sufficiently small r , so H is a good terminal leaf in the fixed-coarse/all-fine decomposition;

(ii) *there is a sequence of failure scales $r_k \downarrow 0$ satisfying*

$$r_k \leq \min\{T^k, p^k\}, \quad p = r_k^{o(1)}, \quad T = r_k^{o(1)}, \quad \frac{T^3}{p} = r_k^{-o(1)}. \quad (389)$$

After translating the cap and dilating directions by T^{-1} , every density or mass normalization introduced by this fixed cap is therefore subpower at the new collision scale. The normalized contact–Maslov graph, the conditional flag amplification, and the subsequent bush/rank-loss routing may be applied on H without a fixed power loss.

In particular the resolution-scale cap from Theorem 8.31 is not itself a terminal normalization obstruction. What remains after this proposition is the aggregation of the resulting nested good-leaf/fine-return stopping tree, not an unpaid scalar density factor at one fixed node.

Proof. If (i) fails, there are arbitrarily small failure scales. Choose the k -th one below $\min\{T^k, p^k\}$. Then

$$\frac{|\log p|}{|\log r_k|} \leq \frac{1}{k}, \quad \frac{|\log T|}{|\log r_k|} \leq \frac{1}{k},$$

which proves (389). The rescaled normalized direction measure has density at most $C_U T^3/p$. Hence this density, the restriction normalization p^{-1} , and every fixed old-flag layer constant are $r_k^{-o(1)}$, exactly as in Theorems 7.74 and 9.47. Restriction changes which physical endpoints are retained but not their conditional flag laws, by Theorem 8.3. Thus all hypotheses used by the fine contact–Maslov return survive with only subpower constants. $\square$

Lemma 8.33 (Quadratic aggregation of terminal stopping leaves). *For one of the finitely many shifted physical r -grids, define*

$$\mathcal{E}_r^J(\lambda) := \int_J \sum_{Q \in \mathcal{Q}_r} ((\Pi_s)_\# \lambda(Q))^2 ds.$$

Let λ have mass $m > 0$. Suppose that for every sufficiently small r there is a decomposition

$$\lambda = \sum_{\gamma \in \mathcal{L}(r)} \lambda_\gamma + \lambda_{\text{tail}}^{(r)}$$

into positive measures, where $\mathcal{L}(r)$ is finite, such that

$$\boxed{\begin{aligned} \mathcal{E}_r^J(\lambda_\gamma) &\leq C(r) \|\lambda_\gamma\|^2 r^{3-o(1)} & (\gamma \in \mathcal{L}(r)), \\ C(r) &= r^{-o(1)}, \\ \|\lambda_{\text{tail}}^{(r)}\| &\leq m r^{3/2-o(1)}. \end{aligned}} \tag{390}$$

Then

$$\boxed{\mathcal{E}_r^J(\lambda) \lesssim m^2 r^{3-o(1)}} \tag{391}$$

If (390) holds for the finite family of shifted grids needed to dominate physical Cr -balls, then the full collision estimate (470), and hence the relative residual target (303), follows.

Proof. For each s, Q , write

$$F_\lambda(s, Q) := (\Pi_s)_\# \lambda(Q).$$

Then $\mathcal{E}_r^J(\lambda) = \|F_\lambda\|_{L^2(ds \times \#Q)}^2$. Minkowski and positivity give

$$\begin{aligned} \mathcal{E}_r^J(\lambda)^{1/2} &\leq \sum_{\gamma \in \mathcal{L}(r)} \mathcal{E}_r^J(\lambda_\gamma)^{1/2} + \mathcal{E}_r^J(\lambda_{\text{tail}}^{(r)})^{1/2} \\ &\lesssim C(r)^{1/2} r^{3/2-o(1)} \sum_{\gamma} \|\lambda_\gamma\| + |J|^{1/2} \|\lambda_{\text{tail}}^{(r)}\| \\ &\lesssim m r^{3/2-o(1)}. \end{aligned}$$

Squaring proves (391). A fixed finite shifted-grid covering dominates every pair with $|\Pi_s(z) - \Pi_s(z')| \leq Cr$ by a same-cube pair at a comparable scale. Summing those grid energies and applying Theorem 7.1 gives the final assertion. $\square$

Corollary 8.34 (The remaining stopping-tree condition is a quantitative tail estimate). *Build a nested stopping tree by applying Theorem 8.32 at every resolution-scale cap. Suppose that at depth L the nonterminal nodes are pairwise disjoint and have total mass $t_L \downarrow 0$, while the finitely many terminal leaves above depth L obey their relative residual estimates. Let $C_L < \infty$ dominate both their residual constants and the reciprocals of their positive masses. By the global bounded-direction-density estimate for $|a - a'| < r$, each such leaf then obeys the full shifted-grid estimate in (390) with constant $O(C_L)$. If there is a depth choice $L = L(r) \rightarrow \infty$ such that*

$$\boxed{t_{L(r)} \leq m r^{3/2-o(1)}, \quad C_{L(r)} = r^{-o(1)},} \tag{392}$$

then the Maslov-bush-to-Frostman bridge Theorem 9.42 closes.

Proof. Take the terminal leaves above depth $L(r)$ as $\mathcal{L}(r)$ and the sum of the nonterminal nodes at that depth as $\lambda_{\text{tail}}^{(r)}$. Their supports form a measurable partition of the root measure up to null sets. For a terminal leaf of mass m_γ , the collision/residual comparison gives the required fixed-angle grid bound, while its near-direction contribution is $O(m_\gamma r^3) \leq O(C_L m_\gamma^2 r^3)$. Apply Theorem 8.33. $\square$

Corollary 8.35 (Absence of a good leaf forces arbitrarily deep alternating endpoint-star chains). *Assume $\dim_H K_b < 4$. Then no positive-direction-mass stopping node can terminate in alternative (i) of Theorem 8.32: after normalization, its relative residual estimate would close by Theorem 6.27. Consequently, unless one of the already paid geometric ledgers occurs, for every integer L there are nested positive-mass selector sets*

$$H^{(0)} \supset H^{(1)} \supset \dots \supset H^{(L)}$$

and direction radii

$$T_0 > T_1 > \dots > T_L > 0, \quad T_{\ell+1} \leq \frac{1}{2} T_\ell,$$

such that every transition retains the conditional old flags and supplies two point-probe centers in the alternating separated windows J and I_{out} . Each returned $H^{(\ell+1)}$ is contained in one $O(T_{\ell+1})$ -direction cap and carries the physical endpoint-star witnesses of the fine Maslov graph which produced it.

Thus the geometric data required by the multiscale endpoint-star side of the optional Maslov secant-to-curve promotion are present to arbitrary finite depth. This statement does not by itself synchronize the continuous center, angular-chart, and polarization labels from one depth to the next. Theorem 8.36 below removes mere scalar scale sparsity by thickening every power-excess failure to a positive logarithmic block. The missing alternative to (392) is therefore coherent selection of the physical endpoint-star data across those blocks, not the existence of enough bad thickness parameters.

Proof. A positive-mass good node, normalized to a probability measure, satisfies the codimension-two residual hypothesis and therefore gives a four-dimensional front subset. Under the contrary dimension assumption, every positive-mass node must have arbitrarily fine failure scales. Apply Theorem 8.32 and then the bush-family/hairbrush route recorded in Theorems 8.30 and 8.31. The returned cap radius tends to zero along the chosen failure sequence, so, after going sufficiently far out, it is at most half the parent radius. Restriction preserves positive mass and the conditional flag laws. Induction gives every finite L . The last paragraph is the scale-density hypothesis explicitly required in the optional Maslov secant-to-curve promotion; finite-depth selection alone does not synchronize the continuous geometric labels. $\square$

Lemma 8.36 (A power-excess residual scale fills a positive logarithmic block). *Let H have mass $p > 0$, fix $0 < \eta < 2$, and suppose*

$$\mathfrak{Z}_{r_n, \geq \tau_0}^{J^+}(\sigma|_H) \geq A_n p^2 r_n^{2-\eta}, \quad A_n \rightarrow \infty. \quad (393)$$

Put

$$\alpha_\eta := \frac{2-\eta}{2-\eta/2} \in (0, 1).$$

Then, for every

$$r_n \leq R \leq r_n^{\alpha_\eta},$$

one has

$$\boxed{\mathfrak{Z}_{R, \geq \tau_0}^{J^+}(\sigma|_H) \geq A_n p^2 R^{2-\eta/2}.} \quad (394)$$

The interval of logarithmic scales has relative length

$$\boxed{\frac{\log(r_n^{\alpha_\eta}/r_n)}{\log(1/r_n)} = 1 - \alpha_\eta = \frac{\eta}{4-\eta} > 0.} \quad (395)$$

Hence a Hausdorff counterexample cannot force only isolated residual-failure scales. At every nonterminal positive-mass node it supplies positive-relative-length logarithmic blocks of normalized Maslov graphs, with exponent still strictly below two.

Proof. The residual is monotone in its thickness:

$$\mathfrak{Z}_{R, \geq \tau_0}^{J^+}(\sigma|_H) \geq \mathfrak{Z}_{r_n, \geq \tau_0}^{J^+}(\sigma|_H) \quad (R \geq r_n).$$

Write $R = r_n^\beta$, where $\alpha_\eta \leq \beta \leq 1$. The definition of α_η gives

$$\beta(2-\eta/2) \geq 2-\eta.$$

Since $0 < r_n < 1$,

$$r_n^{2-\eta} \geq r_n^{\beta(2-\eta/2)} = R^{2-\eta/2}.$$

Insert this in (393). The logarithmic-length calculation is immediate. $\square$

8.5 Coherence across logarithmic blocks

Lemma 8.37 (Finite-dimensional geometric labels synchronize with only subpower loss). *Let $\mathcal{Y}$ be the compact label space consisting of the two bounded physical point-probe centers in J and I_{out} , one of the finitely many angular charts, the old/new polarization planes in the relevant compact Grassmannians, and the angular-scale exponent*

$$\varkappa_R := \frac{\log(1/\tau_R)}{\log(1/R)} \in [0, 1],$$

where $R \leq \tau_R \leq 1$ is the selected direction-shell scale. There are constants $D_{\mathcal{Y}}, C_{\mathcal{Y}} < \infty$ such that $\mathcal{Y}$ is covered by at most

$$C_{\mathcal{Y}} \zeta^{-D_{\mathcal{Y}}}$$

balls of radius ζ .

On the logarithmic block from Theorem 8.36, make any measurable choice $R \mapsto \mathfrak{y}_R \in \mathcal{Y}$ of the geometric output of the flagged Maslov routing. Put $L_n = \log(1/r_n)$ and choose

$$\zeta_n := \exp(-\sqrt{L_n}) = r_n^{o(1)}.$$

Then there are a label ball $B_{\mathcal{Y}}(\mathfrak{y}_n, \zeta_n)$ and a measurable subset $\mathcal{S}_n \subset [r_n, r_n^{\alpha_\eta}]$ such that

$$\boxed{\int_{\mathcal{S}_n} \frac{dR}{R} \gtrsim_{\eta, \mathcal{Y}} \zeta_n^{D_{\mathcal{Y}}} L_n = r_n^{o(1)} L_n, \quad \mathfrak{y}_R \in B_{\mathcal{Y}}(\mathfrak{y}_n, \zeta_n) \quad (R \in \mathcal{S}_n).} \quad (396)$$

After a subsequence, $\mathfrak{y}_n \rightarrow \mathfrak{y}_\infty$, so all retained centers, angular charts, and polarization planes converge on $\mathcal{S}_n$. The loss in (396) is subpower and therefore does not consume the fixed gap between the residual exponent $2 - \eta/2$ and the endpoint exponent 2.

Proof. Cover $\mathcal{Y}$ by the stated number of ζ_n -balls and partition the logarithmic block according to the first ball containing $\mathfrak{y}_R$. Its dR/R -length is $(1 - \alpha_\eta)L_n$. One cell therefore has measure at least

$$\frac{(1 - \alpha_\eta)L_n}{C_{\mathcal{Y}} \zeta_n^{-D_{\mathcal{Y}}}},$$

which is (396). Compactness gives the convergent subsequence. Finally

$$\zeta_n^{D_{\mathcal{Y}}} = \exp(-D_{\mathcal{Y}} \sqrt{L_n}) = r_n^{D_{\mathcal{Y}}/\sqrt{L_n}} = r_n^{o(1)},$$

whereas the exponent gain $\eta/2$ is fixed. $\square$

Proposition 8.38 (Positive-density coherent label or separated label variation). *Start from the logarithmic block*

$$\mathcal{B}_n^0 = [r_n, r_n^{\alpha_\eta}], \quad |\mathcal{B}_n^0|_{\log} := \int_{\mathcal{B}_n^0} \frac{dR}{R}.$$

First restrict to one of the finitely many angular charts on a measurable scale subset $\mathcal{B}_n \subset \mathcal{B}_n^0$ satisfying

$$|\mathcal{B}_n|_{\log} \gtrsim |\mathcal{B}_n^0|_{\log}.$$

Let $R \mapsto \mathfrak{y}_R \in \mathcal{Y}$ be the remaining continuous geometric-label choice from Theorem 8.37, and define the label probability

$$\varpi_n := \frac{1}{|\mathcal{B}_n|_{\log}} (\mathfrak{y}_\bullet)_\# \left(\mathbf{1}_{\mathcal{B}_n} \frac{dR}{R} \right).$$

After passing to a subsequence, $\varpi_n \rightarrow \varpi$. According as ϖ has an atom or is non-atomic, the corresponding one of the following two alternatives is available.

(i) There are $\mathfrak{y}_\infty \in \mathcal{Y}$, $\theta > 0$, radii $\zeta_n \downarrow 0$, and scale sets $\mathcal{S}_n \subset \mathcal{B}_n$ such that

$$\boxed{\int_{\mathcal{S}_n} \frac{dR}{R} \geq \theta |\mathcal{B}_n|_{\log}, \quad d_{\mathcal{Y}}(\mathfrak{y}_R, \mathfrak{y}_\infty) \leq \zeta_n \quad (R \in \mathcal{S}_n).} \quad (397)$$

Thus the center/polarization labels converge, in one fixed angular chart, on a fixed positive proportion of every retained logarithmic block.

(ii) There are compact sets $Y_1, Y_2 \subset \mathcal{Y}$ and constants $\theta, c > 0$ such that

$$\boxed{\text{dist}(Y_1, Y_2) \geq c, \quad \int_{\{R \in \mathcal{B}_n : \mathfrak{y}_R \in Y_i\}} \frac{dR}{R} \geq \theta |\mathcal{B}_n|_{\log} \quad (i = 1, 2)} \quad (398)$$

for all large n . Hence a fixed positive proportion of scales carries genuinely separated physical-center or polarization data.

The first alternative supplies the positive-density finite-dimensional label coherence required on the endpoint-star side of the optional Maslov secant-to-curve promotion. In the second, the corresponding implication is the variation-to-four-cluster projection alternative stated there; it is no longer hidden inside a scale or entropy loss.

Proof. Compactness of $\mathcal{Y}$ gives a weakly convergent subsequence. If ϖ has an atom $\mathfrak{y}_\infty$ of mass $2\theta > 0$, choose a decreasing sequence of continuity radii $\zeta^{(k)} \downarrow 0$ for the balls about $\mathfrak{y}_\infty$. Weak convergence gives

$$\varpi_n(B(\mathfrak{y}_\infty, \zeta^{(k)})) \geq \theta$$

for all n sufficiently large depending on k . A diagonal choice $k = k(n) \rightarrow \infty$ proves (397).

If ϖ has no atoms, its support contains two distinct points. Choose disjoint compact neighborhoods Y_1, Y_2 , separated by some $c > 0$, whose ϖ -masses are both larger than $2\theta > 0$ and whose boundaries have zero ϖ -mass. Weak convergence then gives $\varpi_n(Y_i) \geq \theta$ for $i = 1, 2$, proving (398). $\square$

Corollary 8.39 (The positive-density dichotomy retains the physical endpoint-star measures). *Let $\mathcal{X}$ be the fixed compact localization of the finite physical endpoint/flag tuples used by one routed endpoint star, and let $\Gamma_R \in \mathcal{P}(\mathcal{X})$ be its normalized retained occurrence measure at scale R . Replace the finite-dimensional label space in Theorem 8.38 by*

$$\mathcal{Y}^\sharp := \mathcal{Y} \times \mathcal{P}(\mathcal{X}),$$

where $\mathcal{P}(\mathcal{X})$ has any metric inducing weak convergence. Then the same atomic/non-atomic conclusion holds.

In the atomic alternative, on a fixed positive proportion of each logarithmic block,

$$\boxed{\mathfrak{y}_R \longrightarrow \mathfrak{y}_\infty, \quad \Gamma_R \rightharpoonup \Gamma_\infty,} \quad (399)$$

uniformly in the selected scale sets in the diagonal sense of (397). Thus the actual physical edge-chain measures, not only their center and polarization labels, satisfy the endpoint-measure coherence part of the optional Maslov secant-to-curve promotion on positive-density thickness blocks. Passing from that thickness density to the exact angular-scale-density hypothesis of that problem remains a separate scale-map check. In the non-atomic alternative, two positive-proportion scale families remain a fixed positive distance apart in either geometric labels or weak endpoint-star measure.

Proof. The space $\mathcal{P}(\mathcal{X})$ is compact metrizable because $\mathcal{X}$ is compact metrizable. Hence $\mathcal{Y}^\sharp$ is compact. Apply the proof of Theorem 8.38 verbatim to the pushforward logarithmic scale probabilities on $\mathcal{Y}^\sharp$. No covering-number estimate is used in that proof. $\square$

Corollary 8.40 (The atomic label branch has an exact angular-scale exponent trichotomy). *In the atomic alternative (397), let $\varkappa_\infty \in [0, 1]$ be the limiting angular-scale exponent. Uniformly on the selected positive-density scale sets,*

$$\boxed{\tau_R = R^{\varkappa_\infty + o(1)}, \quad \frac{R}{\tau_R} = R^{1 - \varkappa_\infty + o(1)}. \quad (400)}$$

Consequently:

- (i) if $0 < \varkappa_\infty < 1$, the physical error is $o(\tau_R)$, so this is a genuine noncollapsed angular Darboux scale;
- (ii) if $\varkappa_\infty = 1$, the angular and physical scales are power-comparable up to subpower factors; this is the only exponent compatible with the resolution-scale cap branch, and the two-probe scale lock of Theorems 8.31 and 8.32 forces precisely this endpoint;
- (iii) if $\varkappa_\infty = 0$, then $\tau_R = R^{o(1)}$; this is the sole remaining degenerate thickness-to-angular scale map.

Thus the atomic branch has no unidentified continuum of scale ratios.

Proof. The exponent coordinate is part of the compact label and converges uniformly on the selected sets in (397). Hence

$$\log(1/\tau_R) = (\varkappa_\infty + o(1)) \log(1/R),$$

which is (400). The three alternatives are the three possible positions of $\varkappa_\infty$ in $[0, 1]$. $\square$

Lemma 8.41 (A compatible alternating chain has one limiting Reeb orbit). *Let*

$$H^{(0)} \supset H^{(1)} \supset \dots$$

be one compatible nested branch of the endpoint-star tree in Theorem 8.35. Suppose that $H^{(\ell)}$ is contained in a direction ball of radius $T_\ell \downarrow 0$. For $j = 0, 1$, let

$$(y_\ell^{(j)}, s_\ell^{(j)}) \in \mathbb{R}^3 \times (I_{\text{out}} \cup J)$$

be the two alternating point probes retained on $H^{(\ell+1)}$, and write $\epsilon_\ell^{(j)} \rightarrow 0$ for their physical errors:

$$\boxed{|b(a) + s_\ell^{(j)} a - y_\ell^{(j)}| \leq \epsilon_\ell^{(j)} \quad (a \in H^{(\ell+1)}, j = 0, 1).} \quad (401)$$

After passage to a subsequence there are

$$a_*, b_* \in \mathbb{R}^3, \quad s_*^{(j)} \in I_{\text{out}} \cup J,$$

such that

$$\boxed{s_\ell^{(j)} \longrightarrow s_*^{(j)}, \quad y_\ell^{(j)} \longrightarrow b_* + s_*^{(j)} a_* \quad (j = 0, 1).} \quad (402)$$

Thus every limiting physical probe center on this branch belongs to the Euclidean image

$$\boxed{\mathcal{O}_{a_*, b_*} := \{(b_* + s a_*, s) : s \in \mathbb{R}\}} \quad (403)$$

of one Reeb orbit above the limiting line (a_, b_*) . In particular, continuous transverse drift of the physical centers is not an independent non-atomic label on a compatible infinite branch; only the longitudinal Reeb level can still drift. Moreover, because $\text{dist}(J, I_{\text{out}}) > 0$,*

$$\boxed{\frac{y_\ell^{(1)} - y_\ell^{(0)}}{s_\ell^{(1)} - s_\ell^{(0)}} \longrightarrow a_*}. \quad (404)$$

Proof. Choose $a_\ell \in H^{(\ell+1)}$. If $m > \ell$, then both a_m and a_ℓ belong to the $O(T_\ell)$ -cap containing $H^{(\ell)}$. Hence (a_ℓ) is Cauchy and converges to some a_* . The fixed localization of G_* makes $b(a_\ell)$ bounded. Pass to a subsequence on which

$$b(a_\ell) \rightarrow b_*, \quad s_\ell^{(j)} \rightarrow s_*^{(j)} \quad (j = 0, 1).$$

Substitution of $a = a_\ell$ in (401) gives

$$y_\ell^{(j)} = b(a_\ell) + s_\ell^{(j)} a_\ell + o(1),$$

which proves (402). Subtract the two identities and divide by the uniformly separated denominator $s_\ell^{(1)} - s_\ell^{(0)}$ to obtain (404). $\square$

Proposition 8.42 (Four Reeb times are a Maslov firewall for a coherent endpoint star). *Let $R_n \downarrow 0$, let $R_n \leq \tau_n \leq 1$, and suppose*

$$\boxed{\frac{R_n}{\tau_n} \longrightarrow 0.} \quad (405)$$

For $j = 0, 1, 2, 3$, suppose that the retained physical endpoint-star measures furnish

$$s_{n,j} \rightarrow s_j, \quad V_{n,j} \rightarrow V_\infty \in \text{Gr}(3, 6), \quad q_{n,j} = (\alpha_{n,j}, \beta_{n,j}) \in S^5,$$

where

$$\boxed{\text{dist}(q_{n,j}, V_{n,j}) = o(1), \quad |\beta_{n,j} + s_{n,j} \alpha_{n,j}| \lesssim \frac{R_n}{\tau_n} + o(1).} \quad (406)$$

If s_0, s_1, s_2, s_3 are distinct, then

$$\boxed{P_{V_\infty} \equiv 0, \quad V_\infty \in K_{101} \cup K_{110} \cup K_{200}, \quad \text{rank}(\pi_a|_{V_\infty}) \leq 2.} \quad (407)$$

Consequently this configuration cannot support a noncollapsed three-dimensional horizontal blow-up.

More generally, if the horizontal frame matrices of the coherent planes approach the rank- r locus, then $r + 1$ distinct limiting Reeb times suffice. This is exactly the rank-adapted interpolation of Theorem 6.15.

Proof. After a further subsequence, $q_{n,j} \rightarrow q_j \in S^5$. The two displayed incidence and error-negligibility relations in the proposition give

$$q_j \in V_\infty \cap L_{s_j}, \quad q_j \neq 0.$$

The matrix-pencil/Maslov equivalence (4) yields

$$P_{V_\infty}(s_j) = 0 \quad (0 \leq j \leq 3).$$

This polynomial has degree at most three, so four distinct roots force $P_{V_\infty} \equiv 0$. The front-null classification and direction-rank law used in Theorem 9.51 give (407). The final assertion follows from Theorem 6.15 after letting the incidence and rank errors tend to zero. $\square$

Corollary 8.43 (Finite-scale compactness form of the four-time firewall). *Let*

$$\mathfrak{N} := \{V \in \text{Gr}(3, 6) : P_V \equiv 0\}.$$

For every $\gamma > 0$ and $\eta > 0$ there is $\varepsilon_0 = \varepsilon_0(\gamma, \eta, I^\sharp) > 0$ with the following property. If

$$\max_{i,j} d_{\text{Gr}}(V_i, V_j) \leq \varepsilon_0, \quad |s_i - s_j| \geq \gamma \quad (i \neq j),$$

and unit vectors $q_i = (\alpha_i, \beta_i)$ satisfy

$$\text{dist}(q_i, V_i) \leq \varepsilon_0, \quad |\beta_i + s_i \alpha_i| \leq \varepsilon_0 \quad (0 \leq i \leq 3),$$

then

$$\text{dist}_{\text{Gr}}(V_i, \mathfrak{N}) < \eta \quad (0 \leq i \leq 3).$$

Thus, after fixing the Reeb packet separation and the rank-loss neighborhood, choosing the physical incidence and polarization-cell errors below ε_0 sends every coherent four-time packet to the quantitative front-null/rank-loss ledger.

Proof. Otherwise take a violating sequence with both errors tending to zero while γ, η remain fixed. Compactness of $I^\sharp$, S^5 , and $\text{Gr}(3, 6)$ gives limits with one common plane V , four pairwise γ -separated times, and nonzero vectors in $V \cap L_{s_i}$. Theorem 8.42 gives $V \in \mathfrak{N}$, contradicting the η -separation. The classification in (407) identifies this neighborhood with the stated horizontal rank-loss output. $\square$

Lemma 8.44 (One coherent polarization is front-null or has three shrinking Reeb packets). *Let $I^\sharp$ be a fixed compact Reeb interval, let $V_n \rightarrow V_\infty$ in $\text{Gr}(3, 6)$, and let*

$$q_n = (\alpha_n, \beta_n) \in S^5, \quad s_n \in I^\sharp,$$

satisfy

$$\text{dist}(q_n, V_n) \leq \varepsilon_n, \quad |\beta_n + s_n \alpha_n| \leq \varepsilon_n, \quad \varepsilon_n \downarrow 0.$$

Exactly one of the following holds.

(i) $P_{V_\infty} \equiv 0$, and hence

$$V_\infty \in K_{101} \cup K_{110} \cup K_{200}, \quad \text{rank}(\pi_a|_{V_\infty}) \leq 2.$$

(ii) The real zero set

$$Z_\infty := \{s \in I^\sharp : P_{V_\infty}(s) = 0\}$$

has at most three points and

$$\boxed{\text{dist}(s_n, Z_\infty) \lesssim_{I^\sharp, V_\infty} (\varepsilon_n + d_{\text{Gr}}(V_n, V_\infty))^{1/3}.} \quad (408)$$

If $Z_\infty = \emptyset$, no such approximate incidences exist for all large n .

Proof. Choose orthonormal frame matrices A_n, B_n for V_n , converging after an orthogonal change of frame to matrices A_∞, B_∞ for V_∞ . Project q_n to V_n . In frame coordinates this gives a vector c_n with $|c_n| = 1 + O(\varepsilon_n)$ and

$$|(B_n + s_n A_n)c_n| \lesssim_{I^\sharp} \varepsilon_n.$$

The other two singular values of $B_n + s_n A_n$ are uniformly bounded, so

$$|P_{V_n}(s_n)| = |\det(B_n + s_n A_n)| \lesssim_{I^\sharp} \varepsilon_n.$$

Continuity of the determinant on the compact frame and Reeb-parameter sets yields

$$|P_{V_\infty}(s_n)| \lesssim_{I^\sharp} \varepsilon_n + d_{\text{Gr}}(V_n, V_\infty). \quad (409)$$

If $P_{V_\infty} \equiv 0$, the classification and rank law give (i). Otherwise factor the fixed nonzero polynomial over $\mathbb{C}$. On $I^\sharp$, its nonreal factors and its distance from zero away from Z_∞ are uniformly bounded below. Since its degree is at most three,

$$|P_{V_\infty}(s)| \gtrsim_{I^\sharp, V_\infty} \min\{1, \text{dist}(s, Z_\infty)^3\}.$$

Combine this with (409). If Z_∞ is empty, the same compactness bound contradicts the right side tending to zero. $\square$

Corollary 8.45 (The three Reeb packets return to the existing flagged Maslov routing). *Work on a positive-density logarithmic block with $\varkappa_\infty < 1$. Suppose the atomic endpoint-measure alternative retains a fixed portion of the current determinant-small four-cycle occurrence measure, including its actual cycle-plane label, and that these plane labels converge to V_∞ . Then this coherent-polarization part has no new multiscale endpoint. At the same occurrence-mass scale it returns to one of:*

- (i) a physical Lagrangian point-probe bush;
- (ii) a smaller angular packet or a close-anchor/horizontal-line ledger;
- (iii) a transverse old/new polarization ledger.

In particular a coherent plane together with an at-most-three-level Reeb packet is not a fourth alternative to the local contact-Maslov trichotomy.

Proof. If $P_{V_\infty} \equiv 0$, then $\text{rank}(\pi_a|_{V_\infty}) \leq 2$. The current cycle planes therefore lie in the determinant-small horizontal-rank branch. Apply the arbitrary-threshold route and the lossless finite-flag amplification Theorems 8.5 and 8.7: its fully coherent return is excluded, while its complement gives one of the three stated outputs.

Suppose $P_{V_\infty} \not\equiv 0$. By Theorem 8.44, every retained collision label lies in one of at most three intervals whose common width tends to zero and whose centers are the real roots of P_{V_∞} . Discard empty root packets. If two root centers coalesce, combine their intervals; otherwise their nonzero limiting gaps are bounded below.

Apply the weighted cycle-relative time trichotomy Theorem 7.10 with a time resolution larger than the shrinking packet widths and smaller than the fixed gaps between distinct roots. Its close-anchor output is angular. Its two-interval output returns, without loss of the Z/T occurrence-mass scale, to a shrinking bush or angular packet by Theorem 7.14. In the remaining output, three pairwise separated labels must lie in the three distinct root packets. Their Vandermonde product is therefore bounded below by a positive constant depending only on V_∞ . The weighted continuous anchored rank packet Theorem 7.11 applies. Its nondegenerate determinant complement is already the physical bush output of Theorem 7.8; its determinant-small part is again eliminated or sent to a transverse ledger by Theorems 8.5 and 8.7. These cases exhaust the retained four-cycle occurrence measure. $\square$

Proposition 8.46 (Same-vertex non-atomic polarization variation is already transverse-paid). *Let a normalized fine Maslov graph have edge mass*

$$M = r^{\chi+o(1)}, \quad 0 \leq \chi < 2,$$

and retain the conditional old flags by Theorem 8.3. Suppose a fixed portion of its determinant-small four-cycle occurrence mass has, at the decorated free endpoint, an old polarization normal in a compact set $N_1 \subset \mathbb{P}^2$ and the current new-cycle polarization normal in a compact set $N_2 \subset \mathbb{P}^2$, where

$$\text{dist}(N_1, N_2) \geq c_0 > 0.$$

Then this portion belongs to the transverse old/new polarization ledger; it cannot be an unpaid coherent endpoint-star return. More precisely, after choosing the packet threshold $\vartheta = r^\alpha < c_0/4$, every such decorated endpoint obeys the selector-mass bound

$$\lesssim_{U, c_0} (t_{\text{old}} + r)(t_{\text{new}} + r),$$

and the finite-flag parameter choice in Theorem 8.5 makes the complementary class in which every sampled old flag is coherent with the new plane smaller than the required M^4 rank-loss mass.

Proof. Orient representatives of the projective normals locally. Their separation gives

$$|n_{\text{old}} \times n_{\text{new}}| \gtrsim c_0.$$

The old and new packets meet the same graph product cell at the decorated endpoint. Alternative (i) of Theorem 7.78, used with any $\vartheta < c_0/4$, therefore gives

$$\sigma(\text{possible decorated directions}) \lesssim_{U, c_0} (t_{\text{old}} + r)(t_{\text{new}} + r).$$

For the quantitative four-cycle occurrence, lift the free endpoint to the number k of independent conditional old flags prescribed in Theorem 8.5. If at least one flag lies in N_1 while the new normal lies in N_2 , the preceding transverse estimate applies. If all flags lie within the ϑ -cell of the new normal, their occurrence mass is bounded by

$$\left(\frac{H_i}{\lambda_*} \vartheta \right)^k$$

as in the proof of Theorem 8.4. The inequalities

$$\frac{4\chi}{k} < \alpha < d - \beta$$

make this $o(M^4)$, while $t_{\text{new}} = o(\vartheta)$. Hence a fixed portion of the rank-loss mass cannot remain in the fully coherent class, and every surviving portion is transverse-paid. $\square$

Corollary 8.47 (Residual polarization variation is purely cross-branch). *In the non-atomic polarization alternative of Theorem 8.38, any positive occurrence mass on which the two separated polarization families are joined by a retained old-flag/current-cycle incidence at one decorated endpoint is disposed of by Theorem 8.46. Consequently the only polarization variation not already present in the transverse ledger is segregated variation between distinct stopping-tree branches: no positive-mass decorated endpoint carries both sides of the separation.*

Thus the four-cluster alternative in the optional Maslov secant-to-curve promotion is needed only for this cross-branch remainder. It is not needed to handle local drift of a coherent Maslov polarization, which is already excluded by the conditional multi-flag symplectic comparison.

Proposition 8.48 (Separated polarization branches either decouple or create a transverse Maslov four-cycle). *Let ν_1, ν_2 be two restrictions of one normalized selector occurrence measure, with*

$$\nu_i(G) \geq \theta > 0 \quad (i = 1, 2),$$

and suppose every retained old polarization normal over ν_i lies in a compact set $N_i \subset \mathbb{P}^2$, where

$$\text{dist}(N_1, N_2) \geq c_0 > 0.$$

Let h_r , $0 \leq h_r \leq 1$, be the physical normalized Maslov graph at scale r , with all old conditional flags retained, and put

$$X_r := \iint h_r(z_1, z_2) d\nu_1(z_1) d\nu_2(z_2).$$

Exactly one of the following occurs.

(i) $X_r \leq r^{2-o(1)}$; the cross-branch contribution is already at the quadratic residual endpoint and may be deleted before treating the two diagonal graphs separately.

(ii) Along a subsequence

$$X_r = r^{\chi+o(1)} \quad \text{for some } 0 \leq \chi < 2.$$

Then the cross graph returns, at the same power-excess scale, to a physical Lagrangian bush, a close-anchor/angular or horizontal-line output, or the transverse old/new polarization ledger.

Thus separated cross-branch polarization mixing is not an additional coherent endpoint.

Proof. Only (ii) requires proof. Normalize ν_i to probabilities and symmetrize the bipartite graph on their disjoint union. Because both masses are bounded below by θ , this changes X_r only by a constant. Two applications of Cauchy–Schwarz give alternating bipartite four-cycle mass $\gtrsim_\theta X_r^4$. Apply the arbitrary-threshold Maslov routing Theorem 7.8. Its determinant-nondegenerate output is a physical bush. Its close-anchor and horizontal-line outputs are already among the stated ledgers.

It remains to consider a determinant-small alternating cycle. Its vertices alternate between ν_1 and ν_2 , and all retain their old conditional polarization flags. Let n_{new} be the normal of the current cycle packet. If $\vartheta < c_0/4$, then n_{new} cannot lie within ϑ of both N_1 and N_2 . Hence at least one of the four vertices has an old flag ϑ -transverse to the new cycle polarization. Cyclically re-root the four-cycle so that this vertex is the decorated free endpoint. The rotation costs only a factor four. Theorem 7.78 places that portion in the transverse ledger. Therefore a fixed part of the alternating four-cycle mass reaches one of the listed outputs. $\square$

Corollary 8.49 (Non-atomic polarization is reduced to coherent components without a power loss). *At a fine scale r , choose $\zeta_r \downarrow 0$ with $\zeta_r^{-1} = r^{-o(1)}$, and split the normalized Maslov graph according as the two retained old polarization normals have distance at most $C\zeta_r$ or greater than $C\zeta_r$. Cover the compact polarization space by $r^{-o(1)} \zeta_r$ -cells and sum the far graph over cell pairs. If its total edge mass exceeds $r^{2-o(1)}$, one separated cell pair still has edge exponent strictly below two, and Theorem 8.48 returns it to a physical or transverse output. Otherwise delete the far graph at total quadratic cost. A bounded-overlap net decomposition of the near graph leaves $r^{-o(1)}$ graph pieces on each of which both endpoint polarizations lie within $O(\zeta_r)$ of one common plane.*

Consequently the non-atomic polarization alternative does not require a new four-cluster theorem at the collision-energy level. After a subpower partition it is either already transverse-paid or graph-decoupled into the coherent components handled by Theorem 8.45. The only branch not covered by this conclusion is the resolution endpoint $\varkappa_\infty = 1$, where the physical incidence error is not Maslov-negligible.

Proof. The number of polarization cells and ordered cell pairs is $r^{-o(1)}$. Hence pigeonholing a far contribution loses only a subpower factor and preserves every fixed exponent gap below two. Apply Theorem 8.48 to a separated color class. If no such contribution is excessive, positivity gives a total deleted edge mass $r^{2-o(1)}$. Within each remaining near piece the two old polarization labels lie in one $O(\zeta_r)$ -ball. Compare the current cycle plane with that ball. A separated current plane is same-vertex transverse-paid by Theorem 8.46; a coherent current plane enters Theorem 8.45. The piece edge masses add before normalization, and the subpower net count is absorbed by the fixed exponent gap. $\square$

Corollary 8.50 (Symplectic reduction of the remaining label drift). *Apply the endpoint-measure compactness of Theorem 8.39 along one compatible nested branch, and include in the occurrence label an actual normalized Maslov secant q , its Reeb parameter s , and its polarization plane V . On every exponent branch*

$$\varkappa_\infty < 1$$

one has

$$R/\tau_R = R^{1-\varkappa_\infty+o(1)} \rightarrow 0,$$

so the limiting occurrence measure is supported on the exact incidence relation

$$q \in V \cap L_s, \quad q \neq 0.$$

Then:

- (i) the limiting physical point-probe centers lie on the single Reeb orbit $\mathcal{O}_{a_*, b_*}$ from Theorem 8.41;
- (ii) if the polarization marginal is non-atomic, two separated compact sets of polarization planes carry positive occurrence mass; their same-vertex part is transverse-paid by Theorem 8.46, leaving only cross-branch variation as the polarized four-cluster input;
- (iii) if the polarization marginal has an atom V_∞ , then, on any noncollapsed three-dimensional horizontal branch, the conditional Reeb-time support above V_∞ has at most three points; at finite scale all flags whose planes converge to V_∞ lie in the three shrinking packets of (408).

Indeed four points in the conditional time support allow four disjoint time neighborhoods and four convergent endpoint-star flags, contradicting Theorem 8.42. Thus, below the resolution endpoint $\varkappa_\infty = 1$, the non-atomic alternative of Theorem 8.38 is no longer arbitrary physical-center drift: it is either genuine polarization

variation or a finite, at-most-three-level Reeb packet over one coherent polarization. Whenever that polarization is the retained current cycle-plane label, the latter packet is returned to the existing bush/angular/transverse routes by Theorem 8.45. The endpoint $\varkappa_\infty = 1$ remains separate because (405) fails there; the two-probe scale lock forbids replacing its resolution-size error by an exact tangent Maslov incidence.

Proof. The first assertion is Theorem 8.41. If the polarization marginal has no atoms, the compact-set separation argument in Theorem 8.38 applies to that marginal itself. If it has an atom V_∞ and the conditional time support contains four distinct points, choose disjoint small neighborhoods of those points. Each has positive conditional mass, so weak convergence of the retained occurrence measures supplies one actual normalized flag from each neighborhood. Their plane labels converge to V_∞ , and for $\varkappa_\infty < 1$ their normalized physical incidence errors tend to zero. Theorem 8.42 gives horizontal rank at most two, contrary to the noncollapse hypothesis. Theorem 8.44 supplies the finite-scale packet width. This proves the trichotomy and also explains why the same inference is unavailable at the resolution endpoint. $\square$

Corollary 8.51 (Every below-resolution label branch returns to the existing geometric outputs). *Assume the actual endpoint-star tuple measure, including its current cycle plane and normalized Maslov secants, is retained as in Theorem 8.39. On every positive-density exponent branch with*

$$\varkappa_\infty < 1,$$

the label compactness alternative has no independent output:

- (i) *physical-center drift is confined to one limiting Reeb orbit by Theorem 8.41;*
- (ii) *separated polarization variation is transverse-paid or graph-decoupled by Theorems 8.48 and 8.49;*
- (iii) *over one coherent polarization, four limiting Reeb levels force front-null horizontal rank loss by Theorem 8.42, while at most three levels return to the flagged bush/angular/transverse routing by Theorem 8.45.*

Consequently any endpoint-star chain which avoids all previously paid geometric outputs must, after passing to a further positive-density subsequence, satisfy

$$\varkappa_\infty = 1.$$

Thus the sole remaining multiscale endpoint is the resolution-scale angular cap; no below-resolution Maslov or polarization branch remains.

Proof. For $\varkappa_\infty < 1$,

$$R/\tau_R = R^{1-\varkappa_\infty+o(1)} \rightarrow 0.$$

Hence the physical endpoint-star measures converge to exact Maslov incidences by Theorem 6.2 and (72). Apply the three alternatives just listed. They exhaust, respectively, the physical-center coordinate, the polarization coordinate, and the Reeb-time/current-cycle part of the compact endpoint-star label. Therefore an unpaid branch cannot have $\varkappa_\infty < 1$. The exponent trichotomy Theorem 8.40 leaves only $\varkappa_\infty = 1$. $\square$

8.6 Resolution shells and the density cocycle

Lemma 8.52 (The resolution shell is automatically relative-quadratic). *Let $H \subset U$ have direction mass $p > 0$, and let $\mu_H = (\text{id}, b)_\#(\sigma|_H)$. For $0 < \rho \leq \tau < 1$, the angular-shell residual satisfies*

$$\mathfrak{Z}_{\rho,\tau}^{J^+}(\mu_H) \lesssim_U p\tau^2. \quad (410)$$

Consequently, along the tax-free failure sequence of Theorem 8.32, where

$$p = \rho^{o(1)},$$

every shell with angular exponent

$$\frac{\log(1/\tau)}{\log(1/\rho)} \longrightarrow 1$$

obeys the relative endpoint estimate

$$\mathfrak{Z}_{\rho,\tau}^{J^+}(\mu_H) \lesssim p^2 \rho^{2-o(1)}. \quad (411)$$

In particular a fixed power-excess residual failure cannot be carried by $\varkappa_\infty = 1$.

Proof. Discard the two residual indicators in the definition (115). On the shell $\tau \leq |a - a'| < 2\tau$, the weight is at most τ^{-1} . The bounded Lebesgue direction density gives, for every $a \in H$,

$$\sigma\{a' \in H : |a - a'| < 2\tau\} \lesssim_U \tau^3.$$

Integration in a proves (410). At exponent one,

$$\tau = \rho^{1+o(1)}, \quad \frac{1}{p} = \rho^{-o(1)}.$$

Therefore

$$p\tau^2 = p^2 \rho^2 \left(\frac{\tau}{\rho}\right)^2 \frac{1}{p} = p^2 \rho^{2-o(1)},$$

which is (411). This is incompatible with a lower bound $\gtrsim p^2 \rho^{2-\eta}$ for any fixed $\eta > 0$. $\square$

Corollary 8.53 (Every genuine fine return has Maslov-negligible physical error). *At a resolution cap, decompose any relative power-excess residual failure into the dyadic angular shells of Theorem 7.2. The $r^{-o(1)}$ number of shells costs no fixed exponent. By Theorem 8.52, a selected bad shell cannot have $\varkappa_\infty = 1$. Hence, after a subsequence,*

$$\varkappa_\infty < 1, \quad \frac{\rho}{\tau_\rho} \longrightarrow 0.$$

Thus every actual fine return from Theorem 8.32 lies in the error-negligible contact–Maslov regime and is subject to Theorem 8.51. The resolution cap introduces no additional Maslov endpoint and reduces the proof to the mass aggregation of successive physical bush returns.

Proof. The ultra-near term in Theorem 7.2 is already below the target after the standard choice of its cutoff. If the sum of the remaining $O(\log(2/\rho)) = \rho^{-o(1)}$ shells has a fixed power excess, one shell retains that excess up to a subpower factor. The preceding lemma excludes exponent one for this shell. The displayed scale separation and the last assertion follow from Theorem 6.2. $\square$

Lemma 8.54 (Mass loss and non-Frostman concentration form one density cocycle). *Let a stopping node H have direction mass p and be contained in a direction cap of radius T . Put*

$$D(H) := \frac{p}{T^3}.$$

Suppose one complete fine-Maslov/bush/non-Frostman return first retains a proportion at least cM of the node mass in its maximal physical bush family. At the normalized Frostman-failure scale R , suppose the Reeb-coherent mergers then retain a proportion at least

$$cAR^{3-\delta}$$

and, after the intrinsic Maslov renormalization and alternating Reeb return, place that mass inside a child direction cap of radius at most CTR . Here R and $\widehat{R}$ are not identified; in the returned hairbrush branch one only has the previously proved lower scale lock $\widehat{R} \gtrsim R^{1-d}$. If the child is H' , then

$$\boxed{D(H') \gtrsim_{I,J,U} AMR^{3-\delta} \widehat{R}^{-3} D(H).} \quad (412)$$

For a chain of such returns,

$$\boxed{\prod_{\ell < L} \left(c_{I,J,U} A_\ell M_\ell R_\ell^{3-\delta} \widehat{R}_\ell^{-3} \right) \lesssim_U \frac{1}{D(H^{(0)})}.} \quad (413)$$

Consequently, an infinite branch cannot have

$$A_\ell M_\ell R_\ell^{3-\delta} \widehat{R}_\ell^{-3} \geq C_{I,J,U} > 1$$

at every sufficiently large generation. Apart from density-increasing generations whose multiplicative gains have bounded total product, every infinite survivor must lie in the compensation regime

$$\boxed{M_\ell \lesssim_{I,J,U} A_\ell^{-1} \widehat{R}_\ell^3 R_\ell^{\delta-3}.} \quad (414)$$

Proof. The hypotheses give

$$p' \gtrsim AMR^{3-\delta}p, \quad T' \lesssim T\hat{R}.$$

Divide the first inequality by $(T')^3$ to obtain (412). Iteration gives the lower bound by the product on the left of (413). On the other hand the direction marginal is bounded Lebesgue measure, so every selector subset contained in a T' -cap has mass $\lesssim_U (T')^3$; hence $D(H') \lesssim_U 1$. Comparing the iterated lower and uniform upper bounds proves (413). The last two assertions are immediate. $\square$

Corollary 8.55 (The stopping-tree tail is confined to the compensation regime). *After Theorem 8.53, every nonterminal node first passes through the exact contact–Maslov routing and then through the physical maximal-bush/Frostman alternative. For the non-Frostman children, denote the normalized Frostman-failure radius by R_ℓ , the final relative radius of the returned child cap by $\hat{R}_\ell$, and combine all fixed retained fractions and excess constants into A_ℓ . The mass and radius hypotheses of Theorem 8.54 follow from Theorems 8.8, 8.12, 8.16, 8.18 and 8.30.*

Therefore the part of the normalized stopping tree on which

$$A_\ell M_\ell R_\ell^{3-\delta} \hat{R}_\ell^{-3}$$

is uniformly larger than one has finite branch depth. The density cocycle therefore confines the normalized tail to nodes in the displayed compensation regime. Equivalently, its remaining normalized scalar is the explicit domination of the bad Maslov edge mass M_ℓ by the exact two-scale cap-density debt $A_\ell^{-1} \hat{R}_\ell^3 R_\ell^{\delta-3}$. This is the same $R_\ell/\hat{R}_\ell$ cubic enlargement already isolated in the fine and coarse two-scale frontier estimates. The cocycle is paired with the absolute-flow exhaustion below, which removes this term.

Lemma 8.56 (Lossless exhaustion of a hereditary Markov route). *Let Γ be a finite occurrence measure with distinguished source map π_{src} , and allow any finite or countable collection of conditional probability marks over that source. Suppose that, for every nonzero restriction $\Gamma' \leq \Gamma$, one application of a routing alternative produces an output submeasure $\Theta \leq \Gamma'$ with*

$$\|\Theta\| \geq c\|\Gamma'\|, \quad c > 0,$$

and assigns Θ to finitely or countably many measurable destinations by source restrictions and extensions by probability kernels. Then Γ has a countable destination decomposition

$$\boxed{\Gamma = \sum_{j \geq 1} \Theta_j, \quad \sum_{j \geq 1} \|\Theta_j\| = \|\Gamma\|,} \quad (415)$$

up to a null remainder. If the source marginal of Γ is $f d\sigma$, then the destination source marginals are $f_j d\sigma$ with

$$0 \leq f_j, \quad \sum_j f_j \leq f \quad \text{a.e.}$$

Every inherited mark is unchanged at the retained source point.

Proof. Apply the route to $\Gamma_0 := \Gamma$, remove the produced output, and apply it again to $\Gamma_1 := \Gamma_0 - \Theta_0$. More generally set $\Gamma_{n+1} := \Gamma_n - \Theta_n$. Positivity and the fixed retention give

$$\|\Gamma_n\| \leq (1 - c)^n \|\Gamma\|,$$

so the remainder tends to zero in total variation and the destination measures sum to Γ . A source restriction replaces a density by a smaller density, and a probability kernel has source marginal equal to the measure from which it is extended. Summing the resulting Radon–Nikodym densities proves $\sum_j f_j \leq f$. The same observation shows that an inherited conditional mark is never resampled or moved to another endpoint. $\square$

Lemma 8.57 (Cap-square summation pays fractional sparse edge tails). *Let*

$$d\nu_\gamma = f_\gamma d\sigma, \quad 0 \leq f_\gamma \leq 1, \quad \sum_\gamma f_\gamma \leq 1,$$

be finite selector-direction submeasures, each supported in a direction cap of radius at most T , and put

$$p_\gamma := \|\nu_\gamma\|, \quad \sum_\gamma p_\gamma \leq m,$$

where $m \leq \|\sigma\|$. For measurable kernels $0 \leq h_\gamma \leq 1$,

$$\boxed{\sum_{\gamma} \iint h_{\gamma}(a, a') d\nu_{\gamma}(a) d\nu_{\gamma}(a') \lesssim_U m T^3.} \quad (416)$$

Consequently, if

$$T^3 \leq m r^{2-o(1)},$$

the complete within-node edge tail is at most $m^2 r^{2-o(1)}$. No bound on the total tail mass $\sum_{\gamma} p_{\gamma}$ stronger than m is required.

Proof. Bounded Lebesgue direction density and $f_{\gamma} \leq 1$ give $p_{\gamma} \lesssim_U T^3$, while $h_{\gamma} \leq 1$ gives

$$\iint h_{\gamma}(a, a') d\nu_{\gamma}(a) d\nu_{\gamma}(a') \leq p_{\gamma}^2.$$

Therefore

$$\sum_{\gamma} p_{\gamma}^2 \leq \left(\sup_{\gamma} p_{\gamma} \right) \sum_{\gamma} p_{\gamma} \lesssim_U m T^3.$$

□

9 Cross-generation edge-flow closure

9.1 Absolute occurrences and the edge-Carleson reduction

Proposition 9.1 (Exact edge-Carleson criterion for closing the bush-return tree). *Keep the hypotheses of Theorem 9.42. Suppose the collision-preserving bush-return construction can be saturated to depth $L = L(r)$ so that, for the incoming absolute normalized Maslov edge mass $\mathcal{M}_{\text{in}}(r)$,*

$$\boxed{\mathcal{M}_{\text{in}}(r) \lesssim r^{-o(1)} \left(\mathcal{M}_{\text{paid}}(r) + \mathcal{M}_{\text{cross}}(r) + \sum_{\gamma \in \mathcal{S}_L} p_\gamma T_\gamma^3 \right)}. \quad (417)$$

Here $\mathcal{S}_L$ is the family of sparse terminal hairbrush nodes, T_γ is the radius of the terminal direction cap, and $\mathcal{M}_{\text{paid}} \lesssim m^2 r^{2-o(1)}$, and $\mathcal{M}_{\text{cross}} \lesssim m^2 r^{2-o(1)}$. If every node in $\mathcal{S}_L$ lies in a direction cap of radius T_L and

$$T_L^3 \leq m r^{2-o(1)},$$

then the relative residual estimate (303) follows and the Maslov–Frostman bridge closes. The terminal nodes may be fractional selector submeasures

$$d\nu_\gamma = f_\gamma d\sigma, \quad \sum_{\gamma \in \mathcal{S}_L} f_\gamma \leq 1, \quad p_\gamma := \|\nu_\gamma\|;$$

thus the statement applies to the target-cell/time-packet occurrence partitions below, not only to pairwise disjoint selector sets.

Proof. Bounded direction density and the fractional source domination give

$$\sum_{\gamma \in \mathcal{S}_L} p_\gamma T_\gamma^3 \lesssim_U m T_L^3 \lesssim m^2 r^{2-o(1)}.$$

Insert this and the two assumed edge bounds into (417). The collision/residual comparison Theorem 7.1 gives (303); then Theorems 6.27 and 9.43 close the stated bridge. $\square$

Lemma 9.2 (The incoming Maslov edge flow survives maximal bush extraction linearly). *Let h be a symmetric measurable kernel with $0 \leq h \leq 1$ on a probability space (G, μ) , put*

$$d_h(z) := \int_G h(z, z') d\mu(z'), \quad M := \iint_{G \times G} h d\mu d\mu,$$

and define the incoming edge-flow measure

$$d\lambda_h(z) := d_h(z) d\mu(z).$$

Suppose the maximal bush extraction gives pairwise disjoint leaf sets H_j , with $H := \bigcup_j H_j$, such that

$$\iint_{H^c \times H^c} h d\mu d\mu < \frac{M}{2}.$$

Then

$$\boxed{\lambda_h(H) > \frac{M}{4}, \quad \sum_j \lambda_h(H_j) = \lambda_h(H), \quad 0 \leq \lambda_h \leq \mu.} \quad (418)$$

Moreover, on $\{d_h > 0\}$,

$$d\kappa_z^h(z') := \frac{h(z, z')}{d_h(z)} d\mu(z')$$

is a probability kernel and the old physical Maslov edge measure disintegrates exactly as

$$\boxed{h(z, z') d\mu(z) d\mu(z') = d\lambda_h(z) d\kappa_z^h(z').} \quad (419)$$

Thus the maximal-bush passage retains a fixed fraction of the incoming edge mass, not merely a comparable amount of unweighted vertex mass. The neighbor z' , together with all older conditional Maslov flags, may be carried as a probability mark over the bush endpoint z with no further normalization factor.

Proof. Write

$$E_{11} := \iint_{H \times H} h, \quad E_{10} := \iint_{H \times H^c} h, \quad E_{00} := \iint_{H^c \times H^c} h,$$

where all integrals use $d\mu d\mu$. Symmetry gives

$$M = E_{11} + 2E_{10} + E_{00}, \quad \lambda_h(H) = E_{11} + E_{10} \geq \frac{M - E_{00}}{2} > \frac{M}{4}.$$

Disjointness gives the middle identity in (418), and $d_h \leq 1$ gives the domination $\lambda_h \leq \mu$. The definition of κ_z^h proves both its normalization and (419). Since all other old flags are already conditional probability laws over z , their tensor product with κ_z^h again has total mass one. $\square$

Lemma 9.3 (Dyadic degree layers retain the linear edge flow). *Keep $h, \mu, d_h, \lambda_h, M$ from Theorem 9.2. Fix $A > 2$, a scale $0 < r < 1/2$, put $K := \lceil A \log_2(1/r) \rceil$, and define*

$$D_k := \{2^{-k-1} < d_h \leq 2^{-k}\} \quad (0 \leq k < K).$$

Then

$$\boxed{\lambda_h(\{d_h \leq 2^{-K}\}) \leq 2^{-K} \leq r^A, \quad 2^{-k-1}\mu|_{D_k} < \lambda_h|_{D_k} \leq 2^{-k}\mu|_{D_k}.} \quad (420)$$

Thus all but an $O(r^A)$ part of the incoming edge flow is a disjoint union of only $O_A(\log(1/r))$ pieces on which edge flow and selector mass are pointwise comparable. No single degree layer has to be selected.

Proof. The second assertion is the definition of D_k . Since $d_h \leq 2^{-K}$ on the discarded set and $\mu(G) = 1$, its λ_h -mass is at most $2^{-K} \leq r^A$. The displayed layers and this remainder form a disjoint partition up to null boundaries. $\square$

Proposition 9.4 (A physical Reeb bush transports every comparable incoming edge flow linearly). *Let H be an R -approximate Lagrangian point-probe bush, let ν be a probability selector measure supported on H , and let λ be a finite measure satisfying*

$$\theta d\nu \leq d\lambda \leq 2\theta d\nu \quad (421)$$

for some $\theta > 0$. Choose the outer Reeb interval W of length $|W| \asymp R$ from Theorem 8.17, and define the normalized physical collision kernel

$$g_R(z, z') := \frac{1}{|W|} \int_W \mathbf{1}_{\{|\Pi_u(z) - \Pi_u(z')| \leq C_{I,J,U} R\}} du.$$

After increasing the fixed geometric constant, $g_R = 1$ on $H \times H$. For any angular threshold $0 < \tau < 1$, split

$$g_R = g_R^{\text{ang}} + g_R^{\text{fix}}$$

by $|a - a'| < \tau$ and $|a - a'| \geq \tau$, and put

$$d_{\text{ang}}(z) := \int g_R^{\text{ang}}(z, z') d\nu(z'), \quad d_{\text{fix}}(z) := \int g_R^{\text{fix}}(z, z') d\nu(z').$$

Then $d_{\text{ang}} + d_{\text{fix}} = 1$. Use the measurable partition

$$X_{\text{ang}} := \{d_{\text{ang}} \geq 1/2\}, \quad X_{\text{fix}} := H \setminus X_{\text{ang}}.$$

Thus

$$\boxed{\lambda(X_{\text{ang}}) + \lambda(X_{\text{fix}}) = \lambda(H), \quad d_{\star}(z) \geq \frac{1}{2} \quad (z \in X_{\star}, \star \in \{\text{ang}, \text{fix}\}).} \quad (422)$$

For $z \in X_{\star}$, define

$$dQ_z^{\star}(z') := \frac{g_R^{\star}(z, z')}{d_{\star}(z)} d\nu(z') \quad (423)$$

and use Q_z^{ang} on X_{ang} and Q_z^{fix} on X_{fix} . These are probability kernels, with density at most 2 relative to ν , supported on actual physical $O(R)$ -collisions. In the fixed-angle case it is supported on the normalized coarse Maslov graph of Theorem 8.28; in the angular case it is supported on the angular Darboux branch. If $d\kappa_z(\zeta)$ is any conditional old-neighbor or old-polarization law, then

$$\sum_{\star \in \{\text{ang}, \text{fix}\}} \mathbf{1}_{X_{\star}}(z) d\lambda(z) d\kappa_z(\zeta) dQ_z^{\star}(z')$$

has total mass $\lambda(H)$. Hence the Reeb collision step transports all of the incoming edge flow without replacing it by the quadratic product mass $\lambda(H)^2$.

More precisely, for either $\star$ with $\lambda(X_\star) > 0$, normalize

$$\alpha_\star := \lambda(X_\star)^{-1} \lambda|_{X_\star}$$

and take disjoint source and target copies with probability measure

$$\tilde{\nu}_\star := \frac{1}{2} \alpha_\star^{(0)} + \frac{1}{2} \nu^{(1)}.$$

Writing $dQ_z^\star(z') = q_\star(z, z') d\nu(z')$, define the symmetric two-copy kernel

$$\tilde{h}_\star((z, 0), (z', 1)) = \tilde{h}_\star((z', 1), (z, 0)) := \frac{1}{2} q_\star(z, z'),$$

and set it to zero on equal copies. Then

$$\boxed{0 \leq \tilde{h}_\star \leq 1, \quad e_{\tilde{\nu}_\star}(\tilde{h}_\star) = \frac{1}{4}.} \quad (424)$$

Under (421), the source marginal $\alpha_\star$ is comparable, within a factor two, to the normalized restriction of ν to $X_\star$. Thus (424) is a normalized physical Maslov graph with fixed edge mass and no hidden cap-density factor. In the four-cycle routing, use the target copy for the anchors and the source copy for the free/common-neighbor vertices; the conditional old neighbor and polarization marks therefore remain attached to the transported edge flow.

Proof. The point-probe relation places all projected bush endpoints in one $O(R)$ -ball throughout the interval W , exactly as in the proof of Theorem 8.17; hence $g_R = 1$ after fixing the displayed constant. The angular and fixed-angle kernels partition it, so their degrees sum to one at every source point. On X_{ang} , the angular degree is at least $1/2$ by definition. On X_{fix} , one has

$$d_{\text{fix}} = 1 - d_{\text{ang}} > \frac{1}{2}.$$

This proves (422). Equation (423) is therefore a probability kernel and its density is at most two. The support assertions follow from the two angular pieces of the defining physical collision kernel. Finally integrate first in $\kappa_z(\zeta)$ and then in the appropriate $Q_z^\star$; both are probability laws. The density $q_\star$ is at most two. Hence the two-copy kernel is at most one, and direct integration over its two oriented cross-copy pieces gives

$$e_{\tilde{\nu}_\star}(\tilde{h}_\star) = 2 \cdot \frac{1}{4} \cdot \frac{1}{2} \int_{X_\star} \int_H q_\star(z, z') d\nu(z') d\alpha_\star(z) = \frac{1}{4}.$$

Finally, (421) implies

$$\frac{1}{2} \frac{\nu|_{X_\star}}{\nu(X_\star)} \leq \alpha_\star \leq 2 \frac{\nu|_{X_\star}}{\nu(X_\star)}$$

after harmlessly weakening the exact normalization constants. This proves the last assertion. $\square$

Corollary 9.5 (The two-copy Maslov routing may be oriented toward the edge-flow source). *For either two-copy graph $\tilde{h}_\star$ in (424), the arbitrary-threshold routing of Theorem 7.8 may be performed with both anchors in the target copy and both common-neighbor/free vertices in the source copy. Its bush output therefore has source marginal mass bounded below by a fixed positive constant, while its determinant-small four-cycle output has fixed positive four-cycle mass. In both cases the conditional old-neighbor and polarization marks remain attached to the source vertices.*

Proof. The proof of Theorem 7.8 is bipartite. With $\alpha_\star$ on the source side, ν on the target side, and

$$k(z, z') := \frac{1}{2} q_\star(z, z'),$$

one has $0 \leq k \leq 1$ and

$$\iint k(z, z') d\alpha_\star(z) d\nu(z') = \frac{1}{2}.$$

Apply the first Cauchy–Schwarz in the source variable and the second in the target variable. The resulting four-cycle has the ordered form

$$p^{(1)}, y^{(0)}, q^{(1)}, (y')^{(0)}, p^{(1)}.$$

Thus the anchors p, q lie in the target copy and the common neighbors y, y' lie in the source copy, with total four-cycle content at least 2^{-4} . Splitting by the determinant threshold gives the rank-loss output with a fixed constant. On its complement, the Cramer-rule and Reeb-time pigeonhole argument in the cited proposition selects its Lagrangian bush from the common-neighbor variable, hence from the source copy, again with fixed positive mass. All added marks are conditional probability variables over that source endpoint and integrate to one. $\square$

Corollary 9.6 (One bush-return generation preserves a fixed portion of the incoming Maslov edge). *Let the incoming normalized Maslov kernel satisfy*

$$e(h) = M = r^{\chi+o(1)}, \quad 0 \leq \chi < 2,$$

and let $H = \bigcup_j H_j$ be the maximal physical bush family of Theorem 8.8 in its un-paid alternative. Decompose the edge-flow measure $\lambda_h|_H$ simultaneously by the disjoint bushes H_j and the degree layers of Theorem 9.3, with any fixed $A > 2$. On every nonempty piece normalize the selector measure and apply Theorem 9.4. Then, for all sufficiently small r , the sum of the transported angular or fixed-angle physical occurrence measures is at least

$$\sum_{j,k} \|\Gamma_{j,k}^{(1)}\| = \lambda_h(H) - O(r^A) \gtrsim M. \quad (425)$$

Every $\Gamma_{j,k}^{(1)}$ consists of actual physical Reeb collisions, retains the old Maslov neighbor and all conditional polarization flags, and has transition density bounded by a fixed constant. Thus a complete bush-return generation has no power loss in incoming edge mass.

Proof. Theorem 9.2 gives $\lambda_h(H) > M/4$. The discarded low-degree part has mass at most r^A by Theorem 9.3. Since $A > 2 > \chi$, one has $r^A = o(M)$. On each remaining disjoint piece, the two measures satisfy (421); sum the two branches in (422). The support and flag assertions are those of Theorem 9.4 and (419). $\square$

9.2 Angular and separated-time strict progress

Corollary 9.7 (Superpolynomially small bush-degree pieces carry negligible edge flow). *In Theorem 9.6, fix $A > 2$. Since*

$$N \lesssim M^{-3} = r^{-3\chi+o(1)}$$

and there are $O_A(\log(1/r))$ degree layers, the total edge-flow mass of all pieces $H_j \cap D_k$ having

$$\lambda_h(H_j \cap D_k) < r^{A+7}$$

is at most $r^{A+1-o(1)}$. Every retained piece has selector mass

$$\mu(H_j \cap D_k) \geq \lambda_h(H_j \cap D_k) \geq r^{A+7}.$$

Thus, after a negligible relative-quadratic discard, every bush entering the same-window amplification argument has the polynomial mass lower bound required by Theorem 9.17.

Proof. There are at most

$$O_A(M^{-3} \log(1/r)) \leq r^{-6-o(1)}$$

pieces because $\chi < 2$. Multiply this bound by r^{A+7} . The selector-mass lower bound uses $d_h \leq 1$, hence $\lambda_h \leq \mu$. $\square$

Lemma 9.8 (Adaptive source-cap disintegration preserves the old-neighbor occurrence). *Keep the notation and hypotheses of Theorem 9.4. For any $0 < \tau < 1$, partition the direction chart by half-open cubes $Q \in \mathcal{Q}_\tau$ of side τ , and put*

$$X_{\star,Q} := X_\star \cap \{a_z \in Q\}, \quad \Lambda_{\star,Q} := \lambda(X_{\star,Q}) \quad (\star \in \{\text{ang}, \text{fix}\}).$$

For every nonnull piece normalize

$$\alpha_{\star,Q} := \Lambda_{\star,Q}^{-1} \lambda|_{X_{\star,Q}}$$

and retain the same Markov kernel $Q_z^\star$ from (423). Then

$$\sum_{\star,Q} \Lambda_{\star,Q} = \lambda(H), \quad \sum_{\star,Q} \Lambda_{\star,Q} \alpha_{\star,Q}(dz) \kappa_z(d\zeta) Q_z^\star(dz') = \sum_{\star} \mathbf{1}_{X_\star}(z) \lambda(dz) \kappa_z(d\zeta) Q_z^\star(dz'). \quad (426)$$

Thus no cap is selected and no factor depending on $\#\mathcal{Q}_\tau$ is lost. On each normalized piece the two-copy construction of (424) still has edge mass $1/4$.

Moreover:

- (i) on an angular piece, both the source direction and the Markov target direction lie in one fixed enlargement Q^* of Q , of diameter $O(\tau)$;
- (ii) on a fixed-angle piece, $|a_z - a_{z'}| \geq \tau$, so the normalized physical Maslov error is $O(R/\tau)$, and every physical-bush output of the oriented two-copy routing has its direction support in Q^* .

All old neighbors and all old polarization flags remain conditional probability marks over the source copy on every cap piece.

Proof. The half-open cubes partition the source direction chart. Hence the sets $X_{\star, Q}$ are pairwise disjoint and their union over $\star, Q$ is H , up to null boundaries. Integrating the two probability kernels κ_z and Q_z^* proves (426). The proof of (424) uses only that Q_z^* is a probability kernel with density at most two; restricting and renormalizing its source marginal does not change that calculation.

If $\star = \text{ang}$, the definition of g_R^{ang} gives $|a_z - a_{z'}| < \tau$. Since $a_z \in Q$, both directions lie in a fixed enlargement Q^* . If $\star = \text{fix}$, the opposite angular inequality holds. The physical collision error is $O(R)$, hence division by the angular scale gives $O(R/\tau)$. Finally Theorem 9.5 chooses the free/common-neighbor variables, and therefore its bush output, from the source copy. Those directions lie in Q^* . Conditional marks integrate to one before and after the restriction. $\square$

Corollary 9.9 (Angular fine returns have unconditional occurrence-level strict progress). *Let a parent node be contained in a direction cap of radius T . Prescribe any $\tau = o(T)$. Along the fine-failure sequence of Theorem 8.32, choose the next physical collision scale so far out that the returned bush error R satisfies*

$$R = o(\tau).$$

Then Theorem 9.8 sends the entire incoming old-neighbor occurrence measure, with all polarization flags attached, into a disjoint sum of the following two forms:

- (i) angular children supported in direction caps of radius $O(\tau) = o(T)$;
- (ii) fixed-angle Maslov graphs with vanishing normalized error, whose physical-bush outputs are again supported in direction caps of radius $O(\tau)$.

In particular an angular fine return never has to replace the incoming occurrence by the degree measure of a newly sampled residual graph. The second requirement formerly listed in Theorem 9.33 is therefore satisfied.

Proof. The diagonal choice in (389) permits the new collision scale, and hence the bush error furnished by the arbitrary-threshold Maslov routing, to be chosen below any prescribed positive function of the fixed parent data. Choose it so that $R/\tau \rightarrow 0$, and apply Theorem 9.8. Its mass identity is linear in the absolute weights $\Lambda_{\star, Q}$, so keeping all cap pieces introduces neither a pigeonhole loss nor a cap-count factor. The two support conclusions give the asserted strict cap decrease. $\square$

Lemma 9.10 (The coarse self-collision of one bush is time-local). *Let H be an R -approximate point-probe bush centered at Reeb time s_0 , and let $z = (a, b), z' = (a', b') \in H$ satisfy $|a - a'| \geq \tau > 0$. If*

$$|\Pi_u(z) - \Pi_u(z')| \lesssim R \quad (|u - s_0| \lesssim R),$$

then their closest collision time and transverse residual obey

$$\boxed{|s_*(z, z') - s_0| \lesssim \frac{R}{\tau}, \quad |r(z, z')| \lesssim R.} \quad (427)$$

Consequently the fixed-angle two-copy graph (424) need not create a new separated Reeb probe: its bush output may be the original bush H itself.

Proof. Write $\alpha = a - a'$ and $\beta = b - b'$. At $u = s_0$, the bush relation gives $|\beta + s_0\alpha| \lesssim R$. Since

$$s_* = -\frac{\alpha \cdot \beta}{|\alpha|^2}, \quad \beta + s_0\alpha = r + (s_0 - s_*)\alpha, \quad r \perp \alpha,$$

orthogonal projection onto $\mathbb{R}\alpha$ gives $|s_* - s_0||\alpha| \lesssim R$, and projection onto $\alpha^\perp$ gives $|r| \lesssim R$. Use $|\alpha| \geq \tau$. $\square$

Lemma 9.11 (A separated old Reeb time gives a lossless fiberwise two-probe cap). *Let H be an R -approximate point-probe bush centered at (c, s_0) . Let $\mathcal{G}_{\text{sep}} \subset H \times G$ be an old physical Maslov edge measure such that*

$$s_*(z, \zeta) \in [t_0 - \delta_s, t_0 + \delta_s], \quad |t_0 - s_0| \geq g_0 > 0, \quad |r(z, \zeta)| \leq r$$

on its support. For a target endpoint ζ , let $N_\zeta \subset H$ be its source fiber. Then there is a center $a_\zeta^\#$ such that

$$N_\zeta \subset B\left(a_\zeta^\#, C_{I,U} g_0^{-1}(R + r + \delta_s)\right) \quad (428)$$

for almost every nonempty fiber. Moreover the disintegration over ζ is lossless:

$$\int \left(\int_{N_\zeta} h(z, \zeta) d\mu(z) \right) d\mu(\zeta) = \iint_{\mathcal{G}_{\text{sep}}} h(z, \zeta) d\mu(z) d\mu(\zeta). \quad (429)$$

Thus separated old time supplies a genuinely new, fiberwise direction cap while retaining the entire old-neighbor occurrence measure; no single Reeb label or target endpoint is selected.

Proof. For $z \in N_\zeta$, the collision identity and the time-packet condition give

$$|\Pi_{t_0}(z) - \Pi_{t_0}(\zeta)| \lesssim_{I,U} r + \delta_s.$$

For $z, z' \in N_\zeta$, subtract this relation and the two current bush relations

$$|\Pi_{s_0}(z) - c| \lesssim R, \quad |\Pi_{s_0}(z') - c| \lesssim R.$$

Since

$$(\Pi_{t_0}(z) - \Pi_{t_0}(z')) - (\Pi_{s_0}(z) - \Pi_{s_0}(z')) = (t_0 - s_0)(a_z - a_{z'}),$$

one obtains

$$|a_z - a_{z'}| \lesssim_{I,U} g_0^{-1}(R + r + \delta_s).$$

Choose any measurable representative of each nonempty fiber as its cap center; the usual rational grid selection makes this choice measurable. Equation (429) is Tonelli's theorem applied to the original edge occurrence measure. $\square$

Corollary 9.12 (The moving fiber caps aggregate without a continuous-label loss). *Keep Theorem 9.11 and put*

$$T := C_{I,U} g_0^{-1}(R + r + \delta_s).$$

Write a_ζ for the direction coordinate of the old target endpoint. Partition direction space by a half-open grid of side T , and assign an old edge occurrence (z, ζ) to the unique grid cell Q containing the fiber-cap center $a_\zeta^\#$. Let Γ_Q be the resulting occurrence measures. Then

$$\sum_Q \|\Gamma_Q\| = \|\mathcal{G}_{\text{sep}}\|, \quad \text{supp}(\pi_z)_\# \Gamma_Q \subset B_Q^*, \quad (430)$$

where B_Q^ is a $C_{I,U}T$ -direction cap and the family $\{B_Q^*\}_Q$ has bounded overlap. If the selector measure has total mass m and bounded three-dimensional direction density, the part for which the old target ζ also belongs to the same enlarged cap obeys*

$$\sum_Q \Gamma_Q\{(z, \zeta) : a_\zeta \in B_Q^*\} \lesssim_U m T^3. \quad (431)$$

The complementary occurrence measure is one aggregate cross-cap physical Maslov graph. Therefore the separated-time branch either reaches the cap-square threshold $T^3 \leq m r^{2-o(1)}$, or continues in caps of radius T , with its entire incoming edge mass accounted for.

Proof. Equation (428) shows that if $a_\zeta^\# \in Q$, every source direction belongs to a fixed enlargement B_Q^* of Q . The half-open assignment partitions the original occurrence measure, proving (430). Fixed enlargements of equal grid cells have bounded overlap. Since $0 \leq h \leq 1$, the internal part is bounded by

$$\sum_Q \mu(B_Q^*)^2 \leq \left(\sup_Q \mu(B_Q^*) \right) \sum_Q \mu(B_Q^*) \lesssim_U T^3 m.$$

All remaining occurrences have source and old target in different caps, so their sum, not each individual pair of caps, is the aggregate cross-cap graph. $\square$

9.3 Same-window amplification

Corollary 9.13 (Separated-window old edges satisfy strict progress). *If the old collision-time packet and the current bush center lie in the two fixed separated Reeb windows, then $g_0 \gtrsim_{I,J} 1$. Along every genuinely bad fine return,*

$$R + r + \delta_s = o(T_{\text{parent}})$$

after the time-packet width is chosen between the physical error and the parent cap scale. Hence the child radius in Theorem 9.12 satisfies

$$T = o(T_{\text{parent}}).$$

The separated-time part of Theorem 9.33 is therefore closed: its full old-neighbor occurrence mass is either internal-cap paid, aggregate-cross routed, or placed in strictly smaller child caps.

Proof. The window separation gives the lower bound for g_0 . Theorem 8.53 supplies the physical-error separation, and the weighted time-packet routing permits

$$R + r \ll \delta_s \ll T_{\text{parent}}$$

on a genuine fine shell. Substitute these estimates in the definition of T in Theorem 9.12. Its mass and internal/cross conclusions are unchanged because no time label was selected. $\square$

Lemma 9.14 (A same-window old-neighbor return is quadratic or amplifies the physical bush). *Let H be an R -approximate point-probe bush centered at (c, s_0) , let $q := \mu(H) > 0$, and suppose that on H*

$$\theta \leq d_h(z) \leq 2\theta.$$

Let $\mathcal{G}_{\text{same}} \subset H \times G$ be any part of the old physical Maslov graph on which

$$|s_*(z, \zeta) - s_0| \leq \delta_s, \quad |r(z, \zeta)| \leq r,$$

and write

$$E_{\text{same}} := \iint_{\mathcal{G}_{\text{same}}} h(z, \zeta) d\mu(z) d\mu(\zeta).$$

Let H^+ be the set of target endpoints ζ occurring in $\mathcal{G}_{\text{same}}$. Then $H \cup H^+$ is a $C_{I,U}(R + \delta_s + r)$ -approximate Lagrangian bush centered at (c, s_0) , and

$$\boxed{\mu(H^+) \geq \frac{E_{\text{same}}}{q}.} \quad (432)$$

In particular, if $E_{\text{same}} \geq c_0 \lambda_h(H)$, then for every $B \geq 1$ one has the dichotomy

$$\boxed{\theta \leq Bq \implies \lambda_h(H) \leq 2Bq^2, \quad \theta > Bq \implies \mu(H^+) \geq c_0 Bq.} \quad (433)$$

Thus a large same-window portion cannot simply repeat an unchanged small bush: it is either already quadratic in the bush mass or enlarges that same physical bush by the factor $\gtrsim \theta/q$.

Proof. For $(z, \zeta) \in \mathcal{G}_{\text{same}}$, the Pythagorean collision identity gives

$$|\Pi_{s_*(z, \zeta)}(z) - \Pi_{s_*(z, \zeta)}(\zeta)| = |r(z, \zeta)| \leq r.$$

Since the direction chart is bounded,

$$\begin{aligned} |\Pi_{s_0}(\zeta) - c| &\leq |\Pi_{s_0}(\zeta) - \Pi_{s_*(z, \zeta)}(\zeta)| + |\Pi_{s_*(z, \zeta)}(\zeta) - \Pi_{s_*(z, \zeta)}(z)| \\ &\quad + |\Pi_{s_*(z, \zeta)}(z) - \Pi_{s_0}(z)| + |\Pi_{s_0}(z) - c| \\ &\lesssim_{I,U} \delta_s + r + R. \end{aligned}$$

This is the asserted common point-probe bush relation. Since $0 \leq h \leq 1$,

$$E_{\text{same}} \leq \mu(H) \mu(H^+) = q \mu(H^+),$$

which proves (432). On the degree layer,

$$\theta q \leq \lambda_h(H) \leq 2\theta q.$$

If $\theta \leq Bq$, the upper bound gives the first alternative in (433). If $\theta > Bq$, then

$$\mu(H^+) \geq \frac{E_{\text{same}}}{q} \geq c_0 \frac{\lambda_h(H)}{q} \geq c_0 \theta > c_0 Bq,$$

which proves the second. $\square$

Corollary 9.15 (Same-window edge flow of a disjoint bush family is amplified or square-summable). *Let $\{H_j\}$ be pairwise disjoint physical bushes, put $q_j := \mu(H_j)$, and decompose each H_j into the degree layers $D_{j,k}$ of Theorem 9.3. On a layer for which the same-window part carries at least one half of $\lambda_h(D_{j,k})$, apply Theorem 9.14 with $c_0 = 1/2$. For every $B \geq 1$, either an amplified same-center bush is produced, or the total old edge flow of all non-amplified same-window layers satisfies*

$$\boxed{\mathcal{M}_{\text{same,quad}} \lesssim B \sum_{j,k} \mu(D_{j,k})^2 \leq B \sum_j q_j^2.} \quad (434)$$

Layers on which the same-window part carries less than one half send a fixed fraction of their edge flow to the separated-time branch and hence to Theorem 9.13. Thus the entire old-neighbor occurrence is partitioned into separated-time strict progress, physical bush amplification, and the single square-sum remainder in (434).

Proof. On a non-amplified same-window layer, (433) gives

$$\lambda_h(D_{j,k}) \lesssim B \mu(D_{j,k})^2.$$

The degree layers are disjoint inside each H_j , so

$$\sum_k \mu(D_{j,k})^2 \leq \left(\sum_k \mu(D_{j,k}) \right)^2 \leq q_j^2.$$

Sum in j . The last assertion is the complementary half of the same/separated time split and uses the lossless occurrence disintegration (429). $\square$

Corollary 9.16 (The same-window square sum is paid at the cap threshold). *If, in Theorem 9.15, every H_j is contained in a direction cap of radius T , then*

$$\mathcal{M}_{\text{same,quad}} \lesssim_U B m T^3, \quad m := \sum_j q_j.$$

Consequently this entire same-window remainder is $\lesssim m^2 r^{2-o(1)}$ whenever

$$T^3 \leq B^{-1} m r^{2-o(1)}.$$

Proof. Theorem 8.57 gives the cap-square estimate $\sum_j q_j^2 \lesssim_U m T^3$. Insert it into the preceding same-window square sum. $\square$

Corollary 9.17 (A same-window amplification chain has sublogarithmic depth). *Let $r \downarrow 0$, put*

$$B(r) := \exp \sqrt{\log(1/r)} = r^{-o(1)},$$

and suppose the initial bush mass satisfies $q_0 \geq r^{C_0}$ for one fixed $C_0 < \infty$. Iterate only the amplified alternative of (433), with a fixed retained fraction $c_0 > 0$. Then either a quadratic alternative

$$\lambda_h(H_\ell) \lesssim B(r) q_\ell^2$$

occurs, or after at most

$$\boxed{L_{\text{amp}} \leq \frac{\log(1/q_0) + O(1)}{\log(c_0 B(r))} = O_{C_0, c_0} \left(\sqrt{\log(1/r)} \right) = o(\log(1/r))} \quad (435)$$

steps the physical bush has direction mass bounded below by a fixed positive constant. If the bush errors tend to zero, the latter alternative gives $\dim_H K_b = 4$. Hence, under the contrary dimension hypothesis, every polynomial-mass same-window chain reaches its quadratic branch in sublogarithmic depth and incurs only subpower fixed-constant loss.

Proof. At every amplified step,

$$q_{\ell+1} \geq c_0 B(r) q_\ell.$$

Since $q_\ell \leq 1$, iteration gives

$$(c_0 B(r))^\ell q_0 \leq 1$$

until a fixed positive mass is reached. Taking logarithms and using $\log B(r) = \sqrt{\log(1/r)}$ proves (435).

For a bush of direction mass $q_\ell \geq q_* > 0$, its normalized direction measure has density at most C_U/q_* . It therefore satisfies the uniform direction-Frostman estimate of Theorem 8.10 for every $0 < \delta < 3$. Along vanishing bush errors that theorem gives $(4 - \delta)$ -Frostman measures on K_b for every $\delta > 0$, hence $\dim_H K_b = 4$. $\square$

9.4 Common-target Markovization

Lemma 9.18 (Amplified same-window targets either grow or form a weighted common-target star). *Let $\{S_i\}$ be pairwise disjoint source pieces of selector masses $q_i := \mu(S_i) > 0$, and let $\Gamma_i \leq \mu|_{S_i} \otimes \mu$ be same-window old-neighbor occurrence measures. Write*

$$e_i(\zeta) := \frac{d(\pi_\zeta)_\# \Gamma_i}{d\mu}(\zeta), \quad w_i(\zeta) := \frac{e_i(\zeta)}{q_i}, \quad n(\zeta) := \sum_i w_i(\zeta).$$

Then $0 \leq w_i \leq 1$. Suppose the amplified alternative of Theorem 9.14 holds on every S_i , so that

$$\int w_i d\mu = \frac{\|\Gamma_i\|}{q_i} \geq c_0 B q_i. \quad (436)$$

Put $Q := \sum_i q_i$ and $H^+ := \bigcup_i \{w_i > 0\}$. For all sufficiently large B , at least one of the following two quantitative conclusions is available:

(i) the target union genuinely amplifies the available selector mass,

$$\mu(H^+) \geq \frac{c_0}{2} B^{1/2} Q; \quad (437)$$

(ii) the off-diagonal weighted common-target content satisfies

$$\boxed{\sum_{i \neq j} \int w_i(\zeta) w_j(\zeta) d\mu(\zeta) \gtrsim_{c_0} B^{3/2} Q.} \quad (438)$$

The second quantity is an actual common-old-target occurrence, not an overlap of unweighted supports. More precisely, where $e_i(\zeta) > 0$, define

$$d\Gamma_i(z, \zeta) = e_i(\zeta) dK_{i,\zeta}(z) d\mu(\zeta).$$

Then $K_{i,\zeta}$ is a probability kernel and every summand in (438) lifts losslessly to

$$w_i(\zeta) w_j(\zeta) d\mu(\zeta) dK_{i,\zeta}(z_i) dK_{j,\zeta}(z_j). \quad (439)$$

All old polarization flags over z_i, z_j may be tensored into this lift.

If S_i is an R_i -bush centered at (c_i, s_i) , and the i -th occurrence has collision time within δ_i of s_i and residual at most r , then every common target $\zeta = (a_\zeta, b_\zeta)$ in (439) satisfies

$$|c_i - (b_\zeta + s_i a_\zeta)| \lesssim_{I,U} R_i + \delta_i + r. \quad (440)$$

Thus the large-overlap alternative is precisely a weighted Reeb carrier-line star through actual selector line ζ .

Proof. Because $\Gamma_i \leq \mu|_{S_i} \otimes \mu$, one has

$$0 \leq e_i(\zeta) \leq \mu(S_i) = q_i,$$

and hence $0 \leq w_i \leq 1$. Summing (436) gives

$$W := \int n d\mu \geq c_0 B Q.$$

If (437) fails, Cauchy–Schwarz on H^+ gives

$$\int n^2 d\mu \geq \frac{W^2}{\mu(H^+)} > 2B^{1/2} W.$$

On the other hand $w_i^2 \leq w_i$, so

$$\sum_i \int w_i^2 d\mu \leq W.$$

Subtracting this diagonal term from $\int n^2 d\mu$, and then using $W \geq c_0 B Q$, proves the asserted common-star bound. The definition of $K_{i,\zeta}$ is the disintegration of the actual i -th occurrence; integrating the two probability kernels in the displayed two-star lift returns its weight. Conditional flag laws again integrate to one.

Finally, the same-window estimate in the proof of Theorem 9.14 applies to every $(z_i, \zeta) \in \text{supp } \Gamma_i$ and gives

$$|\Pi_{s_i}(\zeta) - c_i| \lesssim_{I,U} R_i + \delta_i + r.$$

Since $\Pi_{s_i}(\zeta) = b_\zeta + s_i a_\zeta$, this is (440). $\square$

Corollary 9.19 (High common-target multiplicity has a linear different-source Markovization). *In Theorem 9.18, restrict to one dyadic source-mass layer*

$$q \leq q_i < 2q.$$

Let

$$e(\zeta) := \sum_i e_i(\zeta), \quad Y := \{\zeta : n(\zeta) \geq B^{1/2}\}.$$

If the union-growth alternative (437) fails, then

$$\boxed{\sum_i \Gamma_i(S_i \times Y) \geq \frac{1}{4} \sum_i \|\Gamma_i\|.} \quad (441)$$

For $\zeta \in Y$, first disintegrate the incoming i -th edge by $K_{i,\zeta}$, and then choose a second source index and endpoint by

$$P_\zeta(j, dz_j) := \frac{e_j(\zeta)}{e(\zeta)} dK_{j,\zeta}(z_j). \quad (442)$$

This is a probability kernel and, conditionally on an incoming edge from S_i ,

$$P_\zeta\{j = i\} \leq \frac{2}{n(\zeta)} \leq 2B^{-1/2}. \quad (443)$$

Consequently, after discarding the event $j = i$, the measure

$$\boxed{\sum_i \mathbf{1}_Y(\zeta) d\Gamma_i(z_i, \zeta) \sum_{j \neq i} \frac{e_j(\zeta)}{e(\zeta)} dK_{j,\zeta}(z_j)} \quad (444)$$

has mass at least a fixed positive fraction of $\sum_i \|\Gamma_i\|$, for all sufficiently large B . Thus the common-target star is linear in the incoming old-neighbor occurrence mass. Both source endpoints retain independent conditional old-polarization flags, and both of their bush centers satisfy (440) for the same actual selector line ζ .

Proof. On the dyadic layer,

$$qn(\zeta) \leq e(\zeta) < 2qn(\zeta).$$

Failure of (437), together with $W = \int n d\mu \geq c_0 BQ$, gives

$$\int_{\{n < B^{1/2}\}} n d\mu \leq B^{1/2} \mu(H^+) < \frac{c_0}{2} BQ \leq \frac{W}{2}.$$

Hence $\int_Y n d\mu \geq W/2$. Since

$$\sum_i \|\Gamma_i\| = \int e d\mu < 2qW, \quad \int_Y e d\mu \geq q \int_Y n d\mu \geq \frac{qW}{2},$$

one obtains (441). Equation (442) has total mass one. Moreover

$$\frac{e_i(\zeta)}{e(\zeta)} \leq \frac{q_i}{qn(\zeta)} < \frac{2}{n(\zeta)},$$

which proves (443). Integrate the two probability kernels and combine the $1/4$ retained high-target fraction with $1 - 2B^{-1/2}$. Tensoring conditional flag laws changes none of these masses. $\square$

Corollary 9.20 (A fixed number of distinct source flags can be sampled linearly). *Keep Theorem 9.19. For every fixed integer $k \geq 2$, extend each incoming edge over Y by $k - 1$ conditionally independent samples from P_ζ . The probability that the original source index and the $k - 1$ new source indices are not all distinct is at most*

$$\binom{k}{2} 2B^{-1/2}. \quad (445)$$

After restricting to distinct indices, the resulting k -source star therefore retains a fixed fraction of the original incoming old-neighbor occurrence mass. In particular $k = 4$ gives, over one actual selector target ζ , four distinct physical source bushes together with four actual old collision times and four independently retained old polarization flags. This four-flag star is obtained by Markov extension of the original occurrence; it incurs no fourth power of its edge mass.

Proof. At every $\zeta \in Y$, each source-index atom of P_ζ is at most $2B^{-1/2}$ by the different-source probability bound. For each of the $\binom{k}{2}$ pairs of samples, condition on the first member of the pair and use this maximal atom bound for the second. The union bound proves the asserted distinct-source loss. Integrating the $k-1$ probability kernels leaves the incoming occurrence mass unchanged before the collision event is discarded. Old flag laws are additional conditional probability kernels and hence are retained as well. $\square$

9.5 Flagged common-target routing

Lemma 9.21 (Every nondegenerate Maslov bush may retain a lossless origin flag). *In the determinant-nondegenerate branch of Theorem 7.8, write the collision thickness as ϵ_{col} , the horizontal determinant threshold as Δ , and*

$$R_{\text{org}} := C_{U,J}\epsilon_{\text{col}}/\Delta.$$

The physical bush may be chosen so that every one of its source endpoints z carries a conditional origin witness

$$\mathbf{w} = (p, q, z^\dagger, s, V, \mathbf{q}), \quad V \in \text{Gr}(3, 6), \quad \mathbf{q} = (\alpha, \beta) \in S^5,$$

with the following properties:

$$\boxed{\text{dist}(\mathbf{q}, V) = 0, \quad |\beta + s\alpha| \lesssim \epsilon_{\text{col}}/\tau, \quad |s - \bar{s}| \lesssim R_{\text{org}}.} \quad (446)$$

Here τ is the retained fixed-angle lower bound, $\bar{s}$ is the bush-center Reeb time, $(p, z, q, z^\dagger, p)$ is an actual determinant-nondegenerate physical Maslov four-cycle, and

$$V := \text{Span}\{\mathbf{z}_z - \mathbf{z}_p, \mathbf{z}_q - \mathbf{z}_p, \mathbf{z}_{z^\dagger} - \mathbf{z}_p\}, \quad \mathbf{q} := \frac{\mathbf{z}_z - \mathbf{z}_p}{|\mathbf{z}_z - \mathbf{z}_p|},$$

with $\mathbf{z}_x := (a_x, b_x)$. The origin-witness law is a probability kernel over z ; therefore attaching it changes neither the physical bush measure nor any later restriction of that measure.

Proof. Use the notation in the proof of Theorems 6.4 and 7.8. For the selected anchors p, q , let

$$d\lambda_{p,q}(z) = h(p, z)h(q, z) d\nu(z), \quad K(p, q) = \|\lambda_{p,q}\|,$$

and let $\mathcal{G}_{p,q}$ be the determinant-nondegenerate set of ordered common-neighbor pairs $(z, z^\dagger)$. The chosen anchor pair satisfies

$$\frac{(\lambda_{p,q} \otimes \lambda_{p,q})(\mathcal{G}_{p,q})}{K(p, q)} \gtrsim M^3.$$

Partition the selected edge label s_{pz} into half-open intervals I of length comparable to R_{org} , and let A_I be the set of z in the I -packet which have positive $\mathcal{G}_{p,q}$ -degree. Cramer's rule gives

$$|s_{pz} - s_{pz^\dagger}| \lesssim R_{\text{org}} \quad ((z, z^\dagger) \in \mathcal{G}_{p,q}),$$

so a good pair meets only $O(1)$ neighboring time packets. If I^* denotes the union of those neighbors, then

$$\begin{aligned} (\lambda_{p,q} \otimes \lambda_{p,q})(\mathcal{G}_{p,q}) &= \sum_I \int_{A_I} \int \mathbf{1}_{\mathcal{G}_{p,q}}(z, z^\dagger) d\lambda_{p,q}(z^\dagger) d\lambda_{p,q}(z) \\ &\leq \sum_I \lambda_{p,q}(A_I) \lambda_{p,q}(I^*) \\ &\lesssim K(p, q) \sup_I \lambda_{p,q}(A_I). \end{aligned}$$

Choose I attaining the displayed lower bound and call its center $\bar{s}$. Then $\lambda_{p,q}(A_I) \gtrsim M^3$; since $h \leq 1$, also $\nu(A_I) \gtrsim M^3$. Every $z \in A_I$ has a good partner $z^\dagger$. Disintegrating the restricted good-pair measure in $z^\dagger$, and making an arbitrary measurable choice on a null set, gives the asserted probability kernel of witnesses. Thus $H := A_I$ has exactly the same mass lower bound as the usual bush output and every point of H is origin-marked.

The horizontal determinant condition implies that the three displayed phase-space differences are linearly independent, so $V \in \text{Gr}(3, 6)$, and the definition gives $\mathbf{q} \in V$. It also gives $|a_z - a_p| \gtrsim_U \Delta$, so one may take $\tau \gtrsim_U \Delta$; on the already separated fixed-angle branch use its sharper retained τ . Hence $|\mathbf{z}_z - \mathbf{z}_p| \gtrsim \tau$. Dividing the actual contact-edge estimate

$$|(b_z - b_p) + s_{pz}(a_z - a_p)| \lesssim \epsilon_{\text{col}}$$

by this norm proves the second assertion in (446). Finally $s = s_{pz} \in I$, so $|s - \bar{s}| \lesssim R_{\text{org}}$, and the contact edge pz gives

$$|\Pi_{\bar{s}}(z) - \Pi_{\bar{s}}(p)| \leq |\Pi_s(z) - \Pi_s(p)| + |s - \bar{s}| |a_z - a_p| \lesssim_{U,J} R_{\text{org}}.$$

Thus A_I is the asserted point-probe bush centered at $(\Pi_{\bar{s}}(p), \bar{s})$. Integrating the conditional partner law first proves the physical-marginal assertion. $\square$

Lemma 9.22 (An origin-marked bush plane stays quantitatively outside the front-null locus). *For the origin plane V in Theorem 9.21, suppose the phase-space coordinates remain in the fixed bounded chart and the generating horizontal determinant is at least Δ . Then*

$$\boxed{\text{dist}_{\text{Gr}}(V, \mathfrak{N}) \gtrsim_U \Delta, \quad \mathfrak{N} = \{W \in \text{Gr}(3, 6) : P_W \equiv 0\}.} \quad (447)$$

Proof. Let D be the 6×3 matrix whose columns are the three phase-space differences defining V , and let D_a be its horizontal 3×3 block. The horizontal nondegeneracy gives

$$|\det D_a| \geq \Delta.$$

If $Q = D(D^\top D)^{-1/2}$ is an orthonormal frame for V , then its horizontal frame Q_a satisfies

$$|\det Q_a| = \frac{|\det D_a|}{\sqrt{\det(D^\top D)}} \gtrsim_U \Delta,$$

because the three phase-space differences are uniformly bounded. The distance of Q_a from the rank-at-most-two matrices is its least singular value, which is at least $|\det Q_a|$ since the other two singular values are at most one. Every $W \in \mathfrak{N}$ has $\text{rank}(\pi_a|_W) \leq 2$ by (407). The Grassmannian distance controls the distance between orthonormal horizontal frame matrices up to a fixed factor, proving (447). $\square$

Corollary 9.23 (The linear four-source star has the carrier-rooted polarized lift). *Let $\Gamma^{(4)}$ be the linear four-source common-target occurrence of Theorem 9.20. Retain on each source bush the origin flag of Theorem 9.21. Choose two Reeb resolutions*

$$\max_i \{R_{\text{org},i}, \epsilon_{\text{col},i}/\tau_i\} \ll \delta_{\text{edge}} \ll \delta_s \ll \tau.$$

On the same-window part

$$|s_*(z_i, \zeta) - \bar{s}_i| \leq \delta_{\text{edge}},$$

the fourfold Markov extension carries actual triples

$$(\mathbf{q}_i, V_i, s_i), \quad \mathbf{q}_i = (\alpha_i, \beta_i), \quad 0 \leq i \leq 3,$$

such that

$$\boxed{\text{dist}(\mathbf{q}_i, V_i) = 0, \quad |\beta_i + s_i \alpha_i| = o(\delta_s), \quad |s_i - s_*(z_i, \zeta)| = o(\delta_s).} \quad (448)$$

The lift preserves the physical four-source occurrence exactly. Hence the four Reeb roots used by Theorem 8.42 are the origin-cycle roots of the actual source bushes and are synchronized with the common selector carrier; no independently sampled target flag is substituted for them.

Proof. In the maximal-bush construction Theorem 8.8, use the origin-marked refinement Theorem 9.21 whenever the arbitrary-threshold routing produces a physical bush. The proof of maximality is unchanged because the refined bush has the same $\gtrsim M^3$ selector-mass lower bound. Restrictions to maximal bushes, degree layers, adaptive source caps, and same-window old edges preserve the mark: each is a physical restriction, while the later Reeb and common-target extensions are probability kernels. The same applies to every descendant after its next fine residual failure produces a fresh physical Maslov bush.

For the i -th marked source, (446) gives

$$\text{dist}(\mathbf{q}_i, V_i) = 0, \quad |\beta_i + s_i \alpha_i| \lesssim \epsilon_{\text{col},i}/\tau_i, \quad |s_i - \bar{s}_i| \lesssim R_{\text{org},i}.$$

The defining same-window restriction gives

$$|s_*(z_i, \zeta) - \bar{s}_i| \leq \delta_{\text{edge}}.$$

The triangle inequality and the displayed hierarchy prove (448). Finally, tensoring the four conditional origin-witness kernels with the Markov extension integrates to one in the new variables, so its physical marginal is exactly $\Gamma^{(4)}$. $\square$

Lemma 9.24 (Separated origin polarizations are routed at the source endpoint). *Keep one dyadic source-mass layer of Theorem 9.19, retain the origin marks from Theorem 9.23, and also lift every source endpoint z by its actual conditional old flags. Suppose a positive portion of the retained common-target occurrence has two source origin planes lying in Grassmannian cells $\mathcal{V}_1, \mathcal{V}_2 \subset \text{Gr}(3, 6)$ with*

$$\text{dist}(\mathcal{V}_1, \mathcal{V}_2) \geq \vartheta, \quad \vartheta^{-1} = r^{-o(1)}.$$

Then that portion admits an occurrence-preserving sourcewise routing into:

- (i) *transverse old/new polarization, close-anchor, horizontal-line, or coherent rank-loss outputs already assigned to the paid ledgers;*
- (ii) *angular fine-return or physical Lagrangian-bush children retaining their inherited source caps and all common-target marks.*

No flag is moved from a source endpoint to the common target, and no power of the incoming occurrence mass is lost.

Proof. Disintegrate the four-source extension over its distinguished source occurrence. By symmetry among the four source samples, on a fixed fraction of the separated-plane event the distinguished source carries one member of a separated pair. Keep its origin plane, the other three sources, and the common target as conditional probability marks. The resulting physical source marginal is therefore a submeasure of the original incoming occurrence; all such marginals add to at most that occurrence.

On every source bush and degree layer apply Theorem 9.4. Its pointwise angular/fixed split preserves the whole restricted source occurrence, including the three extra source marks and the common target. On the fixed part the symmetric two-copy graph has edge mass $1/4$ by (424). Apply Theorems 8.5 and 9.5 with the source copy as the decorated free side. The old flag has therefore remained at the same source endpoint as the current cycle plane. A transverse plane is paid by Theorems 7.78 and 8.46; the close-anchor/angular, horizontal-line, and coherent rank-loss outputs have their established destinations. The remaining output is the physical source-side Lagrangian bush of Theorem 8.7. It retains the inherited cap and all conditional common-target marks. The angular part is sent to the smaller-cap child of Theorem 9.9.

Every operation above is either a restriction of the distinguished source occurrence or an extension by a probability kernel. Summing the source bushes, degree layers, and polarization cells is therefore linear. The fixed edge mass $1/4$ is taken only after normalizing each restricted source piece, so no power of its absolute mass is introduced. $\square$

Corollary 9.25 (The linear four-source star reaches the polarization/time trichotomy). *On the four-source occurrence from Theorem 9.20, use the carrier-rooted lift of Theorem 9.23, and fix a polarization resolution $\vartheta \downarrow 0$ and a Reeb-time resolution $\delta_s \downarrow 0$. The origin cycles use the same determinant threshold Δ at one routed generation; equivalently first retain its dyadic layer. Fix*

$$\eta_{\text{rk}} < c_U \Delta / 2$$

with the constant from Theorem 9.22, and choose the next fine failure scale so that the normalized incidence error and the $O(\vartheta)$ -coherence diameter are below

$$\varepsilon_0(\delta_s, \eta_{\text{rk}}, I^\sharp)$$

from Theorem 8.43. Up to a subpower polarization-cell decomposition, every four-source occurrence belongs to one of:

- (i) *two of its origin Maslov planes V_i are ϑ -separated;*
- (ii) *all four origin planes are $O(\vartheta)$ -coherent and their four origin Reeb times contain four pairwise δ_s -separated values;*
- (iii) *all four origin planes are $O(\vartheta)$ -coherent and their origin Reeb times lie in at most three intervals of length $O(\delta_s)$.*

By Theorem 9.24, the first alternative is routed at its actual source endpoints into paid outputs or source-side physical bush children, with no power loss in the incoming occurrence. In the second alternative the four origin-cycle Maslov incidences, synchronized with the same selector target, give the Vandermonde input of Theorems 8.42 and 8.43. That corollary places every V_i within η_{rk} of $\mathfrak{N}$, contradicting Theorem 9.22; hence this branch is empty. In the third alternative, at least two source bushes belong to the same shrinking Reeb-time packet and, by (440), merge at one physical point of the common selector line up to error $O(R + \delta_{\text{edge}} + \delta_s + r)$. Consequently the final gate is reduced to the lossless occurrence-level aggregation of the merged packets in (iii); both the transverse branch and the four-separated-time coherent branch are already routed.

Proof. Cover the compact polarization space by a bounded-overlap ϑ -net. The number of cells is subpower for the usual choice $\vartheta^{-1} = r^{-o(1)}$, and all cells are retained in the absolute occurrence sum. Either two retained cell labels are separated or all four lie in one fixed enlargement of a single cell. In the latter case, four real numbers are either pairwise δ_s -separated or are covered by at most three intervals of length $2\delta_s$. This proves the trichotomy without selecting one label cell or one time packet.

For (i), apply Theorem 9.24. For (ii), apply the finite-scale four-time firewall and then Theorem 9.22, using the parameter hierarchy in the statement. In the last alternative choose a time interval containing two source labels. Their common target ζ satisfies (440) for both centers; replacing their two times by the interval representative changes the front point by $O_U(\delta_s)$. The union of their disjoint source subsets is therefore an $O(R + \delta_s + r)$ -point-probe bush at that representative point. $\square$

Lemma 9.26 (A four-sample Reeb test gives a lossless three-packet stopping). *Let*

$$d\Gamma(z, \zeta) = e(\zeta) dP_\zeta(z) d\mu(\zeta)$$

be a finite common-target occurrence measure, where P_ζ is a probability kernel and every source occurrence has a measurable Reeb label $s(z, \zeta) \in I^\sharp$. Fix $\delta_s > 0$. At each target put

$$\mathfrak{q}(\zeta) := P_\zeta^{\otimes 4} \{(z_0, z_1, z_2, z_3) : |s(z_i, \zeta) - s(z_j, \zeta)| > \delta_s \text{ for all } i \neq j\}.$$

There is a measurable stopping decomposition

$$\Gamma = \Gamma_{\text{four}} + \sum_{\ell \geq 0} \sum_{m=1}^3 \Gamma_{\ell, m} \quad (449)$$

up to a null limiting tail, with the following properties.

- (i) *Over every part assigned to Γ_{four} , the four-fold Markov extension contains a pairwise δ_s -separated four-time tuple with conditional probability at least $1/8$.*
- (ii) *For every (ℓ, m) , there is a measurable interval center $t_{\ell, m}(\zeta)$ such that*

$$|s(z, \zeta) - t_{\ell, m}(\zeta)| \leq 2\delta_s \quad \text{on } \text{supp } \Gamma_{\ell, m}.$$

- (iii) *After L stopping rounds, the as-yet unassigned occurrence mass is at most*

$$2^{-L} \|\Gamma\|.$$

Thus a common-target occurrence is exhausted linearly by the four-separated-time output and at most three coherent Reeb packets per stopping round. No target and no time packet is selected.

Proof. We first use the following elementary probability fact. If ν is a probability measure on a real interval and

$$\nu^{\otimes 4} \{|s_i - s_j| > \delta_s \ (i \neq j)\} < \frac{1}{8},$$

then three intervals of radius δ_s cover at least one half of ν . Indeed, if no three such intervals do so, then for every choice of at most three centers the complement of their union has mass greater than $1/2$. Draw four independent samples successively. After the first sample, each of the next three has conditional probability greater than $1/2$ of lying outside the δ_s -neighborhoods of all previous samples. The probability of four pairwise separated samples is therefore greater than $2^{-3} = 1/8$, a contradiction.

Apply this fact to the pushforward of P_ζ by $s(\cdot, \zeta)$. On $\{\mathfrak{q} \geq 1/8\}$, put the current occurrence into Γ_{four} . On the complement, first take the three radius- δ_s intervals supplied by the probability fact. Approximating their centers by a fixed countable rational grid enlarges them to radius $2\delta_s$. Choose the lexicographically first such rational triple whose union has conditional mass at least $1/2$, assign each covered source to the first interval containing its time, and remove the three resulting packet measures. This choice is measurable. Repeat the construction with the conditional remainder. Each low-four-time round removes at least half of the remaining occurrence mass, while every high-four-time part is assigned to Γ_{four} . The remainder after L rounds is therefore at most $2^{-L} \|\Gamma\|$; continuity from above gives a null limiting tail and (449). $\square$

Lemma 9.27 (Target cells turn synchronized Reeb packets into a fractional physical bush family). *Let one of the packet measures $\Gamma_{\ell,m}$ from Theorem 9.26 be supported on physical old edges satisfying*

$$|\Pi_u(z) - \Pi_u(\zeta)| \leq \rho, \quad |s - u| \leq \varepsilon_{\text{sync}}, \quad |s - t_{\ell,m}(\zeta)| \leq 2\delta_s.$$

Partition the bounded selector line chart of the target $\zeta = (a_\zeta, b_\zeta)$ into half-open cells C of diameter η , and quantize $t_{\ell,m}(\zeta)$ into half-open intervals of length δ_s . Assign every occurrence to its unique pair (C, J) , and let $\Gamma_{\ell,m,C,J}$ be the resulting restrictions. Then

$$\boxed{\sum_{C,J} \|\Gamma_{\ell,m,C,J}\| = \|\Gamma_{\ell,m}\|.} \quad (450)$$

For representatives $\zeta_C \in C$ and $t_J \in J$, the source marginal of every nonnull piece is supported on the physical point-probe bush

$$\boxed{|\Pi_{t_J}(z) - \Pi_{t_J}(\zeta_C)| \lesssim_{I,U} \rho + \varepsilon_{\text{sync}} + \delta_s + \eta.} \quad (451)$$

If the incoming source was already restricted by Theorem 9.8 to a direction cap of radius τ , every such bush remains in that same τ -cap. Moreover the sum of all source marginals in (450) is exactly the source marginal of $\Gamma_{\ell,m}$, even when their physical supports overlap. Hence this is a fractional occurrence partition, not an unweighted family requiring a choice of one target cell.

Proof. The half-open target cells and time intervals give a measurable partition, proving (450). For an occurrence assigned to (C, J) , write $s = s(z, \zeta)$ and let u be its actual old-edge collision time. Then

$$\begin{aligned} |\Pi_{t_J}(z) - \Pi_{t_J}(\zeta_C)| &\leq |\Pi_u(z) - \Pi_u(\zeta)| + |s - u| |a_z - a_\zeta| \\ &\quad + |t_J - s| |a_z - a_\zeta| \\ &\quad + |\Pi_{t_J}(\zeta) - \Pi_{t_J}(\zeta_C)| \\ &\lesssim_{I,U} \rho + \varepsilon_{\text{sync}} + \delta_s + \eta. \end{aligned}$$

Here the direction chart and the relevant Reeb interval are bounded, and the target cell controls both a_ζ and b_ζ . This proves (451). Direction-cap support is inherited from the source restriction. Finally, pushforward commutes with the disjoint sum of the occurrence restrictions, which proves the last assertion. $\square$

Corollary 9.28 (The same-window selector-carrier gate closes at occurrence level). *Let r_0 be the fixed root collision scale in the desired edge-Carleson estimate, and let ρ be the later fine failure scale which produces the current physical bush. Put*

$$B_0 := B(r_0) = \exp \sqrt{\log(1/r_0)}.$$

Theorem 9.14 is used here with its old-edge time-packet width renamed δ_{edge} ; the larger symbol δ_s is reserved for the four-sample Maslov/Reeb stopping. Theorem 9.23 supplies a synchronization error

$$\varepsilon_{\text{sync}} := \max_i |s_i - s_*(z_i, \zeta)| = o(\delta_s).$$

Apply the adaptive source-cap disintegration at a radius $\tau = o(T_{\text{parent}})$, with the physical collision error chosen $o(\tau)$. For every polynomial-mass same-window source family, its entire incoming old-neighbor occurrence, up to an $O(r_0^A)$ tail for arbitrary fixed A , is a disjoint absolute sum of:

(i) *quadratic cap pieces of total mass*

$$\lesssim_U B_0 m \tau^3;$$

(ii) *transverse-polarization, excluded four-separated-time, angular, horizontal-line, or front-nondegenerate pieces already assigned to the paid ledgers;*

(iii) *physical point-probe bush children of error*

$$O(\rho + \varepsilon_{\text{sync}} + \delta_s + \eta) = o(\tau)$$

whose source directions remain in $O(\tau)$ -caps and whose occurrence masses add linearly to the unpaid input.

If

$$\tau^3 \leq B_0^{-1} m r_0^{2-o(1)},$$

the first part is at the relative quadratic target. Hence every unpaid same-window occurrence makes strict physical/cap progress.

Proof. In every local lemma used below, relabel its collision thickness r as the present fine scale ρ . First choose τ from the root-scale cap-square target, and then use Theorem 8.32 to choose ρ so far out on the failure sequence that all physical and synchronization errors are $o(\tau)$.

The non-amplified part is Theorems 9.15 and 9.16, because Theorem 9.8 places every oriented source bush in an $O(\tau)$ -cap. This gives (i).

For the amplified part apply Theorem 9.18. In its union-growth alternative, reverse the old edges and use their target marginal. Since the source pieces are disjoint,

$$\sum_i e_i(\zeta) \leq \sum_i q_i \leq 1,$$

so the reversed edge-flow marginal is still dominated by μ , and its total occurrence mass is unchanged. Degree-layering therefore restarts the same physical-bush/Frostman alternative without a power loss. If union growth repeats L times from initial selector mass q_0 , then

$$\left(cB_0^{1/2}\right)^L q_0 \leq 1.$$

For $q_0 \geq r_0^{C_0}$, taking logarithms gives

$$L \lesssim_{C_0} \frac{\log(1/r_0)}{\log B_0} = O_{C_0} \left(\sqrt{\log(1/r_0)} \right).$$

A fixed positive-mass vanishing-error bush family gives the vector-Frostman output by Theorems 8.11 and 9.34 and is recorded in the paid ledger. Otherwise this branch terminates in sublogarithmic depth.

In the common-target alternative, first dyadically layer the source masses; the number of layers is subpower after Theorem 9.7. Theorem 9.19 retains a fixed fraction of the incoming edge mass in a different-source Markov star, and Theorem 9.20 supplies four distinct source flags without taking an edge-mass power. Separated source polarizations are routed at their actual source endpoints by Theorem 9.24. On the coherent part, the carrier-rooted coupling Theorem 9.23 synchronizes the old-edge times with the Maslov planes, and Theorem 9.25 excludes the four-separated-time output: the finite-scale Reeb-front firewall would place every origin plane near the front-null locus, contrary to its determinant-nondegenerate origin cycle.

Apply Theorem 9.26 to the remaining conditional Reeb-time law. Choose $L_{\text{pkt}} = \lceil A \log_2(1/r_0) \rceil$. Its unassigned tail is at most r_0^A times the incoming occurrence, while all removed mass is contained in three time packets per stopping round. This internal stopping has exact mass addition and introduces no factor $C^{L_{\text{pkt}}}$. Finally apply Theorem 9.27 with

$$\rho + \varepsilon_{\text{sync}} + \delta_s + \eta = o(\tau).$$

Every retained packet becomes the physical child in (iii), remains in its inherited $O(\tau)$ -source cap, and all target cells and all three packets are summed rather than selected. This exhausts the common-target alternative and proves the corollary. $\square$

9.6 Cross-generation conservation and deterministic saturation

Proposition 9.29 (Cross-generation edge-flow conservation). *In the bush-return tree of Theorem 9.1, retain the incoming edge-flow disintegration (419) at every nonterminal node. Route every extracted physical bush by the two pointwise branches and the two-copy graph of Theorem 9.4. Assign transverse polarization, horizontal, front-nondegenerate, angular-good, and vector-Frostman outputs to $\mathcal{M}_{\text{paid}}$; assign every occurrence whose old neighbor leaves the current child cap to $\mathcal{M}_{\text{cross}}$. The strict progress needed for this routing is supplied by Theorem 9.13 on separated old-time windows, Theorem 9.9 on angular fine returns, and Theorem 9.28 on the same-window selector-carrier branch. Then, for every fixed $A > 2$ and every finite depth L , the routing may be exhausted so that*

$$\boxed{\mathcal{M}_{\text{in}}(r) \leq \mathcal{M}_{\text{paid}}(r) + \mathcal{M}_{\text{cross}}(r) + \sum_{\gamma \in \mathcal{S}_L} \mathcal{M}_{\gamma}^{\text{same}}(r) + O_A(Lr^A)\mathcal{M}_{\text{in}}(r),} \quad (452)$$

where $\mathcal{M}_{\gamma}^{\text{same}}$ is the terminal old-edge flow whose two endpoints lie in the cap γ . If the terminal source marginal is $d\nu_{\gamma} = f_{\gamma}d\sigma$, has mass p_{γ} , and is supported in a cap of radius T_{γ} , then

$$\boxed{\mathcal{M}_{\gamma}^{\text{same}}(r) \lesssim_U p_{\gamma} T_{\gamma}^3, \quad \sum_{\gamma \in \mathcal{S}_L} f_{\gamma} \leq f_{\text{root}.}} \quad (453)$$

Consequently (417) holds for every $L = r^{-o(1)}$; no factor C^L occurs.

Proof. Represent the node by its ordered old-edge occurrence measure

$$d\Gamma_v(z, \zeta) := h_v(z, \zeta) d\mu_v(z) d\mu_v(\zeta) = d\lambda_{h_v}(z) d\kappa_z^{h_v}(\zeta).$$

The first coordinate is now the source once and for all; no later orientation factor is needed. On every nonzero unused restriction of Γ_v , maximal bush extraction and Theorem 9.2 assign a uniform positive fraction either to a paid output or to disjoint physical bushes. Theorem 8.56 therefore exhausts Γ_v by countably many such pieces. Their source marginals add to λ_{h_v} , and the old endpoint ζ remains the conditional mark $\kappa_z^{h_v}$ throughout.

On every bush piece use the degree partition of Theorem 9.3. The normalized low-degree remainder costs at most r^A times the incoming absolute node weight. Every other layer satisfies (421). Theorem 9.6 and, more precisely, the pointwise angular/fixed partition in Theorem 9.4, transport its entire remaining occurrence mass through actual $O(R)$ -Reeb collisions. No unweighted square of its selector mass is taken.

Before routing either source branch, apply the adaptive source-cap disintegration Theorem 9.8, with a cap radius prescribed below the parent radius and the bush error chosen still smaller. The cap pieces retain the whole absolute occurrence mass and all old marks. For a fixed-angle source piece, the symmetric two-copy graph (424) has edge mass $1/4$. Apply Theorems 8.5 and 9.5 with the target copy as the anchor side and the source copy as the free/common-neighbor side. Its close-anchor, line, and transverse-polarization outputs are paid. Split the retained old-neighbor time relative to the current bush center. The separated-time part enters a genuinely smaller alternating-window child by Theorem 9.13. The same-window part is partitioned, with its absolute occurrence mass unchanged, into paid outputs or genuinely smaller-cap physical children by Theorem 9.28; this includes the amplified common-target branch. The old neighbor, every old polarization, and the origin-cycle marks remain attached by Theorems 8.3, 9.21 and 9.23. Every fixed-proportion output of this routing is exhausted on the unused source occurrence by Theorem 8.56; the two-copy edge mass remains $1/4$ on each remainder. The coherent rank-loss alternative is excluded by the same finite-flag argument because the graph edge mass is fixed. For an angular source piece, Theorem 9.9 places both copies directly in a strictly smaller cap; at a further failure scale it repeats the same lossless source-cap disintegration instead of sampling an unrelated fine residual graph. Consequently every nonterminal node has the absolute identity

$$\mathbf{m}_v = \mathbf{m}_{v,\text{paid}} + \mathbf{m}_{v,\text{cross}} + \sum_{w \in \text{ch}(v)} \mathbf{m}_w + \mathbf{e}_v, \quad 0 \leq \mathbf{e}_v \lesssim_A r^A \mathbf{m}_v,$$

where $\mathbf{m}_v = \|\Gamma_v\|$. The child source marginals are fractional restrictions of the parent marginal, and their Radon–Nikodym densities add pointwise to at most the parent density.

Sum this identity over all nodes above depth L . At each depth the node occurrences are submeasures of a partition of the root occurrence, so

$$\sum_{|v|=k} \mathbf{m}_v \leq \mathcal{M}_{\text{in}}(r), \quad \sum_{|v|<L} \mathbf{e}_v \lesssim_A L r^A \mathcal{M}_{\text{in}}(r).$$

All intermediate child terms telescope with coefficient one. This proves the finite-depth identity and removes the former C^L .

At a terminal cap γ , split the old-neighbor mark according as ζ lies inside or outside that cap. The outside part is $\mathcal{M}_{\text{cross}}$. For the inside part, use the root edge density $h \leq 1$, the source density f_γ , and bounded Lebesgue direction density:

$$\mathcal{M}_\gamma^{\text{same}} \leq \int f_\gamma(z) \mathbf{1}_\gamma(z) \left(\int \mathbf{1}_\gamma(\zeta) d\sigma(\zeta) \right) d\sigma(z) \lesssim_U p_\gamma T_\gamma^3.$$

This proves (453). If $L = r^{-o(1)}$, choose $A > 2$ larger than the fixed terminal exponent reserve. Then $L r^A = r^{A-o(1)}$ is absorbed, and the result is (417). $\square$

Proposition 9.30 (Lossless deterministic saturation of the return tree). *Fix a coarse occurrence layer of source mass $m > 0$, a root collision thickness r , and put*

$$B(r) := \exp \sqrt{\log(1/r)}.$$

At a node in a cap of radius T_v , prescribe

$$\tau_v := \frac{T_v}{C_* B(r)},$$

where C_ dominates the fixed cap enlargement in Theorem 9.8. Every non-paid child can be chosen with*

$$\boxed{T_w \leq B(r)^{-1} T_v \quad (w \in \text{ch}(v)).} \quad (454)$$

The choices preserve the entire absolute occurrence measure and all inherited marks. The least depth at which $T_L^3 \leq mr^{2-o(1)}$ satisfies

$$L(r) = O_m\left(\sqrt{\log(1/r)}\right) = o(\log(1/r)). \quad (455)$$

The assertion is uniform in the number and shape of the children and in every carrier label.

Proof. Since $B(r) \rightarrow \infty$, one has $\tau_v = o(T_v)$. On a positive-mass node which is not a good residual leaf, Theorem 8.32 supplies arbitrarily fine failure scales. Choose the node's fine scale only after τ_v is fixed, so that every physical, synchronization, polarization, and target-cell error is at most $T_v/(C_*^2 B(r))$. The angular, separated-window, and same-window support statements in Theorems 9.9, 9.13 and 9.28 then give (454).

The fine scales are auxiliary certificates attached to individual occurrence nodes; they are not identified with one common generation scale. The tree is countable because it is built from half-open cap, target-cell, degree, and rational time-packet partitions, so all nodewise choices can be made. More importantly, all pieces are retained. The identities in Theorems 8.56, 9.8, 9.26 and 9.27 show that no minimum node mass enters the construction. Each cap normalization is subpower at its own chosen fine scale by (389); such constants are not multiplied across nodes or generations because Theorem 9.29 uses absolute occurrence measures.

Finally $T_L^3 \leq T_0^3 B(r)^{-3L}$. Since m and T_0 belong to the fixed coarse layer, the least admissible integer satisfies

$$L \leq 1 + \frac{\log(T_0^3/m) + (2 + o(1)) \log(1/r)}{3 \log B(r)} = O_m\left(\sqrt{\log(1/r)}\right).$$

Carrier labels are passive conditional marks in every cited mass identity, proving uniformity. $\square$

Corollary 9.31 (Maslov–bush–Frostman bridge). *For every fixed coarse low-reuse occurrence layer G_* , either the relative residual estimate (303) holds uniformly at all sufficiently fine scales, or K_b carries a nonzero $(4 - \delta)$ -Frostman measure for every $0 < \delta < 3$. In particular, under the contrary hypothesis $\dim_H K_b < 4$, the relative residual estimate holds. Thus the physical bush output cannot remain as an unassigned subquadratic Maslov residual.*

Proof. If a positive-mass good leaf occurs, use Theorem 8.11 or the good alternative of Theorem 9.34. Otherwise every positive-mass node has arbitrarily fine residual-failure scales. Theorem 9.30 supplies a simultaneous mass-preserving saturation with $L = O(\sqrt{\log(1/r)})$ and $T_L^3 \leq mr^{2-o(1)}$. Unlike the former diagonal choice, this uses the predetermined contraction $B(r)^{-1}$ and the exact absolute-flow identities, so neither the retained mass nor a minimum node mass enters the depth estimate. Theorem 9.29 supplies (417). Applying the edge–Carleson criterion once more now gives the relative residual estimate and closes the stated bridge. $\square$

Theorem 9.32 (Contact–symplectic closure of the four-dimensional Sticky Takeya selector). *For every compact full-direction Sticky selector in the contactification used in this manuscript, the Euclidean Reeb front $K_b \subset \mathbb{R}^4$ satisfies*

$$\dim_H K_b = 4.$$

Proof. Assume $\dim_H K_b < 4$. The finite Darboux, angular-sector, Reeb-window, and occurrence-layer decompositions lose only fixed or subpower factors. Fix one coarse occurrence layer and suppose that its relative residual target fails along arbitrarily fine scales. Theorem 7.76 converts that failure, with the quadratic layer mass exactly normalized out, into a normalized Maslov graph. The old-flag/new-cycle comparison and finite conditional flag amplification route this graph into the transverse-polarization, horizontal-line, front-nondegenerate, angular, or physical Lagrangian-bush outputs; see Theorems 8.5 to 8.7. The first four outputs obey the required quadratic relative estimates in their assigned ledgers and therefore cannot carry the assumed power-excess failure. For the last output, Theorem 9.31 gives either that same relative estimate or a nonzero $(4 - \delta)$ -Frostman measure for every $0 < \delta < 3$; the latter contradicts $\dim_H K_b < 4$. Hence the relative residual target holds on every fixed coarse layer.

Summing the finite/coarse layers and absorbing all fixed coarse constants at the fine scale as in Theorems 7.74 and 9.47 gives, for every $\eta > 0$,

$$3_\rho^{J^+}(\mu) \lesssim_\eta \rho^{2-\eta} \quad (0 < \rho < 1).$$

The codimension-two residual criterion Theorem 6.27 then yields $\dim_H K_b = 4$, a contradiction. $\square$

9.7 The marked source-hereditary interface

Corollary 9.33 (The polarized selector-carrier strict-progress gate is closed). *Let a returned bush H , centered at (c, s_0) , carry the conserved occurrence measure*

$$d\lambda(z)d\kappa_z^h(\zeta)$$

from (419), and let its physical coarse Reeb collision be represented by the Markov kernels (423). For a fixed root scale r_0 , choose the current fine failure scale ρ after the child-cap target. Its entire incoming occurrence is partitioned, up to an arbitrarily prescribed $O(r_0^A)$ tail, into paid outputs and physical children lying in direction caps $o(T_{\text{parent}})$. In particular, the strict-progress input used in Theorem 9.29 holds with no additional hypothesis.

Proof. The separated old-time part has strict progress by Theorems 9.11 to 9.13. The angular all-fine part also has occurrence-level strict progress by Theorems 9.8 and 9.9: it uses the coarse Reeb Markov transition on every adaptive source cap and never substitutes the degree measure of a newly sampled fine graph.

For the same-window part apply Theorem 9.28. Its non-amplified pieces are paid by the cap-square ledger. In its amplified common-target branch, Theorems 9.19 and 9.20 preserve a fixed fraction of the original occurrence linearly. Separated planes are routed sourcewise by Theorem 9.24. On the coherent part, Theorem 9.23 makes the old-edge Reeb times and the Maslov plane incidences simultaneous; four-separated-time flags are then impossible by the finite-scale Reeb-front firewall and the quantitative separation of every determinant-nondegenerate origin plane from the front-null locus, as recorded in Theorem 9.25. Finally, Theorem 9.26 partitions the remaining conditional time law exactly into three-packet pieces with $O(r_0^A)$ tail, and Theorem 9.27 turns every such piece into a fractional physical bush family in $O(\tau)$ -source caps. Choosing $\rho + \varepsilon_{\text{sync}} + \delta_s + \eta = o(\tau) = o(T_{\text{parent}})$ gives strict cap progress. All cap, target-cell, and time-packet pieces are summed rather than selected, so the absolute incoming occurrence is conserved. $\square$

Lemma 9.34 (Mass-conserving vector-Frostman stopping for a maximal bush family). *Let*

$$\{H_j : 1 \leq j \leq N\}$$

be a finite pairwise disjoint family of ε -bushes. Let λ be a finite measure supported on their disjoint union and assume, for some fixed $C_\lambda < \infty$,

$$\lambda|_{H_j} \leq C_\lambda \sigma|_{H_j}, \quad q := \|\lambda\| > 0.$$

Use the common outer window I_{out} and the vector centers $\mathbf{a}_j(y, s)$ from Theorem 8.11. Fix $0 < \delta < 3$, a lower radius $\rho \geq \varepsilon$, and $C_0 \geq 1$. There is a pairwise disjoint measurable decomposition of a subset carrying at least $q/2$ of the family mass into exactly one of the following forms.

- (i) *A restriction $\lambda_{\text{good}} \leq \lambda$, of mass at least $q/2$, whose normalization satisfies*

$$\sup_{\substack{y \in \mathbb{R}^3 \\ s \in I_{\text{out}}}} \frac{1}{\|\lambda_{\text{good}}\|} \sum_j \lambda_{\text{good}}(H_j \cap B(\mathbf{a}_j(y, s), R)) \leq 2C_0 R^{3-\delta} \quad (\rho \leq R \leq 1).$$

- (ii) *Countably many disjoint Reeb-coherent hairbrush packets $\mathcal{H}_\ell$, with masses S_ℓ , witness points (y_ℓ, s_ℓ) , and radii $\rho \leq R_\ell \leq 1$, such that*

$$\boxed{\sum_\ell S_\ell \geq \frac{q}{2}, \quad S_\ell > C_0 q_\ell R_\ell^{3-\delta}.} \quad (456)$$

Here q_ℓ is the λ -mass remaining immediately before $\mathcal{H}_\ell$ is removed. Every $\mathcal{H}_\ell$ merges into one physical $O(R_\ell)$ -bush centered at (y_ℓ, s_ℓ) , its underlying selector-direction mass is at least $C_\lambda^{-1} S_\ell$, and all old conditional Maslov flags remain attached.

Thus failure of the vector-Frostman output need not be represented by one arbitrarily small hairbrush: a fixed fraction of the maximal-family mass is retained across the full stopping collection.

Proof. Start with $\lambda_0 = \lambda$, of mass $q_0 = q$. Given λ_ℓ , stop with alternative (i) if its normalization satisfies the displayed vector-Frostman inequality with constant $2C_0$. Otherwise choose a triple (y_ℓ, s_ℓ, R_ℓ) whose violating cap mass is at least one half of the supremal violation, and remove the corresponding physical selector subsets from every H_j . Denote their union by $\mathcal{H}_\ell$, its mass by S_ℓ , and the remainder by $\lambda_{\ell+1}$. Then

$$S_\ell > C_0 q_\ell R_\ell^{3-\delta}.$$

The removed selector sets are disjoint by construction.

Continue while the removed mass is below $q/2$. If the procedure stops, the remaining mass is at least $q/2$ and gives (i). If it does not stop and the accumulated removed mass reaches $q/2$, retain the first countable initial segment which does so and obtain (ii). The only other formal possibility is that the remainders decrease to a measure λ_∞ of mass greater than $q/2$ while the removed masses have sum below $q/2$. Continuity from above gives convergence of every fixed cap mass. A strict vector-Frostman violation for λ_∞ would then, after a small enlargement of its ball and rational approximation of (y, s, R) , give a uniformly positive near-supremal violation for all sufficiently large remainders, contradicting convergence of the removed masses to zero. Hence λ_∞ satisfies the bound and gives (i).

For every removed packet, all cap centers are generated by its single witness (y_ℓ, s_ℓ) . The calculation in Theorem 8.16 therefore merges it into one physical bush. The domination by $C_\lambda \sigma$ gives the asserted lower bound for its underlying direction mass. In alternative (i), the proof of Theorem 8.11 applies with the component restrictions of λ_{good} in place of $\sigma|_{H_j}$: push forward $\lambda_{\text{good}} \otimes ds$, and use the displayed vector Frostman bound in the same $O(R)$ -long Reeb window. Thus this weighted good alternative has the same $(4 - \delta)$ -Frostman conclusion. Restrictions preserve the conditional flag laws by Theorem 8.3. $\square$

Principle 9.35 (Finite-scale routing of ideal reuse into the flagged contact-star trichotomy). *At the coarse scale R_n , Theorems 8.16 and 8.17 already give a genuine physical bush and full product collision mass. Theorem 8.28 therefore gives, on its fixed-angle part, a normalized thickened Maslov graph of constant edge mass; no ideal-to-physical conversion or cell-density factor is needed there. The issue below concerns only a strictly finer Maslov scale. Theorem 8.30 sends that constant-edge graph through the arbitrary-threshold route and returns, unless a paid ledger occurs, an $O(1)$ -member bush family carrying a fixed fraction of the normalized hairbrush mass. Thus a single intrinsic-scale Maslov return is lossless. Theorem 8.31 then uses the required alternation of the two separated Reeb windows to show that a second merged hairbrush lies in one shrinking direction cap. Hence fixed-angle hairbrush renormalization does not iterate indefinitely; its only return is the angular Darboux chart at a scale comparable to the inherited physical error. Theorem 8.32 freezes that actual positive-mass cap. It is either already a good all-fine-scales residual leaf, or every cap-normalization constant becomes subpower along a new, much finer failure sequence and the same flagged contact-Maslov route restarts without a power tax.*

The measure in (366) is an ideal Monge-type coupling: for each source direction a and target label k , it uses the single transported direction $\Psi_{jk,u}(a)$. Theorem 9.37 shows that its actual selector endpoints retain their old conditional Maslov flags. Theorem 8.24 shows that, under uniform differentiation-scale separation, the lossless benchmark would be to find

$$C\varepsilon_n \leq \tilde{r}_n = o(R_n)$$

such that

$$\boxed{\sum_{j \neq k} \int_{W_n} \iint_{A_{n,j} \times A_{n,k}} \mathbf{1}_{\{|\Pi_u(z(a)) - \Pi_u(z(a'))| \leq \tilde{r}_n\}} d\sigma(a) d\sigma(a') du} \gtrsim_{I,J,U} \tilde{r}_n^3 \mathcal{R}_n. \quad (457)$$

Here $\tilde{r}_n^3$ is the transverse three-dimensional thickening of the Monge graph.

Uniform differentiation is not assumed. Theorem 8.25 proves at every $\varepsilon_n \leq r \leq R_n$ the unconditional replacement: either physical product collision retains $\omega r^3 \mathcal{R}_n$, reverse Reeb contact-star reuse is high, or the complete energy is bounded by $BN_n \omega R_n^4$. Theorem 8.26 chooses B, ω so that the third case is absent whenever $S_n \gg R_n^3$. Thus the unknown differentiation threshold has been eliminated. On the fixed-angle portion, Theorem 8.27 divides out only the genuine $O_{\tau_0}(r)$ collision-time window and returns the physical product branch to a normalized $O(r)$ -Maslov graph. Its edge mass is

$$\widetilde{M}_{n,r} \gtrsim \omega_n (r/R_n)^2,$$

and the old conditional polarizations remain attached without a second normalization loss. The complementary small-angle portion is the Darboux branch. Theorem 8.29 compares this edge mass with the next merged-cap density tax and records the exact remaining two-scale quotient; no relation between the two scales is inserted by hand. The remaining scalar in the product branch is exactly

$$\omega_n^{-1} \lesssim_{I,J,U,G_*} A_n^{-1} M_n^{-4} R_n^\delta,$$

while the complementary branch is an actual high-reuse contact star. After the original occurrence weights are restored, these are precisely the two inputs to Theorem 7.53; its outputs are angular collapse, a tetrahedral Reeb-front seed, or one contact-polarized $E \oplus E$ packet. The last output is handled by the old/new flag comparison. No label count is identified with direction mass before this weighted regularization.

Corollary 9.36 (Dyadic label reuse has an exact clustered-star mass threshold). *Restrict the target labels of one reuse star from (368) to a dyadic cap-mass layer*

$$p \leq \sigma(A_{n,k}) < 2p, \quad |s_{n,j_0} - s_{n,k}| \geq \delta_s \quad (k \in \mathcal{N}),$$

where the near-equal center-level pairs have already been routed separately, and suppose $\#\mathcal{N} \geq B$. Then either two distinguished centers $a_{j_0 k}^{\text{hom}}$ are η -separated, or

$$\boxed{Bp \lesssim_{I,J,U,\delta_s} (R_n + \eta)^2.} \quad (458)$$

Hence, at the level of the ideal coupling, failure to obtain separated homothety centers is an explicit scalar cap-mass loss, not an unlabeled moderate-recurrence remainder. Passage to the physical weighted star still requires (457).

Proof. If no two distinguished centers are η -separated, they lie in one η -ball. Apply Theorem 8.21; its left side is at least $\sum_{k \in \mathcal{N}} \sigma(A_{n,k}) \geq Bp$. $\square$

Corollary 9.37 (The local re-collision energy retains the old Maslov flags losslessly). *Write $z(a) := (a, b(a))$ for the selector point over direction a . For every $k \geq 1$, every pair $j \neq k'$, and every $u \in W_n$, the corresponding summand in (363) is unchanged after decorating both physical selector endpoints independently by k old flags:*

$$\boxed{\int_{A_{n,j} \cap \Psi_{jk',u}^{-1}(A_{n,k'})} \left(\prod_{\ell=1}^k d\kappa_{z(a)}(\mathbf{f}_\ell) \right) \left(\prod_{\ell=1}^k d\kappa_{z(\Psi_{jk',u}(a))}(\mathbf{g}_\ell) \right) d\sigma(a)} \quad (459)$$

$$= \sigma\left(A_{n,j} \cap \Psi_{jk',u}^{-1}(A_{n,k'})\right).$$

Consequently neither the coherent cap restriction nor passage to the exact re-collision energy discards the conditional polarization normals used by the Maslov routing.

Proof. Each κ_z is a probability measure. Integrate the flag variables first, exactly as in Theorem 8.3. The cap restriction changes the retained physical endpoints but not their conditional laws, by Theorem 9.41. $\square$

Corollary 9.38 (The coherent hairbrush has an exact density–re-collision dichotomy). *For the maximal families in alternative (ii) of Theorem 8.8, let M_n denote the associated normalized collision edge mass and set*

$$\Theta_n := A_n q_n R_n^{-\delta}.$$

After passing to a subsequence, one of the following holds:

(i) $\Theta_n \rightarrow \infty$, and the off-diagonal local re-collision energy satisfies

$$\boxed{\sum_{j \neq k} \int_{W_n} |u - s_{n,j}|^3 \sigma\left(A_{n,j} \cap \Psi_{jk,u}^{-1}(A_{n,k})\right) du \gtrsim_{I,J} A_n^2 q_n^2 R_n^{4-2\delta}.} \quad (460)$$

(ii) $\Theta_n \lesssim 1$, and therefore

$$\boxed{q_n \lesssim A_n^{-1} R_n^\delta, \quad M_n \lesssim_{G_*} A_n^{-1} R_n^\delta.} \quad (461)$$

The first alternative is precisely an input to the fixed-angle contact-homothety recurrence ledger; only its balanced sign-reversing subcase may be sent to the two-time rigidity theorem. The second records, without hiding it in an angular label, the remaining small bush-mass debt.

Proof. By (364), $S_n/R_n^3 > \Theta_n$. In the first alternative the first term in (363) eventually dominates the diagonal term, and substitution of $S_n > A_n q_n R_n^{3-\delta}$ proves (460). In the second alternative the definition of Θ_n gives the first inequality in (461); the second follows from $q_n \gtrsim_{G_*} M_n$. $\square$

Corollary 9.39 (A Reeb-coherent bad cap family selects one labeled bush with an explicit M^{-4} normalization debt). *Assume the second alternative of Theorem 8.8, so that*

$$q_n := \sum_j \sigma(H_{n,j}) \gtrsim_{G_*} M_n, \quad N_n \lesssim M_n^{-3}.$$

If the Reeb-coherent cap concentration (341) holds, then some index j_n satisfies

$$p_n := \sigma(H_{n,j_n} \cap B(a_{n,j_n}, R_n)) \gtrsim_{G_*} A_n M_n^4 R_n^{3-\delta}. \quad (462)$$

After translating that cap and dilating directions by R_n^{-1} , its normalized direction measure has density

$$D_{n,\text{cap}} \lesssim_{G_*} \frac{R_n^3}{p_n} \lesssim_{G_*} \frac{R_n^\delta}{A_n M_n^4}. \quad (463)$$

The selected component retains one actual Lagrangian-bush center, its Reeb labels, the old conditional Maslov flags, and the common outer-wing hairbrush point (y_n, s_n^{out}) .

Proof. The left side of (341) is a sum of N_n nonnegative cap masses. One is therefore at least the sum divided by N_n . Use $q_n \gtrsim M_n$ and $N_n \lesssim M_n^{-3}$ to obtain (462). The rescaling Jacobian gives $D_{n,\text{cap}} \lesssim R_n^3/p_n$, proving (463). All labels belong to the selected bush and are unchanged by this restriction. $\square$

Corollary 9.40 (The physical bush output has one exact remaining failure mode). *For any sequence of approximate Lagrangian point-probe bushes satisfying the geometric hypotheses of Theorem 8.10, fix $0 < \delta < 3$. If no subsequence satisfies its uniform direction-Frostman hypothesis with this δ , then there are a subsequence, numbers $A_n \rightarrow \infty$, and radii*

$$\varepsilon_n \leq R_n \downarrow 0,$$

and direction balls $B(a_n, R_n)$ for which

$$\eta_n(B(a_n, R_n)) > A_n R_n^{3-\delta}. \quad (464)$$

Thus the non-Frostman bush output is an actual angular Darboux concentration of the normalized bush-leaf measure. It must be passed to the angular-collapse/Maslov-pencil analysis with its bush center and old flags retained.

Proof. This is the contrapositive of (336). If one constant C_δ worked after a subsequence, Theorem 8.10 would apply. Otherwise take $A_n \rightarrow \infty$ and make the corresponding diagonal choice. Since the left side of (464) is at most one,

$$R_n^{3-\delta} < A_n^{-1},$$

so $R_n \rightarrow 0$. $\square$

Lemma 9.41 (Exact normalization ledger for a non-Frostman bush cap). *Let $H \subset U$ be a bush leaf set of selector-direction mass*

$$q := \sigma(H) > 0, \quad \eta := q^{-1}\sigma|_H.$$

Suppose that for some direction cap $B(a_0, R)$,

$$m_R := \eta(B(a_0, R)) > AR^{3-\delta}.$$

Let

$$S_R(a) := \frac{a - a_0}{R}, \quad \eta_R := (S_R)_\# \left(\frac{\sigma|_{H \cap B(a_0, R)}}{qm_R} \right).$$

Then η_R is a probability measure on a fixed unit ball, absolutely continuous with respect to three-dimensional Lebesgue measure, and

$$\left\| \frac{d\eta_R}{d\mathcal{L}^3} \right\|_\infty \lesssim_U \frac{R^3}{qm_R} \lesssim_U \frac{R^\delta}{qA}. \quad (465)$$

If the original bush error is $\varepsilon \leq R$, its error in the rescaled direction chart is $O(\varepsilon/R)$. Moreover conditioning on the cap does not alter the pointwise old-flag law κ_z ; consequently the prescribed-normal estimate (317) remains available at every retained physical endpoint.

Proof. The unnormalized cap has direction mass qm_R . Under the dilation S_R , Lebesgue density acquires the Jacobian factor R^3 . Division by qm_R gives the first bound in (465); the assumed lower bound for m_R gives the second. The bush-error statement is the same affine rescaling. Finally κ_z is the conditional distribution of flag marks over a fixed physical endpoint z . Restricting the physical marginal to a direction cap changes which z 's are retained but does not change that conditional distribution at any retained z . $\square$

Principle 9.42 (Maslov–bush–to–Frostman bridge: reduction and discharge). Assume a fixed coarse low-reuse occurrence layer G_* violates the relative residual target (303) along arbitrarily fine scales. The maximal M -mass family of point-probe bushes from Theorem 8.8, with its old Maslov flags and Reeb windows still attached, reduces the two contact-overlap sources in Theorem 8.9 to either of the following equivalent closing outputs:

- (i) the weighted conditional tangential estimate (487) after all bush outputs are aggregated;
- (ii) on one fixed positive-measure set $G \subset J$, probability measures ν_s supported on $K_{b,s}$ satisfying, for every $\varepsilon > 0$,

$$\nu_s(B(y, R)) \lesssim_\varepsilon R^{3-\varepsilon} \quad (s \in G, 0 < R < 1).$$

The second output closes by Theorem 6.24; the first closes by Theorems 6.27 and 9.43. The required step is therefore a scale-coherent aggregation theorem for the actual Reeb-escaped Lagrangian bushes, not another exclusion of coherent polarization and not a categorical symplectic invariant. For a strict fine-scale treatment, after Theorems 8.11, 8.16, 8.19 and 9.40, it is enough to show that the fine-scale fixed-angle ledger pays the merged cap-density debt (356); the componentwise M^{-4} debt (463) is avoidable. The non-Frostman input is now exactly the Reeb-coherent hairbrush of Theorem 8.12; its scale is constrained by Theorem 8.13. Theorem 8.16 merges all coherent caps into one actual physical bush, and Theorem 8.18 closes whenever the normalized directions of these merged bushes have a uniform Frostman subsequence. In the remaining single-bush non-Frostman branch, by Theorems 8.14, 8.15, 8.22 and 9.37, the retained labeled caps form an exact Ψ -cocycle orbit, their canonical physical blow-up returns to the Lagrangian pencil, their whole mass enters the preceding re-collision integral, and all old conditional Maslov flags survive. What remains is to split that localized integral into the near-angle Darboux branch, the balanced sign-reversing two-time subcase controlled by Theorems 5.4 and 5.8, and the genuinely moderate fixed-angle recurrence that must pay the explicit density debt (356). No generic moderate overlap is being promoted to a small symmetric-difference defect. Theorem 9.38 isolates the alternative small-mass debt from the genuinely off-diagonal recurrence input. On the latter input, Theorems 8.20 and 8.21 remove the clustered-center star unless its cap mass already lies in a rank-one direction tube. The remaining fixed-angle star therefore has separated homothety centers before the old/new flag-polarization split is invoked. Theorem 8.23 identifies the high-density energy with a common Reeb contact-label reuse integral, while Theorems 8.25 to 8.27 and 9.35 remove the unknown differentiation threshold: at every finite scale they give either a physical product collision with the explicit cell-density factor ω_n , or high reverse contact-star reuse. In the fixed-angle physical branch the collision measure is now an honest normalized thickened Maslov graph with edge mass $\bar{M}_{n,r} \gtrsim \omega_n(r/R_n)^2$; the small-angle complement is routed to the Darboux chart. At $r = R_n$, the full coarse collision instead gives constant normalized edge mass by Theorem 8.28. Theorem 8.30 then bypasses both density factors for one complete generation: it returns a bounded-cardinality bush family carrying a fixed normalized mass fraction, which either satisfies vector Frostman or forms another common outer-point hairbrush. The latter alternates back to the original separated Reeb window, so Theorem 8.31 places its entire direction set in one $O(\hat{R}_n)$ -cap. Thus the strict fine-scale scalar obstruction is (384), while the intrinsic-scale formulation leaves no repeated fixed-angle chain. Its remaining endpoint is the nonplanar angular Darboux blow-up at the scale locked by (388). Since an error-free noncollapsed tangent limit would require the opposite scale separation, the planar exclusion Theorem 9.51 cannot simply be cited here. The remaining endpoint is handled nodewise by Theorem 8.32: the cap is either a good residual leaf or a tax-free fine return. The remaining global endpoint is therefore the weighted aggregation of the nested good-leaf/fine-return stopping tree. Theorem 8.33 shows that terminal-leaf cross terms create no additional incidence loss; by Theorem 8.34, it is enough to prove the quantitative tail/leaf-constant condition (392). If that condition fails under a null-front hypothesis, Theorem 8.35 supplies alternating physical endpoint-star chains to arbitrary finite depth. The exact remaining substitute is to upgrade the positive-logarithmic failure blocks from Theorem 8.36. Theorem 8.37 already gives a common shrinking label cell for the physical centers, angular charts, and polarization planes with only subpower loss. Thus the stronger label-measure dichotomy Theorem 8.38 leaves exactly two cases: a fixed positive proportion of scales with convergent center/polarization labels, or two positive-proportion scale families with separated labels. The edge-flow argument below discharges both cases by producing the corresponding geometric gain or, equivalently, direct Carleson tail decay. Theorem 8.39 retains the actual physical endpoint-star measures in this alternative. In the atomic case, Theorem 8.40 further reduces the thickness-to-angle map to $0 < \varkappa_\infty < 1$, the resolution endpoint $\varkappa_\infty = 1$, or the sole scale-degenerate endpoint $\varkappa_\infty = 0$. The contact-symplectic information now removes two more apparent degrees of freedom. Theorem 8.41 shows that, on a compatible nested branch, all alternating physical probe centers converge on the Euclidean image of one Reeb orbit. On every branch with $\varkappa_\infty < 1$, the retained physical flags have vanishing normalized incidence error; hence Theorems 8.42 and 8.50 show that four distinct Reeb levels over one coherent polarization force the front-null Kähler locus and horizontal rank at most two. Thus the remaining noncollapsed part below the resolution endpoint is reduced

to genuine polarization variation or an at-most-three-level Reeb packet over one polarization; Theorem 8.44 gives those three packets the finite-scale width $O((R/\tau_R + d_{\text{Gr}}(V_R, V_\infty))^{1/3})$. Theorem 8.45 sends the latter, when it is the retained current cycle-plane label, back to the existing bush/angular/transverse alternatives. A further non-Frostman bush return is then precisely the resolution cap already isolated above. Thus the genuinely new below-resolution endpoint is polarization variation. Theorem 8.46 and Theorem 8.47 remove its same-vertex part by the old/new symplectic transverse ledger, while Theorems 8.48 and 8.49 either charge its cross-branch graph or decouple it into coherent components. At $\varkappa_\infty = 1$ the inherited coarse error is resolution-sized, so the exact-root argument is deliberately not invoked. However Theorems 8.52 and 8.53 show that this lowest angular shell is already relative-quadratic: every genuine fine residual failure has $\rho/\tau_\rho \rightarrow 0$ and therefore re-enters the exact contact–Maslov routing. The final endpoint is the mass aggregation of successive physical bush returns. Theorem 8.54 combines the loss M_ℓ , the Frostman-failure radius R_ℓ , and the returned child-cap radius $\widehat{R}_\ell$ into the exact density multiplier $A_\ell M_\ell R_\ell^{3-\delta} \widehat{R}_\ell^{-3}$. Since Lebesgue direction density is bounded, generations with a multiplier uniformly above one have finite depth. Theorem 8.55 confines the final tail to $M_\ell \lesssim A_\ell^{-1} \widehat{R}_\ell^3 R_\ell^{\delta-3}$. Theorem 9.34 removes the former one-cap selection loss: either a positive fraction of the maximal family is vector-Frostman, or a disjoint collection of Reeb-coherent hairbrushes retains total mass $\gtrsim m_* M$. Theorem 9.2 retains the stronger linear edge mass as the conditional flow measure λ_h , and Theorem 9.4 transports it through the physical Reeb re-collision graph without the false p^2 normalization. Its coarse return can nevertheless be time-local by Theorem 9.10. The shared-target obstruction is closed at the conserved-occurrence level by Theorems 9.21, 9.23 and 9.33; consequently Theorems 9.29 and 9.31 close the edge-Carleson estimate and the present bridge. Theorem 9.36 exposes the exact scalar loss if the distinguished centers do not separate. Thus this displayed reduction is discharged, rather than retained as an additional hypothesis.

Proposition 9.43 (Exact factor budget in the final coherent polarization). *Let λ be the restriction of the low-reuse selector measure to one fixed-angle, cell-locally coherent $E \oplus E$ class, put $m = \|\lambda\|$, and use the coordinates (x, q, y, p) of (159). Suppose its q -marginal satisfies*

$$\lambda_q(I) \leq D(|I| + r) \quad \text{for every interval } I. \quad (466)$$

Let

$$d\Lambda_r(z, z', s) := \mathbf{1}_{\{|N_s(z, z')| \leq r\}} d\lambda(z) d\lambda(z') ds.$$

Then

$$\|\Lambda_r\| \lesssim_J Dmr \log(2/r). \quad (467)$$

If, in addition,

$$D \lesssim mr^{-o(1)} \quad (468)$$

and the collision-preserved tangential shading satisfies

$$\int \mathbf{1}_{\{|T_s(z, z')| \leq r\}} d\Lambda_r(z, z', s) \lesssim r^{2-o(1)} \|\Lambda_r\|, \quad (469)$$

then

$$\int_J \int \int \mathbf{1}_{\{|\Pi_s(z) - \Pi_s(z')| \leq r\}} d\lambda(z) d\lambda(z') ds \lesssim m^2 r^{3-o(1)}. \quad (470)$$

Equivalently, the relative Maslov residual bound (303) holds on this coherent class.

Proof. For intervals of length at least r , (466) gives $\lambda_q(I) \leq 2D|I|$. Thus (467) is Theorem 7.20 with $W = r$; intervals shorter than r occur only in the first shell of that proof and are absorbed by the $+r$ convention. The kernel factorization (331), followed by (469), bounds the full collision integral by

$$r^{2-o(1)} \|\Lambda_r\| \lesssim Dmr^{3-o(1)}.$$

Use (468) to obtain (470). Finally apply the collision/residual equivalence (121) after dividing the collision integral by its universal window factor r . $\square$

Lemma 9.44 (Occurrence-density transfer exposes the sharp ledger offset tax). *Retain the low alternative in Theorem 7.71, and assume, as in its proof, that this alternative carries a fixed portion of the first moment. Write*

$$H_{\text{lo}} := \{0 < \mathcal{N}_{\text{flag}}, P_i^{(H)} \leq L\}, \quad A_{\text{lo}} := \int_{H_{\text{lo}}} \mathcal{N}_{\text{flag}} d\nu_i.$$

Then

$$\boxed{A_{\text{lo}} \gtrsim K_i H_i P_i.} \quad (471)$$

After discarding occurrence values carrying $o(A_{\text{lo}})$ and taking a joint logarithmic dyadic layer in $\mathcal{N}_{\text{flag}}$ and $P_i^{(H)}$, there are $\lambda_* > 0$, $0 < P_* \leq L$, and

$$G_* \subset \left\{ z \in H_{\text{lo}} : \lambda_* \leq \mathcal{N}_{\text{flag}}(z) < 2\lambda_*, \quad P_* \leq P_i^{(H)}(z) < 2P_* \right\}$$

such that, with $m_* := \nu_i(G_*)$,

$$\boxed{\lambda_* m_* \asymp \int_{G_*} \mathcal{N}_{\text{flag}} d\nu_i = K_i H_i P_i W^{o(1)}, \quad \lambda_* \leq 4H_i P_*} \quad (472)$$

For every interval I in the coherent normal coordinate q ,

$$\boxed{(\nu_i|_{G_*})_q(I) \leq \lambda_*^{-1} \int_{G_* \cap \{q \in I\}} \mathcal{N}_{\text{flag}} d\nu_i} \quad (473)$$

Consequently the relative normal estimate needed in (466)–(468) follows on this layer from the single weighted slab estimate

$$\boxed{\int_{G_* \cap \{q \in I\}} \mathcal{N}_{\text{flag}} d\nu_i \lesssim \left(\int_{G_*} \mathcal{N}_{\text{flag}} d\nu_i \right) (|I| + r) r^{-o(1)} \quad \text{for every } I.} \quad (474)$$

The pointwise low-reuse cap alone proves only an absolute estimate. Indeed bounded direction density and (293) give

$$\int_{G_* \cap \{q \in I\}} \mathcal{N}_{\text{flag}} d\nu_i \lesssim_U H_i P_* (|I| + r), \quad (475)$$

so after (472) its relative-density loss is bounded by the layer-sharp tax

$$\boxed{\mathfrak{g}_{\text{rel}} := \frac{P_*}{K_i P_i} W^{-o(1)}} \quad (476)$$

The weighted dense/sparse argument below completes the coarse low old-flag reuse bound by proving (474), or gain the reciprocal factor $\mathfrak{g}_{\text{rel}}^{-1}$ in the conditional tangential estimate before the normal and tangential factors are multiplied.

Proof. The first assertion is exactly the low-half case in the proof of Theorem 7.71. Dyadically decompose the nonzero values of $\mathcal{N}_{\text{flag}}$ and $P_i^{(H)}$; values below sufficiently small fixed powers of W carry $o(A_{\text{lo}})$, and the remaining number of joint layers is $W^{-o(1)}$. One joint layer therefore gives the lower bound in (472); its upper bound uses (287), while its pointwise upper bound uses (293). On G_* , $1 \leq \lambda_*^{-1} \mathcal{N}_{\text{flag}}$, which proves (473). Substitution of (474) and (472) yields the required relative marginal estimate.

For the final assertion, the map $a \mapsto q = n \cdot a$ sends a q -interval of length $|I|$ to a direction slab of measure $O_U(|I| + r)$ after scale- r enlargement. Multiply this by the pointwise bound $\mathcal{N}_{\text{flag}} \leq 2H_i P_i^{(H)} \leq 4H_i P_*$ to obtain (475). Dividing its coefficient by the total weighted mass in (472) gives $P_*/(K_i P_i)$ up to subpower factors, which is (476). $\square$

Proposition 9.45 (Exact cancellation of the relative offset tax). *Use the occurrence layer G_* from Theorem 9.44, put $\lambda := \nu_i|_{G_*}$, $m_* := \|\lambda\|$, and let Λ_r be its normal-window pair measure. The absolute slab estimate (475) gives an admissible normal-offset constant*

$$\boxed{D_* \lesssim_U \frac{H_i P_*}{\lambda_*} \lesssim m_* \frac{P_*}{K_i P_i} W^{-o(1)}} \quad (477)$$

Consequently

$$\boxed{\|\Lambda_r\| \lesssim m_*^2 \frac{P_*}{K_i P_i} r W^{-o(1)}} \quad (478)$$

If the collision-preserved tangential shading satisfies the tax-cancelling estimate

$$\boxed{\int \mathbf{1}_{\{|T_s(z, z')| \leq r\}} d\Lambda_r(z, z', s) \lesssim \frac{K_i P_i}{P_*} r^{2-o(1)} \|\Lambda_r\|,} \quad (479)$$

then the full coherent collision bound (470), and hence the relative residual target (303), holds on G_* . Thus relative q -equidistribution is sufficient but not necessary: the existing contact-polarized dense/sparse route may instead close the branch by producing exactly the factor $K_i P_i / P_*$ in (479); its sparse alternative remains a physical Lagrangian-bush exit.

Proof. Divide (475) by λ_* and use (473); this gives the first bound in (477). The second follows from $\lambda_* m_* \gtrsim K_i H_i P_i W^{o(1)}$. Apply Theorem 7.20 at scale r to obtain (478). Multiplying it by the conditional fraction in (479) cancels $P_*/(K_i P_i)$ and gives

$$\int_J \int_J \int \mathbf{1}_{\{|\Pi_s(z) - \Pi_s(z')| \leq r\}} d\lambda(z) d\lambda(z') ds \lesssim m_*^2 r^{3-o(1)}.$$

Now use Theorem 7.1 exactly as in Theorem 9.43. $\square$

Proposition 9.46 (The final collision lifts exactly to two old-flag occurrences). *Keep the notation of Theorem 9.44. Write $\mathfrak{f} = (y, f)$ for a rooted flag and define the restricted occurrence measure*

$$d\rho_*(\mathfrak{f}, z) := \mathbf{1}_{G_*}(z) w_f(z) d\Lambda(y, f) d\nu_i(z). \quad (480)$$

Its physical marginal and total mass are

$$(\pi_z)_\# \rho_* = \mathbf{1}_{G_*} \mathcal{N}_{\text{flag}} d\nu_i, \quad \|\rho_*\| = \int_{G_*} \mathcal{N}_{\text{flag}} d\nu_i \asymp \lambda_* m_*. \quad (481)$$

Its rooted-flag marginal retains the actual packet weights and obeys only the honest bound

$$(\pi_{\mathfrak{f}})_\# \rho_* = \left(\int_{G_*} w_f(z) d\nu_i(z) \right) d\Lambda(y, f) \leq 2H_i d\Lambda(y, f). \quad (482)$$

For every nonnegative kernel $K(z, z', s)$,

$$\int_J \int_J \int_{G_* \times G_*} K(z, z', s) d\nu_i(z) d\nu_i(z') ds \asymp \lambda_*^{-2} \int_J \int_J \int K(z, z', s) d\rho_*(\mathfrak{f}, z) d\rho_*(\mathfrak{f}', z') ds. \quad (483)$$

In particular, with the normal kernel retained, the tax-cancelling target (479) is equivalent, up to constants and subpower losses, to

$$\int \mathbf{1}_{\{|N_s(z, z')| \leq r\}} \mathbf{1}_{\{|T_s(z, z')| \leq r\}} d\rho_* d\rho'_* ds \lesssim \frac{K_i P_i}{P_*} r^{2-o(1)} \int \mathbf{1}_{\{|N_s(z, z')| \leq r\}} d\rho_* d\rho'_* ds. \quad (484)$$

This is the collision-preserved old-flag/new-cycle six-point estimate with all occurrence weights still present. No unweighted selector measure and no categorical symplectic invariant has been substituted for it.

Proof. The first identity follows from the definition of $\mathcal{N}_{\text{flag}}$; the second is Tonelli. Equation (482) uses $\int w_f d\nu_i < 2H_i$. Since $\lambda_* \leq \mathcal{N}_{\text{flag}} < 2\lambda_*$ on G_* , multiplying two copies of this comparison and integrating against K proves (483). Apply it first to the product of the normal and tangential kernels and then to the normal kernel alone; the common factor λ_*^{-2} cancels, yielding (484). $\square$

Corollary 9.47 (Fine collision scales absorb every coarse flag-power tax). *Let $W_n \downarrow 0$ be coarse old-flag scales, let $G_{*,n}$ be occurrence layers as above, and choose the new collision scales r_n by Theorem 7.74. All retained quantities have been placed on fixed W_n -power layers. In particular, for some fixed $C < \infty$,*

$$W_n^{o(1)} \lesssim \frac{P_{*,n}}{K_{i,n} P_{i,n}} \leq W_n^{-C+o(1)}.$$

Therefore

$$\frac{P_{*,n}}{K_{i,n} P_{i,n}} = r_n^{-o(1)}, \quad \frac{K_{i,n} P_{i,n}}{P_{*,n}} = r_n^{o(1)}. \quad (485)$$

The normal constant in (477) consequently satisfies

$$D_{*,n} \lesssim m_{*,n} r_n^{-o(1)}. \quad (486)$$

Hence the relative normal-offset gate (468) is automatic on the coarse/fine scale pairs. At those pairs the tax-cancelling target (484) is equivalent, at subpower precision, to the single collision-preserved conditional tangential estimate

$$\boxed{\int \mathbf{1}_{\{|N_s| \leq r_n\}} \mathbf{1}_{\{|T_s| \leq r_n\}} d\rho_{*,n} d\rho'_{*,n} ds \lesssim r_n^{2-o(1)} \int \mathbf{1}_{\{|N_s| \leq r_n\}} d\rho_{*,n} d\rho'_{*,n} ds.} \quad (487)$$

Thus no fixed power inherited from the old flag generation remains in the final fine-scale numerical ledger. If one coarse W, H, G_* is frozen instead, the same conclusions hold, with constants depending on that coarse layer, uniformly at the exponent level for all sufficiently small r . This fixed-coarse/all-fine form, rather than the diagonal sequence alone, is the one that can feed the Hausdorff Frostman passage.

Proof. The lower bound on the ratio follows, up to a subpower factor, from (472), $\lambda_* \leq 4H_i P_*$, and $m_* \leq 1$. Its upper bound follows from the fixed power dyadic ranges of K_i, P_i, P_* . By (302),

$$\frac{|\log W_n|}{|\log r_n|} \leq \frac{1}{n}.$$

Taking logarithms proves (485). Substitute the first part into (477) to obtain (486). The second part is absorbed into the $r_n^{-o(1)}$ allowance in (484), which gives (487). For a fixed coarse layer, replace the ratio $|\log W_n|/|\log r_n|$ by a fixed numerator divided by $|\log r| \rightarrow \infty$. $\square$

Lemma 9.48 (A q -separated anchor plane is transverse to the coherent polarization). *Let $E_* \subset \mathbb{R}^3$ have unit normal n_* , and write $\mathbf{q}(a) = n_* \cdot a$. If a two-plane E contains a nonzero vector u with*

$$|\mathbf{q}(u)| \geq R_q$$

and u lies in a fixed bounded set, then, for either unit normal n_E of E ,

$$\boxed{|n_E \times n_*| \gtrsim_U R_q.} \quad (488)$$

For the plane E in Theorem 7.40, every retained rank-loss cycle from Theorem 7.39 satisfies this conclusion.

Proof. The norm of the restriction of $a \mapsto n_* \cdot a$ to E is

$$\sup_{\substack{e \in E \\ |e|=1}} |n_* \cdot e| = |n_E \times n_*|.$$

Testing it on $u/|u|$, and using the boundedness of $|u|$, proves (488). In the anchored packet take $u = a_{y_0} - a_p$. This is one of the four retained edges, so (210) gives $|\mathbf{q}(u)| \gtrsim R_q$. $\square$

Proposition 9.49 (The polarized rank packet returns to the transverse Maslov ledger). *Assume $1/2 \leq \chi < 3/5$, choose α as in (209), and put $R_q = W^\alpha$. Choose*

$$\boxed{\alpha < d < 1, \quad 0 < \beta < d - \alpha, \quad \Delta = W^d, \quad \theta_{\text{hr}} = W^\beta.} \quad (489)$$

Then $W/\Delta \rightarrow 0$, as required by the arbitrary-threshold Maslov routing, while every polarized-plane rank packet obtained from the surviving four-cycles has

$$\boxed{t_{\text{dir}} \lesssim W + W^{d-\beta} = o(W^\alpha) = o(R_q), \quad |n_E \times n_*| \gtrsim R_q.} \quad (490)$$

Thus this output cannot remain in the coherent polarization class at angular resolution $o(R_q)$; it is a genuinely transverse Maslov packet.

More precisely, if a reused direction cell is also carried by an affine E_ -packet of thickness w_* , then its actual direction mass is bounded by*

$$\boxed{\sigma(H_{\text{cell}}) \lesssim_U \frac{w_* t_{\text{dir}}}{R_q}.} \quad (491)$$

Consequently the weighted high-reuse summation in Theorem 7.16 holds here with w^2/ϑ replaced by $w_ t_{\text{dir}}/R_q$. The formerly coherent polarized-plane output is therefore returned to the already isolated transverse-polarization ledger with the additional vanishing factor*

$$\boxed{\frac{t_{\text{dir}}}{R_q} \lesssim W^{1-\alpha} + W^{d-\beta-\alpha} = o(1).} \quad (492)$$

Proof. The first inequality in (490) is (215); the strict inequalities in (489) make both powers 1 and $d - \beta$ larger than α . The angle bound is Theorem 9.48.

For completeness, the unequal-width version of Theorem 7.15 follows from the same coordinates: the two normal coordinates range over intervals of lengths $O(w_*)$ and $O(t_{\text{dir}})$, and their Jacobian is $|n_E \times n_*| \gtrsim R_q$. Bounded horizontal density gives (491). The proof of Theorem 7.16 then applies verbatim, and division of (490) by R_q gives (492). $\square$

Corollary 9.50 (The final Hausdorff loss is one residual estimate). *The fixed-angle and angular analyses have the same remaining quantitative endpoint: for every $\eta > 0$, prove*

$$3_\rho^{J^+}(\mu) \lesssim_\eta \rho^{2-\eta} \quad (493)$$

after summing the fixed-angle shells and their Darboux-rescaled angular counterparts. By Theorem 6.27, this estimate closes the full Hausdorff conclusion directly. By Theorem 7.1, every violation of (493) is a quantitatively excessive Maslov collision shell. The preceding stage machinery routes the part above its geometric thresholds to a Lagrangian bush, angular collapse, or a persistent nonplanar Reeb-front seed. The precise remainder is the union of that persistent seed branch and the diffuse exponent-gap window (126); both must be upgraded to the quadratic normal-slice estimate. On the latter window, Theorem 7.7 already forces a divergent staged density in the two symplectic normal coordinates, while Theorem 7.8 sends its collision graph to a shrinking bush or determinant-small cycles at every useful Δ . In the separated-anchor, separated-time part, Theorem 7.12 gives the explicit saving (149). The remaining content is the mass-preserving weighted close-anchor refinement and the final summation of the rank-packet cap to the quadratic normal-slice estimate; the time-packet part is closed by Theorem 7.14, and another scalar fourth-root estimate cannot supply the remaining summation. the preceding flag construction retain this summation as an old-flag/new-cycle occurrence measure. the preceding stage uses the lossless conditional eight-flag lift Theorem 8.3 to close its fully coherent rank-loss part, but Theorem 8.7 shows that the other alternative is a genuine physical point-probe bush. Converting those bushes into scale-coherent transverse measures is precisely Theorem 9.42.

Theorem 9.51 (Planar angular-collapse exclusion). *Suppose a longitudinally saturated angular-collapse limit has all its secant vectors in one three-plane V . Then*

$$\boxed{P_V \equiv 0, \quad \|\omega|_V\|_{\text{op}} = 1, \quad V \in K_{101} \cup K_{110} \cup K_{200}, \quad \text{rank}(\pi_a|_V) \leq 2.} \quad (494)$$

In particular such a planar limit cannot carry a noncollapsed three-dimensional direction blow-up. Hence the rectifiable and quantitatively planar angular-collapse branches of a full-direction selector are excluded.

Proof. By (72), for every $s \in J$ the plane V contains a nonzero vector of L_s . The matrix-pencil/Maslov equivalence (4) gives $P_V(s) = 0$ on J , so the cubic polynomial vanishes identically. The complex-evaluation identity, the preceding classification, and the front-null direction-rank law (15) give the four conclusions in (494). A noncollapsed three-dimensional horizontal blow-up would make $\pi_a|_V$ have rank three, which is impossible. $\square$

Lemma 9.52 (Polarized form of the characteristic collision directions). *Let V be front-null.*

- (i) *In the aligned strata K_{101} and K_{110} , choose the gauge*

$$u = (e_1, 0), \quad Ju = (0, e_1), \quad w = (c, d), \quad c, d \perp e_1.$$

At every parameter at which the pencil has rank two,

$$\boxed{\kappa_V(s) = \mathbb{P} \text{span}\{u - sJu\}.$$

Thus the entire generic Maslov collision curve lies in the unique complex line $\mathcal{C}_V = \mathbb{C}u$; the symplectic-null direction w is invisible to the generic collision pairs.

- (ii) *In K_{110} , the rank is two for every real s , while the cokernel line also moves with projective degree one. Hence this is the only stratum with simultaneous moving right- and left-incidence data.*
- (iii) *In K_{101} , writing $w = (\mu e_2, \nu e_2)$, the third pencil column is $(\nu + s\mu)e_2$. There is at most one finite exceptional parameter $s_* = -\nu/\mu$ at which the rank drops to one; at that parameter the null direction w enters $V \cap L_{s_*}$. This is the only aligned mechanism that can add a third collision direction at one Reeb level.*

(iv) In $\mathcal{K}_{200}$, there is a fixed physical output plane $W_V \subset \mathbb{R}^3$ such that

$$\boxed{\operatorname{im}(B + sA) \subset W_V \quad (s \in \mathbb{R}),}$$

and hence one fixed projective cokernel normal $n_V \in \mathbb{P}((\mathbb{R}^3)^*)$. Its right characteristic curve has degree two.

Proof. In the aligned gauge the pencil is

$$B + sA = [se_1, e_1, d + sc].$$

Whenever it has rank two, its right kernel is generated by $(1, -s, 0)$, which represents $u - sJu$ in V . For $\mathcal{K}_{110}$, the vectors c, d are independent, so $d + sc \neq 0$ and the rank never drops. In $\mathcal{K}_{101}$, substitute $c = \mu e_2, d = \nu e_2$. The exceptional statement follows immediately. The final item is Theorems 4.8 and 4.9 together with the characteristic-degree theorem of the preceding stage. $\square$

Proposition 9.53 (Density-bearing Maslov–Alberti branch closes). *Suppose there is a positive- μ -mass set $E \subset G_b$ on which*

$$0 < \liminf_{r \downarrow 0} \frac{\mu(B(x, r))}{r^3} \leq \limsup_{r \downarrow 0} \frac{\mu(B(x, r))}{r^3} < \infty,$$

and $\mu|_E$ admits three independent Alberti representations. Then E has a positive-mass countably three-rectifiable part. Its horizontal projection has positive Lebesgue measure, and the affine Reeb sweep of that part has positive four-dimensional measure. Hence it cannot occur in a null-front counterexample.

Proof. The density and independent-representation hypotheses give countable three-rectifiability by the rectifiability and Alberti-representation results of Bate and Bate–Li [2, 3]. Because the graph projection sends $\mu|_E$ to Lebesgue measure on a positive-measure direction set, the area formula gives a positive-measure rectifiable subset on which the horizontal tangent map has rank three. On that subset the leading coefficient $\det A$ of the Reeb-front polynomial is nonzero. The rectifiable Reeb-sweep argument of Theorems 4.4 and 4.11 then gives positive four-volume. $\square$

Proof of Theorem 1.2. By Theorem 3.1, pass to one marked line in every direction. The selected line graph still has packing dimension three and its Euclidean front is contained in the original front. The contact–symplectic closure theorem Theorem 9.32 applies to this selector and gives dimension four for its Reeb front. The containing front therefore has Hausdorff dimension four as well. $\square$

10 Discussion

The reduction uses three manifestations of the same contact–symplectic structure. The evaluation map turns Reeb motion into Euclidean line motion. The matrix pencil $B + sA$ measures infinitesimal front rank and identifies total collapse with Maslov trapping. Finally, the rescaled collision equation is the same pencil equation, so the polarization attached to a collision survives the passage from local geometry to the multiscale edge flow.

Two safeguards are important. First, being three-dimensional in line space does not make a rough selector Lagrangian or rectifiable. Second, a high-degree graph vertex is not credited as a Euclidean bush until the contact evaluation produces a common physical point-probe. These are the places where a purely combinatorial reformulation would lose the geometry needed for the Hausdorff endpoint. The compensation region is closed at the level at which it actually occurs: the ordered, absolute edge-flow measure. The same identity supplies the weighted marked finite-scale estimate. What it does not supply is any typed Hall/Radon–Nikodym comparison; no such algebraic conclusion is claimed here.

A Measurable disintegration and passive marks

This appendix collects the measure-theoretic facts used in the main iteration; the disintegration conventions are standard, as in [5]. They are grouped by function rather than by the stage at which they first arose.

Lemma A.1 (Restriction–kernel calculus). *Let $d\Gamma(x, m) = d\lambda(x)dK_x(m)$, where K_x is a probability kernel. If $0 \leq f \leq 1$ and $Q_{x, m}$ is another probability kernel, then*

$$d\tilde{\Gamma}(x, m, n) = f(x, m)d\lambda(x)dK_x(m)dQ_{x, m}(n)$$

has source marginal dominated by λ . For a countable disjoint family of such restrictions, the source Radon–Nikodym densities add pointwise to at most one.

Proof. Integrating first in n removes $Q_{x,m}$. Integrating in m leaves the density $x \mapsto \int f(x, m) dK_x(m)$, which lies in $[0, 1]$. Additivity for disjoint restrictions follows from monotone convergence. $\square$

Lemma A.2 (Countable adaptive cap partition). *Let X be a separable metric space and let $t(x) > 0$ be measurable. There is a countable half-open partition $X = \bigsqcup_j E_j$ and centers x_j such that $E_j \subset B(x_j, Ct(x))$ on a measurable layer where t is comparable to $t(x_j)$. Applying the partition to the source coordinate of an occurrence measure preserves its total mass and every conditional mark.*

Proof. Split into dyadic layers of t . On each layer choose a countable maximal net and order its balls. Subtract all earlier balls from the current one. The resulting half-open cells are disjoint, measurable, and cover the layer. The last assertion is Theorem A.1 with indicator densities. $\square$

Remark A.3. *The affine fibre label and the complete carrier history are represented by a Dirac kernel $x \mapsto \delta_{(\mathfrak{f}(x), \mathfrak{Q}(x))}$. Since its mass is one, it survives every restriction and kernel extension. This is the precise meaning of a passive mark in the proof.*

B Uniform finite-scale parameter hierarchy

Fix a physical thickness δ , a bush exponent $0 < d_0 < 1/3$, and

$$B(\delta) = \exp \sqrt{\log(1/\delta)}.$$

At a node of radius T_v , first prescribe $\tau_v = T_v/(C_* B(\delta))$. Only then choose the available failure scale so that the physical, synchronization, polarization, target-cell, and time-packet errors are bounded by $T_v/(C_*^2 B(\delta))$. This order of choices is important: the fine scale is a local certificate, whereas T_v is the scale carried by the absolute flow.

Before termination, $T_v^3 > \delta^{2-o(1)}$, while the physical bush error is $R_{\text{bush}} = O(\delta^{1-d_0})$. Hence

$$\frac{R_{\text{bush}}}{\tau_v} \lesssim B(\delta) \delta^{1/3-d_0-o(1)} = o(1).$$

Thus every local geometric error fits inside the prescribed child cap without passing below the physical thickness. Since $T_w \leq B(\delta)^{-1} T_v$, after

$$L = O(\sqrt{\log(1/\delta)})$$

generations one has $T_L^3 \leq \delta^{2-o(1)}$. At that frontier,

$$\sum_{\gamma} p_{\gamma} T_{\gamma}^3 \leq \delta^{2-o(1)} \sum_{\gamma} p_{\gamma},$$

and the accumulated degree tail is $L\delta^A = \delta^{A-o(1)}$. Both estimates are uniform in the number of children, their shapes, and every passive carrier mark.

References

- [1] V. I. Arnol'd, *Singularities of Caustics and Wave Fronts*, Mathematics and Its Applications (Soviet Series), vol. 62, Kluwer Academic Publishers, Dordrecht, 1990.
- [2] D. Bate, *Structure of measures in Lipschitz differentiability spaces*, J. Amer. Math. Soc. **28** (2015), no. 2, 421–482.
- [3] D. Bate and S. Li, *Characterizations of rectifiable metric measure spaces*, Ann. Sci. Éc. Norm. Supér. (4) **50** (2017), no. 1, 1–37.
- [4] T. Berger and S. Trenn, *The quasi-Kronecker form for matrix pencils*, SIAM J. Matrix Anal. Appl. **33** (2012), no. 2, 336–368.
- [5] V. I. Bogachev, *Measure Theory*, vols. I–II, Springer, Berlin, 2007.
- [6] J. Bourgain, *Besicovitch type maximal operators and applications to Fourier analysis*, Geom. Funct. Anal. **1** (1991), no. 2, 147–187.
- [7] A. Cannas da Silva, *Lectures on Symplectic Geometry*, Lecture Notes in Mathematics, vol. 1764, Springer, Berlin, 2001.

- [8] H. Federer, *Geometric Measure Theory*, Grundlehren der mathematischen Wissenschaften, vol. 153, Springer-Verlag, New York, 1969.
- [9] H. Geiges, *An Introduction to Contact Topology*, Cambridge Studies in Advanced Mathematics, vol. 109, Cambridge University Press, Cambridge, 2008.
- [10] I. Gohberg, P. Lancaster, and L. Rodman, *Matrix Polynomials*, Classics in Applied Mathematics, vol. 58, Society for Industrial and Applied Mathematics, Philadelphia, 2009.
- [11] L. Guth, H. Wang, and J. Zahl, *A streamlined proof of the Kakeya set conjecture in $\mathbb{R}^3$* , arXiv:2601.14411 (2026).
- [12] N. H. Katz, I. Laba, and T. Tao, *An improved bound on the Minkowski dimension of Besicovitch sets in $\mathbb{R}^3$* , Ann. of Math. (2) **152** (2000), no. 2, 383–446.
- [13] N. H. Katz and T. Tao, *New bounds for Kakeya problems*, J. Anal. Math. **87** (2002), 231–263.
- [14] A. S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics, vol. 156, Springer-Verlag, New York, 1995.
- [15] P. Mattila, *Geometry of Sets and Measures in Euclidean Spaces: Fractals and Rectifiability*, Cambridge Studies in Advanced Mathematics, vol. 44, Cambridge University Press, Cambridge, 1995.
- [16] P. Van Dooren, *The computation of Kronecker’s canonical form of a singular pencil*, Linear Algebra Appl. **27** (1979), 103–140.
- [17] H. Wang and J. Zahl, *Sticky Kakeya sets and the sticky Kakeya conjecture*, J. Amer. Math. Soc. **39** (2026), no. 2, 515–585.
- [18] T. Wolff, *An improved bound for Kakeya type maximal functions*, Rev. Mat. Iberoamericana **11** (1995), no. 3, 651–674.